

Mathematics

A Comprehensive Review | Study Edition

Formulas, definitions, context, and cut-out flashcards
spanning high-school and university mathematics, physics, engineering,
finance, and machine learning.

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How to use this guide

Every idea in this guide is a **concept block** with a unique **concept ID**. The ID is built from the section code and a number, so **C1.7** is the seventh concept of Calculus I. Each block follows the same anatomy, shown at right.

Cross references. When a term or variable depends on another idea, its ID appears beside it in a small box, e.g. A1.11. In the PDF these boxes are clickable links.

Flashcards. The final pages form a grid of 12 cards per sheet: a *front* page (concept name and ID) immediately followed by its *back* page (the formula). Print those pages double-sided at **100% / actual size** and choose **flip on long edge**. The back page is mirrored left-to-right, so each answer lands exactly behind its prompt; cut along the dashed lines.

Section codes. A1 Algebra I • A2 Algebra II • G Geometry • T Trigonometry • C1 Calculus I • C2 Calculus II • LA Linear Algebra • V Vector Calculus • P Physics • S Statistics & Probability • O Optimization & Simulation • ML Machine Learning • EE Electrical Engineering • FN Finance.

ID

Concept title

Formula in standard notation

TERMS

Every symbol defined, with ID links to prerequisite concepts

CONTEXT

Origin, general use, and a data-science use case

Figure (where visual)

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01 Algebra I

The grammar of mathematics: operating on real numbers, manipulating expressions, and modeling with lines and parabolas. Nearly every later concept is built from these moves.

A1.1 Properties of Real Numbers & Order of Operations

$$a + b = b + a \quad ab = ba \quad (a + b) + c = a + (b + c) \quad a(b + c) = ab + ac$$

$$\underbrace{() \rightarrow}_{\text{Parentheses}} \quad \underbrace{x^n \rightarrow}_{\text{Exponents}} \quad \underbrace{\times \div \rightarrow}_{\text{left to right}} \quad \underbrace{+ - \rightarrow}_{\text{left to right}}$$

STATED-EQUATION a plus b equals b plus a ; a b equals b a ; the quantity a plus b , plus c , equals a plus the quantity b plus c ; a times the quantity b plus c equals a b plus a c . Order: parentheses, exponents, multiplication and division left to right, then addition and subtraction left to right.

MEMORY AID PEMDAS: "Please Excuse My Dear Aunt Sally" (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

TERMS

a, b, c – any real numbers

Commutative – order of operands does not matter

Associative – grouping does not matter

Distributive – multiplication spreads over addition

Exponents – repeated multiplication [A1.2](#)

CONTEXT

- **Origin.** The field axioms, formalized in the 19th century by Peacock and Hamilton.
- **Use.** Simplifying any expression unambiguously.
- **Data science.** Floating-point addition is *not* associative, so the order of a large sum (e.g. a GPU reduction) can change the last digits of a result.
- **Physical application.** Adding the weights of groceries in any order gives the same total; the distributive law is how you compute the cost of 3 bags each holding a \$4 and a \$2 item: $3(4 + 2) = 18$.

A1.2 Exponent Rules

$$a^m \cdot a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \quad a^0 = 1 \quad a^{-n} = \frac{1}{a^n}$$

STATED-EQUATION a to the m times a to the n equals a to the m plus n . a to the m over a to the n equals a to the m minus n . a to the m , all to the n , equals a to the m times n . a b to the n equals a to the n b to the n . a over b to the n equals a to the n over b to the n . a to the zero equals one. a to the minus n equals one over a to the n .

MEMORY AID Multiply $\rightarrow$ add exponents; power of a power $\rightarrow$ multiply exponents.

TERMS

a, b – bases; nonzero when the exponent is 0 or negative

m, n – exponents (integers here; fractions in [A1.3](#))

a^n – a multiplied by itself n times

CONTEXT

- **Origin.** Modern superscript notation comes from Descartes, *La Géométrie* (1637).
- **Use.** Scientific notation, compound growth [A2.6](#), simplifying rational expressions.
- **Data science.** Products of many probabilities underflow; taking logs [A2.7](#) turns $\prod p_i$ into $\sum \log p_i$, which is why models maximize the log-likelihood.
- **Physical application.** Scientific notation in physics: $(3 \times 10^8)(2 \times 10^3) = 6 \times 10^{11}$ multiplies by adding exponents.

A1.3 Radicals & Rational Exponents

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m \quad \sqrt{ab} = \sqrt{a} \sqrt{b} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad \frac{c}{\sqrt{a}} = \frac{c\sqrt{a}}{a}$$

STATED-EQUATION a to the m over n equals the n -th root of a to the m , which equals the n -th root of a , raised to the m . The square root of a b equals root a times root b . The square root of a over b equals root a over root b . c over root a equals c root a over a .

MEMORY AID “Power over root”: in $a^{m/n}$ the top is the power, the bottom is the root.

TERMS

n – index (root); $\sqrt{}$ means $n = 2$

a – radicand; $a \geq 0$ when n is even (for real results)

m – power, obeying the exponent rules [A1.2](#)

Rationalize – remove radicals from a denominator

CONTEXT

- **Origin.** The radical sign $\sqrt{}$ first appeared in Christoff Rudolff’s *Coss* (1525).
- **Use.** Undoing powers; geometric lengths [G.3](#).
- **Data science.** Root-mean-square error $\sqrt{\frac{1}{n} \sum e_i^2}$ and the standard deviation [S.2](#) both use a square root to return to the original units.
- **Physical application.** The side of a square garden with area 50 m^2 is $\sqrt{50} = 5\sqrt{2} \approx 7.07 \text{ m}$.

A1.4 Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}}$$

STATED-EQUATION m equals y two minus y one over x two minus x one, which is Δy over Δx : rise over run.

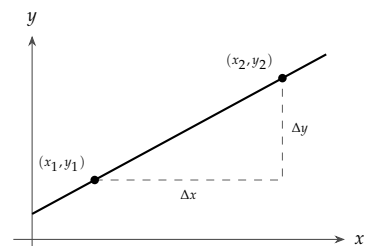
MEMORY AID “Rise over run” — y changes on top, like rising up.

TERMS

$(x_1, y_1), (x_2, y_2)$ — two distinct points on the line
 m — slope: change in y per unit change in x
 Δ — “change in”; m undefined when $\Delta x = 0$

CONTEXT

- **Origin.** Coordinate geometry of Descartes and Fermat (17th c.).
- **Use.** Any constant rate: speed, cost per unit, growth per month.
- **Data science.** A regression coefficient [S.19] is a slope: the expected change in the target per one-unit change in a feature. The derivative [C.14] is a slope in the limit.
- **Physical application.** A road that rises 6 m over 100 m of horizontal distance has a 6% grade, slope 0.06.



A1.5 Equations of a Line

$$\underbrace{y = mx + b}_{\text{slope-intercept}}$$

$$\underbrace{y - y_1 = m(x - x_1)}_{\text{point-slope}}$$

$$\underbrace{Ax + By = C}_{\text{standard}}$$

STATED-EQUATION Slope-intercept form: y equals m x plus b . Point-slope form: y minus y one equals m times x minus x one. Standard form: A x plus B y equals C .

TERMS

m — slope [A1.4]
 b — y -intercept (value of y when $x = 0$)
 (x_1, y_1) — any known point on the line
 A, B, C — constants, usually integers with $A \geq 0$

CONTEXT

- **Origin.** Descartes and Fermat’s analytic geometry.
- **Use.** Modeling anything that changes at a constant rate.
- **Data science.** A linear model $\hat{y} = \beta_0 + \beta_1 x$ is slope-intercept form; a single neuron computes $w \cdot x + b$ before its activation.
- **Physical application.** A taxi fare of \$3 plus \$2 per mile is the line $y = 2x + 3$: intercept is the flag drop, slope is the per-mile rate.

A1.6 Parallel & Perpendicular Lines

$$\text{parallel: } m_1 = m_2 \qquad \text{perpendicular: } m_1 \cdot m_2 = -1 \iff m_2 = -\frac{1}{m_1}$$

STATED-EQUATION Parallel lines: m one equals m two. Perpendicular lines: m one times m two equals minus one, so m two equals the negative reciprocal of m one.

MEMORY AID Perpendicular slopes: “flip it and switch the sign.”

TERMS

m_1, m_2 — slopes of the two lines [A1.4]
 Negative reciprocal — flip the fraction and change the sign
 Exception — a vertical line is perpendicular to a horizontal one

CONTEXT

- **Origin.** Euclid’s parallel postulate; the slope test follows from the Pythagorean theorem [G.3].
- **Use.** Constructing right angles, shortest-distance problems.
- **Data science.** Perpendicular (orthogonal) directions carry non-overlapping information: PCA axes are orthogonal, and OLS residuals are orthogonal to fitted values [S.19]. See the dot product [T.12].
- **Physical application.** Floor tiles laid along parallel lines share one slope; a wall meeting the floor at a right angle has the negative-reciprocal slope in a tilted drawing.

A1.7 Absolute Value Equations & Inequalities

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \quad |x - a| < r \iff a - r < x < a + r \quad |x - a| > r \iff x < a - r \text{ or } x > a + r$$

STATED-EQUATION The absolute value of x equals x when x is at least zero, and minus x when x is negative. The absolute value of x minus a less than r means a minus r less than x less than a plus r . Greater than r means x less than a minus r , or x greater than a plus r .

MEMORY AID “Less than” is a sandwich (and); “greater than” goes two ways (or).

TERMS

$|x|$ – distance from x to 0; always ≥ 0

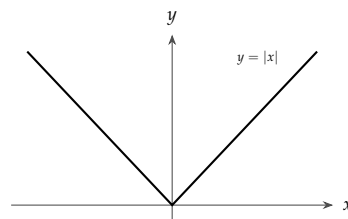
a – center of the interval

r – radius or tolerance ($r > 0$)

Flip rule – “less than” $\rightarrow$ and; “greater than” $\rightarrow$ or

CONTEXT

- **Origin.** Notation $|x|$ introduced by Weierstrass (1841).
- **Use.** Tolerances, error bounds, distances on a number line.
- **Data science.** Mean absolute error and L1 (Lasso) regularization use $|\cdot|$; it penalizes outliers less than squaring does, and its corner at 0 drives weights exactly to zero.
- **Physical application.** A bolt specified as $10 \text{ mm} \pm 0.05 \text{ mm}$ must satisfy $|x - 10| \leq 0.05$.



A1.8 Systems of Linear Equations

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad \text{solve by substitution, elimination, or matrices} \quad \boxed{\text{A2.14}}$$

STATED-EQUATION A system of two linear equations: a one x plus b one y equals c one, and a two x plus b two y equals c two. Solve by substitution, elimination, or matrices.

TERMS

a_i, b_i – coefficients; c_i constants

One solution – lines intersect (different slopes)

No solution – parallel lines A1.6

Infinite – same line (equations are multiples)

CONTEXT

- **Origin.** Elimination appears in the Chinese *Nine Chapters* (c. 200 BCE); formalized by Gauss.
- **Use.** Mixture, pricing, and break-even problems.
- **Data science.** Least-squares regression solves the normal equations $X^T X \beta = X^T y$, a (large) linear system.
- **Physical application.** Mixing a 10% and a 30% saline solution to get 2 L of 25% solution is a two-equation system in the two volumes.

A1.9 Polynomial Products & Special Products

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2 \quad (a + b)(a - b) = a^2 - b^2$$

$$(a + b)(c + d) = ac + ad + bc + bd \quad (\text{FOIL})$$

STATED-EQUATION The quantity a plus b , squared, equals a squared plus two a b plus b squared. a minus b , squared, equals a squared minus two a b plus b squared. a plus b times a minus b equals a squared minus b squared. a plus b times c plus d equals a c plus a d plus b c plus b d .

MEMORY AID FOIL: First, Outer, Inner, Last.

TERMS

Polynomial – sum of terms $c x^k$ with whole-number k

Degree – largest exponent present

Like terms – same variable and power; combine them

Distributive law – the engine of every expansion [A1.1](#)

CONTEXT

- **Origin.** Al-Khwarizmi’s geometric “completing the square” (c. 820).
- **Use.** Expanding, simplifying, and setting up factoring [A1.10](#).
- **Data science.** $\|x - y\|^2 = \|x\|^2 - 2x \cdot y + \|y\|^2$ lets libraries compute all pairwise distances with one matrix product (k-NN, k-means).
- **Physical application.** A square patio of side $x + 1$ m has area $x^2 + 2x + 1$: the big square, two strips, and a corner square.

A1.10 Factoring

$$ab + ac = a(b + c) \quad x^2 + (p + q)x + pq = (x + p)(x + q) \quad a^2 - b^2 = (a - b)(a + b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

STATED-EQUATION a b plus a c equals a times b plus c . x squared plus p plus q x plus p q equals x plus p times x plus q . a squared minus b squared equals a minus b times a plus b . a cubed plus b cubed equals a plus b times a squared minus a b plus b squared. a cubed minus b cubed equals a minus b times a squared plus a b plus b squared.

MEMORY AID Sum/difference of cubes signs: SOAP — Same, Opposite, Always Positive.

TERMS

GCF – greatest common factor, pulled out first

p, q – numbers whose sum is the middle coefficient and product the constant

Zero product – if $AB = 0$ then $A = 0$ or $B = 0$

Reverse of – special products [A1.9](#)

CONTEXT

- **Origin.** The inverse of expansion; systematized in Renaissance algebra.
- **Use.** Solving polynomial equations and simplifying rational expressions [A2.5](#).
- **Data science.** Factoring out common terms cuts computation: Horner’s rule evaluates degree- n polynomials with only n multiplications.
- **Physical application.** Factoring $h(t) = -4.9t^2 + 19.6t = -4.9t(t - 4)$ shows a ball thrown up at 19.6 m/s lands after 4 s.

A1.11 Quadratic Formula & Discriminant

$$ax^2 + bx + c = 0 \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \Delta = b^2 - 4ac$$

STATED-EQUATION a x squared plus b x plus c equals zero implies x equals negative b plus or minus the square root of b squared minus four a c , all over two a . The discriminant, delta, equals b squared minus four a c .

MEMORY AID Sung to “Pop Goes the Weasel”: “ x equals negative b , plus or minus the square root, of b squared minus four a c , all over two a .”

TERMS

a, b, c – coefficients, $a \neq 0$

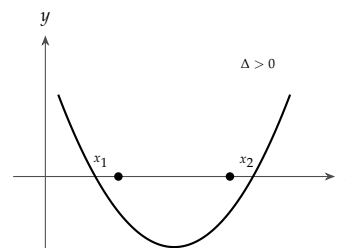
Δ – discriminant: > 0 two real roots; $= 0$ one repeated root; < 0 two complex roots [A2.3](#)

Roots – x -intercepts of the parabola

Derived by – completing the square [A1.12](#)

CONTEXT

- **Origin.** Babylonian tablets (c. 2000 BCE); general rule by Brahmagupta (628 CE).
- **Use.** Projectile motion, area, break-even points.
- **Data science.** Quadratic discriminant analysis draws curved class boundaries; finding where two Gaussian class densities cross means solving a quadratic.
- **Physical application.** When a projectile at height $h(t) = -4.9t^2 + v_0t + h_0$ hits the ground, t comes from the quadratic formula.



A1.12 Vertex Form of a Quadratic

$$y = a(x - h)^2 + k \quad h = -\frac{b}{2a}, \quad k = f(h)$$

STATED-EQUATION y equals a times the quantity x minus h squared, plus k . The vertex x -coordinate h equals negative b over two a , and k equals f of h .

TERMS

(h, k) – vertex: minimum if $a > 0$, maximum if $a < 0$

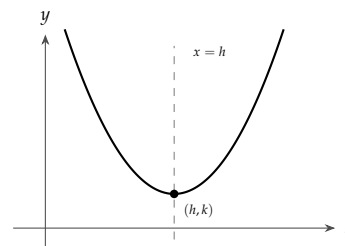
$x = h$ – axis of symmetry

a – stretch and direction, same a as in [A1.11](#)

Completing the square – $x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$

CONTEXT

- **Origin.** Completing the square, Al-Khwarizmi (c. 820).
- **Use.** Finding a maximum height, minimum cost, or best price without calculus.
- **Data science.** Mean squared error in one parameter is a parabola; its vertex is the sample mean. Extrema in general: [C1.12](#).
- **Physical application.** The peak height of a thrown ball is the vertex of its parabolic path.



A1.13 Direct & Inverse Variation

$$\underbrace{y = kx}_{\text{direct}} \qquad \underbrace{y = \frac{k}{x}}_{\text{inverse}} \qquad \underbrace{y = kxz}_{\text{joint}}$$

STATED-EQUATION Direct variation: y equals k x . Inverse variation: y equals k over x . Joint variation: y equals k x z .

TERMS

k – constant of variation, $k \neq 0$

Direct – doubling x doubles y (a line through the origin [A1.5])

Inverse – doubling x halves y (a hyperbola)

CONTEXT

- **Origin.** Proportion theory in Euclid's *Elements*, Book V.
- **Use.** Ohm's law, unit pricing, speed and travel time.
- **Data science.** Ideal parallel speed-up: runtime $\approx k/n$ for n workers. The standard error of a mean varies inversely with $\sqrt{n}$ [S.13].
- **Physical application.** At fixed temperature, gas pressure varies inversely with volume (Boyle's law, $P = k/V$); Hooke's law $F = kx$ is direct variation.

A1.14 Ratios, Proportions & Percent Change

$$\text{percent change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\% \qquad \frac{a}{b} = \frac{c}{d} \iff ad = bc$$

STATED-EQUATION Percent change equals new minus old, over old, times one hundred percent. a over b equals c over d if and only if a d equals b c .

MEMORY AID Percent change: "difference over original."

TERMS

new, old – final and reference values (old $\neq 0$)

$a:b$ – ratio; $\frac{a}{b} = \frac{c}{d}$ is a proportion

Cross-multiplication – $ad = bc$

Asymmetry – $+p\%$ then $-p\%$ does not return to the start

CONTEXT

- **Origin.** Latin *per centum*; Roman taxes were levied in hundredths.
- **Use.** Discounts, interest, scaling recipes and maps.
- **Data science.** "Lift" in an A/B test is a percent change in conversion rate; relative error and month-over-month growth metrics use the same formula.
- **Physical application.** Scaling a recipe from 4 servings to 6: $\frac{2 \text{ cups}}{4} = \frac{x}{6}$, so $x = 3$ cups.

A1.15 Linear Inequalities

$$a < b \implies a + c < b + c \qquad a < b, c > 0 \implies ac < bc \qquad a < b, c < 0 \implies ac > bc$$

STATED-EQUATION If a is less than b , then a plus c is less than b plus c . If a is less than b and c is positive, a c is less than b c . If c is negative, the inequality flips: a c is greater than b c .

MEMORY AID "Multiply or divide by a negative, flip the sign."

TERMS

a, b, c – real numbers

Flip rule – multiplying or dividing by a negative reverses the sign

Compound – $p < x < q$ means both $x > p$ and $x < q$

Related – absolute value inequalities [A1.7]

CONTEXT

- **Origin.** Thomas Harriot introduced the symbols $<$ and $>$ (1631).
- **Use.** Budgets, capacity limits, feasible ranges.
- **Data science.** Linear programming and SVMs are defined by linear inequality constraints, e.g. the margin condition $y_i(w \cdot x_i + b) \geq 1$.
- **Physical application.** An elevator rated for 1000 kg with a 75 kg operator carries passengers of 80 kg each: $75 + 80n \leq 1000$, so $n \leq 11$.

A1.16 Sets, Set Operations & Interval Notation

$$A \cup B \text{ (or)} \quad A \cap B \text{ (and)} \quad A^c \text{ (not)} \quad A \setminus B \quad |A \cup B| = |A| + |B| - |A \cap B| \quad (A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c \quad [a, b) = \{x : a \leq x < b\}$$

STATED-EQUATION *A union B means elements in A or B. A intersect B means elements in both. A complement means elements not in A. A minus B means elements in A but not in B. The size of A union B equals the size of A plus the size of B minus the size of A intersect B. De Morgan's laws: the complement of a union is the intersection of the complements, and vice versa. The interval bracket a comma b parenthesis is the set of x with a less than or equal to x, less than b.*

MEMORY AID Square bracket = "included" (closed); parenthesis = "not included" (open). De Morgan: "break the bar, change the sign."

TERMS

$\in, \subseteq$ – "is an element of", "is a subset of"

$\emptyset$ – empty set

$|A|$ – cardinality (number of elements)

Venn diagram – circles for sets; overlaps for intersections

Number sets – $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

Inclusion-exclusion – the same idea as $P(A \cup B)$ [§5](#)

CONTEXT

- **Origin.** Georg Cantor (1874); De Morgan (1847); Venn diagrams by John Venn (1880).
- **Use.** The language of all mathematics: domains, solution sets, events.
- **Data science.** SQL UNION, INTERSECT, EXCEPT and joins are set operations; Jaccard similarity $|A \cap B| / |A \cup B|$ compares sets of items.
- **Physical application.** Sorting laboratory samples into "acidic" and "contains metal" categories, with an overlap region for both.

02 Algebra II

Functions as objects: composing, inverting, and transforming them; exponential and logarithmic models; sequences, series, and the matrix language that underlies all of modern data work.

A2.1 Functions: Composition & Inverses

$$(f \circ g)(x) = f(g(x)) \qquad f^{-1}(f(x)) = f(f^{-1}(x)) = x$$

STATED-EQUATION f composed with g of x equals f of g of x . f inverse of f of x equals f of f inverse of x , which equals x .

MEMORY AID Composition works inside-out: apply g first, then f .

TERMS

Domain – set of allowed inputs; *range*: set of outputs

$f \circ g$ – apply g first, then f

f^{-1} – inverse: swap x and y , solve for y

One-to-one – passes the horizontal line test; required for an inverse

CONTEXT

- **Origin.** Leibniz coined “function” (1673); Euler introduced $f(x)$ notation (1734).
- **Use.** Chaining processes, unit conversions, undoing operations.
- **Data science.** A trained pipeline is a composition (scaler, then encoder, then model), and a deep network is a composition of layers. Predicting on a log target requires the inverse, e^y [A2.7].
- **Physical application.** Converting Celsius to Fahrenheit, $F = \frac{9}{5}C + 32$, and its inverse $C = \frac{5}{9}(F - 32)$ undo each other.

A2.2 Transformations of Functions

$$y = a f(b(x - h)) + k$$

STATED-EQUATION y equals a times f of b times x minus h , plus k .

MEMORY AID Inside the parentheses acts on x and does the opposite (minus h shifts right); outside acts on y and does what it says.

TERMS

h – horizontal shift (right if $h > 0$)

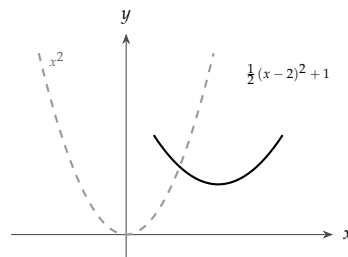
k – vertical shift (up if $k > 0$)

a – vertical stretch by $|a|$; reflect over x -axis if $a < 0$

b – horizontal compression by $1/|b|$; reflect over y -axis if $b < 0$

CONTEXT

- **Origin.** Systematized with graphing of function families in the 19th century.
- **Use.** Sketching any member of a family from one parent graph.
- **Data science.** Standardization $\frac{x-\mu}{\sigma}$ [6.4] is a shift and a scale; location-scale distribution families (normal, logistic) are one parent density transformed.
- **Physical application.** Shifting a sound wave in time by h delays it; multiplying by a makes it louder.



A2.3 Complex Numbers

$$i^2 = -1 \quad z = a + bi \quad \bar{z} = a - bi \quad |z| = \sqrt{a^2 + b^2}$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i \quad \frac{z}{w} = \frac{z\bar{w}}{w\bar{w}}$$

STATED-EQUATION i squared equals minus one. z equals a plus b i ; its conjugate, $\bar{z}$, equals a minus b i ; its modulus equals the square root of a squared plus b squared. a plus b i times c plus d i equals a c minus b d , plus a d plus b c times i . z over w equals z $\bar{w}$ over w $\bar{w}$.

TERMS

a – real part, $\text{Re}(z)$; b : imaginary part, $\text{Im}(z)$

$\bar{z}$ – complex conjugate

$|z|$ – modulus: distance from 0 [G.3]

Polar form – $z = re^{i\theta}$ [T.13]

CONTEXT

- **Origin.** Cardano (1545) and Bombelli (1572), while solving cubics; Euler named i .
- **Use.** Roots of polynomials with $\Delta < 0$ [A1.11]; AC circuits.
- **Data science.** The Fast Fourier Transform decomposes a time series into complex exponentials to expose seasonality; non-symmetric matrices can have complex eigenvalues.
- **Physical application.** AC circuits use complex impedance $Z = R + iX$; the modulus $|Z|$ sets the current amplitude.

A2.4 Remainder, Factor & Rational Root Theorems

$$P(x) = (x - c)Q(x) + P(c) \quad P(c) = 0 \iff (x - c) \text{ is a factor} \quad x = \pm \frac{p}{q}, \quad p \mid a_0, \quad q \mid a_n$$

STATED-EQUATION P of x equals x minus c times Q of x , plus P of c . P of c equals zero if and only if x minus c is a factor. Possible rational roots are plus or minus p over q , where p divides a naught and q divides a n .

TERMS

$P(x)$ – polynomial $a_n x^n + \dots + a_1 x + a_0$

$Q(x)$ – quotient after dividing by $x - c$

a_n, a_0 – leading coefficient and constant term

$p \mid a_0$ – p divides a_0 evenly

FTA – a degree- n polynomial has n complex roots [A2.3]

CONTEXT

- **Origin.** Descartes (1637) for the factor theorem; Gauss proved the Fundamental Theorem of Algebra (1799).
- **Use.** Finding exact roots of higher-degree polynomials.
- **Data science.** Eigenvalues are roots of the characteristic polynomial $\det(A - \lambda I) = 0$ [A2.14], which drive PCA and stability analysis.
- **Physical application.** Checking whether a design polynomial for a beam's deflection vanishes at a support point is a remainder-theorem evaluation.

A2.5 Rational Functions & Asymptotes

$$f(x) = \frac{P(x)}{Q(x)} \quad \text{VA: } x = c \text{ where } Q(c) = 0, P(c) \neq 0 \quad \text{end behavior: } \begin{cases} \text{HA } y = 0 & \deg P < \deg Q \\ \text{HA } y = \frac{a_n}{b_n} & \deg P = \deg Q \\ \text{slant (oblique) asymptote} & \deg P = \deg Q + 1 \end{cases}$$

STATED-EQUATION f of x equals P of x over Q of x . A vertical asymptote at x equals c where Q of c is zero and P of c is not. If the degree of P is less than the degree of Q , the horizontal asymptote is y equals zero; if equal, y equals the ratio of leading coefficients; if one more, a slant asymptote.

MEMORY AID Degrees: “Bigger on Bottom, $y = 0$; Exactly Even, divide the leads; Bigger on Top by one, slant.”

TERMS

VA – vertical asymptote at a zero of Q that does not cancel

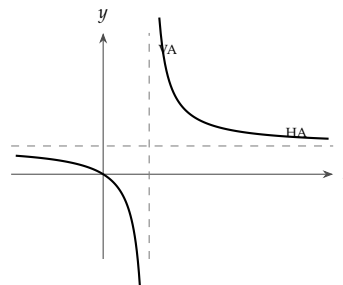
Hole – a factor common to P and Q cancels

HA – horizontal asymptote: end behavior (limits at infinity) [C1.1](#)

a_n, b_n – leading coefficients of P and Q

CONTEXT

- **Origin.** Ratios of polynomials studied since the 17th century; asymptote from Greek *asymptotos*, “not falling together”.
- **Use.** Average cost, concentration, rates.
- **Data science.** Learning curves and saturation (Michaelis–Menten) models $\frac{ax}{b+x}$ level off at an asymptote; average cost per prediction $\frac{C_0 + cn}{n} \rightarrow c$.
- **Physical application.** The average cost of producing x items, $\frac{500 + 2x}{x}$, levels off at the horizontal asymptote \$2 per item.



A2.6 Exponential Growth & Decay

$$A = P \left(1 + \frac{r}{n}\right)^{nt} \quad A = Pe^{rt} \quad y = y_0 e^{kt} \quad t_{1/2} = \frac{\ln 2}{|k|}$$

STATED-EQUATION A equals P times the quantity one plus r over n , raised to the n t . A equals P e to the r t . y equals y naught e to the k t . The half-life equals natural log of two over the absolute value of k .

MEMORY AID “Pert”: continuous compounding is $A = Pe^{rt}$.

TERMS

P – principal (initial amount); A : amount after time t

r – annual rate as a decimal; n : compounding periods per year

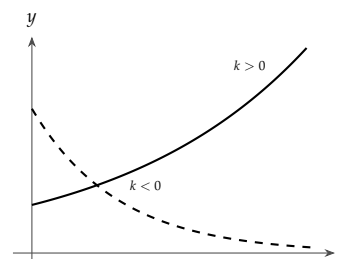
$e \approx 2.71828 = \lim(1 + \frac{1}{n})^n$ [C1.2](#)

k – growth ($k > 0$) or decay ($k < 0$) constant

$t_{1/2}$ – half-life (or doubling time for growth) [A2.8](#)

CONTEXT

- **Origin.** Jacob Bernoulli discovered e studying continuous compound interest (1683).
- **Use.** Interest, population growth, radioactive decay, cooling.
- **Data science.** Learning-rate decay, exponentially weighted moving averages, early epidemic curves, and user-retention decay all follow e^{kt} .
- **Physical application.** Carbon-14 has a half-life of about 5730 years, so $k = \ln 2 / 5730$ dates ancient wood.



A2.7 Logarithms: Definition & Properties

$$\log_b x = y \iff b^y = x$$

$$\log_b(xy) = \log_b x + \log_b y \quad \log_b \frac{x}{y} = \log_b x - \log_b y \quad \log_b x^p = p \log_b x \quad \log_b 1 = 0$$

STATED-EQUATION Log base b of x equals y if and only if b to the y equals x . Log of x y equals log x plus log y . Log of x over y equals log x minus log y . Log of x to the p equals p log x . Log base b of one equals zero.

MEMORY AID “A log is an exponent”: $\log_b x$ asks “ b to what power gives x ?”

TERMS

b – base, $b > 0$, $b \neq 1$

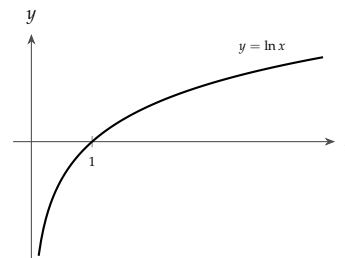
x – argument, $x > 0$

$\ln$ – natural log, base e [A2.6]; log alone often means base 10

Inverse of – the exponential, b^x [A2.1]

CONTEXT

- **Origin.** John Napier (1614) to speed astronomical multiplication; Henry Briggs produced base-10 tables.
- **Use.** pH, decibels, earthquake magnitudes, solving for exponents.
- **Data science.** Log-likelihood and log-loss (cross-entropy); log-transforming right-skewed features (income, counts); entropy in bits uses $\log_2$.
- **Physical application.** The decibel and Richter scales are logarithmic: each +10 dB is 10 times the sound intensity.



A2.8 Change of Base & Exponential Equations

$$\log_b x = \frac{\ln x}{\ln b} = \frac{\log_{10} x}{\log_{10} b} \quad b^x = c \implies x = \frac{\ln c}{\ln b}$$

STATED-EQUATION Log base b of x equals natural log of x over natural log of b , which equals log base ten of x over log base ten of b . If b to the x equals c , then x equals natural log of c over natural log of b .

TERMS

b – original base; $\ln$ is any convenient base [A2.7]

Strategy – isolate the exponential, take logs of both sides

Doubling time – $t = \frac{\ln 2}{\ln(1+r)}$

CONTEXT

- **Origin.** A direct consequence of the power rule in [A2.7].
- **Use.** Solving for time in growth problems; evaluating any base on a calculator.
- **Data science.** Converting nats to bits (divide by $\ln 2$); the depth of a balanced tree or binary search on n items is $\log_2 n$.
- **Physical application.** Solving $2^t = 1000$ for bacterial doublings: $t = \ln 1000 / \ln 2 \approx 9.97$ doublings.

A2.9 Arithmetic Sequences & Series

$$a_n = a_1 + (n - 1)d \quad S_n = \frac{n(a_1 + a_n)}{2} = \frac{n}{2}[2a_1 + (n - 1)d]$$

STATED-EQUATION *a sub n equals a one plus n minus one times d. The sum S n equals n times a one plus a n, over two, which equals n over two times the quantity two a one plus n minus one d.*

MEMORY AID Arithmetic sum: “average of first and last, times the number of terms” (Gauss’s trick).

TERMS

a_1 – first term; a_n : n th term

d – common difference

S_n – sum of the first n terms (in Σ form [A2.11](#))

Graph – terms lie on a line of slope d [A1.4](#)

CONTEXT

- **Origin.** The (possibly apocryphal) story of young Gauss summing 1 to 100 by pairing terms.
- **Use.** Linear schedules, stacked seating, salary steps.
- **Data science.** $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ is why comparing every pair of n records costs $O(n^2)$; linear learning-rate warm-up is arithmetic.
- **Physical application.** Theater seating with 20 seats in row 1 and 2 more per row: row n has $20 + 2(n - 1)$ seats.

A2.10 Geometric Sequences & Series

$$a_n = a_1 r^{n-1} \quad S_n = \frac{a_1(1 - r^n)}{1 - r} \quad S_\infty = \sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1 - r}, \quad |r| < 1$$

STATED-EQUATION *a sub n equals a one times r to the n minus one. S n equals a one times one minus r to the n, over one minus r. The infinite sum equals a one over one minus r, when the absolute value of r is less than one.*

TERMS

a_1 – first term

r – common ratio (each term divided by the previous)

S_∞ – infinite sum; converges only if $|r| < 1$ [C2.10](#)

Relation – exponential growth in discrete steps [A2.6](#)

CONTEXT

- **Origin.** Zeno’s paradoxes; sum formula in Euclid’s *Elements*, Book IX.
- **Use.** Loans, annuities, bouncing balls, present value.
- **Data science.** Discounted return in reinforcement learning $\sum_t \gamma^t r_t$; with constant reward r it equals $\frac{r}{1-\gamma}$. EMA weights form a geometric series summing to 1.
- **Physical application.** A ball rebounding to 60% of its previous height travels a finite total distance, a geometric series.

A2.11 Sigma Notation & Summation Formulas

$$\sum_{i=1}^n c = cn \quad \sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$\sum_{i=1}^n (c a_i + b_i) = c \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

STATED-EQUATION *The sum from i equals one to n of c equals c n. The sum of i equals n times n plus one over two. The sum of i squared equals n times n plus one times two n plus one, over six. The sum of i cubed equals the square of n times n plus one over two. Sums split and constants factor out.*

TERMS

i – index of summation; n : upper limit

c – constant (does not depend on i)

Linearity – sums split and constants factor out

Series types – arithmetic [A2.9](#), geometric [A2.10](#)

CONTEXT

- **Origin.** Euler introduced Σ for sums (1755).
- **Use.** Compactly expressing totals and averages.
- **Data science.** Every mean, loss, and gradient is a sum; linearity of sums becomes linearity of expectation [S.9](#); integrals are limits of sums [C1.17](#).
- **Physical application.** Stacking cannonballs in a square pyramid of n layers uses $\sum i^2$ balls.

A2.12 Binomial Theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

STATED-EQUATION a plus b to the n equals the sum from k equals zero to n of n choose k , a to the n minus k , b to the k . n choose k equals n factorial over k factorial times n minus k factorial.

MEMORY AID Coefficients come from Pascal's triangle: 1; 1 1; 1 2 1; 1 3 3 1; ...

TERMS

$\binom{n}{k}$ – binomial coefficient, “ n choose k ” [S.8]

$n!$ – factorial $n(n-1) \cdots 1$, with $0! = 1$

Pascal's triangle – each entry is the sum of the two above it

Approximation – $(1+x)^n \approx 1+nx$ for small x

CONTEXT

- **Origin.** Known to al-Karaji and Yang Hui; Pascal's *Traité* (1654); Newton extended it to any exponent.
- **Use.** Expanding powers, probability, approximations.
- **Data science.** Each term $\binom{n}{k} p^k q^{n-k}$ of $(p+q)^n$ is a binomial probability [S.10], the model for conversions out of n trials.
- **Physical application.** Engineers approximate $(1+x)^n \approx 1+nx$ for small x , e.g., thermal expansion of a cube: $(1+\alpha\Delta T)^3 \approx 1+3\alpha\Delta T$.

A2.13 Matrix Operations

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix} \quad (AB)^T = B^T A^T$$

STATED-EQUATION The i j entry of A B equals the sum over k of a i k times b k j . For two-by-two matrices: first row is a e plus b g , a f plus b h ; second row is c e plus d g , c f plus d h . A B transpose equals B transpose A transpose.

MEMORY AID “Row times column”: slide across the row of A and down the column of B .

TERMS

$m \times n$ – m rows, n columns

Compatibility – $(m \times n)(n \times p) = (m \times p)$

a_{ij} – entry in row i , column j

A^T – transpose: rows become columns

Non-commutative – $AB \neq BA$ in general; I is the identity

CONTEXT

- **Origin.** Arthur Cayley's matrix algebra (1858); Sylvester coined “matrix”.
- **Use.** Solving systems [A1.8], geometric transformations [G.13].
- **Data science.** A dataset is an $n \times p$ matrix; a dense neural layer is $XW + b$; embedding lookups and attention are matrix products.
- **Physical application.** Rotating and then scaling a robot arm's coordinates is a product of two transformation matrices.

A2.14 Determinants, Inverses & Cramer's Rule

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det A = ad - bc \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad x = \frac{\det A_x}{\det A}, \quad y = \frac{\det A_y}{\det A}$$

STATED-EQUATION For A with entries a b c d , the determinant equals a d minus b c . A inverse equals one over a d minus b c times the matrix d , minus b , minus c , a . Cramer's rule: x equals $\det A_x$ over $\det A$, y equals $\det A_y$ over $\det A$.

MEMORY AID Two-by-two inverse: “swap the diagonal, negate the off-diagonal, divide by the determinant.”

TERMS

$\det A$ – determinant: signed area scale factor

Singular – $\det A = 0$: no inverse, system has no unique solution [A1.8]

A_x, A_y – A with the x (or y) column replaced by the constants

$AA^{-1} = I$ – identity matrix [A2.13]

CONTEXT

- **Origin.** Leibniz (1693) and Gabriel Cramer (1750).
- **Use.** Solving small systems, testing invertibility, computing areas.
- **Data science.** A near-zero $\det(X^T X)$ signals multicollinearity; $\det \Sigma$ appears in the multivariate normal density; normalizing flows use Jacobian determinants.
- **Physical application.** The determinant is the area scale factor: a transformation with $\det = 2$ doubles the area of any shape drawn on it.

A2.15 Conic Sections

$$\begin{array}{cc}
 \underbrace{(x-h)^2 + (y-k)^2 = r^2}_{\text{circle}} & \underbrace{(x-h)^2 = 4p(y-k)}_{\text{parabola}} \\
 \underbrace{\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1}_{\text{ellipse: } c^2 = a^2 - b^2} & \underbrace{\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1}_{\text{hyperbola: } c^2 = a^2 + b^2}
 \end{array}$$

STATED-EQUATION Circle: x minus h squared plus y minus k squared equals r squared. Parabola: x minus h squared equals four p times y minus k . Ellipse: x minus h squared over a squared plus y minus k squared over b squared equals one, with c squared equals a squared minus b squared. Hyperbola: the same with a minus sign, and c squared equals a squared plus b squared.

MEMORY AID Ellipse subtracts ($c^2 = a^2 - b^2$), hyperbola adds ($c^2 = a^2 + b^2$).

TERMS

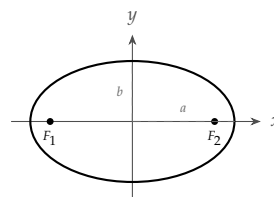
(h, k) – center (or vertex of the parabola)

r – radius; a, b : semi-axes

c – center-to-focus distance

p – vertex-to-focus distance of a parabola

Completing the square – converts general form to standard [A1.12](#)



CONTEXT

- **Origin.** Apollonius of Perga (c. 200 BCE) named them as slices of a cone; Kepler showed planetary orbits are ellipses.
- **Use.** Orbits, reflectors, antennas, lenses.
- **Data science.** Contours of a bivariate normal density are ellipses (constant Mahalanobis distance); the L2 constraint in ridge regression is a circle, L1 a diamond.
- **Physical application.** Satellite dishes are parabolic so incoming signals reflect to the focus; planets orbit in ellipses with the Sun at one focus.

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03 Geometry

Shape, size, and space: angles, triangles, circles, solids, and coordinate transformations. Geometry supplies the intuition behind distance, similarity, and dimensionality in data.

G.1 Angle Relationships

$$\underbrace{\alpha + \beta = 90^\circ}_{\text{complementary}} \quad \underbrace{\alpha + \beta = 180^\circ}_{\text{supplementary}} \quad \underbrace{\angle 1 = \angle 3}_{\text{vertical}} \quad \underbrace{\angle_{\text{corr}} = \angle_{\text{corr}}, \angle_{\text{alt int}} = \angle_{\text{alt int}}}_{\text{parallel lines, transversal}}$$

STATED-EQUATION Complementary angles: alpha plus beta equals ninety degrees. Supplementary: alpha plus beta equals one hundred eighty degrees. Vertical angles are equal. With parallel lines cut by a transversal, corresponding angles are equal and alternate interior angles are equal.

MEMORY AID Complementary makes a Corner (90°); Supplementary makes a Straight line (180°).

TERMS

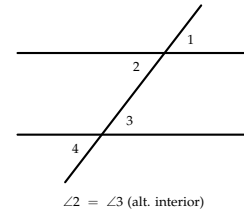
Vertical angles – opposite angles at an intersection

Transversal – a line crossing two parallel lines [A1.6](#)

Corresponding – same position at each intersection; equal

Alternate interior – opposite sides, between the parallels; equal

Same-side interior – supplementary (sum 180°)



CONTEXT

- **Origin.** Euclid's *Elements*, Book I (c. 300 BCE).
- **Use.** Surveying, construction, proofs.
- **Data science.** The angle between two feature vectors underlies cosine similarity [T.12](#); angle 0° means identical direction, 90° means unrelated.
- **Physical application.** A ramp at 20° to the ground meets a vertical wall at 70° (complementary angles).

G.2 Triangle Angle Sum & Triangle Inequality

$$A + B + C = 180^\circ \quad \angle_{\text{ext}} = \angle_{\text{remote}_1} + \angle_{\text{remote}_2} \quad |a - b| < c < a + b$$

STATED-EQUATION A plus B plus C equals one hundred eighty degrees. An exterior angle equals the sum of the two remote interior angles. The absolute value of a minus b is less than c, which is less than a plus b.

TERMS

A, B, C – interior angles

a, b, c – side lengths (a opposite A , etc.)

Exterior angle – formed by extending one side

Triangle inequality – any side is shorter than the other two combined

CONTEXT

- **Origin.** Euclid, *Elements* I.20 and I.32.
- **Use.** Checking whether three lengths can form a triangle; finding missing angles.
- **Data science.** The triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$ defines a metric; ball trees and other nearest-neighbor indexes use it to prune whole regions of a search.
- **Physical application.** Truss bridges use triangles because three fixed side lengths fix the angles: the shape cannot deform.

G.3 Pythagorean Theorem

$$a^2 + b^2 = c^2 \quad \text{converse: } c^2 < a^2 + b^2 \Rightarrow \text{acute}, \quad c^2 > a^2 + b^2 \Rightarrow \text{obtuse}$$

STATED-EQUATION a squared plus b squared equals c squared. If c squared is less than a squared plus b squared, the triangle is acute; if greater, obtuse.

MEMORY AID Common triples: 3–4–5, 5–12–13, 8–15–17, 7–24–25.

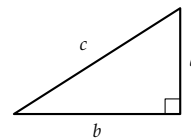
TERMS

a, b – legs of a right triangle

c – hypotenuse (side opposite the right angle; the longest side)

Converse – classifies any triangle by its longest side c

Triples – integer solutions: (3, 4, 5), (5, 12, 13), (8, 15, 17)



CONTEXT

- **Origin.** Named for Pythagoras (6th c. BCE); Babylonian tablet Plimpton 322 lists triples a millennium earlier.
- **Use.** Distances, diagonals, construction squareness checks.
- **Data science.** Euclidean (L2) distance $\sqrt{\sum (x_i - y_i)^2}$ [G.6] is the theorem in n dimensions, the default metric in k-NN and k-means.
- **Physical application.** A 3–4–5 rope loop lets builders lay out a perfect right-angle corner without a square.

G.4 Special Right Triangles

$$45^\circ\text{--}45^\circ\text{--}90^\circ : x : x : x\sqrt{2} \quad 30^\circ\text{--}60^\circ\text{--}90^\circ : x : x\sqrt{3} : 2x$$

STATED-EQUATION A forty-five forty-five ninety triangle has sides x, x, x root two. A thirty sixty ninety triangle has sides x, x root three, two x .

MEMORY AID 30–60–90: “1, root 3, 2” — the short leg is half the hypotenuse.

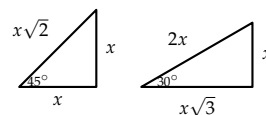
TERMS

x – length of the shortest side (or each leg)

$x\sqrt{2}$ – hypotenuse of the isosceles right triangle

$x\sqrt{3}$ – side opposite 60°

$2x$ – hypotenuse of the 30–60–90



CONTEXT

- **Origin.** Half of a square and half of an equilateral triangle, via [G.3].
- **Use.** Exact trigonometric values on the unit circle [T.3] without a calculator.
- **Data science.** A 45° rotation [G.13] of standardized features maps (x, y) to sums and differences scaled by $\frac{\sqrt{2}}{2}$, common in rotated-coordinate tricks.
- **Physical application.** A square tile’s diagonal is side $\times \sqrt{2}$; an equilateral roof truss has height $\frac{\sqrt{3}}{2}$ of its side.

G.5 Similar Triangles & Scale Factor

$$\triangle ABC \sim \triangle DEF \implies \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k \quad \frac{\text{Area}_1}{\text{Area}_2} = k^2 \quad \frac{\text{Vol}_1}{\text{Vol}_2} = k^3$$

STATED-EQUATION If triangle $A B C$ is similar to triangle $D E F$, then $A B$ over $D E$ equals $B C$ over $E F$ equals $A C$ over $D F$ equals k . The area ratio is k squared, and the volume ratio is k cubed.

MEMORY AID Length k , area k^2 , volume k^3 .

TERMS

$\sim$ – “is similar to”: same shape, possibly different size

k – scale factor between corresponding lengths

Tests – AA, SAS, SSS similarity

Proportions – solved by cross-multiplication [A1.14](#)

CONTEXT

- **Origin.** Thales measured the height of the Great Pyramid from its shadow (c. 600 BCE).
- **Use.** Maps, models, indirect measurement.
- **Data science.** Scaling laws: doubling image resolution multiplies pixel count (and convolution compute) by $k^2 = 4$; doubling a 3-D grid multiplies voxels by 8.
- **Physical application.** Doubling every dimension of a cup multiplies its volume (and mass of drink) by 8 but its surface by only 4.

G.6 Distance & Midpoint Formulas

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

STATED-EQUATION d equals the square root of x two minus x one squared plus y two minus y one squared. The midpoint M equals x one plus x two over two, y one plus y two over two. In n dimensions, d equals the square root of the sum of p i minus q i squared.

TERMS

$(x_1, y_1), (x_2, y_2)$ – two points

d – straight-line distance, from [G.3](#)

M – midpoint: average of coordinates

$\mathbf{p}, \mathbf{q}$ – points in n dimensions (feature vectors)

CONTEXT

- **Origin.** Descartes’ coordinates applied to the Pythagorean theorem.
- **Use.** Navigation, mapping, geometry proofs.
- **Data science.** k-NN and k-means measure similarity by Euclidean distance; the k-means centroid generalizes the midpoint (the mean of the points in a cluster).
- **Physical application.** The straight-line distance between two GPS points on a local map is the distance formula.

G.7 Polygon Angle Sums & Diagonals

$$\begin{aligned} \text{interior sum} &= (n - 2) \cdot 180^\circ & \text{each (regular)} &= \frac{(n - 2) \cdot 180^\circ}{n} \\ \text{exterior sum} &= 360^\circ & \text{diagonals} &= \frac{n(n - 3)}{2} \end{aligned}$$

STATED-EQUATION The interior angle sum equals n minus two times one hundred eighty degrees. Each interior angle of a regular polygon equals n minus two times one eighty, over n . The exterior angles sum to three hundred sixty degrees. The number of diagonals equals n times n minus three, over two.

TERMS

n – number of sides (and vertices)

Regular – all sides and angles equal

Derivation – split into $n - 2$ triangles from one vertex [G.2](#)

Exterior angle – supplement of the interior angle [G.1](#)

CONTEXT

- **Origin.** Triangulation arguments in Greek geometry.
- **Use.** Tiling, architecture, mesh design.
- **Data science.** The diagonal count is $\binom{n}{2} - n$ [S.8](#); all pairs minus the edges. The same counting sizes a complete graph of n nodes in network analysis.
- **Physical application.** Honeycomb cells are regular hexagons with 120° angles that tile the plane with no gaps.

G.8 Area of Plane Figures

$$\underbrace{lw}_{\text{rectangle}}$$

$$\underbrace{bh}_{\text{parallelogram}}$$

$$\underbrace{\frac{1}{2}bh}_{\text{triangle}}$$

$$\underbrace{\frac{1}{2}(b_1 + b_2)h}_{\text{trapezoid}}$$

$$\underbrace{\sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}}_{\text{Heron}}$$

STATED-EQUATION Rectangle: lw . Parallelogram: bh . Triangle: one half bh . Trapezoid: one half times b one plus b two, times h . Heron's formula: the square root of s times s minus a times s minus b times s minus c , where s equals a plus b plus c over two.

TERMS

b, h – base and perpendicular height

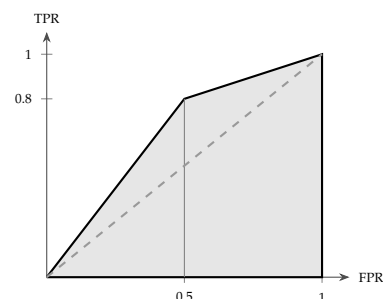
b_1, b_2 – parallel sides of a trapezoid

s – semi-perimeter

SAS area – $\frac{1}{2}ab \sin C$ [T.7]

CONTEXT

- **Origin.** Egyptian Rhind papyrus (c. 1650 BCE); Heron of Alexandria (c. 60 CE).
- **Use.** Land, materials, floor plans.
- **Data science.** ROC AUC is usually computed with the trapezoid rule, summing trapezoids between successive (FPR, TPR) points, a preview of numeric integration [G.1.17].
- **Physical application.** Estimating the paint needed for a gable wall: rectangle plus triangle area.



G.9 Circles: Circumference, Area, Arcs & Sectors

$$C = 2\pi r = \pi d \quad A = \pi r^2 \quad s = r\theta = \frac{\theta^\circ}{360^\circ} \cdot 2\pi r \quad A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{\theta^\circ}{360^\circ} \cdot \pi r^2$$

STATED-EQUATION C equals two pi r , which equals pi d . A equals pi r squared. Arc length s equals r theta, or theta degrees over three sixty times two pi r . Sector area equals one half r squared theta, or theta over three sixty times pi r squared.

MEMORY AID “Cherry pie is delicious” ($C = \pi d$); “apple pies are too” ($A = \pi r^2$).

TERMS

r, d – radius and diameter ($d = 2r$)

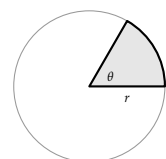
$\pi \approx 3.14159$, ratio of circumference to diameter

θ – central angle in radians [T.2] (or degrees, θ°)

s – arc length

CONTEXT

- **Origin.** Archimedes bounded π with 96-sided polygons (c. 250 BCE).
- **Use.** Wheels, pipes, rotation, circular tracks.
- **Data science.** Pie and donut charts encode values as sector angles; people misjudge angles, one reason bar charts are preferred for precise comparisons.
- **Physical application.** A pizza slice with a 45° angle is one-eighth of the pizza's area.



G.10 Circle Theorems

$$\angle_{\text{inscribed}} = \frac{1}{2} \angle_{\text{central}} \quad \text{angle in a semicircle} = 90^\circ \quad \text{tangent} \perp \text{radius} \quad \underbrace{ab = cd}_{\text{intersecting chords}}$$

STATED-EQUATION An inscribed angle equals one half the central angle on the same arc. An angle inscribed in a semicircle is ninety degrees. A tangent is perpendicular to the radius. Intersecting chords: $a b$ equals $c d$.

TERMS

Inscribed angle – vertex on the circle, subtending an arc
 Central angle – vertex at the center, same arc
 Thales' theorem – inscribed angle on a diameter is right
 a, b, c, d – segments of two chords crossing inside the circle

CONTEXT

- **Origin.** Euclid, *Elements* Book III; Thales (c. 600 BCE).
- **Use.** Constructions, optics, locating centers.
- **Data science.** “Tangent $\perp$ radius” generalizes to gradients: the gradient is perpendicular to the level curve of a loss, so gradient descent crosses contours at right angles [C1.13].
- **Physical application.** A carpenter finds the center of a round table by drawing two chords and their perpendicular bisectors.

G.11 Prisms & Cylinders

$$V = Bh \quad \underbrace{V = lwh, \quad SA = 2(lw + lh + wh)}_{\text{rectangular prism}} \quad \underbrace{V = \pi r^2 h, \quad SA = 2\pi r^2 + 2\pi rh}_{\text{cylinder}}$$

STATED-EQUATION Volume equals base area times height. Rectangular prism: V equals $l w h$, surface area equals two times $l w$ plus $l h$ plus $w h$. Cylinder: V equals πr squared h , surface area equals two πr squared plus two $\pi r h$.

TERMS

B – area of the base [G.3]
 h – height between the parallel bases
 l, w – length and width
 SA – surface area: total area of all faces

CONTEXT

- **Origin.** Cavalieri's principle (1635): equal cross-sections at every height give equal volumes.
- **Use.** Containers, tanks, packaging.
- **Data science.** Tensor memory is a “box volume”: a batch of shape $32 \times 224 \times 224 \times 3$ in float32 needs $32 \cdot 224 \cdot 224 \cdot 3 \cdot 4$ bytes.
- **Physical application.** A soda can with $r = 3.3$ cm and $h = 12.2$ cm holds $\pi(3.3)^2(12.2) \approx 417$ cm³, about 355 mL of drink plus headspace.

G.12 Pyramids, Cones & Spheres

$$\underbrace{V = \frac{1}{3} Bh}_{\text{pyramid}} \quad \underbrace{V = \frac{1}{3} \pi r^2 h, \quad SA = \pi r^2 + \pi r \ell}_{\text{cone}} \quad \underbrace{V = \frac{4}{3} \pi r^3, \quad SA = 4\pi r^2}_{\text{sphere}}$$

STATED-EQUATION Pyramid: V equals one third $B h$. Cone: V equals one third πr squared h , surface area equals πr squared plus $\pi r \ell$. Sphere: V equals four thirds πr cubed, surface area equals four πr squared.

MEMORY AID “A cone is one-third of the cylinder” with the same base and height.

TERMS

B, h – base area and height [G.11]
 ℓ – slant height, $\ell = \sqrt{r^2 + h^2}$ [G.3]
 $\frac{1}{3}$ – a point-topped solid holds one third of its prism
 r – radius

CONTEXT

- **Origin.** Archimedes proved the sphere is $\frac{2}{3}$ of its circumscribed cylinder and asked for it on his tombstone.
- **Use.** Storage, physics, astronomy.
- **Data science.** Curse of dimensionality: in n dimensions the fraction of a ball's volume within radius $1 - \epsilon$ is $(1 - \epsilon)^n$, so almost all volume sits near the surface.
- **Physical application.** Volume of a cup: a tapered cup is a frustum, $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$; a cup 10 cm tall with rim radius 4 cm and base radius 3 cm holds $\frac{1}{3} \pi (10)(16 + 12 + 9) \approx 387$ mL.

G.13 Coordinate Transformations

$$\underbrace{(x, y) \mapsto (x + a, y + b)}_{\text{translation}}$$

$$\underbrace{(x, y) \mapsto (y, x)}_{\text{reflect in } y=x}$$

$$\underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}}_{\text{rotation by } \theta \text{ counterclockwise}}$$

$$\underbrace{(kx, ky)}_{\text{dilation}}$$

STATED-EQUATION Translation: x, y maps to x plus a , y plus b . Reflection in the line y equals x swaps the coordinates. Rotation by θ counterclockwise: x prime, y prime equals the matrix cosine θ , minus sine θ ; sine θ , cosine θ , times x, y . Dilation: kx, ky .

TERMS

a, b – horizontal and vertical shift

θ – rotation angle [T.1](#)

Matrix form – matrix–vector product [A2.13](#)

Reflections – x -axis: $(x, -y)$; y -axis: $(-x, y)$

k – scale factor [G.5](#)

CONTEXT

- **Origin.** Felix Klein’s Erlangen program (1872) defined geometries by the transformations they preserve.
- **Use.** Computer graphics, robotics, symmetry.
- **Data science.** Image augmentation applies flips, rotations, and shifts; PCA rotates the axes to align with variance; point-cloud registration finds the best rigid rotation.
- **Physical application.** Video-game and robotics engines rotate objects with exactly this matrix every frame.

04 Trigonometry

Angles become functions: ratios in right triangles extend to the unit circle, to periodic waves, and to vectors. This is the mathematics of rotation, cycles, and similarity.

T.1 Right-Triangle Trigonometric Ratios

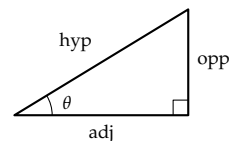
$$\begin{aligned}\sin \theta &= \frac{\text{opp}}{\text{hyp}} & \cos \theta &= \frac{\text{adj}}{\text{hyp}} & \tan \theta &= \frac{\text{opp}}{\text{adj}} = \frac{\sin \theta}{\cos \theta} \\ \csc \theta &= \frac{1}{\sin \theta} & \sec \theta &= \frac{1}{\cos \theta} & \cot \theta &= \frac{1}{\tan \theta}\end{aligned}$$

STATED-EQUATION Sine theta equals opposite over hypotenuse. Cosine theta equals adjacent over hypotenuse. Tangent theta equals opposite over adjacent, which equals sine over cosine. Cosecant is one over sine, secant is one over cosine, cotangent is one over tangent.

MEMORY AID SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent.

TERMS

θ – an acute angle of a right triangle
opp, adj – legs opposite and adjacent to θ
hyp – hypotenuse [6.3]
Mnemonic – SOH-CAH-TOA
Reciprocals – cosecant, secant, cotangent



CONTEXT

- **Origin.** Hipparchus' chord tables (c. 140 BCE); the sine (*jya*) from Aryabhata (499 CE).
- **Use.** Heights, slopes, navigation, engineering.
- **Data science.** Resolving a vector into components, $x = r \cos \theta$, $y = r \sin \theta$ [T.11]; sine and cosine features encode cyclical variables [T.3].
- **Physical application.** Surveyors find a building's height from a distance and an angle of elevation: height = $d \tan \theta$.

T.2 Radians, Degrees & Angular Speed

$$\pi \text{ rad} = 180^\circ \quad \theta_{\text{rad}} = \theta^\circ \cdot \frac{\pi}{180^\circ} \quad s = r\theta \quad \omega = \frac{\theta}{t} \quad v = r\omega$$

STATED-EQUATION *Pi radians equals one hundred eighty degrees. Theta in radians equals theta in degrees times pi over one eighty. Arc length s equals r theta. Angular speed ω equals theta over t . Linear speed v equals r omega.*

MEMORY AID Degrees to radians: multiply by $\frac{\pi}{180}$ ("pi on top to go to radians").

TERMS

Radian – angle whose arc equals the radius

s – arc length [G.9]

ω – angular speed (radians per unit time)

v – linear speed of a point at radius r

CONTEXT

- **Origin.** Roger Cotes (1714); the word "radian" is due to James Thomson (1873).
- **Use.** Calculus requires radians: $\frac{d}{dx} \sin x = \cos x$ only in radians [C1.8].
- **Data science.** Cyclical feature engineering: hour h maps to the angle $\frac{2\pi h}{24}$ before taking sine and cosine.
- **Physical application.** A bicycle wheel of radius 0.35 m spinning at 10 rad/s moves the bike at $v = r\omega = 3.5$ m/s.

T.3 Unit Circle & Exact Values

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

STATED-EQUATION *The point at angle theta on the unit circle is cosine theta, sine theta. At zero, pi over six, pi over four, pi over three, pi over two, sine is zero, one half, root two over two, root three over two, one; cosine is the same list in reverse.*

MEMORY AID Sine values are $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cosine is the reverse. Signs by quadrant: "All Students Take Calculus" (All, Sin, Tan, Cos positive).

TERMS

Unit circle – radius 1 centered at the origin, $x^2 + y^2 = 1$ [A2.15]

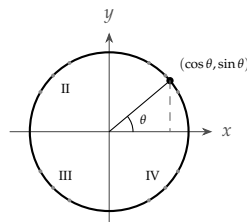
Reference angle – acute angle to the x -axis

Signs – "All Students Take Calculus": QI all +, QII sin, QIII tan, QIV cos

Source of values – special triangles [G.4]

CONTEXT

- **Origin.** Euler's circle definitions (18th c.) extended trigonometry to every angle.
- **Use.** Evaluating trig functions of any angle exactly.
- **Data science.** Encoding a cyclical feature as $(\cos \theta, \sin \theta)$ places 23:00 next to 00:00 and December next to January, which a raw integer cannot do.
- **Physical application.** The height of a Ferris-wheel seat is $r \sin \theta$ above the axle as the wheel turns.



T.4 Pythagorean & Fundamental Identities

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 & 1 + \tan^2 \theta &= \sec^2 \theta & 1 + \cot^2 \theta &= \csc^2 \theta \\ \sin(-\theta) &= -\sin \theta & \cos(-\theta) &= \cos \theta & \tan(-\theta) &= -\tan \theta\end{aligned}$$

STATED-EQUATION Sine squared theta plus cosine squared theta equals one. One plus tangent squared equals secant squared. One plus cotangent squared equals cosecant squared. Sine of minus theta equals minus sine theta; cosine of minus theta equals cosine theta; tangent of minus theta equals minus tangent theta.

MEMORY AID Divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ to get the other two. Cosine is the only even one.

TERMS

$\sin^2 \theta$ – means $(\sin \theta)^2$

Derivation – Pythagorean theorem [6.3] on the unit circle [T.3]

Odd / even – sine and tangent are odd; cosine is even

Other two – divide the first by $\cos^2 \theta$ or $\sin^2 \theta$

CONTEXT

- **Origin.** Follows from Pythagoras once trig is defined on the circle.
- **Use.** Simplifying expressions, verifying identities, trig integrals [C2.2].
- **Data science.** Because $\sin^2 + \cos^2 = 1$, every cyclical encoding $(\cos \theta, \sin \theta)$ has norm 1: no hour is artificially “larger” than another to a distance-based model.
- **Physical application.** In a pendulum or spring, $\sin^2 + \cos^2 = 1$ is why kinetic plus potential energy stays constant.

T.5 Sum & Difference Formulas

$$\begin{aligned}\sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta & \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}\end{aligned}$$

STATED-EQUATION Sine of alpha plus or minus beta equals sine alpha cosine beta plus or minus cosine alpha sine beta. Cosine of alpha plus or minus beta equals cosine alpha cosine beta minus or plus sine alpha sine beta. Tangent of alpha plus or minus beta equals tangent alpha plus or minus tangent beta, over one minus or plus tangent alpha tangent beta.

MEMORY AID Sine keeps the sign and mixes (sin cos + cos sin); cosine keeps the pairs and flips the sign.

TERMS

α, β – any two angles

$\pm, \mp$ – use the top signs together, or the bottom signs together

Special cases – double angles [T.6] when $\alpha = \beta$

CONTEXT

- **Origin.** Ptolemy’s theorem on chords in the *Almagest* (c. 150 CE).
- **Use.** Exact values such as $\sin 75^\circ$; simplifying wave sums.
- **Data science.** Rotary positional embeddings (RoPE) in transformers rotate queries and keys; angle addition makes their dot product depend only on relative position $m - n$.
- **Physical application.** Two sound waves that differ slightly in frequency combine into audible “beats” (sum-to-product identities).

T.6 Double-Angle & Half-Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}} \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

STATED-EQUATION Sine two theta equals two sine theta cosine theta. Cosine two theta equals cosine squared minus sine squared, which equals two cosine squared minus one, which equals one minus two sine squared. Tangent two theta equals two tangent theta over one minus tangent squared. Sine of theta over two equals plus or minus the square root of one minus cosine theta over two; cosine of theta over two equals plus or minus the square root of one plus cosine theta over two.

TERMS

Double angle – $\alpha = \beta = \theta$ in [T.5](#)

Power reduction – $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$, $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

$\pm$ – sign chosen by the quadrant of $\frac{\theta}{2}$ [T.3](#)

CONTEXT

- **Origin.** Special cases of Ptolemy's sum formulas.
- **Use.** Integrating $\sin^2 x$ [C.2.2](#); exact values like $\sin 22.5^\circ$.
- **Data science.** Power reduction explains why squaring a seasonal signal produces a component at *twice* the frequency, visible as a harmonic in a periodogram.
- **Physical application.** A projectile's range $R = \frac{v^2 \sin 2\theta}{g}$ is largest at $\theta = 45^\circ$ because $\sin 2\theta$ peaks there.

T.7 Law of Sines & Triangle Area

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \quad \text{Area} = \frac{1}{2} ab \sin C$$

STATED-EQUATION a over sine A equals b over sine B equals c over sine C , which equals two R . The area equals one half $a b$ sine C .

TERMS

a, b, c – sides opposite angles A, B, C [G.2](#)

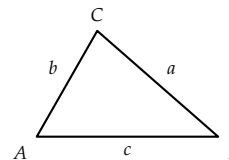
R – radius of the circumscribed circle

Use when – you know AAS, ASA, or SSA (ambiguous case)

Area – generalizes $\frac{1}{2}bh$ [G.8](#)

CONTEXT

- **Origin.** Stated in full by Nasir al-Din al-Tusi (13th c.).
- **Use.** Surveying and navigation by triangulation.
- **Data science.** Geospatial localization: a device's position is triangulated from bearings to known towers or beacons.
- **Physical application.** Triangulation: two observers a known distance apart measure angles to a ship to locate it.



T.8 Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cos C \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

STATED-EQUATION c squared equals a squared plus b squared minus two a b cosine C . Cosine C equals a squared plus b squared minus c squared, over two a b .

MEMORY AID It is the Pythagorean theorem with a correction: when $C = 90^\circ$, $\cos C = 0$.

TERMS

C – angle between sides a and b (opposite c)

Use when – you know SAS or SSS

$C = 90^\circ$ – reduces to the Pythagorean theorem [G.3]

CONTEXT

- **Origin.** Geometric form in Euclid II.12–13; trigonometric form by al-Kashi (15th c.).
- **Use.** Distances across obstacles, navigation, robotics.
- **Data science.** $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$: for unit embeddings, Euclidean distance and cosine similarity [T.12] carry the same ranking, $d^2 = 2 - 2\cos\theta$.
- **Physical application.** The distance between two ships leaving port on different headings is found by the law of cosines.

T.9 Graphs of Sinusoids

$$y = A \sin(B(x - C)) + D \quad \text{amplitude} = |A| \quad \text{period} = \frac{2\pi}{|B|} \quad \text{phase shift} = C \quad \text{midline: } y = D$$

STATED-EQUATION y equals A sine of B times x minus C , plus D . The amplitude is the absolute value of A . The period is two pi over the absolute value of B . The phase shift is C . The midline is y equals D .

TERMS

A – amplitude: half the peak-to-trough height

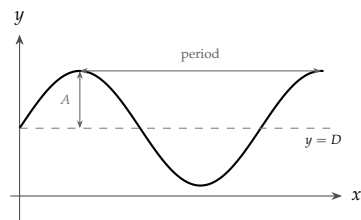
B – angular frequency; frequency = $\frac{|B|}{2\pi}$

C, D – horizontal and vertical shifts [A2.2]

Period – length of one full cycle

CONTEXT

- **Origin.** Joseph Fourier (1822): any periodic signal is a sum of sinusoids.
- **Use.** Tides, sound, alternating current, circadian rhythms.
- **Data science.** Seasonality in time-series models (e.g. Fourier terms in Prophet) uses sin and cos with periods of 7 or 365.25 days.
- **Physical application.** Household AC voltage is $170 \sin(2\pi \cdot 60 t)$ volts in the US: amplitude 170 V, period $\frac{1}{60}$ s.



T.10 Inverse Trigonometric Functions

$$y = \arcsin x \iff \sin y = x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad y = \arccos x \in [0, \pi] \quad y = \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

STATED-EQUATION y equals arcsine x if and only if sine y equals x , with y between minus pi over two and pi over two. Arccosine x lies between zero and pi. Arctangent x lies strictly between minus pi over two and pi over two.

TERMS

$\sin^{-1} x$ – same as $\arcsin x$; not $\frac{1}{\sin x}$

Restricted domain – makes the function one-to-one [A2.1]

$\text{atan2}(y, x)$ – full-circle angle using signs of both coordinates

CONTEXT

- **Origin.** The $\sin^{-1}$ notation is due to John Herschel (1813).
- **Use.** Recovering angles from ratios or measurements.
- **Data science.** $\arccos(\text{cosine similarity})$ gives angular distance, a true metric; atan2 recovers a phase or hour from a $(\cos, \sin)$ encoding.
- **Physical application.** The angle of a ramp that rises 1 m over 12 m is $\arctan(1/12) \approx 4.8^\circ$ (the ADA limit).

T.11 Polar Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad r = \sqrt{x^2 + y^2} \quad \tan \theta = \frac{y}{x} \quad (\theta = \text{atan2}(y, x))$$

STATED-EQUATION x equals r cosine theta; y equals r sine theta. r equals the square root of x squared plus y squared. Tangent theta equals y over x ; use a tan two of y, x to get the right quadrant.

TERMS

r – distance from the origin (the pole) [G.6]

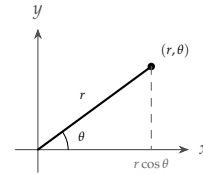
θ – angle from the positive x -axis [T.2]

Quadrant – choose θ from the signs of x, y [T.10]

Polar area – see [C2.15]

CONTEXT

- **Origin.** Used by Newton and Jacob Bernoulli (1691).
- **Use.** Circular motion, radar, antenna patterns, spirals.
- **Data science.** The Box–Muller transform generates normal samples from a random radius and angle; radar charts plot feature profiles in polar form.
- **Physical application.** Radar reports a plane's position as a distance and a bearing, which are polar coordinates.



T.12 Vectors, Dot Product & Cosine Similarity

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \quad \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$$

STATED-EQUATION The norm of v equals the square root of v one squared plus v two squared through v n squared. u dot v equals the sum of u i v i , which equals the norm of u times the norm of v times cosine theta. Cosine theta equals u dot v over the norms. The projection of u onto v equals u dot v over the norm of v squared, times v .

TERMS

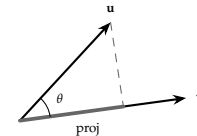
$\|\mathbf{v}\|$ – magnitude (L2 norm), from [G.6]

$\mathbf{u} \cdot \mathbf{v}$ – dot (inner) product, a scalar

θ – angle between the vectors [T.8]

Orthogonal – $\mathbf{u} \cdot \mathbf{v} = 0$, i.e. $\theta = 90^\circ$ [A1.6]

proj – component of $\mathbf{u}$ along $\mathbf{v}$



CONTEXT

- **Origin.** Vector analysis of Gibbs and Heaviside (1880s), from Hamilton's quaternions.
- **Use.** Work $W = \mathbf{F} \cdot \mathbf{d}$; components of forces.
- **Data science.** Cosine similarity ranks embeddings and TF-IDF documents; every linear model scores $\mathbf{w} \cdot \mathbf{x}$; projection is one-feature regression through the origin.
- **Physical application.** Work done by a force is a dot product: $W = \mathbf{F} \cdot \mathbf{d}$, pulling a sled at an angle.

T.13 Euler's Formula & De Moivre's Theorem

$$e^{i\theta} = \cos \theta + i \sin \theta \quad z = re^{i\theta} = r(\cos \theta + i \sin \theta) \quad z^n = r^n e^{in\theta} \quad e^{i\pi} + 1 = 0$$

STATED-EQUATION *e to the i theta equals cosine theta plus i sine theta. z equals r e to the i theta. z to the n equals r to the n e to the i n theta. e to the i pi plus one equals zero.*

TERMS

r – modulus $|z|$ [A2.3]

θ – argument (angle) of z [T.11]

i – imaginary unit, $i^2 = -1$

Multiplication – multiply moduli, add angles (a rotation)

Proof idea – compare Taylor series [C2.13]

CONTEXT

- **Origin.** Leonhard Euler (1748); Abraham de Moivre (1707).
- **Use.** Powers and roots of complex numbers; circuit and signal analysis.
- **Data science.** The discrete Fourier transform $X_k = \sum_n x_n e^{-2\pi i k n / N}$ powers spectral analysis of time series; RoPE treats embedding pairs as complex numbers and rotates them.
- **Physical application.** Electrical engineers write AC signals as $V e^{i\omega t}$ so that phase shifts become simple multiplications.

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05

Calculus I

Limits, derivatives, and the first integrals. Calculus I is the mathematics of change and accumulation, and the derivative is the engine inside every gradient-trained model.

C1.1 Limits & Limit Laws

$$\lim_{x \rightarrow a} f(x) = L \quad \lim_{x \rightarrow a} [f \pm g] = \lim_{x \rightarrow a} f \pm \lim_{x \rightarrow a} g \quad \lim_{x \rightarrow a} [fg] = \lim_{x \rightarrow a} f \cdot \lim_{x \rightarrow a} g \quad \lim_{x \rightarrow a} \frac{f}{g} = \frac{\lim_{x \rightarrow a} f}{\lim_{x \rightarrow a} g} \quad (\lim_{x \rightarrow a} g \neq 0)$$

$$\lim_{x \rightarrow a} f(x) \text{ exists} \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

STATED-EQUATION The limit as x approaches a of f of x equals L . The limit of a sum or difference is the sum or difference of the limits; the limit of a product is the product of the limits; the limit of a quotient is the quotient of the limits, provided the denominator's limit is not zero. The limit exists if and only if the left-hand and right-hand limits are equal.

TERMS

a – point approached (need not be in the domain)

L – limiting value

a^-, a^+ – from the left, from the right

ε - δ – for every $\varepsilon > 0$ there is $\delta > 0$ with $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

At infinity – leading terms dominate [A2.5](#)

CONTEXT

- **Origin.** Cauchy (1821); rigorous ε - δ form by Weierstrass (1860s).
- **Use.** Foundation of derivatives [C1.4](#) and integrals [C1.17](#).
- **Data science.** Consistency of an estimator means $\hat{\theta}_n \rightarrow \theta$ as $n \rightarrow \infty$; “training converged” is a claim about a limit of the loss sequence.
- **Physical application.** Instantaneous speed is a limit of average speeds over shorter and shorter time intervals.

C1.2 Special Limits & the Number e

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \quad \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n = e^r$$

STATED-EQUATION The limit as x goes to zero of sine x over x equals one. Of one minus cosine x over x equals zero. Of e to the x minus one over x equals one. The limit as n goes to infinity of one plus one over n , to the n , equals e ; of one plus r over n , to the n , equals e to the r .

TERMS

x – in radians [T.2](#)

e – Euler’s number ≈ 2.71828 [A2.6](#)

Squeeze theorem – proves the sine limit by trapping it between $\cos x$ and 1

Consequence – $(1 - \frac{1}{n})^n \rightarrow e^{-1} \approx 0.368$

CONTEXT

- **Origin.** Jacob Bernoulli (1683) for e ; the sine limit from the geometry of the unit circle.
- **Use.** Derivatives of $\sin x$ and e^x [C1.8](#); continuous compounding.
- **Data science.** Bootstrap: a given row is left out of a resample with probability $(1 - \frac{1}{n})^n \approx e^{-1}$, so about 36.8% of rows are “out-of-bag” in random forests.
- **Physical application.** The small-angle approximation $\sin \theta \approx \theta$ (radians) simplifies the pendulum equation.

C1.3 Continuity & the Intermediate Value Theorem

$$f \text{ continuous at } a \iff \lim_{x \rightarrow a} f(x) = f(a) \quad \text{IVT: } f \text{ continuous on } [a, b], N \text{ between } f(a), f(b) \implies \exists c \in (a, b) : f(c) = N$$

STATED-EQUATION f is continuous at a if and only if the limit as x approaches a of f of x equals f of a . Intermediate value theorem: if f is continuous on a to b , and N is between f of a and f of b , then there is a c in a to b with f of c equal to N .

TERMS

Conditions – $f(a)$ defined, limit exists [C1.1] and they agree

Removable – a hole; the limit exists but $\neq f(a)$

Jump – one-sided limits differ

Infinite – vertical asymptote [A2.5]

CONTEXT

► **Origin.** Bernard Bolzano (1817).

► **Use.** Guaranteeing that equations have solutions.

► **Data science.** The bisection method (and many threshold searches, such as finding where precision equals recall) relies on IVT; decision-tree predictions are step functions, discontinuous, hence gradient-free.

► **Physical application.** If it was 10°C at 6 a.m. and 20°C at noon, the temperature was exactly 15°C at some moment in between.

C1.4 Definition of the Derivative

$$f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{central difference: } f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

STATED-EQUATION f prime of x , or $d y d x$, equals the limit as h goes to zero of f of x plus h minus f of x , over h . The central difference approximates f prime of x by f of x plus h minus f of x minus h , over two h .

TERMS

h – small change in x

Difference quotient – slope of a secant line [A1.4]

$f'(x)$ – slope of the tangent: instantaneous rate of change

$\frac{dy}{dx}$ – Leibniz notation; f' is Lagrange's

Differentiable – implies continuous [C1.3]

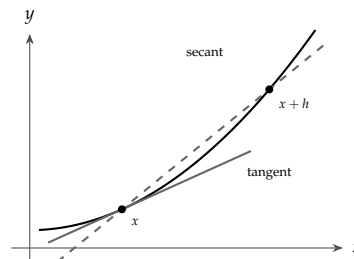
CONTEXT

► **Origin.** Newton's "fluxions" (1666) and Leibniz's publication (1684).

► **Use.** Velocity, marginal cost, any instantaneous rate.

► **Data science.** Gradient checking: compare backprop gradients with a central-difference estimate to catch implementation bugs.

► **Physical application.** A car's speedometer shows the derivative of position with respect to time.



C1.5 Power, Constant & Sum Rules

$$\frac{d}{dx} c = 0 \quad \frac{d}{dx} x^n = n x^{n-1} \quad \frac{d}{dx} [c f(x)] = c f'(x) \quad \frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$$

STATED-EQUATION The derivative of a constant is zero. The derivative of x to the n is $n x$ to the n minus one. The derivative of $c f$ is $c f$ prime. The derivative of f plus or minus g is f prime plus or minus g prime.

MEMORY AID Power rule: "bring the power down, subtract one."

TERMS

c – constant

n – any real exponent [A1.2]; e.g. $\sqrt{x} = x^{1/2}$ [A1.3]

Linearity – derivatives split across sums and pull out constants

CONTEXT

► **Origin.** Derived from the limit definition [C1.4]; Leibniz (1684).

► **Use.** Differentiating any polynomial on sight.

► **Data science.** The gradient of squared error with respect to a prediction is $\frac{d}{dy} (y - \hat{y})^2 = -2(y - \hat{y})$: the residual, scaled.

► **Physical application.** From $h(t) = -4.9t^2 + 20t$, velocity is $h'(t) = -9.8t + 20$, zero at the top of the throw.

C1.6 Product & Quotient Rules

$$\frac{d}{dx}[fg] = f'g + fg' \qquad \frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$$

STATED-EQUATION The derivative of fg equals f prime g plus fg prime. The derivative of f over g equals f prime g minus fg prime, over g squared.

MEMORY AID Quotient: “low d-high minus high d-low, over the square of what’s below.”

TERMS

f, g – differentiable functions of x
 $g \neq 0$ – required for the quotient
 Mnemonic – “low d-high minus high d-low, over low squared”

CONTEXT

- **Origin.** Leibniz (1684).
- **Use.** Derivatives of products and ratios of functions.
- **Data science.** The quotient rule gives the sigmoid identity $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ used in logistic regression, and sensitivities of ratio metrics such as CTR.
- **Physical application.** The rate of change of power $P = IV$ when both current and voltage change uses the product rule.

C1.7 Chain Rule

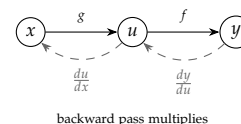
$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x) \qquad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

STATED-EQUATION The derivative of f of g of x equals f prime of g of x times g prime of x . dy/dx equals dy/du times du/dx .

MEMORY AID “Derivative of the outside (inside left alone) times derivative of the inside.”

TERMS

g – inner function; $u = g(x)$
 f – outer function; $y = f(u)$
 Composition – $f \circ g$ [A2.1]
 Longer chains – multiply one derivative per link



CONTEXT

- **Origin.** Leibniz (1676).
- **Use.** Differentiating nested functions: powers of polynomials, $e^{g(x)}$, $\ln g(x)$.
- **Data science.** Backpropagation is the chain rule run backward over a network’s computation graph, multiplying local derivatives from the loss to each weight.
- **Physical application.** If a balloon’s radius grows with time, its volume’s rate is $\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt}$.

C1.8 Derivatives of Trig, Exponential & Log Functions

$$\begin{array}{ccccccc} \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \cos x = -\sin x & \frac{d}{dx} \tan x = \sec^2 x & \frac{d}{dx} \arctan x = \frac{1}{1+x^2} & \frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} e^x = e^x & \frac{d}{dx} a^x = a^x \ln a & \frac{d}{dx} \ln x = \frac{1}{x} & \frac{d}{dx} \log_a x = \frac{1}{x \ln a} \end{array}$$

STATED-EQUATION The derivative of sine is cosine; of cosine, minus sine; of tangent, secant squared; of arctangent, one over one plus x squared; of arcsine, one over the square root of one minus x squared. The derivative of e to the x is e to the x ; of a to the x , a to the x natural log a ; of natural log x , one over x ; of log base a of x , one over x natural log a .

MEMORY AID Derivatives of “co” functions (cos, cot, csc) get a minus sign.

TERMS

Trig – x in radians [T.2]; follows from [C1.2]

e^x – its own derivative [A2.6]

$\ln x$ – natural log [A2.7], $x > 0$

With inner functions – combine with the chain rule [C1.7]

CONTEXT

- **Origin.** Euler’s *Introductio* (1748) and *Institutiones* (1755).
- **Use.** Growth models, oscillations, and log-transformed quantities.
- **Data science.** The log-loss gradient: $\frac{d}{dp}(-\ln p) = -\frac{1}{p}$, huge when a confident prediction is wrong; with the sigmoid it simplifies to $\hat{p} - y$.
- **Physical application.** Radioactive decay $N = N_0 e^{-kt}$ has rate $-kN$: the derivative of an exponential is proportional to itself.

C1.9 Implicit Differentiation

$$\frac{d}{dx}[y^n] = n y^{n-1} \frac{dy}{dx} \quad \text{Differentiate both sides in } x, \text{ treating } y = y(x), \text{ then solve for } \frac{dy}{dx}$$

STATED-EQUATION The derivative of y to the n with respect to x is $n y$ to the n minus one, times dy/dx . Differentiate both sides with respect to x , treating y as a function of x , then solve for dy/dx .

TERMS

$y(x)$ – y defined implicitly by an equation in x, y

Key step – every y -term gets a factor $\frac{dy}{dx}$ by the chain rule [C1.7]

Result – usually depends on both x and y

CONTEXT

- **Origin.** Leibniz and the Bernoulli brothers (1690s).
- **Use.** Slopes of circles and other curves that are not functions; related rates [C1.10].
- **Data science.** The implicit function theorem lets frameworks differentiate through the solution of an optimization or equilibrium (implicit layers, hyperparameter gradients).
- **Physical application.** The slope of a circular track $x^2 + y^2 = r^2$ at any point is $-x/y$, found implicitly.

C1.10 Related Rates

$$\text{relate the quantities} \rightarrow \text{differentiate with respect to } t \rightarrow \text{substitute known values} \quad \text{e.g. } A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

STATED-EQUATION Relate the quantities, differentiate with respect to t , then substitute the known values. For example, A equals πr squared gives dA/dt equals two πr dr/dt .

TERMS

t – time; every changing quantity is a function of t

$\frac{dr}{dt}$ – rate of change of r

Method – implicit differentiation in t [C1.9]

Order – differentiate first, substitute numbers after

CONTEXT

- **Origin.** Newton’s fluxions were rates with respect to time.
- **Use.** Ladders, tanks, shadows, expanding circles.
- **Data science.** Metrics that are products change by the product rule [C1.8]: if revenue $R = pN$, then $\frac{dR}{dt} = N \frac{dp}{dt} + p \frac{dN}{dt}$, splitting growth into price and volume effects.
- **Physical application.** As oil spreads in a circle, the area grows faster the larger the slick, because $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

C1.11 Tangent Lines & Linear Approximation

$$y - f(a) = f'(a)(x - a) \quad f(x) \approx L(x) = f(a) + f'(a)(x - a) \quad dy = f'(x) dx$$

STATED-EQUATION The tangent line is y minus f of a equals f prime of a times x minus a . Near a , f of x is approximately L of x , which equals f of a plus f prime of a times x minus a . The differential dy equals f prime of x times dx .

TERMS

a – point of tangency

$L(x)$ – linearization: point-slope line [A1.5](#)

dy, dx – differentials

Accuracy – best near a ; first term of Taylor [C2.13](#)

CONTEXT

- **Origin.** Newton and Leibniz; the first-order Taylor polynomial.
- **Use.** Quick estimates and error propagation.
- **Data science.** Gradient descent relies on $L(\theta - \eta \nabla L) \approx L(\theta) - \eta \|\nabla L\|^2$; the delta method uses linearization to get standard errors of transformed estimates.
- **Physical application.** Engineers estimate that a 1% error in a sphere's radius gives about a 3% error in its volume ($dV/V = 3 dr/r$).

C1.12 Critical Points, Extrema & Concavity

$$\text{critical: } f'(c) = 0 \text{ or undefined} \quad f''(c) > 0 \Rightarrow \text{local min} \quad f''(c) < 0 \Rightarrow \text{local max} \quad f'' \text{ changes sign} \Rightarrow \text{inflection}$$

STATED-EQUATION A critical point is where f prime of c equals zero or is undefined. If f double prime of c is positive, it is a local minimum; if negative, a local maximum. Where f double prime changes sign, there is an inflection point.

MEMORY AID Concave up holds water like a cup (a min); concave down spills like a frown (a max).

TERMS

First-derivative test – f' goes $+$ $\rightarrow$ $-$: max; $-$ $\rightarrow$ $+$: min

Second-derivative test – sign of f'' at c ; inconclusive if 0

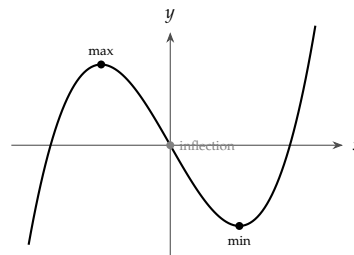
Concave up – $f'' > 0$ (holds water); down: $f'' < 0$

Convex function – $f'' \geq 0$ everywhere

Absolute extrema – also check endpoints of a closed interval

CONTEXT

- **Origin.** Fermat's method of adequality for maxima and minima (1638).
- **Use.** Finding peaks, troughs, and changes in trend.
- **Data science.** Convex losses (MSE, log-loss for linear models) have no bad local minima; deep networks are non-convex and full of saddle points where $f' = 0$ but no extremum exists.
- **Physical application.** The highest point of a thrown ball is where its velocity (first derivative) is zero and acceleration is negative.



C1.13 Optimization & Gradient Descent

solve $f'(x^*) = 0$, verify with [C1.12](#) and endpoints

$$x_{k+1} = x_k - \eta f'(x_k) \quad \theta_{k+1} = \theta_k - \eta \nabla L(\theta_k)$$

STATED-EQUATION Solve f prime of x star equals zero, then verify with the second derivative and endpoints. Gradient descent: x k plus one equals x k minus eta f prime of x k ; θ k plus one equals θ k minus eta times the gradient of L at θ k .

TERMS

Objective – quantity to maximize or minimize

Constraint – condition used to reduce to one variable

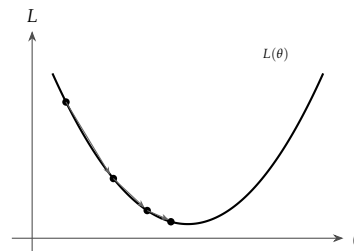
η – learning rate (step size)

∇L – gradient: vector of partial derivatives

Divergence – if η is too large, iterates overshoot

CONTEXT

- **Origin.** Augustin-Louis Cauchy proposed gradient descent in 1847.
- **Use.** Maximizing area or profit, minimizing cost or material.
- **Data science.** Training nearly every ML model: stochastic gradient descent, Adam, and friends are refinements of this update rule.
- **Physical application.** A can holding a fixed volume uses the least metal when its height equals its diameter.



C1.14 Mean Value Theorem

$$f \text{ continuous on } [a, b], \text{ differentiable on } (a, b) \implies \exists c \in (a, b) : f'(c) = \frac{f(b) - f(a)}{b - a}$$

STATED-EQUATION If f is continuous on a to b and differentiable on the open interval, then there is a c in between with f prime of c equal to f of b minus f of a , over b minus a .

TERMS

Right side – average rate of change (secant slope [A1.4](#))

$f'(c)$ – instantaneous rate (tangent slope [C1.4](#))

Rolle's theorem – special case $f(a) = f(b)$, so $f'(c) = 0$

Lipschitz – $|f(x) - f(y)| \leq L|x - y|$ with $L = \max |f'|$

CONTEXT

- **Origin.** Michel Rolle (1691); general form by Lagrange (1797).
- **Use.** Speed-trap logic; proving error bounds.
- **Data science.** Lipschitz bounds from the MVT set safe step sizes (gradient descent converges for $\eta < 2/L$ on smooth convex losses) and certify model robustness to input perturbations.
- **Physical application.** Average-speed cameras rely on the MVT: if your average speed exceeded the limit, your instantaneous speed did too at some moment.

C1.15 L'Hôpital's Rule

$$\text{If } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \text{ or } \pm \infty, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ (if the right side exists)}$$

STATED-EQUATION If the limits of f and g as x approaches a are both zero or both infinite, then the limit of f over g equals the limit of f prime over g prime, provided that limit exists.

MEMORY AID Differentiate top and bottom separately — not the quotient rule.

TERMS

Indeterminate – $\frac{0}{0}, \frac{\infty}{\infty}$; rewrite $0 \cdot \infty, \infty - \infty, 1^\infty$ first

f', g' – differentiate separately, *not* the quotient rule [C1.6](#)

a – finite or $\pm \infty$ [C1.1](#)

CONTEXT

- **Origin.** Johann Bernoulli taught it to the Marquis de L'Hôpital, who published it (1696).
- **Use.** Evaluating indeterminate limits.
- **Data science.** Comparing growth rates: $\frac{\ln n}{n} \rightarrow 0$ ranks $O(\log n)$ below $O(n)$; $x \ln x \rightarrow 0$ justifies $0 \log 0 = 0$ in entropy.
- **Physical application.** Evaluating $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$ in small-angle optics or the damped-oscillator limit uses L'Hôpital's rule.

C1.16 Antiderivatives & Basic Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \quad \int \frac{1}{x} dx = \ln|x| + C \quad \int e^x dx = e^x + C \quad \int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C \quad \int \cos x dx = \sin x + C \quad \int \sec^2 x dx = \tan x + C \quad \int \frac{dx}{1+x^2} = \arctan x + C$$

STATED-EQUATION The integral of x to the n dx equals x to the n plus one over n plus one, plus C , for n not equal to minus one. The integral of one over x is natural log of the absolute value of x plus C . The integral of e to the x is e to the x plus C ; of a to the x , a to the x over natural log a . The integral of sine is minus cosine; of cosine, sine; of secant squared, tangent; of one over one plus x squared, arctangent; each plus C .

MEMORY AID “Add one to the power, divide by the new power”; don’t forget $+C$.

TERMS

F – antiderivative of f : $F' = f$

C – constant of integration

$\int$ – Leibniz’s elongated S for *summa*

Check – differentiate the answer [C1.8]

CONTEXT

- **Origin.** Leibniz introduced $\int$ (1675); Newton’s inverse method of fluxions.
- **Use.** Recovering position from velocity, total from rate.
- **Data science.** A CDF is the antiderivative of its PDF, $F'(x) = f(x)$ [S.12]; cumulative metrics are running integrals of rates.
- **Physical application.** Integrating acceleration gives velocity, and integrating velocity gives position.

C1.17 Riemann Sums & the Definite Integral

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x \quad \Delta x = \frac{b-a}{n} \quad \int_a^b f = - \int_b^a f$$

STATED-EQUATION The integral from a to b of f of x dx equals the limit as n goes to infinity of the sum from i equals one to n of f of x i star times delta x , where delta x equals b minus a over n . Swapping the limits changes the sign.

TERMS

Δx – width of each subinterval

x_i^* – sample point: left, right, or midpoint

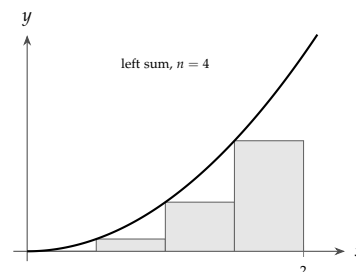
Σ – summation [A2.11]

Meaning – signed area between f and the x -axis

Trapezoid rule – average left and right sums [G.8]

CONTEXT

- **Origin.** Archimedes’ method of exhaustion; formalized by Riemann (1854).
- **Use.** Area, distance, total accumulation.
- **Data science.** Expected values of continuous variables are integrals [S.9]; Monte Carlo integration averages random samples when dimensions are high.
- **Physical application.** Estimating the distance a car traveled from speed readings every second is a Riemann sum.



C1.18 Fundamental Theorem of Calculus

$$\text{Part 1: } \frac{d}{dx} \int_a^x f(t) dt = f(x)$$

$$\text{Part 2: } \int_a^b f(x) dx = F(b) - F(a) = \left[F(x) \right]_a^b$$

STATED-EQUATION Part one: the derivative of the integral from a to x of f of t dt equals f of x . Part two: the integral from a to b of f of x dx equals F of b minus F of a .

TERMS

F – any antiderivative of f [C1.16]
 t – dummy variable of integration
 Link – differentiation and integration are inverses
 Continuity – f continuous on $[a, b]$ [C1.3]

CONTEXT

- **Origin.** Isaac Barrow, Newton, and Leibniz (1660s–1680s).
- **Use.** Evaluating definite integrals exactly.
- **Data science.** Probabilities from a CDF: $P(a \leq X \leq b) = F(b) - F(a)$ [S.12]; the density is the derivative of the CDF.
- **Physical application.** Total water that flowed into a tank equals the integral of the inflow rate, found from any antiderivative.

C1.19 Integration by u -Substitution

$$\int f(g(x)) g'(x) dx = \int f(u) du \quad u = g(x), \quad du = g'(x) dx \quad \int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

STATED-EQUATION The integral of f of g of x times g prime of x dx equals the integral of f of u du , with u equals g of x and du equals g prime of x dx . For definite integrals, change the limits to g of a and g of b .

MEMORY AID Look for an “inside function” whose derivative is also present.

TERMS

u – inner function, chosen so du appears in the integrand
 Reverse of – the chain rule [C1.7]
 Definite – change the limits to u -values

CONTEXT

- **Origin.** The chain rule read backward (Leibniz).
- **Use.** Integrals of composite functions.
- **Data science.** Change of variables for densities: if $Y = g(X)$, then $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$, which turns a normal into a log-normal.
- **Physical application.** Computing the work to stretch a nonlinear spring often requires a substitution.

C1.20 Newton’s Method for Roots

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

stop when $|x_{n+1} - x_n| < \text{tolerance}$

STATED-EQUATION x n plus one equals x n minus f of x n over f prime of x n . Stop when successive estimates differ by less than the tolerance.

MEMORY AID “Follow the tangent line down to the axis, then repeat.”

TERMS

x_n – current estimate of the root
 Tangent line – linearization [C1.11] its x -intercept is x_{n+1}
 Convergence – quadratic near a simple root: correct digits roughly double each step
 Failure – $f'(x_n) \approx 0$, bad starting point, or oscillation
 Optimization version – apply to f' : [O.9]

CONTEXT

- **Origin.** Newton (1669), Raphson (1690); Simpson gave the calculus form (1740).
- **Use.** Calculators computing square roots, solving equations with no closed form.
- **Data science.** Solving for an internal rate of return or implied volatility; the core of Newton/IRLS fitting in logistic regression [ML.7].
- **Physical application.** Engineers solve the Colebrook equation for pipe friction [P.10] with Newton’s method because it has no algebraic solution.

06

Calculus II

Integration techniques, applications of the integral, infinite series, parametric and polar curves, and first differential equations. Calculus II is where approximation (Taylor series) and accumulation meet the models used every day in statistics and machine learning.

C2.1 Integration by Parts

$$\int u dv = uv - \int v du \qquad \int_a^b u dv = [uv]_a^b - \int_a^b v du$$

STATED-EQUATION The integral of $u dv$ equals uv minus the integral of $v du$. For definite integrals: uv evaluated from a to b , minus the integral from a to b of $v du$.

MEMORY AID LIATE picks u : Log, Inverse trig, Algebraic, Trig, Exponential. “Ultraviolet voodoo”: $uv - \int v du$.

TERMS

u – part to differentiate; choose by LIATE (Log, Inverse trig, Algebraic, Trig, Exponential)

dv – part to integrate [C1.16]

$du, v - du = u' dx, v = \int dv$

Reverse of – the product rule [C1.6]

Tabular method – repeated parts when u is a polynomial

CONTEXT

- **Origin.** Brook Taylor (1715), from integrating the product rule.
- **Use.** Integrals of products such as xe^x , $x \sin x$, $\ln x$.
- **Data science.** Moments of distributions: $E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$ for the exponential [S.20], computed by parts.
- **Physical application.** The work done by a force that varies like $F = xe^{-x}$ along a path is integrated by parts.

C2.2 Trig Integrals & Trig Substitution

$$\sin^2 x = \frac{1 - \cos 2x}{2} \qquad \cos^2 x = \frac{1 + \cos 2x}{2} \qquad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \qquad \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$

$$\sqrt{a^2 - x^2} \Rightarrow x = a \sin \theta \qquad \sqrt{a^2 + x^2} \Rightarrow x = a \tan \theta \qquad \sqrt{x^2 - a^2} \Rightarrow x = a \sec \theta$$

STATED-EQUATION Sine squared x equals one minus cosine two x , over two; cosine squared x equals one plus cosine two x , over two. The integral of dx over a squared plus x squared equals one over a arctangent of x over a , plus C . The integral of dx over the square root of a squared minus x squared equals arcsine of x over a , plus C . For root a squared minus x squared use x equals a sine theta; root a squared plus x squared, x equals a tangent theta; root x squared minus a squared, x equals a secant theta.

MEMORY AID Match the root to the identity: $a^2 - x^2 \rightarrow \sin$, $a^2 + x^2 \rightarrow \tan$, $x^2 - a^2 \rightarrow \sec$.

TERMS

Power reduction – from double-angle identities [T.6]

Substitution – uses Pythagorean identities [T.4] to remove the root

Odd powers – save one factor of $\sin x$ or $\cos x$, convert the rest, then

u -sub [C1.19]

Back-substitute – with a reference triangle [T.1]

$\tan^{-1}, \sin^{-1}$ – inverse trig [T.10]

CONTEXT

- **Origin.** Developed by Euler and contemporaries in the 1700s.
- **Use.** Areas of circles and ellipses, arc lengths, physics of oscillation.
- **Data science.** The Cauchy distribution integrates to 1 because $\int_{-\infty}^\infty \frac{dx}{\pi(1+x^2)} = \frac{1}{\pi} [\tan^{-1} x]_{-\infty}^\infty = 1$; its heavy tails make the mean undefined.
- **Physical application.** The average power of an AC signal uses $\sin^2 = \frac{1}{2}$, which is why $V_{\text{rms}} = V_{\text{peak}} / \sqrt{2}$.

C2.3 Partial Fractions

$$\frac{px + q}{(x - a)(x - b)} = \frac{A}{x - a} + \frac{B}{x - b} \quad \frac{r(x)}{(x - a)^2} = \frac{A}{x - a} + \frac{B}{(x - a)^2} \quad \frac{r(x)}{x^2 + c} = \frac{Ax + B}{x^2 + c} \quad \int \frac{dx}{x - a} = \ln |x - a| + C$$

STATED-EQUATION $px + q$ over x minus a times x minus b equals A over x minus a plus B over x minus b . A repeated factor x minus a squared gives A over x minus a plus B over x minus a squared. An irreducible quadratic factor gets a numerator $Ax + B$. The integral of dx over x minus a equals natural log of the absolute value of x minus a , plus C .

TERMS

Requirement – degree of numerator < degree of denominator; otherwise divide first [A2.4]

Factors – from factoring the denominator [A1.10]

A, B – constants found by clearing denominators and plugging in roots (Heaviside cover-up)

Irreducible quadratic – numerator $Ax + B$; integrates to $\ln$ and $\tan^{-1}$ [C2.2]

CONTEXT

► **Origin.** Johann Bernoulli and Leibniz (1702).

► **Use.** Integrating rational functions; inverse Laplace transforms in engineering.

► **Data science.** Solving the logistic equation [C2.16]: $\int \frac{dy}{y(1-y)} = \ln \left| \frac{y}{1-y} \right| + C$, the *logit*, which is exactly the link function of logistic regression.

► **Physical application.** Solving for the concentration in a two-step chemical reaction leads to partial-fraction integrals.

C2.4 Improper Integrals

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \quad \int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx \quad (f \text{ unbounded at } a) \quad \int_1^\infty \frac{dx}{x^p} = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \leq 1 \end{cases}$$

STATED-EQUATION The integral from a to infinity of f of x equals the limit as b goes to infinity of the integral from a to b . If f is unbounded at a , take the limit as t approaches a from the right. The integral from one to infinity of dx over x to the p equals one over p minus one if p is greater than one, and diverges if p is at most one.

MEMORY AID p -test: “ p bigger than one converges.”

TERMS

Type 1 – infinite interval of integration

Type 2 – integrand has a vertical asymptote [A2.5]

Converges – the limit [C1.1] exists and is finite

p -test – benchmark for comparison; mirrors p -series [C2.10]

Comparison – $0 \leq f \leq g$ and $\int g$ converges $\Rightarrow \int f$ converges

CONTEXT

► **Origin.** Formalized by Cauchy (1823).

► **Use.** Total accumulated quantities over unbounded time or space.

► **Data science.** Every density must satisfy $\int_{-\infty}^\infty f(x) dx = 1$ [S.12]; a Pareto tail $\propto x^{-(\alpha+1)}$ has a finite mean only when $\alpha > 1$, which is why heavy-tailed revenue data breaks averages.

► **Physical application.** The escape energy from Earth’s gravity is $\int_R^\infty \frac{GMm}{r^2} dr = \frac{GMm}{R}$, a finite improper integral.

C2.5 Area Between Curves

$$A = \int_a^b [f(x) - g(x)] dx \quad (f \geq g \text{ on } [a, b]) \quad A = \int_c^d [x_{\text{right}}(y) - x_{\text{left}}(y)] dy$$

STATED-EQUATION The area between curves equals the integral from a to b of f of x minus g of x , dx , with f above g . With horizontal slices, integrate x right minus x left with respect to y .

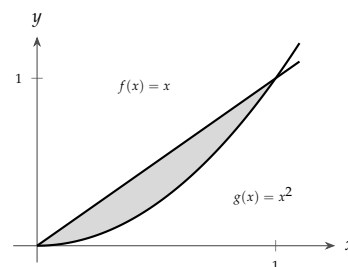
MEMORY AID “Top minus bottom” (or “right minus left”).

TERMS

$f(x)$ – upper curve; $g(x)$ lower curve
 a, b – intersection points: solve $f(x) = g(x)$
 Crossing curves – split the interval where they switch, or integrate $|f - g|$
 Built on – the definite integral [C1.18]

CONTEXT

- **Origin.** Archimedes’ method of exhaustion; integral form from Leibniz.
- **Use.** Consumer surplus, net displacement, profit regions.
- **Data science.** Model lift: the Gini coefficient is twice the area between the ROC curve and the diagonal, $\text{Gini} = 2 \cdot \text{AUC} - 1$ [G.8].
- **Physical application.** The area of a lens or a river cross-section between two bank profiles is an area between curves.



C2.6 Volumes of Revolution

$$\text{Disk: } V = \pi \int_a^b [R(x)]^2 dx \quad \text{Washer: } V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx \quad \text{Shell: } V = 2\pi \int_a^b x h(x) dx$$

STATED-EQUATION Disk method: V equals pi times the integral of R of x squared dx . Washer method: V equals pi times the integral of R squared minus r squared. Shell method: V equals two pi times the integral of x times h of x dx .

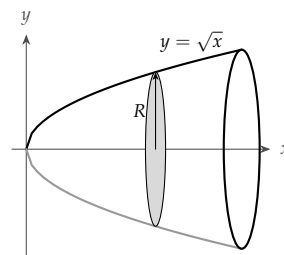
MEMORY AID Washer = “big disk minus small disk”; shell = “circumference $\times$ height $\times$ thickness.”

TERMS

$R(x), r(x)$ – outer and inner radius measured from the axis
 πR^2 – area of a circular slice [G.9]
 $h(x)$ – height of a cylindrical shell; $2\pi x$ is its circumference
 Slicing – $V = \int A(x) dx$ for any known cross-section $A(x)$ [G.11]

CONTEXT

- **Origin.** Cavalieri’s principle (1635); Kepler measured wine-barrel volumes this way.
- **Use.** Tanks, machine parts, and any solid with rotational symmetry.
- **Data science.** The same slice-and-sum idea computes probabilities over 2-D regions by integrating a joint density one strip at a time.
- **Physical application.** Volume of a cup: rotate its side profile $y = R(x)$ about the axis and integrate $\pi R(x)^2$; a cup whose radius grows linearly from 3 to 4 cm over 10 cm holds $\pi \int_0^{10} (3 + 0.1x)^2 dx \approx 387$ mL.



C2.7 Arc Length

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx \qquad L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

STATED-EQUATION L equals the integral from a to b of the square root of one plus f prime of x squared, dx . For a parametric curve, L equals the integral of the square root of dx squared plus dy squared, dt .

TERMS

$f'(x)$ – derivative [C1.4]

Derivation – sum of tiny hypotenuses $\sqrt{dx^2 + dy^2}$ [G.3], [G.6]

Parametric – $x(t), y(t)$ [C2.14]

Polar – $L = \int \sqrt{r^2 + (dr/d\theta)^2} d\theta$ [T.11]

CONTEXT

- **Origin.** First rectifications of curves by Neile and van Heuraet (1659).
- **Use.** Cable and road lengths, distance travelled along a path.
- **Data science.** Distance along a GPS trace, or the total path length of an optimizer's trajectory, is a discrete version of this integral.
- **Physical application.** The length of a sagging power cable (a catenary) between two towers is found with the arc-length integral.

C2.8 Average Value of a Function

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx \qquad \text{Mean Value Theorem for Integrals: } \exists c \in [a, b] : f(c) = f_{\text{avg}}$$

STATED-EQUATION f average equals one over b minus a times the integral from a to b of f of x dx . Mean value theorem for integrals: there is a c in the interval with f of c equal to the average.

MEMORY AID "Average = total over length."

TERMS

$b - a$ – length of the interval

$\int_a^b f$ – total accumulation [C1.18]

Discrete analog – the arithmetic mean [S.1]

MVT – requires f continuous [C1.3]; compare [C1.14]

CONTEXT

- **Origin.** A consequence of the Fundamental Theorem of Calculus.
- **Use.** Average temperature, average velocity, average power.
- **Data science.** If $X \sim \text{Uniform}(a, b)$ then $E[f(X)] = \frac{1}{b-a} \int_a^b f(x) dx$ [S.20]; Monte Carlo integration estimates it by averaging f at random points.
- **Physical application.** The average temperature over a day is the integral of the temperature curve divided by 24 hours.

C2.9 Sequences

$$\{a_n\}_{n=1}^{\infty} \qquad \lim_{n \rightarrow \infty} a_n = L \qquad \lim_{n \rightarrow \infty} r^n = 0 \ (|r| < 1) \qquad \text{bounded and monotonic} \implies \text{convergent}$$

STATED-EQUATION A sequence a_n converges to L if the limit as n goes to infinity of a_n equals L . r to the n goes to zero when the absolute value of r is less than one. A bounded, monotonic sequence converges.

MEMORY AID Growth order: logs $\ll$ powers $\ll$ exponentials $\ll$ factorials.

TERMS

a_n – n th term; a function on the positive integers

Converges – terms approach a finite limit [C1.1]

Squeeze – if $b_n \leq a_n \leq c_n$ and $b_n, c_n \rightarrow L$, then $a_n \rightarrow L$

Growth order – $\ln n \ll n^p \ll b^n \ll n! \ll n^n$

Examples – arithmetic [A2.9], geometric [A2.10]

CONTEXT

- **Origin.** Made rigorous by Cauchy and Bolzano (early 1800s).
- **Use.** Iterative algorithms, recurrences, compound interest.
- **Data science.** Learning-rate schedules $\eta_n = \frac{\eta_0}{1+kn}$ and estimator sequences $\hat{\theta}_n$ are sequences; "converges" is a statement about their limit.
- **Physical application.** The amount of a drug in the bloodstream just after each daily dose forms a sequence that converges to a steady level.

C2.10 Series & Basic Convergence Tests

$$S_N = \sum_{n=1}^N a_n \quad \sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} S_N \quad \lim_{n \rightarrow \infty} a_n \neq 0 \Rightarrow \text{diverges} \quad \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges} \iff p > 1 \quad \sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad (|r| < 1)$$

$$\text{Integral test: } \sum f(n) \text{ and } \int_1^{\infty} f(x) dx \text{ converge together} \quad \text{Comparison: } 0 \leq a_n \leq b_n, \sum b_n < \infty \Rightarrow \sum a_n < \infty$$

STATED-EQUATION S_N equals the sum of the first N terms; the series equals the limit of S_N as N goes to infinity. If the limit of a_n is not zero, the series diverges. The p -series, sum of one over n to the p , converges if and only if p is greater than one. The geometric series, sum of a times r to the n from zero, equals a over one minus r when the absolute value of r is less than one. Integral test and comparison test as stated.

MEMORY AID "If the terms don't go to zero, the series won't go."

TERMS

S_N – N th partial sum [A2.11]

n th-term test – can only prove divergence

p -series – $p = 1$ is the harmonic series, which diverges

Geometric – see [A2.10]

Integral test – f positive, decreasing, continuous [C2.4]

Limit comparison – $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0 \Rightarrow$ same behavior

CONTEXT

- **Origin.** Nicole Oresme proved the harmonic series diverges (~1350); Euler found $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ (1734).
- **Use.** Deciding whether an infinite sum has a finite value.
- **Data science.** Robbins–Monro conditions for SGD: step sizes need $\sum \eta_n = \infty$ yet $\sum \eta_n^2 < \infty$; $\eta_n = \frac{1}{n}$ works because of the harmonic and $p = 2$ series.
- **Physical application.** Zeno's paradox resolved: walking half the remaining distance forever covers $\frac{1}{2} + \frac{1}{4} + \dots = 1$ total.

C2.11 Ratio, Root & Alternating Series Tests

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad \text{or} \quad L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \quad \begin{cases} L < 1 & \text{converges absolutely} \\ L > 1 & \text{diverges} \\ L = 1 & \text{inconclusive} \end{cases}$$

$$\text{Alternating: } \sum (-1)^n b_n \text{ converges if } b_{n+1} \leq b_n \text{ and } b_n \rightarrow 0 \quad |S - S_N| \leq b_{N+1}$$

STATED-EQUATION L equals the limit of the absolute value of a_n plus one over a_n , or the limit of the n -th root of the absolute value of a_n . If L is less than one the series converges absolutely; greater than one, it diverges; equal to one, the test is inconclusive. An alternating series converges if its terms decrease to zero, and the error after N terms is at most b_{N+1} plus one.

TERMS

Ratio test – best with factorials and exponentials

Root test – best when a_n has the form $(\cdot)^n$

Absolute – $\sum |a_n|$ converges [C2.10]

Conditional – $\sum a_n$ converges but $\sum |a_n|$ does not

b_{N+1} – first omitted term bounds the error

CONTEXT

- **Origin.** Ratio test by d'Alembert (1768); alternating test by Leibniz (1705).
- **Use.** Radius of convergence of power series [C2.12]; error bounds.
- **Data science.** Factorials beat exponentials, so $\sum \frac{\lambda^k}{k!}$ converges and Poisson probabilities sum to 1 [S.11]; alternating bounds give guaranteed accuracy for truncated approximations.
- **Physical application.** Truncating the series for $\sin x$ in a calculator: the alternating-series bound certifies the error.

C2.12 Power Series & Radius of Convergence

$$\sum_{n=0}^{\infty} c_n(x-a)^n \quad R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| \quad \text{converges for } |x-a| < R \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1)$$

STATED-EQUATION A power series is the sum of c_n times x minus a to the n . The radius R equals the limit of the absolute value of c_n over c_{n+1} plus one; the series converges for the absolute value of x minus a less than R . One over one minus x equals the sum of x to the n for the absolute value of x less than one.

TERMS

c_n – coefficients; a the center

R – radius of convergence, found with the ratio test [C2.11]

Endpoints – test $x = a \pm R$ separately

Term by term – differentiate [C1.5] or integrate [C1.16] inside $|x-a| < R$

Geometric source – $\frac{1}{1-x}$ is a geometric series [A2.10]

CONTEXT

- **Origin.** Newton’s power series (1665); convergence theory by Abel and Cauchy.
- **Use.** Representing functions as “infinite polynomials.”
- **Data science.** Probability generating functions $G(s) = \sum_k P(X = k) s^k$ encode a distribution; $G'(1) = E[X]$ [S.9].
- **Physical application.** Physicists expand potentials in power series to approximate forces near equilibrium.

C2.13 Taylor & Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots \quad |R_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \quad \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

STATED-EQUATION f of x equals the sum of f n -th derivative at a over n factorial, times x minus a to the n : f of a plus f prime of a times x minus a plus f double prime of a over two factorial times x minus a squared, and so on. The remainder is at most M times the absolute value of x minus a to the n plus one, over n plus one factorial. e to the x equals the sum of x to the n over n factorial; sine x equals the sum of minus one to the n x to the two n plus one over two n plus one factorial; cosine x uses the even powers; natural log of one plus x equals the sum of minus one to the n plus one x to the n over n .

MEMORY AID sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

TERMS

$f^{(n)}(a)$ – n th derivative at the center [C1.5]

$n!$ – factorial [S.8]

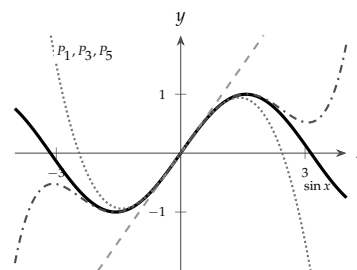
Maclaurin – Taylor series with $a = 0$

M – bound on $|f^{(n+1)}|$ between a and x (Lagrange remainder)

Degree 1 – the linearization [C1.11]

CONTEXT

- **Origin.** Brook Taylor (1715); Colin Maclaurin (1742); Gregory and Newton earlier.
- **Use.** How calculators compute $\sin$, e^x , $\ln$; physics approximations.
- **Data science.** Newton’s method and XGBoost use a 2nd-order Taylor expansion of the loss (gradient and Hessian); the delta method uses a 1st-order expansion to get standard errors of transformed estimates [S.13].
- **Physical application.** The pendulum period formula $T \approx 2\pi\sqrt{L/g}$ comes from the first Taylor term $\sin \theta \approx \theta$.



C2.14 Parametric Equations

$$x = f(t), \quad y = g(t) \qquad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \left(\frac{dx}{dt} \neq 0 \right) \qquad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

STATED-EQUATION x equals f of t , y equals g of t . dy/dx equals dy/dt over dx/dt , when dx/dt is not zero. The second derivative d^2y/dx^2 equals the derivative with respect to t of dy/dx , divided by dx/dt .

TERMS

t – parameter (often time); gives the curve a direction

Slope formula – from the chain rule [C1.7]

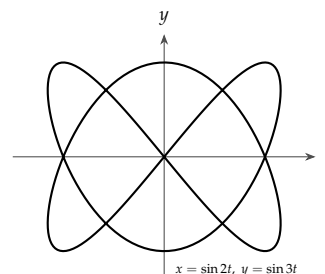
Eliminate t – solve for t or use $\sin^2 + \cos^2 = 1$ [T.4]

Arc length – see [C2.7]

Horizontal tangent – $dy/dt = 0$; vertical: $dx/dt = 0$

CONTEXT

- **Origin.** Used by Euler and in the study of projectile and planetary motion.
- **Use.** Trajectories, animation, and robot paths.
- **Data science.** A ROC curve is parametric in the threshold τ : $(\text{FPR}(\tau), \text{TPR}(\tau))$; precision–recall curves work the same way.
- **Physical application.** A projectile's path $x = v_0 t \cos \theta$, $y = v_0 t \sin \theta - \frac{1}{2} g t^2$ is parametric in time.



C2.15 Polar Area & Calculus

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta \qquad \frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

STATED-EQUATION A equals one half the integral from α to β of r squared $d\theta$. The slope dy/dx equals $dr/d\theta$ sine θ plus r cosine θ , over $dr/d\theta$ cosine θ minus r sine θ .

MEMORY AID Polar area is a sum of thin pizza slices: $\frac{1}{2} r^2 d\theta$.

TERMS

$r = f(\theta)$ – polar curve [T.11]

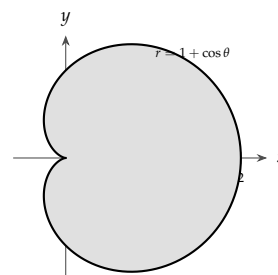
$\frac{1}{2} r^2 d\theta$ – area of a thin sector [G.9]

α, β – angle limits; one full loop or one petal

Slope – treat $x = r \cos \theta$, $y = r \sin \theta$ as parametric [C2.14]

CONTEXT

- **Origin.** Area by sectors goes back to Archimedes' spiral; calculus form from Newton and the Bernoullis.
- **Use.** Antenna radiation patterns, orbits, rose and cardioid curves.
- **Data science.** Radar (spider) charts encode metrics as polar radii; their enclosed area depends on axis order, a known pitfall for model-comparison dashboards.
- **Physical application.** The area swept by a planet's radius line (Kepler's second law) is a polar area integral.



C2.16 Differential Equations: Separable, Growth & Logistic

$$\frac{dy}{dx} = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x) dx \quad \frac{dy}{dt} = ky \Rightarrow y = y_0 e^{kt}$$

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right) \Rightarrow P(t) = \frac{K}{1 + Ae^{-rt}}, \quad A = \frac{K - P_0}{P_0} \quad \frac{dT}{dt} = -k(T - T_a)$$

STATED-EQUATION If $\frac{dy}{dx}$ equals g of x times h of y , then the integral of $\frac{dy}{h(y)}$ equals the integral of $g(x)$ dx . $\frac{dy}{dt}$ equals ky gives y equals y_0 naught e to the k t . The logistic equation $\frac{dP}{dt}$ equals rP times one minus P over K gives P of t equals K over one plus Ae to the minus r t , with A equals K minus P naught over P naught. Newton's law of cooling: $\frac{dT}{dt}$ equals minus k times T minus T_a .

MEMORY AID Separable: "get all the y 's with dy , all the x 's with dx , then integrate both sides."

TERMS

k – growth ($k > 0$) or decay ($k < 0$) rate [A2.6]

y_0, P_0 – initial value (initial condition)

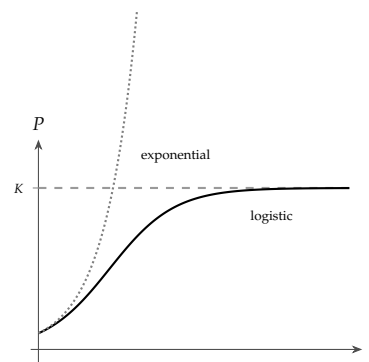
r – intrinsic growth rate; K carrying capacity

Solving – separate, integrate [C1.16], partial fractions for logistic [C2.3]

T_a – ambient temperature (Newton's law of cooling)

CONTEXT

- **Origin.** Logistic model by Pierre-François Verhulst (1838); separation of variables by Leibniz and Bernoulli.
- **Use.** Population, radioactive decay, cooling, epidemics (SIR).
- **Data science.** The logistic solution is the sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ of logistic regression and neural nets; adoption curves are fit with it; neural ODEs learn $\frac{dy}{dt}$ directly.
- **Physical application.** Newton's law of cooling predicts how fast coffee cools; logistic growth models bacteria in a dish reaching its capacity [P.14].



C2.17 Euler's Method for Differential Equations

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0 : \quad y_{n+1} = y_n + h f(t_n, y_n) \quad t_{n+1} = t_n + h \quad \text{global error} = O(h)$$

STATED-EQUATION Given $\frac{dy}{dt}$ equals f of t and y , with y at t naught equal to y naught: y n plus one equals y n plus h times f of t n and y n , and t n plus one equals t n plus h . The global error is of order h .

MEMORY AID "New value = old value + step $\times$ slope."

TERMS

h – step size

$f(t_n, y_n)$ – slope from the differential equation [C2.16]

Idea – follow the tangent line for one step [C1.11]

$O(h)$ – halving h roughly halves the error (first-order method)

Better methods – Runge-Kutta 4 (error $O(h^4)$), adaptive step sizes

CONTEXT

- **Origin.** Leonhard Euler, *Institutionum calculi integralis* (1768).
- **Use.** Simulating any ODE that has no closed-form solution.
- **Data science.** Epidemic (SIR) and population simulations; a residual network layer $\mathbf{x}_{k+1} = \mathbf{x}_k + f(\mathbf{x}_k)$ is one Euler step, the idea behind neural ODEs.
- **Physical application.** Game and physics engines advance position and velocity each frame with Euler-type steps: $x \leftarrow x + v \Delta t$, $v \leftarrow v + a \Delta t$.

C2.18 Numerical Integration: Trapezoid & Simpson's Rules

$$T_n = \frac{h}{2} [f_0 + 2f_1 + 2f_2 + \cdots + 2f_{n-1} + f_n] \quad S_n = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n] \quad (n \text{ even}) \quad h = \frac{b-a}{n}$$

STATED-EQUATION The trapezoid rule: T_n equals h over two times the quantity f naught plus two f one plus two f two, and so on, plus two f n minus one plus f n . Simpson's rule: S_n equals h over three times f naught plus four f one plus two f two plus four f three, alternating, plus four f n minus one plus f n , for even n . h equals b minus a over n .

MEMORY AID Simpson's weights: 1, 4, 2, 4, 2, $\dots$, 4, 1.

TERMS

$f_i = f(a + ih)$, function values at equally spaced points

Trapezoid – straight lines between points (compare [6.8], [C1.17]); error $O(h^2)$

Simpson – parabolas through triples of points [A1.12]; error $O(h^4)$, exact for cubics

h – spacing

CONTEXT

- **Origin.** Trapezoid rule known to Babylonian astronomers; Simpson's rule by Thomas Simpson (1743), earlier by Kepler.
- **Use.** Integrating tabulated or measured data.
- **Data science.** `numpy.trapz` computes ROC AUC and cumulative metrics from sampled curves [6.8].
- **Physical application.** Estimating the total distance of a run from speed readings every 10 seconds.

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07 Linear Algebra

Vectors, bases, and the matrices that transform them. Linear algebra is the working language of data science: every dataset is a matrix, every model a transformation, and every embedding a vector in some basis.

LA.1 Vectors & Linear Combinations

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n) \quad c\mathbf{v} = (cv_1, \dots, cv_n) \quad \mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k$$

STATED-EQUATION $\mathbf{u}$ plus $\mathbf{v}$ is the vector of componentwise sums. c times $\mathbf{v}$ scales every component by c . $\mathbf{w}$ equals c one $\mathbf{v}$ one plus c two $\mathbf{v}$ two, and so on through c k $\mathbf{v}$ k : a linear combination of the vectors $\mathbf{v}$ one through $\mathbf{v}$ k .

MEMORY AID Add tip-to-tail; scaling stretches (and a negative scalar flips).

TERMS

$\mathbf{v} \in \mathbb{R}^n$ – an ordered list of n numbers; an arrow from the origin [T.12](#)

c_i – scalars (weights) of the combination

$\mathbf{0}$ – zero vector

Geometry – addition is the parallelogram rule

Matrix form – $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = V\mathbf{c}$ with the $\mathbf{v}_i$ as columns [A2.13](#)

CONTEXT

- **Origin.** Grassmann (1844) and Hamilton; vectors in physics popularized by Gibbs and Heaviside (1880s).
- **Use.** Forces, velocities, displacements, colors (RGB), coordinates.
- **Data science.** A row of features is a vector; a linear model's prediction $\mathbf{w} \cdot \mathbf{x}$ is a weighted combination; word-embedding arithmetic (king – man + woman $\approx$ queen) is vector addition.
- **Physical application.** Two tugboats pulling a ship: the ship moves along the vector sum of their forces.

LA.2 Span, Linear Independence, Basis & Dimension

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k\} \quad \text{independent} \iff (c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0} \Rightarrow c_1 = \dots = c_k = 0) \quad \dim V = \# \text{ vectors in any basis}$$

STATED-EQUATION The span of $\mathbf{v}$ one through $\mathbf{v}$ k is the set of all their linear combinations. The vectors are linearly independent if and only if the only combination equal to the zero vector is the one with every coefficient zero. The dimension of V is the number of vectors in any basis.

MEMORY AID A basis is a “minimal complete set”: enough vectors to reach everything, none redundant. n vectors in $\mathbb{R}^n$ form a basis $\iff \det \neq 0$.

TERMS

Span – everything reachable by combinations [LA.1](#)

Dependent – some vector is a combination of the others (redundant)

Basis – independent set that spans the space

Standard basis – $\mathbf{e}_1 = (1, 0, \dots)$, $\mathbf{e}_2 = (0, 1, \dots)$, ...

Dimension – number of basis vectors; $\dim \mathbb{R}^n = n$

Test – square matrix of the vectors has $\det \neq 0$ [A2.14](#)

CONTEXT

- **Origin.** Formalized by Peano (1888) and Steinitz (1913).
- **Use.** Coordinates, degrees of freedom, solving systems.
- **Data science.** Multicollinearity is linear dependence among features; one-hot encoding with all categories plus an intercept is dependent (the “dummy-variable trap”).
- **Physical application.** A point on a flat table needs 2 coordinates (dimension 2); a drone in the air needs 3.

LA.3 Change of Basis

$$P = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n] \quad \mathbf{v} = P[\mathbf{v}]_B \quad [\mathbf{v}]_B = P^{-1}\mathbf{v} \quad [A]_B = P^{-1}AP$$

STATED-EQUATION P is the matrix whose columns are the new basis vectors $\mathbf{b}$ one through $\mathbf{b}_n$. $\mathbf{v}$ equals P times the coordinates of $\mathbf{v}$ in basis B . The coordinates of $\mathbf{v}$ in B equal P inverse times $\mathbf{v}$. The matrix of the transformation A in basis B equals P inverse A P .

MEMORY AID “ P takes you from the new basis to the old (standard) one; P^{-1} takes you back.”

TERMS

$B = \{\mathbf{b}_i\}$ – new basis [LA.2]

$[\mathbf{v}]_B$ – coordinates of $\mathbf{v}$ relative to B

P – change-of-basis matrix; invertible because its columns are independent [A2.14]

$P^{-1}AP$ – same transformation described in new coordinates (a similar matrix)

Best basis – eigenvectors make A diagonal [LA.10]

CONTEXT

- **Origin.** Cayley and Sylvester’s matrix theory (1850s).
- **Use.** Rotating coordinate frames, simplifying transformations.
- **Data science.** PCA is a change of basis to the principal axes [ML.19]; whitening, Fourier and wavelet transforms re-express data in a more useful basis.
- **Physical application.** Engineers describe forces on a ramp in tilted axes (along and perpendicular to the slope) instead of horizontal and vertical.

LA.4 Linear Transformations & Their Matrices

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v}) \quad T(\mathbf{x}) = A\mathbf{x} \quad A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)] \quad (S \circ T)(\mathbf{x}) = BA\mathbf{x}$$

STATED-EQUATION T of a $\mathbf{u}$ plus b $\mathbf{v}$ equals a T of $\mathbf{u}$ plus b T of $\mathbf{v}$. T of $\mathbf{x}$ equals A $\mathbf{x}$. The columns of A are T applied to the standard basis vectors $\mathbf{e}$ one through $\mathbf{e}_n$. Doing T then S corresponds to the product BA .

MEMORY AID “Columns are where the basis vectors land.”

TERMS

Linear – preserves addition and scaling; so $T(\mathbf{0}) = \mathbf{0}$

$\mathbf{e}_i$ – standard basis [LA.2]

Examples – rotation [G.13], reflection, scaling, projection, shear

Composition – matrix product, applied right to left [A2.1], [A2.13]

$|\det A|$ – factor by which areas/volumes scale [A2.14]

CONTEXT

- **Origin.** Grassmann; matrices as transformations by Cayley (1858).
- **Use.** Computer graphics, robotics, signal processing.
- **Data science.** Each neural-network layer applies a linear map $W\mathbf{x}$ followed by a nonlinearity [ML.14]; image augmentation uses rotation, shear and scaling matrices.
- **Physical application.** Stretching a rubber sheet evenly in one direction is a linear transformation; every drawn square becomes a parallelogram.

LA.5 Gaussian Elimination & Row Reduction

$$A \mid \mathbf{b} \xrightarrow{\text{row operations}} (\text{reduced}) \text{ row echelon form} \quad R_i \leftrightarrow R_j, \quad R_i \rightarrow cR_i \ (c \neq 0), \quad R_i \rightarrow R_i + cR_j$$

STATED-EQUATION Row reduce the augmented matrix A given $\mathbf{b}$ to row echelon form. The three row operations: swap two rows; multiply a row by a nonzero c ; add c times one row to another.

MEMORY AID Solutions: a pivot in every column $\rightarrow$ unique; a row $[0 \ \cdots \ 0 \mid c \neq 0] \rightarrow$ none; a free variable $\rightarrow$ infinitely many.

TERMS

Augmented matrix – coefficients with the right-hand side attached [A1.8]

Pivot – first nonzero entry of a row after elimination

Back substitution – solve from the bottom row up

RREF – pivots equal 1 with zeros above and below

Free variable – column without a pivot

Cost – about $\frac{2}{3}n^3$ operations (LU factorization)

CONTEXT

- **Origin.** Chinese text *Nine Chapters* (~200 BCE); named after Gauss (1810).
- **Use.** Solving any linear system, computing inverses and ranks.
- **Data science.** Linear-regression solvers, circuit and network analysis, and constraint systems behind LP [O.2].
- **Physical application.** Balancing a chemical equation such as $a\text{CH}_4 + b\text{O}_2 \rightarrow c\text{CO}_2 + d\text{H}_2\text{O}$ is a linear system in a, b, c, d .

LA.6 Rank, Column Space, Null Space & Rank–Nullity

$$\text{Col } A = \text{span}\{\text{columns}\} \quad \text{Null } A = \{x : Ax = 0\} \quad \text{rank } A = \dim \text{Col } A \quad \text{rank } A + \dim \text{Null } A = n$$

STATED-EQUATION The column space of A is the span of its columns. The null space of A is the set of x with Ax equal to zero. The rank of A is the dimension of the column space. Rank plus nullity equals n , the number of columns.

TERMS

Rank – number of pivots [LA.5]; number of independent columns (or rows)

$Ax = b$ solvable – $\iff b \in \text{Col } A$

Nullity – number of free variables

Full rank – rank = $\min(m, n)$; square full-rank $\iff$ invertible [A2.14]

n – number of columns (unknowns)

CONTEXT

- **Origin.** Rank introduced by Frobenius (1879); rank–nullity is the fundamental theorem of linear algebra (Strang’s name).
- **Use.** Counting independent equations and degrees of freedom.
- **Data science.** A rank-deficient design matrix means coefficients are not identifiable; low-rank structure is exploited by recommender systems and LoRA fine-tuning of LLMs.
- **Physical application.** A camera projecting 3-D space onto a 2-D image has a null space: every point along a line of sight maps to the same pixel.

LA.7 Orthogonality, Projections & Least Squares

$$u \perp v \iff u \cdot v = 0 \quad \text{proj}_a b = \frac{a \cdot b}{a \cdot a} a \quad A^T A \hat{x} = A^T b \quad \hat{b} = A(A^T A)^{-1} A^T b \quad Q^T Q = I$$

STATED-EQUATION u is perpendicular to v if and only if u dot v equals zero. The projection of b onto a equals a dot b over a dot a , times a . The normal equations: A transpose A hat equals A transpose b . The projection of b onto the column space is A times A transpose A inverse times A transpose b . An orthonormal matrix Q satisfies Q transpose Q equals the identity.

MEMORY AID Least squares: the error $b - A\hat{x}$ is perpendicular to every column of A .

TERMS

Orthogonal – dot product zero [T.12]

Orthonormal – orthogonal unit vectors; $Q^{-1} = Q^T$

Projection – closest point in a subspace

Normal equations – least-squares solution [S.19]

Residual – $b - A\hat{x}$, orthogonal to $\text{Col } A$ [LA.6]

CONTEXT

- **Origin.** Least squares by Legendre (1805) and Gauss (1809); projections formalized by Hilbert and von Neumann.
- **Use.** Fitting data, shadows, signal denoising.
- **Data science.** OLS regression is the projection of y onto the column space of X ; orthogonal features make coefficients independent of one another.
- **Physical application.** The shadow of a pole on flat ground at noon is its orthogonal projection onto the ground.

LA.8 Gram–Schmidt & QR Factorization

$$u_1 = v_1 \quad u_k = v_k - \sum_{j=1}^{k-1} \frac{u_j \cdot v_k}{u_j \cdot u_j} u_j \quad q_k = \frac{u_k}{\|u_k\|} \quad A = QR$$

STATED-EQUATION u_1 equals v_1 . Each next u_k equals v_k minus its projections onto all the previous u_j . Normalize: q_k equals u_k over its norm. The result factors A as Q times R , with Q orthonormal and R upper triangular.

MEMORY AID “Subtract the shadows, then normalize.”

TERMS

v_k – original independent vectors [LA.2]

Projection – [LA.7]

q_k – orthonormal basis for the same span

Q – orthonormal columns; R upper triangular

Use – solve least squares stably: $R\hat{x} = Q^T b$

CONTEXT

- **Origin.** Gram (1883) and Schmidt (1907); Laplace used it earlier.
- **Use.** Building orthonormal bases, numerically stable solving.
- **Data science.** `numpy.linalg.lstsq` and `statsmodels` use QR or SVD instead of forming $X^T X$, which would square the condition number [LA.13].
- **Physical application.** Setting up perpendicular axes on a tilted surveying site from two measured, non-perpendicular directions.

LA.9 Eigenvalues & Eigenvectors

$$A\mathbf{v} = \lambda\mathbf{v} \ (\mathbf{v} \neq \mathbf{0}) \quad \det(A - \lambda I) = 0 \quad \text{tr } A = \sum_i \lambda_i \quad \det A = \prod_i \lambda_i \quad 2 \times 2: \lambda^2 - (\text{tr } A)\lambda + \det A = 0$$

STATED-EQUATION $A\mathbf{v}$ equals $\lambda\mathbf{v}$, for a nonzero vector $\mathbf{v}$. The eigenvalues solve the characteristic equation: the determinant of A minus λI equals zero. The trace of A equals the sum of the eigenvalues, and the determinant equals their product. For two-by-two: $\lambda^2 - (\text{tr } A)\lambda + \det A = 0$.

MEMORY AID “Eigen” is German for “own”: an eigenvector keeps its own direction; λ is the stretch factor.

TERMS

λ – eigenvalue: how much $\mathbf{v}$ is stretched (negative flips it)

$\mathbf{v}$ – eigenvector; solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ [LA.6]

Characteristic polynomial – $\det(A - \lambda I)$ [A2.14]

Trace – sum of the diagonal entries

Triangular A – eigenvalues are the diagonal entries

Complex λ – indicate rotation [A2.3]

CONTEXT

- **Origin.** Euler and Lagrange (rotating bodies); Cauchy (1829); “eigen” by Hilbert (1904).
- **Use.** Vibration modes, stability, quantum mechanics, Markov chains.
- **Data science.** PCA [ML.19], Google’s PageRank (the dominant eigenvector of the link matrix), spectral clustering, and stability of RNN training.
- **Physical application.** A bridge’s natural vibration frequencies are eigenvalues; soldiers break step on bridges so they don’t excite one.

LA.10 Diagonalization & Matrix Powers

$$A = PDP^{-1} \quad A^k = PD^kP^{-1} \quad D = \text{diag}(\lambda_1, \dots, \lambda_n) \quad A = A^\top \Rightarrow A = Q\Lambda Q^\top \text{ (spectral theorem)}$$

STATED-EQUATION A equals PDP inverse, where the columns of P are eigenvectors and D is the diagonal matrix of eigenvalues. A to the k equals P D to the k P inverse. If A is symmetric, A equals $Q\Lambda Q$ transpose with Q orthonormal: the spectral theorem.

MEMORY AID In the eigenbasis, a matrix just stretches each axis; powers become powers of numbers.

TERMS

P – eigenvector matrix [LA.9]; a change of basis [LA.3]

D – diagonal eigenvalue matrix

Diagonalizable – n independent eigenvectors

Spectral theorem – real symmetric matrices have real λ and orthonormal eigenvectors [LA.7]

Steady state – Markov chain: eigenvector with $\lambda = 1$ [S.6]

CONTEXT

- **Origin.** Jacobi (1846) for symmetric matrices; Jordan (1870) in general.
- **Use.** Solving linear recurrences and systems of ODEs, long-run behavior.
- **Data science.** PageRank and Markov-chain steady states; the long-run share of customers in each state of a churn model.
- **Physical application.** Coupled pendulums or masses on springs decouple into independent “normal modes” in the eigenbasis.

LA.11 Symmetric, Positive Definite Matrices & Quadratic Forms

$$q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j \quad A \succ 0 \iff \mathbf{x}^\top A \mathbf{x} > 0 \ \forall \mathbf{x} \neq \mathbf{0} \iff \text{all } \lambda_i > 0 \iff \text{all leading principal minors} > 0$$

STATED-EQUATION The quadratic form q of $\mathbf{x}$ equals $\mathbf{x}$ transpose A $\mathbf{x}$, the sum over i and j of $a_{ij} x_i x_j$. A is positive definite if and only if $\mathbf{x}$ transpose A $\mathbf{x}$ is positive for every nonzero $\mathbf{x}$, if and only if all eigenvalues are positive, if and only if all leading principal minors are positive.

MEMORY AID Positive definite = a bowl (unique minimum); indefinite = a saddle [VS.5].

TERMS

$A = A^\top$ – symmetric

$\succ 0, \succeq 0$ – positive definite, positive semidefinite (all $\lambda \geq 0$)

Leading principal minors – determinants of top-left $1 \times 1, 2 \times 2, \dots$ blocks (Sylvester’s criterion)

Indefinite – eigenvalues of both signs

Examples – covariance matrices [S.18] and $X^\top X$ are PSD

CONTEXT

- **Origin.** Sylvester (1852); quadratic forms studied by Lagrange and Gauss.
- **Use.** Energy functions, stability, second-derivative tests.
- **Data science.** Convexity of losses [O.8], valid kernels and covariance matrices must be PSD; Cholesky factorization of a PD matrix samples correlated Gaussians.
- **Physical application.** The stiffness matrix of a stable structure is positive definite: every deformation stores positive energy.

LA.12 Singular Value Decomposition (SVD)

$$A = U\Sigma V^T \quad \sigma_i = \sqrt{\lambda_i(A^T A)} \quad A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T \quad \|A\|_2 = \sigma_1$$

STATED-EQUATION A equals U Σ V transpose. The singular values σ_i are the square roots of the eigenvalues of A transpose A . The best rank- k approximation keeps the k largest terms, σ_i u_i v_i transpose. The spectral norm of A equals the largest singular value.

MEMORY AID Every matrix is “rotate, stretch, rotate” (V^T , Σ , U).

TERMS

U, V – orthonormal matrices of left/right singular vectors [LA.7]

Σ – diagonal, $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$

Works for – any $m \times n$ matrix (unlike eigen-decomposition)

Eckart–Young – A_k minimizes $\|A - B\|$ over rank- k matrices B

Rank – number of nonzero σ_i [LA.6]

CONTEXT

► **Origin.** Beltrami and Jordan (1870s); Eckart–Young (1936); Golub–Kahan algorithm (1965).

► **Use.** Compression, pseudo-inverses, noise reduction.

► **Data science.** PCA via SVD [ML.19], latent semantic analysis, matrix-factorization recommenders (the Netflix Prize), and image compression.

► **Physical application.** A photo stored with only its top 50 singular values looks almost the same at a fraction of the size.

LA.13 Norms, Trace & Condition Number

$$\|\mathbf{x}\|_1 = \sum_i |x_i| \quad \|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2} \quad \|\mathbf{x}\|_\infty = \max_i |x_i| \quad \|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{tr}(A^T A)} \quad \kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

STATED-EQUATION The L_1 norm is the sum of absolute values. The L_2 norm is the square root of the sum of squares. The infinity norm is the largest absolute entry. The Frobenius norm is the square root of the sum of squared entries, which equals the square root of the trace of A transpose A . The condition number κ equals the largest singular value over the smallest.

MEMORY AID Large κ = “wobbly”: tiny changes in the data cause big changes in the answer.

TERMS

Norm – length: positive, scales, triangle inequality [T.12]

L_1 vs L_2 – lasso vs ridge penalties [ML.5]

Trace – sum of diagonal = $\sum \lambda_i$ [LA.9]

σ – singular values [LA.12]

κ – lose about $\log_{10} \kappa$ digits of accuracy when solving $A\mathbf{x} = \mathbf{b}$

CONTEXT

► **Origin.** Minkowski (1896) for L_p norms; condition number by Turing (1948).

► **Use.** Measuring error size, numerical stability.

► **Data science.** Multicollinear features make $X^T X$ ill-conditioned (huge κ), so coefficients swing wildly; ridge regression lowers κ .

► **Physical application.** City-block (Manhattan) distance is the L_1 norm; a crow’s flight is the L_2 norm.

LA.14 Matrix Algebra Rules & Identities

$$\begin{aligned}
 AI = IA = A \quad A(BC) &= (AB)C \quad A(B+C) = AB + AC \quad AB \neq BA \text{ (in general)} \quad (A^\top)^\top = A \quad (A+B)^\top = A^\top + B^\top \\
 (AB)^\top &= B^\top A^\top \quad (AB)^{-1} = B^{-1}A^{-1} \quad (A^\top)^{-1} = (A^{-1})^\top \quad (cA)^{-1} = \frac{1}{c}A^{-1} \quad AA^{-1} = A^{-1}A = I \\
 \det(AB) &= \det A \det B \quad \det(A^\top) = \det A \quad \det(A^{-1}) = \frac{1}{\det A} \quad \det(cA) = c^n \det A \quad \text{tr}(AB) = \text{tr}(BA) \quad \text{tr}(A+B) = \text{tr} A + \text{tr} B
 \end{aligned}$$

STATED-EQUATION A times the identity equals the identity times A equals A . Multiplication is associative and distributive, but AB generally does not equal BA . The transpose of a transpose is A ; the transpose of a sum is the sum of transposes. The transpose of AB equals B transpose A transpose, and the inverse of AB equals B inverse A inverse. The inverse of A transpose equals the transpose of A inverse. The inverse of cA is one over c times A inverse. A times A inverse equals the identity. The determinant of AB equals $\det A$ times $\det B$; $\det$ of A transpose equals $\det A$; $\det$ of A inverse equals one over $\det A$; $\det$ of cA equals c to the n times $\det A$ for an n -by- n matrix. The trace of AB equals the trace of BA , and trace is additive.

MEMORY AID “Socks and shoes”: to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

TERMS

I – identity matrix: 1s on the diagonal, 0s elsewhere

$A^\top$ – transpose: rows become columns [A2.13]

A^{-1} – inverse; exists iff $\det A \neq 0$ [A2.14]

n – size of the square matrix

Trace – sum of diagonal entries = $\sum \lambda_i$ [LA.9]

Not commutative – order matters: rotate then scale $\neq$ scale then rotate [LA.4]

CONTEXT

- **Origin.** Arthur Cayley, “A Memoir on the Theory of Matrices” (1858).
- **Use.** Simplifying and rearranging every matrix equation.
- **Data science.** Deriving the OLS solution, writing neural-network layers as products, and the “trace trick” $\mathbf{x}^\top A \mathbf{x} = \text{tr}(A \mathbf{x} \mathbf{x}^\top)$ used in expectations of quadratic forms.
- **Physical application.** In robotics, a chain of joint transformations $T = T_1 T_2 T_3$ must be multiplied in order; reversing it moves the arm somewhere else.

LA.15 Special Matrices

$$\begin{aligned}
 \text{diagonal } D \quad \text{triangular } (\det = \prod a_{ii}) \quad \text{symmetric } A &= A^\top \quad \text{skew } A^\top = -A \\
 \text{orthogonal } Q^\top Q &= I \quad \text{idempotent } P^2 = P \quad \text{nilpotent } N^k = 0 \quad \text{permutation}
 \end{aligned}$$

STATED-EQUATION A diagonal matrix has nonzero entries only on the diagonal. A triangular matrix’s determinant is the product of its diagonal entries. Symmetric means A equals A transpose; skew-symmetric means A transpose equals minus A . Orthogonal means Q transpose Q equals the identity. Idempotent means P squared equals P . Nilpotent means some power N to the k equals zero. A permutation matrix reorders rows.

MEMORY AID Orthogonal matrices are “rigid”: they rotate or reflect without stretching, so their inverse is just the transpose.

TERMS

Diagonal – inverse is $1/d_{ii}$; powers are easy [LA.10]

Triangular – eigenvalues on the diagonal [LA.9]; solved by back substitution [LA.5]

Symmetric – real eigenvalues, orthogonal eigenvectors [LA.10]

Orthogonal – $\det = \pm 1$; preserves lengths and angles [LA.7]

Idempotent – projection matrices [LA.7]

Permutation – an identity with rows reordered; also orthogonal

Sparse – mostly zeros; stored and multiplied efficiently

CONTEXT

- **Origin.** Named and classified by Cayley, Sylvester, Hermite and Frobenius (19th century).
- **Use.** Exploiting structure makes computation faster and more stable.
- **Data science.** Covariance matrices are symmetric PSD [LA.11]; PCA loadings form an orthogonal matrix [ML.19]; the OLS hat matrix $H = X(X^\top X)^{-1}X^\top$ is idempotent; user–item matrices are sparse.
- **Physical application.** A rotation matrix describing a spacecraft’s orientation is orthogonal: its inverse (to rotate back) is its transpose.

LA.16 Inner, Outer, Hadamard, Kronecker & Block Products

$$\mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i \quad \mathbf{u} \mathbf{v}^\top = [u_i v_j] \quad (A \odot B)_{ij} = a_{ij} b_{ij} \quad A \otimes B = \begin{bmatrix} a_{11} B & \cdots \\ \vdots & \ddots \end{bmatrix} \quad A \mathbf{x} = \sum_j x_j \mathbf{a}_j \quad AB = \sum_k \mathbf{a}_k \mathbf{b}_k^\top$$

STATED-EQUATION The inner product $\mathbf{u}^\top \mathbf{v}$ is a scalar, the sum of $u_i v_i$. The outer product $\mathbf{u} \mathbf{v}^\top$ is a matrix with entries $u_i v_j$. The Hadamard product multiplies entries elementwise. The Kronecker product replaces each entry a_{ij} with the block a_{ij} times B . $A \mathbf{x}$ is a combination of the columns of A weighted by the entries of $\mathbf{x}$. AB is the sum of outer products of columns of A with rows of B .

MEMORY AID Inner product “collapses” ($1 \times n$ times $n \times 1$ is 1×1); outer product “expands” ($n \times 1$ times $1 \times m$ is $n \times m$).

TERMS

Inner – dot product [T.12]

Outer – always rank 1 [LA.6]; SVD is a sum of these [LA.12]

Hadamard $\odot$ – same-shape elementwise product ($\mathbf{A} * \mathbf{B}$ in NumPy)

Kronecker $\otimes$ – size $(mp) \times (nq)$ for $m \times n$ and $p \times q$

Column view – $A \mathbf{x}$ is a linear combination [LA.1]

Block matrices – multiply blocks like entries when sizes match

CONTEXT

- **Origin.** Kronecker (1850s); Hadamard (1890s); the column and outer-product views popularized by Strang.
- **Use.** Organizing large computations, structured models.
- **Data science.** Attention scores $QK^\top$ and gating $(\sigma(\cdot) \odot \mathbf{h})$ in LSTMs use these products; NumPy’s `@` is matrix product while `*` is Hadamard, a common bug.
- **Physical application.** In quantum mechanics, the state of two combined systems is a Kronecker (tensor) product.

LA.17 Matrix Calculus Identities

$$\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a} \quad \nabla_{\mathbf{x}}(\mathbf{x}^\top A \mathbf{x}) = (A + A^\top) \mathbf{x} \quad (= 2A \mathbf{x} \text{ if symmetric}) \quad \nabla_{\mathbf{x}} \|\mathbf{x}\|^2 = 2\mathbf{x}$$

$$\nabla_{\beta} \|\mathbf{y} - X\beta\|^2 = -2X^\top (\mathbf{y} - X\beta) \quad \frac{\partial \operatorname{tr}(AB)}{\partial A} = B^\top \quad \frac{\partial \ln \det X}{\partial X} = X^{-\top}$$

STATED-EQUATION The gradient of a transpose $\mathbf{x}$ is $\mathbf{a}$. The gradient of the quadratic form $\mathbf{x}^\top A \mathbf{x}$ is A plus A transpose, times $\mathbf{x}$, which is two $A \mathbf{x}$ when A is symmetric. The gradient of the squared norm of $\mathbf{x}$ is two $\mathbf{x}$. The gradient of the squared residual norm, $\mathbf{y}$ minus $X\beta$, is minus two X transpose times $\mathbf{y}$ minus $X\beta$. The derivative of the trace of AB with respect to A is B transpose. The derivative of the log determinant of X is X inverse transpose.

MEMORY AID Matrix calculus mirrors scalar calculus: $\frac{d}{dx}(ax) = a$, $\frac{d}{dx}(ax^2) = 2ax$; just watch transposes and shapes.

TERMS

$\nabla_{\mathbf{x}}$ – gradient with respect to a vector [V.2]

Shapes – gradient has the same shape as the variable

Quadratic form – [LA.11]

Residual – $\mathbf{y} - X\beta$ [S.19]

Setting to zero – gives the normal equations $X^\top X\beta = X^\top \mathbf{y}$ [LA.7]

Log-determinant – appears in Gaussian log-likelihoods [S.21]

CONTEXT

- **Origin.** Developed for statistics and econometrics (Dwyer 1948; Magnus and Neudecker 1988).
- **Use.** Deriving estimators and gradients in closed form.
- **Data science.** Deriving OLS and ridge solutions [ML.5], Gaussian MLE for covariance matrices, and backpropagation formulas for linear layers [ML.14].
- **Physical application.** The stiffness-weighted energy $\frac{1}{2} \mathbf{x}^\top K \mathbf{x}$ of a structure has gradient $K \mathbf{x}$, the restoring force vector.

LA.18 Pseudoinverse & Inverse-Update Identities

$$A^+ = (A^T A)^{-1} A^T \text{ (full column rank)} \quad A^+ = V \Sigma^+ U^T \quad (A + \mathbf{u}\mathbf{v}^T)^{-1} = A^{-1} - \frac{A^{-1} \mathbf{u} \mathbf{v}^T A^{-1}}{1 + \mathbf{v}^T A^{-1} \mathbf{u}} \quad (A + U C V)^{-1} = A^{-1} - A^{-1} U (C^{-1} + V A^{-1} U)^{-1} V A^{-1}$$

STATED-EQUATION For a matrix with full column rank, the Moore-Penrose pseudoinverse A^+ plus equals A transpose A inverse, times A transpose. In general, A^+ plus equals V Sigma plus U transpose, inverting the nonzero singular values. The Sherman-Morrison formula: the inverse of A plus u v transpose equals A inverse minus A inverse u v transpose A inverse, over one plus v transpose A inverse u . The Woodbury identity generalizes it to a low-rank update $U C V$.

MEMORY AID A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

TERMS

A^+ – pseudoinverse: gives the least-squares (minimum-norm) solution $\mathbf{x} = A^+ \mathbf{b}$ [LA.7]

Σ^+ – replace each nonzero σ_i by $1/\sigma_i$ and transpose [LA.12]

$\mathbf{u}\mathbf{v}^T$ – rank-1 update [LA.16]

Woodbury – inverse of a low-rank update

Cost – [0.17]

CONTEXT

► **Origin.** Moore (1920), Penrose (1955); Sherman–Morrison (1949–50); Woodbury (1950).

► **Use.** Solving non-square and singular systems; fast updates.

► **Data science.** `numpy.linalg.pinv` and `lstsq`; recursive least squares and online regression update inverses one data point at a time; Gaussian-process and Kalman-filter computations use Woodbury.

► **Physical application.** GPS receivers solve an overdetermined system of satellite ranges with the pseudoinverse (least squares).

08

Multivariable & Vector Calculus

Calculus in several dimensions: partial derivatives, gradients, multiple integrals, and the fields and flux theorems of physics. Gradients drive every optimizer; Jacobians and Hessians appear throughout machine learning.

V.1 Partial Derivatives

$$\frac{\partial f}{\partial x} = f_x = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \quad \frac{\partial^2 f}{\partial y \partial x} = f_{xy} \quad f_{xy} = f_{yx} \text{ (Clairaut, if continuous)}$$

STATED-EQUATION The partial derivative of f with respect to x , f_x , is the limit as h goes to zero of f of x plus h , y minus f of x , y , over h , holding y fixed. The mixed partial f_{xy} means differentiate by x , then by y . Clairaut's theorem: f_{xy} equals f_{yx} when they are continuous.

MEMORY AID "Freeze the other variables, then differentiate as usual."

TERMS

$f(x, y)$ – function of several variables (a surface)
 ∂ – "partial": other variables held constant
 f_x – slope of the surface in the x -direction [C1.4]
Mixed partials – order usually does not matter
Rules – all single-variable rules apply [C1.5]–[C1.8]

CONTEXT

- **Origin.** Euler, Clairaut and d'Alembert (1740s); the ∂ symbol by Legendre and Jacobi.
- **Use.** Rates of change of quantities depending on several inputs.
- **Data science.** Each partial derivative of the loss with respect to a weight is one entry of the gradient that backpropagation computes [ML.14].
- **Physical application.** The ideal-gas pressure $P = nRT/V$ has $\partial P / \partial T = nR/V$: how pressure rises with temperature at fixed volume.

V.2 Gradient & Directional Derivative

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \quad D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} \quad (\|\mathbf{u}\| = 1) \quad \max_{\mathbf{u}} D_{\mathbf{u}} f = \|\nabla f\| \quad \nabla f \perp \text{level sets}$$

STATED-EQUATION The gradient of f , ∇f , is the vector of partial derivatives. The directional derivative in the unit direction $\mathbf{u}$ equals the gradient dot $\mathbf{u}$. The largest directional derivative equals the length of the gradient, in the gradient's direction. The gradient is perpendicular to the level curves.

MEMORY AID The gradient points straight uphill; go the opposite way to descend [C1.13].

TERMS

∇ – "del" or "nabla" operator
 $\mathbf{u}$ – unit direction vector [T.12]
 $D_{\mathbf{u}} f$ – rate of change moving in direction $\mathbf{u}$
 $\|\nabla f\|$ – steepest slope
Level set – $f = c$ (contour line)

CONTEXT

- **Origin.** Hamilton introduced ∇ (1837); gradient named by Maxwell's era (1870s).
- **Use.** Steepest ascent, contour maps, normal vectors.
- **Data science.** Gradient descent on a loss surface [ML.16]; saliency maps use ∇_x of the output to show which pixels matter.
- **Physical application.** Heat flows opposite the temperature gradient [P.13]; water runs down the negative gradient of terrain height.

V.3 Multivariable Chain Rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad \frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \quad \frac{d}{dt} f(\mathbf{x}(t)) = \nabla f \cdot \mathbf{x}'(t)$$

STATED-EQUATION $\frac{dz}{dt}$ equals partial z partial x times $\frac{dx}{dt}$, plus partial z partial y times $\frac{dy}{dt}$. Similarly for partial z partial s . In vector form, the derivative of f of $\mathbf{x}$ of t equals the gradient of f dot $\mathbf{x}'$ of t .

MEMORY AID Sum over every path from the input to the output: multiply along each path, add the paths.

TERMS

Paths – each intermediate variable contributes one term

Single-variable case – [C1.7](#)

Vector form – gradient dot velocity [V2](#)

Computation graph – backpropagation applies this rule node by node

[ML.14](#)

CONTEXT

- **Origin.** Leibniz and Euler; systematized by Jacobi.
- **Use.** Related rates with several variables, thermodynamics.
- **Data science.** Backpropagation is the multivariable chain rule on a computation graph; automatic differentiation (PyTorch, JAX) implements it.
- **Physical application.** Temperature felt by a moving car: $\frac{dT}{dt} = \frac{\partial T}{\partial x} \frac{dx}{dt} + \frac{\partial T}{\partial t}$, change along the road plus change over time.

V.4 The Jacobian

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} \quad \mathbf{f}(\mathbf{x} + \mathbf{h}) \approx \mathbf{f}(\mathbf{x}) + J\mathbf{h} \quad dx dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

STATED-EQUATION The Jacobian J is the matrix of all first partial derivatives: row i , column j is partial f_i partial x_j . $\mathbf{f}$ of $\mathbf{x}$ plus $\mathbf{h}$ is approximately $\mathbf{f}$ of $\mathbf{x}$ plus $J\mathbf{h}$. Under a change of variables, $dx dy$ equals the absolute value of the Jacobian determinant times $du dv$.

MEMORY AID Rows are outputs, columns are inputs; $|\det J|$ is the local area-stretch factor.

TERMS

$\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ – vector-valued function

J – $m \times n$; best linear approximation [C1.11](#), [LA.4](#)

$\det J$ – area/volume scale factor [A2.14](#)

Gradient – Jacobian of a scalar function is $\nabla f^\top$ [V2](#)

Chain rule – $J_{g \circ f} = J_g J_f$ [V3](#)

CONTEXT

- **Origin.** Carl Jacobi (1841).
- **Use.** Coordinate changes, robotics kinematics, nonlinear systems.
- **Data science.** Normalizing flows use $\log |\det J|$ to transform densities; backprop multiplies Jacobians layer by layer.
- **Physical application.** A robot arm's Jacobian converts joint rotation speeds into the speed of its hand.

v.5 Hessian & the Second-Derivative Test

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad D = \det H = f_{xx}f_{yy} - f_{xy}^2 \quad \begin{cases} D > 0, f_{xx} > 0 & \text{local min} \\ D > 0, f_{xx} < 0 & \text{local max} \\ D < 0 & \text{saddle point} \\ D = 0 & \text{inconclusive} \end{cases}$$

STATED-EQUATION The Hessian H is the matrix of second partial derivatives. D , its determinant, equals $f_{xx}f_{yy} - f_{xy}^2$. If D is positive and f_{xx} is positive, it is a local minimum; if D is positive and f_{xx} negative, a local maximum; if D is negative, a saddle point; if D is zero, the test is inconclusive.

MEMORY AID Hessian positive definite $\rightarrow$ bowl (min); negative definite $\rightarrow$ dome (max); mixed $\rightarrow$ saddle [LA.11].

TERMS

Critical point – $\nabla f = 0$ [V.2]

H – symmetric (Clairaut [V.1]); curvature in every direction

D – product of the eigenvalues of H [LA.9]

Saddle – up in one direction, down in another

One variable – reduces to the f'' test [C1.12]

CONTEXT

- **Origin.** Otto Hesse (1840s); second-derivative test from Lagrange.
- **Use.** Classifying extrema of surfaces; stability.
- **Data science.** Newton's method and second-order optimizers use H [O.9]; high-dimensional loss surfaces are full of saddle points rather than local minima.
- **Physical application.** A mountain pass is a saddle: the lowest point along the ridge but the highest point along the valley path.

v.6 Double & Triple Integrals

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy \quad \iint_R 1 dA = \text{area} \quad \iiint_E f dV$$

STATED-EQUATION The double integral of f over R equals the iterated integral from a to b of the integral from c to d of $f(x, y) dy dx$; by Fubini's theorem the order can be swapped. The double integral of one gives the area. Triple integrals work the same way over a solid E .

MEMORY AID Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

TERMS

dA – area element $dx dy$

Iterated integral – two single integrals [C1.18]

Non-rectangular regions – inner limits are functions, e.g. $0 \leq y \leq x$ [C2.5]

dV – volume element

Applications – volume, mass $\iint \rho dA$, averages [C2.8]

CONTEXT

- **Origin.** Euler (1769); Fubini (1907).
- **Use.** Mass, center of mass, total charge, volume under a surface.
- **Data science.** Joint probability: $P((X, Y) \in R) = \iint_R f(x, y) dA$; marginals integrate out one variable [S.9].
- **Physical application.** Total rainfall over a region is the double integral of rainfall depth per area.

V.7 Polar, Cylindrical & Spherical Coordinates

$$dA = r dr d\theta \quad dV_{\text{cyl}} = r dz dr d\theta \quad dV_{\text{sph}} = \rho^2 \sin \phi d\rho d\phi d\theta \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

STATED-EQUATION In polar coordinates, dA equals $r dr d\theta$. In cylindrical, dV equals $r dz dr d\theta$. In spherical, dV equals $\rho^2 \sin \phi d\rho d\phi d\theta$. The Gaussian integral of e^{-x^2} over the whole real line equals the square root of π .

MEMORY AID “Don’t forget the extra r ”: it is the Jacobian [V.4].

TERMS

r, θ – polar radius and angle [T.11]

z – height (cylindrical)

ρ, ϕ, θ – distance from origin, angle from $+z$ axis, azimuth

Extra factors – Jacobian determinants [V.4]

Gaussian integral – square it and switch to polar: $\iint_{\pi} e^{-(x^2+y^2)} dA =$

CONTEXT

- **Origin.** Polar integration by Euler; the Gaussian integral by Laplace (1774) and Gauss.
- **Use.** Circular, cylindrical and spherical symmetry: pipes, planets, antennas.
- **Data science.** The Gaussian integral is why the normal density has the constant $\frac{1}{\sqrt{2\pi}}$ [S.12].
- **Physical application.** The mass of the atmosphere, a spherical shell, is computed in spherical coordinates.

V.8 Vector Fields & Line Integrals

$$\mathbf{F}(x, y, z) = (P, Q, R) \quad \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad \int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$$

STATED-EQUATION A vector field F assigns the vector P, Q, R to each point. The line integral of F along C equals the integral from a to b of F at $\mathbf{r}(t)$, dotted with $\mathbf{r}'(t)$. The scalar line integral of f with respect to arc length equals the integral of f at $\mathbf{r}(t)$ times the speed, the norm of $\mathbf{r}'(t)$.

TERMS

Vector field – an arrow at every point (wind, force, flow)

$\mathbf{r}(t)$ – parametrized path [C2.14]

$\mathbf{F} \cdot d\mathbf{r}$ – work along a tiny step [P.4], [T.12]

ds – arc length [C2.7]

Orientation – reversing the path flips the sign

CONTEXT

- **Origin.** Developed by Green, Gauss, and Maxwell for electromagnetism (1820s–1870s).
- **Use.** Work, circulation, fluid flow along a curve.
- **Data science.** Gradient fields visualize optimization (loss landscapes as arrows); score-based generative models learn the vector field $\nabla \log p(\mathbf{x})$.
- **Physical application.** The work to push a cart along a winding track against a varying wind is a line integral.

V.9 Conservative Fields & Potentials

$$\mathbf{F} = \nabla \varphi \quad \int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A) \quad \oint_C \mathbf{F} \cdot d\mathbf{r} = 0 \quad \text{test (2-D, simply connected): } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

STATED-EQUATION F is conservative if F equals the gradient of a potential φ . Then the line integral from A to B equals $\varphi(B)$ minus $\varphi(A)$, independent of path, and every closed-loop integral is zero. Test in two dimensions: partial P partial y equals partial Q partial x .

MEMORY AID The Fundamental Theorem of Calculus for line integrals: only the endpoints matter [C1.18].

TERMS

φ – potential function (negative of potential energy in physics)

Path independence – same answer for every route

$\oint$ – integral around a closed loop

Test – “cross-partials equal” [V.1]; equivalently $\nabla \times \mathbf{F} = \mathbf{0}$ [V.10]

CONTEXT

- **Origin.** Potential theory by Lagrange (1773), Laplace and Green (1828).
- **Use.** Gravity, electrostatics, energy conservation.
- **Data science.** Only conservative fields have a scalar “loss” behind them; a field with circulation (like GAN training dynamics) can cycle instead of converging.
- **Physical application.** Lifting a box to a shelf takes the same gravitational work whether you use the stairs or the elevator.

V.10 Divergence & Curl

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$$

STATED-EQUATION The divergence of $\mathbf{F}$, $\text{del dot } \mathbf{F}$, equals partial P partial x plus partial Q partial y plus partial R partial z . The curl of $\mathbf{F}$, $\text{del cross } \mathbf{F}$, is the determinant with i, j, k in the first row, the partial derivative operators in the second, and P, Q, R in the third: R_y minus Q_z , P_z minus R_x , Q_x minus P_y .

MEMORY AID Divergence = “spreading out” (source > 0 , sink < 0); curl = “spinning” (right-hand rule).

TERMS

$\nabla \cdot$ – divergence: net outflow per volume (a scalar)

$\nabla \times$ – curl: rotation axis and strength (a vector)

Incompressible – $\nabla \cdot \mathbf{v} = 0$ (continuity [P.9](#))

Irrotational – $\nabla \times \mathbf{F} = \mathbf{0}$ [V.9](#)

Identities – $\nabla \times \nabla \phi = \mathbf{0}$; $\nabla \cdot (\nabla \times \mathbf{F}) = 0$

CONTEXT

- **Origin.** Maxwell (1873) and Gibbs/Heaviside’s vector notation (1880s).
- **Use.** Fluid dynamics, electromagnetism (Maxwell’s equations), weather.
- **Data science.** Physics-informed neural networks enforce $\nabla \cdot \mathbf{v} = 0$ for fluids; optical-flow fields in video are analyzed with divergence (zoom) and curl (rotation).
- **Physical application.** Water draining from a bathtub swirls (curl) while flowing into the drain, a sink with negative divergence.

V.11 Green’s Theorem

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad \text{Area}(D) = \frac{1}{2} \oint_C (x dy - y dx)$$

STATED-EQUATION The counterclockwise line integral around C of $P dx$ plus $Q dy$ equals the double integral over the enclosed region D of partial Q partial x minus partial P partial y . The area of D equals one half the loop integral of $x dy$ minus $y dx$.

MEMORY AID Circulation around the boundary = total spin inside. Counterclockwise is positive.

TERMS

C – closed curve, counterclockwise

D – region enclosed by C

$Q_x - P_y$ – 2-D curl [V.10](#)

Area formula – choose $P = -\frac{y}{2}$, $Q = \frac{x}{2}$

Relation – 2-D case of Stokes [V.13](#)

CONTEXT

- **Origin.** George Green (1828), a self-taught miller.
- **Use.** Computing circulation and areas from boundaries.
- **Data science.** The shoelace formula for polygon area (GIS, computational geometry) is Green’s theorem applied to straight edges.
- **Physical application.** A planimeter, a mechanical tool, measures the area of a map region just by tracing its border.

V.12 Flux & the Divergence Theorem

$$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS \quad \iint_{S \text{ closed}} \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \nabla \cdot \mathbf{F} dV$$

STATED-EQUATION The flux of $\mathbf{F}$ through a surface S is the surface integral of $\mathbf{F}$ dot the unit normal $\mathbf{n}$, dS . The divergence theorem: the outward flux through a closed surface equals the triple integral of the divergence of $\mathbf{F}$ over the enclosed solid E .

MEMORY AID What flows out through the skin = what is produced inside.

TERMS

S – surface; closed surface bounds a solid E

$\mathbf{n}$ – outward unit normal

$\mathbf{F} \cdot \mathbf{n}$ – component crossing the surface [T.12](#)

Divergence – [V.10](#)

Also called – Gauss’s theorem or Ostrogradsky’s theorem

CONTEXT

- **Origin.** Lagrange (1762), Gauss (1813), Ostrogradsky (1826).
- **Use.** Conservation laws: mass, charge, heat.
- **Data science.** Conservation constraints in physics-informed models and flux balances in hydrology and climate data assimilation.
- **Physical application.** Gauss’s law: the electric flux through any closed surface equals the enclosed charge divided by ϵ_0 .

V.13 Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS \quad (\text{Green: } S \text{ flat in the plane})$$

STATED-EQUATION The circulation of $\mathbf{F}$ around the boundary curve C equals the flux of the curl of $\mathbf{F}$ through any surface S bounded by C . Green's theorem is the special case of a flat surface in the plane.

MEMORY AID Right-hand rule: curl your fingers along C , your thumb gives $\mathbf{n}$.

TERMS

C – boundary of S with matching orientation

$\nabla \times \mathbf{F}$ – curl [V.10]

Any surface – same answer for every surface with the same boundary

Special case – Green's theorem [V.11]

Big picture – FTC [C1.18], Green, Stokes, divergence: “boundary integral = interior integral of a derivative”

CONTEXT

- **Origin.** Stated by Kelvin (1850), set as an exam question by Stokes (1854).
- **Use.** Electromagnetism (Faraday's and Ampère's laws), aerodynamics (lift from circulation).
- **Data science.** Discrete versions underpin graph Laplacians and Hodge decompositions used in ranking and flow data on networks.
- **Physical application.** A changing magnetic flux through a coil induces a voltage around the wire (Faraday), which is how generators work.

V.14 The Laplacian & Classic PDEs

$$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \quad u_t = \alpha \nabla^2 u \quad u_{tt} = c^2 \nabla^2 u \quad \nabla^2 u = 0$$

STATED-EQUATION The Laplacian of f , $\text{del squared } f$, is the divergence of the gradient: the sum of the unmixed second partial derivatives. The heat equation: u_t equals α times the Laplacian of u . The wave equation: u_{tt} equals c squared times the Laplacian of u . Laplace's equation: the Laplacian of u equals zero.

MEMORY AID The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

TERMS

∇^2 – Laplacian, also written Δ

Heat equation – diffusion [P.19]

Wave equation – vibrating strings, sound, light [P.7]

Harmonic – $\nabla^2 u = 0$: steady-state temperature, electrostatic potential

Discrete Laplacian – $f(x+h) - 2f(x) + f(x-h)$ over h^2 [C1.4]

CONTEXT

- **Origin.** Laplace (1780s) on gravitational potential; Fourier's heat equation (1822); d'Alembert's wave equation (1747).
- **Use.** Heat, waves, electrostatics, fluid flow.
- **Data science.** The graph Laplacian powers spectral clustering and graph neural networks; the Laplacian filter detects edges in images.
- **Physical application.** The steady temperature across a metal plate with fixed edge temperatures solves Laplace's equation.

09

Physics & Natural Phenomena

Mechanics, fluids, heat transfer, and the dimensionless numbers that tell you which physics dominates. Calculus turns these laws into predictions; the same equations govern weather, oceans, cooking, and the cooling of electronics. Take $g = 9.81 \text{ m/s}^2$ throughout.

P.1 Units & Dimensional Analysis

$$= \frac{\text{kg} \cdot \text{m}}{\text{s}^2} = \text{N} \quad [\text{energy}] = \text{N} \cdot \text{m} = \text{J} \quad [\text{power}] = \frac{\text{J}}{\text{s}} = \text{W} \quad \text{Buckingham } \pi : \# \text{ dimensionless groups} = n - k$$

STATED-EQUATION The unit of force is kilograms times meters per second squared, called a newton. A joule is a newton-meter, and a watt is a joule per second. The Buckingham pi theorem: the number of dimensionless groups equals the number of variables, n , minus the number of base dimensions, k .

MEMORY AID “Units first”: if the units of both sides do not match, the equation is wrong.

TERMS

SI base units – meter (m), kilogram (kg), second (s), kelvin (K), ampere (A), mole (mol)

$[\cdot]$ – “dimensions of”

n – number of physical variables in the problem

k – number of independent base dimensions (e.g. M, L, T)

Dimensionless group – a ratio with no units, like Re [P.10](#) or Bi [P.15](#)

Conversions – multiply by 1 in a useful form, e.g. $\frac{1000 \text{ m}}{1 \text{ km}}$ [A1.14](#)

CONTEXT

- **Origin.** Fourier (1822) stated dimensional homogeneity; Buckingham (1914) the π theorem.
- **Use.** Checking equations, converting units, and building scale models (wind tunnels, ship tanks).
- **Data science.** Unit-consistent feature engineering (normalizing by physically meaningful scales), and dimensionless features that generalize across system sizes.
- **Physical application.** Engineers test a small model ship in a tank at the same Froude number as the full ship, so the waves match.

P.2 Kinematics: Constant Acceleration

$$v = \frac{dx}{dt} \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2} \quad v = v_0 + at \quad x = x_0 + v_0t + \frac{1}{2}at^2 \quad v^2 = v_0^2 + 2a\Delta x$$

STATED-EQUATION Velocity is dx/dt ; acceleration is d^2x/dt^2 , the second derivative of position. v equals v naught plus a t . x equals x naught plus v naught t plus one half a t squared. v squared equals v naught squared plus two a Δx .

MEMORY AID “Velocity is the slope of position; acceleration is the slope of velocity” (and area under v - t is distance).

TERMS

x, x_0 – position and initial position (m)

v, v_0 – velocity and initial velocity (m/s); derivative [C1.4]

a – acceleration (m/s²); free fall $a = -g$

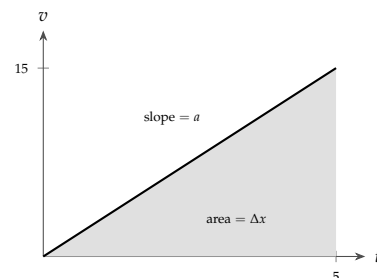
t – time (s)

Δx – displacement $x - x_0$

Derivation – integrate a twice [C1.16]; the $\frac{1}{2}$ comes from $\int t dt$

CONTEXT

- **Origin.** Galileo (1638) with rolling balls on ramps; calculus form by Newton.
- **Use.** Braking distances, projectile motion, elevator design.
- **Data science.** Sensor fusion and Kalman filters use these equations as the motion model to track vehicles and phones from noisy GPS and accelerometer data.
- **Physical application.** Stopping distance grows with v^2 : doubling your speed quadruples the braking distance.



P.3 Newton's Laws & Friction

$$\text{1st: } \sum \mathbf{F} = \mathbf{0} \Leftrightarrow \mathbf{v} = \text{const} \quad \text{2nd: } \sum \mathbf{F} = m\mathbf{a} \quad \text{3rd: } \mathbf{F}_{AB} = -\mathbf{F}_{BA} \quad W = mg \quad f_k = \mu_k N$$

STATED-EQUATION First law: net force zero if and only if velocity is constant. Second law: the sum of the forces equals $m a$. Third law: the force of A on B equals minus the force of B on A . Weight equals $m g$. Kinetic friction equals μ_k times the normal force N .

MEMORY AID “ F equals $m a$ ” — every problem starts with a free-body diagram: draw all forces, then sum them.

TERMS

$\sum \mathbf{F}$ – net (vector) force in newtons [T.12]

m – mass (kg); a acceleration [P.2]

$W = mg$ – weight, the force of gravity near Earth

N – normal force from a surface (equals mg on level ground)

μ_k, μ_s – kinetic and static friction coefficients; static friction $f_s \leq \mu_s N$

On an incline – components $mg \sin \theta$ along, $mg \cos \theta$ into the surface [T.1]

CONTEXT

- **Origin.** Isaac Newton, *Principia* (1687).
- **Use.** Designing structures, vehicles, and machines.
- **Data science.** Physics engines for simulation and robotics; physics-informed ML adds $F = ma$ as a constraint in the loss.
- **Physical application.** A heavy box and a light box on the same ramp slide identically when friction is proportional to weight, because m cancels.

P.4 Work, Energy & Power

$$W = Fd \cos \theta = \int \mathbf{F} \cdot d\mathbf{x} \quad KE = \frac{1}{2}mv^2 \quad PE = mgh \quad KE_1 + PE_1 = KE_2 + PE_2 \quad P = \frac{W}{t} = Fv$$

STATED-EQUATION *Work equals F d cosine theta, the integral of F dot d x. Kinetic energy equals one half m v squared. Potential energy equals m g h. Kinetic plus potential energy before equals kinetic plus potential energy after. Power equals work over time, which equals F v.*

TERMS

W – work (J); θ angle between force and displacement (dot product [T.12])

KE – kinetic energy (J)

PE – gravitational potential energy; h height (m)

Conservation – holds when friction is negligible; otherwise add heat

P – power ($W = J/s$)

Work–energy theorem – net work = ΔKE (from $F = ma$ and the chain rule [C1.7])

CONTEXT

► **Origin.** Leibniz's *vis viva* (1686); energy conservation by Joule, Mayer, and Helmholtz (1840s).

► **Use.** Roller coasters, hydroelectric dams, engine ratings.

► **Data science.** Energy-based models and Hamiltonian Monte Carlo [Q.16] use “potential energy” = $-\log p(x)$ and “kinetic energy” to explore distributions.

► **Physical application.** A 60 W bulb left on for 10 hours uses 0.6 kWh; your electricity bill is an energy (work) bill.

P.5 Momentum & Impulse

$$\mathbf{p} = m\mathbf{v} \quad \mathbf{J} = \int \mathbf{F} dt = \mathbf{F}_{\text{avg}} \Delta t = \Delta \mathbf{p} \quad m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 = m_1 \mathbf{v}'_1 + m_2 \mathbf{v}'_2$$

STATED-EQUATION *Momentum p equals m v. Impulse J equals the integral of F d t, the average force times delta t, which equals the change in momentum. m one v one plus m two v two equals m one v one prime plus m two v two prime.*

TERMS

$\mathbf{p}$ – momentum (kg·m/s), a vector

$\mathbf{J}$ – impulse; area under force–time curve [C1.18]

Conservation – total momentum is constant when no external force acts [P.3]

Elastic – kinetic energy also conserved

Perfectly inelastic – objects stick: $v' = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$ (a weighted mean [S.1])

CONTEXT

► **Origin.** Descartes (1644) and Huygens; Newton's second law is originally $\mathbf{F} = \frac{d\mathbf{p}}{dt}$.

► **Use.** Airbags, crash testing, rocket propulsion.

► **Data science.** “Momentum” in optimizers is named by analogy: a velocity term accumulates past gradients [ML.16].

► **Physical application.** Airbags and crumple zones lengthen Δt so the same Δp produces a much smaller force on your body.

P.6 Circular Motion & Gravitation

$$a_c = \frac{v^2}{r} = \omega^2 r \quad F_c = \frac{mv^2}{r} \quad F = \frac{G m_1 m_2}{r^2} \quad g = \frac{GM}{R^2} \quad v_{\text{orbit}} = \sqrt{\frac{GM}{r}}$$

STATED-EQUATION *Centripetal acceleration equals v squared over r, which equals omega squared r. Centripetal force equals m v squared over r. Gravitational force equals G m one m two over r squared. Little g equals G M over R squared. Orbital speed equals the square root of G M over r.*

MEMORY AID Gravity is an inverse-square law: double the distance, one quarter the force.

TERMS

a_c – centripetal acceleration, pointing to the center

ω – angular speed (rad/s) [T.2]

G – $6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$

M, R – Earth: $5.972 \times 10^{24} \text{ kg}$, $6.371 \times 10^6 \text{ m}$

r – distance between centers

Orbit – gravity supplies the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$

CONTEXT

► **Origin.** Huygens (1673) for circular motion; Newton's universal gravitation (1687).

► **Use.** Satellites, banked curves, centrifuges, planetary science.

► **Data science.** Inverse-square falloff inspires distance-decay weights in spatial models and gravity models of trade and migration flows.

► **Physical application.** A car on a curve relies on tire friction to supply mv^2/r ; that is why icy curves are dangerous.

P.7 Oscillations & Waves

$$x(t) = A \cos(\omega t + \phi) \quad \omega = \sqrt{\frac{k}{m}}, \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad T_{\text{pendulum}} \approx 2\pi \sqrt{\frac{L}{g}} \quad v = f\lambda, \quad f = \frac{1}{T}$$

STATED-EQUATION x of t equals A cosine of ωt plus ϕ . ω equals the square root of k over m ; the period T equals two π over ω , which is two π root m over k . A pendulum's period is approximately two π root L over g . Wave speed v equals frequency f times wavelength λ ; f equals one over T .

TERMS

A – amplitude; ϕ phase [T9]

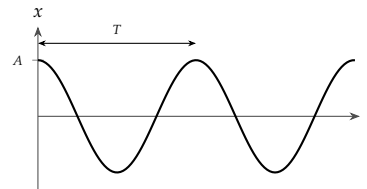
k – spring constant (N/m), from Hooke's law $F = -kx$

ω – angular frequency (rad/s) [T2]

T, f – period (s) and frequency (Hz)

λ – wavelength (m)

Origin of the solution – $m\ddot{x} = -kx$ is solved by sinusoids [C1.8]



CONTEXT

- **Origin.** Hooke (1678); Galileo noticed pendulum isochronism (1602); Huygens built the pendulum clock (1656).
- **Use.** Clocks, suspensions, musical instruments, seismology.
- **Data science.** Fourier features and spectral analysis decompose signals into sinusoids; vibration data from machines feeds predictive-maintenance models.
- **Physical application.** A grandfather clock's 1 m pendulum swings with a period of about 2 s, one tick per second.

P.8 Pressure, Hydrostatics & Buoyancy

$$P = \frac{F}{A} \quad P = P_0 + \rho gh \quad F_B = \rho_{\text{fluid}} g V_{\text{displaced}} \quad \text{floating: } \frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

STATED-EQUATION Pressure equals force over area. Pressure at depth h equals P naught plus $\rho g h$. The buoyant force equals the fluid density times g times the displaced volume. For a floating object, the submerged fraction equals the object's density over the fluid's density.

MEMORY AID Archimedes: “the fluid pushes up with the weight of the fluid you push aside.”

TERMS

P – pressure (Pa = N/m²); P_0 surface pressure (1 atm = 101,325 Pa)

ρ – density (kg/m³); water 1000, seawater $\approx$ 1025

h – depth below the surface (m)

F_B – buoyant force (N)

$V_{\text{displaced}}$ – volume of fluid pushed aside [G.11], [G.12]

Pascal's principle – pressure applied to a confined fluid is transmitted everywhere (hydraulics)

CONTEXT

- **Origin.** Archimedes (~250 BCE); Pascal (1653); Stevin (1586).
- **Use.** Dams, submarines, ship design, blood pressure, hydraulic lifts.
- **Data science.** Ocean-sensor data (pressure $\rightarrow$ depth) and weather-station barometric pressure are standard engineered features in climate models.
- **Physical application.** Only about 90% of an iceberg is underwater because ice is 917 kg/m³ and seawater 1025 kg/m³.

P.9 Continuity & Bernoulli's Equation

$$Q = Av, \quad A_1 v_1 = A_2 v_2 \quad P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2 \quad v_{\text{out}} = \sqrt{2gh}$$

STATED-EQUATION Volumetric flow rate Q equals area times velocity, so A one v one equals A two v two. P one plus one half ρv one squared plus $\rho g h$ one equals P two plus one half ρv two squared plus $\rho g h$ two. Torricelli's law: the outflow speed equals the square root of two $g h$.

MEMORY AID "Fast fluid, low pressure" — squeeze the pipe, speed goes up, pressure goes down.

TERMS

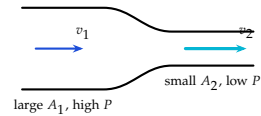
Q – volumetric flow rate (m^3/s)

A – cross-sectional area πr^2 [G.9]

Continuity – conservation of mass for an incompressible fluid

Bernoulli – energy per volume [E.4] along a streamline: pressure + kinetic + potential

Assumptions – steady, incompressible, inviscid (no friction [P.10])



CONTEXT

- **Origin.** Daniel Bernoulli, *Hydrodynamica* (1738); Euler wrote the general equations (1757).
- **Use.** Venturi flow meters, carburetors, pitot tubes on aircraft, plumbing.
- **Data science.** Flow sensors (Venturi and pitot readings) are converted to velocity with Bernoulli before feeding pipeline anomaly-detection models.
- **Physical application.** Putting your thumb over a garden hose shrinks A , so the water speeds up and sprays farther.

P.10 Viscosity & the Reynolds Number

$$\tau = \mu \frac{du}{dy} \quad \text{Re} = \frac{\rho v L}{\mu} = \frac{v L}{\nu} = \frac{\text{inertial forces}}{\text{viscous forces}} \quad \text{pipe: Re} < 2300 \text{ laminar, Re} > 4000 \text{ turbulent} \quad Q = \frac{\pi r^4 \Delta P}{8 \mu L}$$

STATED-EQUATION Shear stress τ equals μ times du/dy . The Reynolds number, Re , equals $\rho v L$ over μ , which equals $v L$ over ν : inertial forces over viscous forces. In a pipe, Reynolds below about twenty-three hundred is laminar and above about four thousand is turbulent. Poiseuille's law: flow rate Q equals πr to the fourth ΔP over eight μL .

MEMORY AID "Re is big, flow goes wild" — high Reynolds means turbulence; low means smooth, syrupy flow.

TERMS

μ – dynamic viscosity ($\text{Pa}\cdot\text{s}$); water $\approx 10^{-3}$, honey ≈ 10

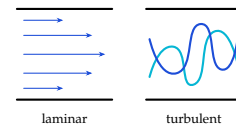
$\nu = \mu/\rho$ – kinematic viscosity (m^2/s)

τ – shear stress; du/dy velocity gradient [C.1.4]

L – characteristic length (pipe diameter, wing chord)

v – characteristic speed

r^4 – doubling pipe radius gives $16\times$ the flow [A1.2]



CONTEXT

- **Origin.** Stokes (1851); Osborne Reynolds' dye experiments (1883); Poiseuille (1840s) studying blood flow.
- **Use.** Pipeline design, aircraft, blood flow, microfluidics.
- **Data science.** Re is the classic example of a dimensionless feature: flows with the same Re look alike at any scale, so models trained on it generalize.
- **Physical application.** Stirring honey stays smooth ($\text{Re} \ll 1$) while a river is turbulent (Re in the millions); the smoke from a candle goes from smooth to swirling as it speeds up.

P.11 Drag & Terminal Velocity

$$F_D = \frac{1}{2} \rho C_D A v^2 \quad v_t = \sqrt{\frac{2mg}{\rho C_D A}} \quad \text{Stokes (Re} \ll 1\text{): } F_D = 6\pi\mu r v \quad P_{\text{drag}} = F_D v \propto v^3$$

STATED-EQUATION Drag force equals one half rho C D A v squared. Terminal velocity equals the square root of two m g over rho C D A. For very small Reynolds number, Stokes drag equals six pi mu r v. The power to overcome drag, F D times v, is proportional to v cubed.

TERMS

ρ – fluid density (air $\approx 1.2 \text{ kg/m}^3$)
 C_D – drag coefficient (sphere ≈ 0.47 , skydiver ≈ 1)
 A – frontal area (m^2)
 v_t – speed where drag equals weight, so $a = 0$ [P.3]
 Stokes regime – viscous drag for tiny particles [P.10]
 v^3 power – why fuel use rises quickly with highway speed

CONTEXT

- **Origin.** Newton’s drag law (1687); Rayleigh; Stokes (1851).
- **Use.** Vehicle efficiency, parachutes, sedimentation, sports ballistics.
- **Data science.** Fuel-economy and range models for EVs use a v^2 drag term as a physics-based feature alongside learned corrections.
- **Physical application.** Raindrops fall at only a few m/s instead of hundreds, because drag stops them accelerating.

P.12 Heat, Temperature & Phase Change

$$Q = mc \Delta T \quad Q = mL \quad T_K = T_C + 273.15 \quad T_F = \frac{9}{5} T_C + 32$$

STATED-EQUATION Heat Q equals $m c$ delta T . Heat for a phase change equals m times L . Kelvin equals Celsius plus two seventy-three point one five. Fahrenheit equals nine fifths Celsius plus thirty-two.

MEMORY AID Water: $c \approx 4186 \text{ J/(kg}\cdot\text{K)}$; melting ice takes as much heat as warming the water by about 80°C .

TERMS

Q – heat added (J)
 c – specific heat capacity ($\text{J/(kg}\cdot\text{K)}$); water 4186
 ΔT – temperature change (same size in K and $^\circ\text{C}$)
 L – latent heat (J/kg); ice melting 3.34×10^5 , water boiling 2.26×10^6
 Kelvin – absolute scale; 0 K is absolute zero
 Conversion – linear functions [A1.5]

CONTEXT

- **Origin.** Joseph Black (1760s) distinguished heat from temperature and found latent heat; Kelvin scale (1848).
- **Use.** Cooking, HVAC, climate, industrial processes.
- **Data science.** Energy-demand forecasting uses heating and cooling degree-days, which are built on $Q = mc\Delta T$ reasoning.
- **Physical application.** Coastal cities have milder climates because water’s large c makes oceans slow to heat and cool.

P.13 Conduction: Fourier’s Law & Thermal Resistance

$$q = -k \frac{dT}{dx} \quad \dot{Q} = \frac{kA \Delta T}{L} \quad R_{\text{cond}} = \frac{L}{kA} \quad \dot{Q} = \frac{\Delta T}{R_{\text{total}}}, \quad R_{\text{series}} = R_1 + R_2 + \dots$$

STATED-EQUATION Heat flux q equals minus k times dT/dx . Heat flow $\dot{Q}$ equals $kA \Delta T$ over L . Conduction resistance equals L over kA . Heat flow equals ΔT over the total resistance, and resistances in series add.

MEMORY AID Heat flows like current: ΔT is the “voltage,” R the “resistor” [P.20].

TERMS

q – heat flux (W/m^2); minus sign: heat flows downhill in T
 k – thermal conductivity ($\text{W/(m}\cdot\text{K)}$); copper 400, glass 0.8, air 0.026
 $\dot{Q}$ – heat flow rate (W)
 A, L – area and thickness
 R – thermal resistance (K/W)
 dT/dx – temperature gradient [C1.4]

CONTEXT

- **Origin.** Joseph Fourier, *Théorie analytique de la chaleur* (1822), which also introduced Fourier series.
- **Use.** Insulation, building energy, CPU heat sinks, cookware.
- **Data science.** Building-energy models combine thermal-resistance networks with regression on smart-meter data to estimate insulation quality.
- **Physical application.** A metal spoon in hot soup gets hot fast; a wooden one does not, because k differs by a factor of about 100.

P.14 Convection & Newton's Law of Cooling

$$\dot{Q} = hA (T_s - T_\infty) \quad T(t) = T_\infty + (T_0 - T_\infty) e^{-t/\tau} \quad \tau = \frac{\rho V c}{hA} = \frac{mc}{hA}$$

STATED-EQUATION Convective heat flow equals hA times the surface temperature minus the fluid temperature. T of t equals T infinity plus the quantity T naught minus T infinity, times e to the minus t over τ . The time constant τ equals $\rho V c$ over hA , which is mc over hA .

MEMORY AID “Hot things cool fastest when they are hottest”: the rate is proportional to the temperature gap.

TERMS

h – convection coefficient ($W/(m^2 \cdot K)$): still air 2–25, moving air 25–250, water 50–20,000

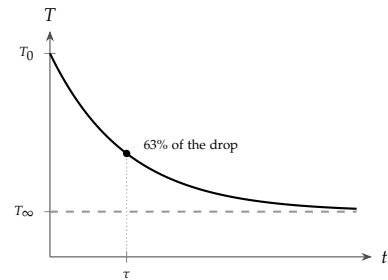
T_s, T_∞ – surface and surrounding-fluid temperatures

Natural convection – fluid moves because warm fluid is less dense and rises [P.16]

Forced convection – a fan or pump drives the flow

τ – time constant; the solution of $\frac{dT}{dt} = -\frac{1}{\tau}(T - T_\infty)$ [C2.16]

Valid when – temperature is uniform inside, $Bi < 0.1$ [P.15]



CONTEXT

- **Origin.** Newton's law of cooling (1701); boundary-layer theory by Prandtl (1904).
- **Use.** Cooling electronics, HVAC, weather, forensic time-of-death estimates.
- **Data science.** Exponential-decay fits (with T_∞ and τ as parameters) are fit by nonlinear least squares [G.9] to sensor data for thermal digital twins.
- **Physical application.** Blowing on hot soup increases h (forced convection), so it cools faster.

P.15 Biot Number & Lumped Capacitance

$$Bi = \frac{hL_c}{k_{\text{solid}}} = \frac{\text{internal conduction resistance}}{\text{surface convection resistance}} \quad L_c = \frac{V}{A_s} \quad Bi < 0.1 \Rightarrow \text{lumped model valid} \quad Fo = \frac{\alpha t}{L_c^2}$$

STATED-EQUATION The Biot number, Bi , equals hL_c over k of the solid: internal conduction resistance over surface convection resistance. The characteristic length L_c equals volume over surface area. If Biot is less than zero point one, the lumped model is valid. The Fourier number, Fo , equals αt over L_c squared.

MEMORY AID “Biot small, same temperature all” — below 0.1, the inside keeps up with the surface.

TERMS

h – convection coefficient [P.14]

k_{solid} – conductivity of the object (not the fluid) [P.13]

L_c – characteristic length; sphere $r/3$, long cylinder $r/2$, plate $\frac{\text{thickness}}{2}$

Lumped capacitance – treat the object as one temperature, giving the exponential in [P.14]

$\alpha = \frac{k}{\rho c}$ – thermal diffusivity (m^2/s)

Fo – dimensionless time: how far heat has diffused [P.19]

CONTEXT

- **Origin.** Jean-Baptiste Biot (1804), a contemporary of Fourier.
- **Use.** Heat treating metals, food processing, battery thermal management.
- **Data science.** Choosing between a cheap lumped (one-parameter) model and an expensive spatial simulation is a model-complexity decision [ML.4]; Bi tells you which is justified.
- **Physical application.** A thick steak (large Bi) sears outside while the center stays rare; a thin slice cooks through evenly.

P.16 Natural Convection: Nu, Pr, Gr & Ra

$$\text{Nu} = \frac{hL}{k_{\text{fluid}}} \quad \text{Pr} = \frac{\nu}{\alpha} \quad \text{Gr} = \frac{g\beta\Delta T L^3}{\nu^2} \quad \text{Ra} = \text{Gr} \cdot \text{Pr} = \frac{g\beta\Delta T L^3}{\nu\alpha} \quad \text{Ra}_c \approx 1708$$

STATED-EQUATION The Nusselt number equals hL over the fluid's k . The Prandtl number equals ν over α . The Grashof number equals $g\beta\Delta T L^3$ over ν^2 . The Rayleigh number equals Grashof times Prandtl, which is $g\beta\Delta T L^3$ over $\nu\alpha$. The critical Rayleigh number for a fluid layer heated from below is about seventeen hundred eight.

MEMORY AID Ra is the “Reynolds number of buoyancy”: above ≈ 1708 , a heated layer starts to roll into convection cells.

TERMS

Nu – convective vs. conductive heat transfer (Nu = 1: pure conduction)

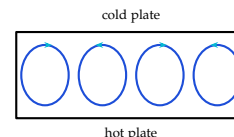
Pr – momentum vs. heat diffusivity (air ≈ 0.71 , water ≈ 7)

Gr – buoyancy vs. viscous forces (natural-convection analog of Re^2 [P.10])

β – thermal expansion coefficient; ideal gas $\beta = 1/T$ in kelvin [P.18]

ν, α – kinematic viscosity and thermal diffusivity

Correlations – Nu = C Raⁿ fit on log–log axes [A2.7], [S.19]



CONTEXT

- **Origin.** Nusselt (1915); Rayleigh (1916) analyzing Bénard’s hexagonal convection cells (1900); Grashof, Prandtl.
- **Use.** Room heating, cooling of electronics, double-pane windows, mantle convection, weather.
- **Data science.** Empirical correlations $\text{Nu} = C \text{Ra}^n$ are power-law regressions; fitting them in log space is a classic linear-regression exercise.
- **Physical application.** Warm air rising from a radiator, thunderstorm updrafts, and the churning pattern in a pot of heated miso soup are all Rayleigh–Bénard convection.

P.17 Thermal Radiation

$$P = \varepsilon\sigma AT^4 \quad P_{\text{net}} = \varepsilon\sigma A (T^4 - T_{\text{surr}}^4) \quad \lambda_{\text{max}}T = 2.898 \times 10^{-3} \text{ m} \cdot \text{K} \quad \sigma = 5.670 \times 10^{-8} \frac{\text{W}}{\text{m}^2\text{K}^4}$$

STATED-EQUATION Radiated power equals epsilon sigma A T to the fourth. Net power equals epsilon sigma A times T to the fourth minus T surroundings to the fourth. Wien’s law: lambda max times T equals two point eight nine eight times ten to the minus three meter-kelvins.

MEMORY AID “Fourth power”: double the absolute temperature, sixteen times the radiation. Always use kelvin.

TERMS

ε – emissivity, 0 to 1 (skin ≈ 0.97 , polished metal ≈ 0.05)

σ – Stefan–Boltzmann constant

T – absolute temperature in kelvin [P.12]

λ_{max} – peak wavelength (Wien’s law; inverse variation [A1.13])

No medium needed – radiation crosses vacuum, unlike conduction and convection

CONTEXT

- **Origin.** Stefan (1879, experiment) and Boltzmann (1884, theory); Wien (1893); Planck (1900) launched quantum theory from this problem.
- **Use.** Thermal cameras, solar panels, climate energy balance, furnaces.
- **Data science.** Satellite land-surface-temperature products are inverted from infrared radiance with these laws before being used as features in crop and climate models.
- **Physical application.** You feel the Sun’s warmth across 150 million km of vacuum; the Sun ($\approx 5778 \text{ K}$) peaks near 500 nm, green light.

P.18 Ideal Gas Law & Thermodynamics

$$PV = nRT \quad \Delta U = Q - W \quad W = \int P dV \quad \eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \quad \Delta S \geq 0 \text{ (isolated system)}$$

STATED-EQUATION $P V$ equals $n R T$. The change in internal energy equals heat added minus work done by the system. Work equals the integral of $P dV$. Carnot efficiency equals one minus T cold over T hot. The entropy of an isolated system never decreases.

MEMORY AID “You can’t win, you can’t break even, you can’t get out of the game” (first, second, third laws).

TERMS

P, V – pressure (Pa) and volume (m^3)
 n – moles; $R = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$
 T – absolute temperature (K)
 ΔU – change in internal energy (first law)
 η – maximum efficiency of any heat engine
 S – entropy (second law); links to information entropy [ML.3]

CONTEXT

- **Origin.** Boyle (1662), Charles, Gay-Lussac; Clapeyron (1834); Carnot (1824); Clausius (1850s).
- **Use.** Engines, refrigerators, weather balloons, tire pressure.
- **Data science.** Boltzmann’s $S = k \ln W$ inspired Shannon entropy and the “temperature” in softmax and simulated annealing [ML.15], [O.13].
- **Physical application.** Tire pressure drops on a cold morning: at fixed V , $P \propto T$.

P.19 Diffusion & the Heat Equation

$$J = -D \frac{dC}{dx} \quad \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} \quad \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad L_{\text{diff}} \approx \sqrt{2Dt}$$

STATED-EQUATION Fick’s law: the flux J equals minus D times dC/dx . The rate of change of concentration, partial C partial t , equals D times the second partial of C with respect to x . The heat equation has the same form with α . The typical diffusion distance is about the square root of two $D t$.

MEMORY AID Diffusion distance grows like $\sqrt{t}$: to go twice as far takes four times as long.

TERMS

J – flux (amount per area per time)
 D – diffusion coefficient (m^2/s); small molecule in water $\sim 10^{-9}$, in air $\sim 10^{-5}$
 C – concentration
 ∂ – partial derivative (other variables held fixed)
 α – thermal diffusivity [P.15]
 $\sqrt{2Dt}$ – SD of a random walk [S.2], [S.13]

CONTEXT

- **Origin.** Fick (1855), by analogy with Fourier’s heat law [P.13]; Einstein linked diffusion to random walks (1905).
- **Use.** Drug delivery, pollution spread, semiconductor doping, cooking.
- **Data science.** Diffusion models for image generation add Gaussian noise step by step and learn to reverse it; the forward process is the heat equation.
- **Physical application.** Sugar sitting in unstirred tea takes hours to spread: stirring (convection) is far faster than diffusion.

P.20 Electric Circuits: Ohm’s Law & Power

$$V = IR \quad P = IV = I^2 R = \frac{V^2}{R} \quad R_{\text{series}} = R_1 + R_2 + \cdots \quad \frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$$

STATED-EQUATION Voltage equals current times resistance. Power equals $I V$, which equals I squared R , which equals V squared over R . Series resistances add. One over the parallel resistance equals one over R one plus one over R two, and so on.

MEMORY AID “VIR” triangle: cover the one you want — $V = IR$, $I = V/R$, $R = V/I$.

TERMS

V – voltage (V), the electric “pressure”
 I – current (A), charge per second
 R – resistance (Ω)
 P – power (W)
 Parallel – total is less than the smallest resistor (harmonic-mean style [ML.8])
 Analogy – same math as heat flow [P.13] and pipe flow [P.10]

CONTEXT

- **Origin.** Georg Ohm (1827); Kirchhoff’s laws (1845).
- **Use.** Household wiring, electronics, batteries, power grids.
- **Data science.** Data-center power budgeting ($P = IV$ per rack) and GPU energy-per-training-run estimates.
- **Physical application.** A 1500 W kettle on 120 V draws 12.5 A, which is why kitchens use 15–20 A circuits.

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10

Statistics & Probability

Describing data, quantifying uncertainty, and drawing conclusions from samples. This is the daily toolkit of data science: every A/B test, confidence band, and regression summary is built from the concepts below.

S.1 Measures of Center

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \mu = \frac{1}{N} \sum_{i=1}^N x_i \quad \bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i} \quad \text{median} = \begin{cases} x_{(\frac{n+1}{2})}, & n \text{ odd} \\ \frac{x_{(n/2)} + x_{(n/2+1)}}{2}, & n \text{ even} \end{cases}$$

STATED-EQUATION $\bar{x}$ equals one over n times the sum of x_i . μ equals one over N times the sum of x_i . The weighted mean equals the sum of $w_i x_i$ over the sum of w_i . The median is the middle value of the sorted data, or the average of the two middle values when n is even.

TERMS

$\bar{x}$, μ – sample mean, population mean
 n , N – sample size, population size
 $\sum$ – summation notation [A2.11]
 w_i – weights (importance, frequency, or inverse variance)
 $x_{(k)}$ – k th value after sorting (order statistic)
 Mode – most frequent value; the only center for categorical data

CONTEXT

- **Origin.** The arithmetic mean dates to antiquity; the median was popularized by Galton (1880s).
- **Use.** Summarizing a typical value.
- **Data science.** Median is robust to outliers (median house price, latency p50); the mean minimizes squared error and the median minimizes absolute error, which is why MSE and MAE losses predict different things.
- **Physical application.** The center of mass of objects on a seesaw is a weighted mean of their positions, weighted by mass.

S.2 Variance & Standard Deviation

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \sigma = \sqrt{\sigma^2}, \quad s = \sqrt{s^2} \quad \sigma^2 = \frac{1}{N} \sum x_i^2 - \mu^2$$

STATED-EQUATION σ^2 equals one over N times the sum of x_i minus μ , squared. s^2 equals one over n minus one times the sum of x_i minus $\bar{x}$, squared. σ and s are the square roots. Also, σ^2 equals the mean of the squares minus the square of the mean.

MEMORY AID Divide by $n - 1$ for a Sample (“s for sample, subtract one”).

TERMS

σ^2 , σ – population variance and standard deviation
 s^2 , s – sample variance and standard deviation
 μ , $\bar{x}$ – means [S.1]
 $n - 1$ – Bessel’s correction: makes s^2 unbiased (one degree of freedom used by $\bar{x}$)
 Units – σ has the data’s units; σ^2 has squared units

CONTEXT

- **Origin.** Standard deviation named by Karl Pearson (1894); variance by R. A. Fisher (1918).
- **Use.** Measuring spread, risk, and consistency.
- **Data science.** Feature scaling divides by s [S.4]; the bias–variance trade-off; volatility of a metric determines how long an A/B test must run [S.13].
- **Physical application.** Machining tolerance: a process with SD 0.01 mm produces far more consistent parts than one with SD 0.05 mm.

S.3 Quartiles, IQR & Outliers

$$\text{IQR} = Q_3 - Q_1 \quad \text{outlier if } x < Q_1 - 1.5\text{IQR} \text{ or } x > Q_3 + 1.5\text{IQR} \quad \text{min, } Q_1, \text{ median, } Q_3, \text{ max}$$

STATED-EQUATION The IQR equals Q three minus Q one. A value is an outlier if it is below Q one minus one point five IQR or above Q three plus one point five IQR. The five-number summary: minimum, Q one, median, Q three, maximum.

TERMS

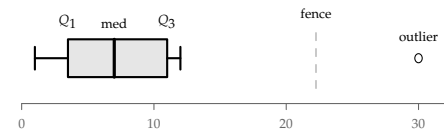
Q_1, Q_3 – 25th and 75th percentiles: medians of the lower and upper halves [S.1]

IQR – spread of the middle 50%; robust to outliers

Fences – 1.5IQR beyond the box (Tukey's rule)

Whiskers – extend to the most extreme non-outlier values

Percentile p – value below which $p\%$ of data fall



CONTEXT

- **Origin.** John Tukey introduced the box plot in *Exploratory Data Analysis* (1977).
- **Use.** Comparing distributions and spotting unusual values quickly.
- **Data science.** Outlier filtering in data cleaning, robust scaling (`RobustScaler` uses median and IQR), and latency SLOs defined by percentiles (p95, p99).
- **Physical application.** Quality engineers flag a batch whose tensile strength falls outside the 1.5 IQR fences.

S.4 Z-Scores & Standardization

$$z = \frac{x - \mu}{\sigma} \quad z = \frac{x - \bar{x}}{s} \quad x = \mu + z\sigma$$

STATED-EQUATION z equals x minus μ over σ ; for a sample, x minus $\bar{x}$ over s . Conversely, x equals μ plus z sigma.

MEMORY AID “How many SDs from the mean?”

TERMS

z – number of standard deviations from the mean

μ, σ – population mean [S.1] and SD [S.2]

$\bar{x}, s$ – sample versions

Result – standardized data have mean 0 and SD 1

With normal data – look up $\Phi(z)$ for probabilities [S.12]

CONTEXT

- **Origin.** Standardization grew out of Gauss's and Pearson's work on the normal curve.
- **Use.** Comparing values on different scales (SAT vs ACT).
- **Data science.** `StandardScaler` before regularized regression, k -NN, SVMs, PCA, and gradient descent; $|z| > 3$ is a quick anomaly flag.
- **Physical application.** Comparing a hot day in Phoenix and in Seattle: standardize each temperature by its own city's mean and SD.

S.5 Basic Probability Rules

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}} \quad 0 \leq P(A) \leq 1 \quad P(A^c) = 1 - P(A) \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

STATED-EQUATION The probability of A equals favorable outcomes over total outcomes, between zero and one. The probability of not A equals one minus P of A . The probability of A or B equals P of A plus P of B minus P of A and B .

MEMORY AID “Or” adds (minus the overlap); “at least one” = $1 -$ “none.”

TERMS

Sample space S – set of all outcomes; $P(S) = 1$

A^c – complement: “not A ”

$A \cup B$ – A or B (or both)

$A \cap B$ – A and B

Mutually exclusive – $A \cap B = \emptyset$, so $P(A \cap B) = 0$

Counting – equally likely outcomes counted with [S.8]

CONTEXT

- **Origin.** Pascal and Fermat's letters on gambling (1654); axioms by Kolmogorov (1933).
- **Use.** Games of chance, reliability, risk.
- **Data science.** “At least one” via complements: the chance that at least one of many A/B metrics looks significant by luck [S.15].
- **Physical application.** The chance that at least one of four independent smoke detectors works is $1 - P(\text{all fail})$.

S.6 Conditional Probability & Independence

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(A \cap B) = P(A | B) P(B) \quad A, B \text{ independent} \iff P(A \cap B) = P(A) P(B) \iff P(A | B) = P(A)$$

STATED-EQUATION P of A given B equals P of A and B over P of B . P of A and B equals P of A given B times P of B . A and B are independent if and only if P of A and B equals P of A times P of B , which is the same as P of A given B equals P of A .

MEMORY AID “Given” shrinks the sample space to B .

TERMS

$P(A | B)$ – probability of A given that B occurred
 $P(B)$ – must be > 0 ; B becomes the new sample space
 Multiplication rule – chain events together
 Independent – knowing B tells nothing about A
 Not the same as – mutually exclusive [S.5]

CONTEXT

- **Origin.** Formalized by de Moivre (1718) and Bayes (1763).
- **Use.** Sequential events, drawing without replacement, reliability.
- **Data science.** Naive Bayes assumes features are conditionally independent given the class; funnel conversion rates are conditional probabilities $P(\text{buy} | \text{click})$.
- **Physical application.** Reliability of components in series: if each part works independently with probability 0.99, three in series work with probability 0.99^3 .

S.7 Bayes' Theorem

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)} \quad P(B) = \sum_i P(B | A_i) P(A_i) = P(B | A) P(A) + P(B | A^c) P(A^c)$$

STATED-EQUATION P of A given B equals P of B given A times P of A , over P of B . P of B equals the sum over i of P of B given A_i times P of A_i , which equals P of B given A P of A plus P of B given not A P of not A .

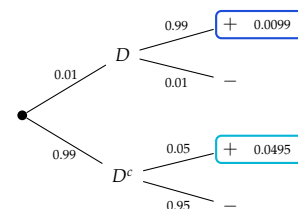
MEMORY AID Posterior $\propto$ likelihood $\times$ prior.

TERMS

$P(A)$ – prior: belief before seeing evidence
 $P(B | A)$ – likelihood of the evidence if A is true
 $P(B)$ – evidence (marginal), by the law of total probability
 $P(A | B)$ – posterior: updated belief [S.6]
 A_i – partition of the sample space
 Base-rate fallacy – ignoring a small prior

CONTEXT

- **Origin.** Rev. Thomas Bayes (published 1763); generalized by Laplace (1774).
- **Use.** Medical testing, forensic evidence, updating beliefs.
- **Data science.** Naive Bayes spam filters, Bayesian A/B testing, and interpreting classifier precision when the positive class is rare.
- **Physical application.** Interpreting a positive result from a sensitive but imperfect sensor (e.g., a gas-leak detector) requires the base rate of real leaks.



S.8 Counting: Permutations & Combinations

$$n! = n(n-1)(n-2) \cdots 1, \quad 0! = 1 \quad {}_nP_r = \frac{n!}{(n-r)!} \quad \binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!} \quad \text{multiplication principle: } m \cdot n \text{ ways}$$

STATED-EQUATION n factorial equals n times n minus one down to one; zero factorial equals one. nPr equals n factorial over n minus r factorial. n choose r equals n factorial over r factorial times n minus r factorial. The multiplication principle: m ways then n ways gives m times n .

MEMORY AID Permutation: Position matters. Combination: Committee (order doesn't matter).

TERMS

$n!$ – factorial: arrangements of n distinct objects
 ${}_nP_r$ – ordered selections of r from n (order matters)
 $\binom{n}{r}$ – unordered selections (“ n choose r ”); binomial coefficient [A2.12]
 Relationship – ${}_nP_r = \binom{n}{r} r!$
 Subsets – a set of n items has 2^n subsets

CONTEXT

- **Origin.** Combinatorics in Indian mathematics (Pingala, ~200 BCE); Pascal's triangle (1654).
- **Use.** Probabilities of equally likely outcomes [S.5]; lotteries and card games.
- **Data science.** Size of search spaces: 2^p feature subsets, $\binom{p}{2}$ pairwise interactions, grid-search combinations of hyperparameters.
- **Physical application.** The number of distinct orderings of 5 lab samples in a queue is $5! = 120$.

S.9 Random Variables: Expected Value & Variance

$$E[X] = \mu = \sum_x x p(x) \quad E[X] = \int_{-\infty}^{\infty} x f(x) dx \quad \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

$$E[aX + b] = aE[X] + b \quad \text{Var}(aX + b) = a^2 \text{Var}(X) \quad E[X + Y] = E[X] + E[Y] \quad \text{Var}(X + Y) = \text{Var } X + \text{Var } Y \text{ (independent)}$$

STATED-EQUATION The expected value E of X equals the sum of $x p$ of x , or the integral of $x f$ of $x d x$. The variance equals the expected value of X minus μ squared, which equals E of X squared minus E of X , squared. E of aX plus b equals aE of X plus b . The variance of aX plus b equals a squared times variance of X . Expectations add. Variances add for independent variables.

MEMORY AID Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2 .

TERMS

X – random variable: a numeric outcome of a random process
 $p(x)$ – probability mass function (discrete); $\sum p(x) = 1$
 $f(x)$ – probability density (continuous) [S.12]; integral [C1.18]
 Linearity – expectation always adds, even for dependent variables
 Dependent case – add $2 \text{Cov}(X, Y)$ [S.18]

CONTEXT

- **Origin.** Christiaan Huygens (1657) on the value of a gamble.
- **Use.** Fair prices, insurance premiums, long-run averages.
- **Data science.** Expected revenue per user, expected loss (risk) minimized by ML training, and the variance of an ensemble average shrinking as models are added.
- **Physical application.** The expected energy of a gas molecule is an expected value over the Boltzmann distribution of speeds.

S.10 Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n \quad \mu = np \quad \sigma^2 = np(1 - p)$$

STATED-EQUATION P of X equals k equals n choose k , p to the k , times one minus p to the n minus k . The mean is np , and the variance is np times one minus p .

MEMORY AID BINS conditions: Binary outcomes, Independent trials, fixed Number, Same p .

TERMS

n – number of independent trials [S.6]

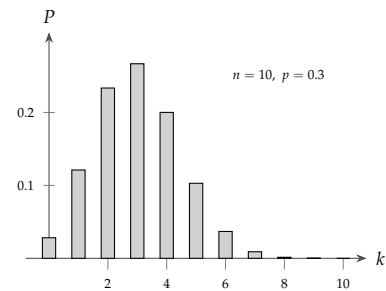
k – number of successes

p – probability of success on each trial

$\binom{n}{k}$ – ways to place k successes [S.8]

Conditions – fixed n , two outcomes, independent, constant p

Large n – approximately normal [S.12]



CONTEXT

- **Origin.** Jacob Bernoulli, *Ars Conjectandi* (1713).
- **Use.** Quality control, polling, free-throw percentages.
- **Data science.** Conversions out of n visitors in an A/B test; the Bernoulli likelihood is the basis of logistic regression's log-loss.
- **Physical application.** The number of defective chips in a batch of 100 with a 2% defect rate is binomial with $n = 100$, $p = 0.02$.

S.11 Poisson Distribution

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots \quad \mu = \lambda \quad \sigma^2 = \lambda$$

STATED-EQUATION P of X equals k equals lambda to the k times e to the minus lambda, over k factorial. The mean and variance both equal lambda.

MEMORY AID Poisson: mean = variance = λ .

TERMS

λ – average number of events per interval (rate $\times$ time)

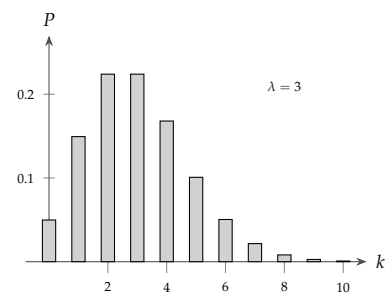
k – number of events observed

e – Euler's number [A2.6]

$k!$ – factorial [S.8]

Approximation – binomial with large n , small p , $\lambda = np$ [S.10]

Sum to 1 – by the series for e^λ [C2.13]



CONTEXT

- **Origin.** Siméon-Denis Poisson (1837); famously fit to Prussian cavalry deaths from horse kicks (Bortkiewicz, 1898).
- **Use.** Arrivals, defects, calls per hour, rare events.
- **Data science.** Modeling event counts (clicks, support tickets, server errors per minute); Poisson regression for count targets; alerting when counts are improbably high.
- **Physical application.** The number of radioactive decays detected by a Geiger counter per second is Poisson.

S.12 Normal Distribution & the CDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad X \sim N(\mu, \sigma^2) \quad F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \quad P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$$

STATED-EQUATION f of x equals one over sigma root two pi, times e to the minus x minus μ squared over two sigma squared. X is normal with mean μ and variance sigma squared. The CDF F of x is the probability X is at most x , the integral of f from minus infinity to x . P of a less than or equal X less than or equal b equals Φ of z b minus Φ of z a .

MEMORY AID 68–95–99.7: within 1, 2, 3 SDs.

TERMS

μ, σ – mean and standard deviation [S.2]

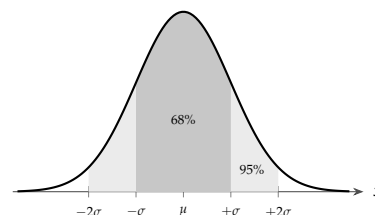
pdf $f(x)$ – density: height of the curve; area = probability

CDF $F(x)$ – accumulated area [C1.18]; $f = F'$

$\Phi(z)$ – standard normal CDF, with z from [S.4]

Empirical rule – 68%, 95%, 99.7% within 1, 2, 3 σ

e – [A2.6]; total area = 1 [C2.4]



CONTEXT

- **Origin.** De Moivre (1733) as a binomial limit; Gauss (1809) for measurement error.
- **Use.** Heights, measurement error, test scores, sums of many small effects.
- **Data science.** Gaussian noise in linear regression, Gaussian processes, weight initialization in neural nets, and the sampling distribution of almost every estimator [S.13].
- **Physical application.** Measurement errors on a precision scale are approximately normal, which is why repeated weighings are averaged.

S.13 Central Limit Theorem & Standard Error

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad (n \text{ large}) \quad SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

STATED-EQUATION $\bar{X}$ bar is approximately normal with mean μ and variance sigma squared over n , for large n . The standard error of $\bar{x}$ equals sigma over root n . The standard error of $\hat{p}$ equals the square root of p times one minus p over n . z equals $\bar{x}$ bar minus μ over sigma over root n .

MEMORY AID “Root- n law”: four times the data halves the error.

TERMS

$\bar{X}$ – sample mean, itself a random variable [S.9]

σ – population SD [S.2]; use s if unknown

n – sample size; rule of thumb $n \geq 30$

SE – standard deviation of a statistic’s sampling distribution

$\sqrt{n}$ – halving SE requires $4\times$ the data

Result – holds for almost any population shape

CONTEXT

- **Origin.** De Moivre (1733), Laplace (1810); named by Pólya (1920).
- **Use.** Justifies normal-based inference for averages and proportions.
- **Data science.** Why A/B tests, confidence intervals [S.14], and bootstrap estimates work; sample-size planning for experiments; SGD gradient noise shrinks like $1/\sqrt{\text{batch size}}$.
- **Physical application.** Averaging 100 noisy thermometer readings cuts the random error by a factor of $\sqrt{100} = 10$.

S.14 Confidence Intervals

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \quad \bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}} \quad \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad n = \left(\frac{z^* \sigma}{E} \right)^2$$

STATED-EQUATION $\bar{x}$ bar plus or minus z star sigma over root n ; $\bar{x}$ bar plus or minus t star with n minus one degrees of freedom, times s over root n ; $\hat{p}$ hat plus or minus z star times the square root of $\hat{p}$ times one minus $\hat{p}$ over n . Required sample size n equals z star sigma over E , squared.

MEMORY AID Estimate $\pm$ (critical value) $\times$ (standard error).

TERMS

z^* – critical value: 1.645 (90%), 1.96 (95%), 2.576 (99%) [S.12]

t_{n-1}^* – critical value from the t distribution [S.16]

$\hat{p}$ – sample proportion

E – margin of error (half-width) = critical value $\times$ SE [S.13]

Meaning – 95% of intervals built this way contain the true parameter

CONTEXT

- **Origin.** Jerzy Neyman (1937).
- **Use.** Reporting estimates with uncertainty; polling margins of error.
- **Data science.** Error bars on dashboards, uplift estimates from experiments, and sample-size calculators; always report the interval, not just the point estimate.
- **Physical application.** Reporting a lab measurement as 9.81 ± 0.02 m/s² is a confidence interval.

S.15 Hypothesis Testing, p -Values & Errors

$$H_0 : \mu = \mu_0 \text{ vs } H_a : \mu \neq \mu_0 \quad z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad p = P(|Z| \geq |z| \mid H_0) \quad \text{reject } H_0 \text{ if } p < \alpha$$

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) \quad \beta = P(\text{fail to reject} \mid H_a \text{ true}) \quad \text{power} = 1 - \beta$$

STATED-EQUATION H naught: μ equals μ naught, versus H a: μ not equal to μ naught. z equals $\bar{x}$ bar minus μ naught over sigma over root n . The p -value is the probability of a test statistic at least this extreme if H naught is true. Reject H naught if p is less than alpha. Alpha is P of rejecting H naught when it is true; beta is P of failing to reject when H a is true; power equals one minus beta.

MEMORY AID “If p is low, the null must go.” Type I = false alarm; Type II = missed detection.

TERMS

H_0, H_a – null (no effect) and alternative hypotheses

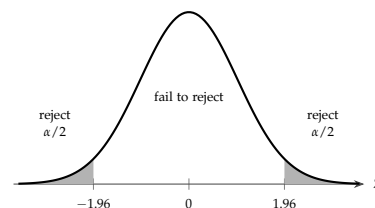
Test statistic – standardized distance from H_0 [S.4], [S.13]

p -value – probability of data this extreme if H_0 is true

α – significance level; Type I (false positive) rate

β – Type II (false negative) rate

Power – chance of detecting a real effect; rises with n



CONTEXT

- **Origin.** Fisher’s significance tests (1925); Neyman–Pearson framework (1933).
- **Use.** Scientific claims, quality control, clinical trials.
- **Data science.** A/B testing; multiple comparisons inflate false positives, so use Bonferroni (α/m) or FDR control; a p -value is *not* the probability H_0 is true (see [S.7]).
- **Physical application.** A factory tests whether a machine’s mean fill volume has drifted from 500 mL.

S.16 *t*-Tests

$$\text{one-sample: } t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

$$\text{Welch two-sample: } t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$\text{paired: } t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

STATED-EQUATION One-sample: t equals x bar minus μ naught over s over root n , with n minus one degrees of freedom. Welch two-sample: t equals x bar one minus x bar two over the square root of s one squared over n one plus s two squared over n two. Paired: t equals d bar over s d over root n .

TERMS

s – sample SD [S.2], replacing unknown σ
 df – degrees of freedom; $t \rightarrow N(0,1)$ as $df \rightarrow \infty$
 t distribution – like the normal but with heavier tails [S.12]
 Welch – does not assume equal variances (the safe default)
 $\bar{d}, s_d$ – mean and SD of paired differences
 Decision – compare p to α [S.15]

CONTEXT

- **Origin.** William Sealy Gosset, writing as “Student,” at the Guinness brewery (1908); Welch (1947).
- **Use.** Comparing means with small samples and unknown σ .
- **Data science.** Comparing a continuous metric (revenue per user, session length) between control and treatment; paired tests for before/after on the same users or models on the same folds.
- **Physical application.** Comparing the mean strength of concrete from two suppliers with small samples uses a two-sample t -test.

S.17 Chi-Square Tests

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \text{goodness of fit: } df = k - 1 \quad \text{independence: } E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}, \quad df = (r - 1)(c - 1)$$

STATED-EQUATION Chi squared equals the sum of observed minus expected, squared, over expected. Goodness of fit has k minus one degrees of freedom. For independence, the expected count is row total times column total over grand total, with r minus one times c minus one degrees of freedom.

TERMS

O, E – observed and expected counts (need $E \geq 5$)
 k – number of categories
 r, c – rows and columns of a contingency table
 Large χ^2 – data far from H_0 ; small p -value [S.15]
 Independence – E uses $P(A \cap B) = P(A)P(B)$ [S.6]

CONTEXT

- **Origin.** Karl Pearson (1900).
- **Use.** Testing fairness of dice, genetics ratios, survey cross-tabs.
- **Data science.** Conversion-rate A/B tests as 2×2 tables, sample-ratio-mismatch checks, and χ^2 feature selection for categorical features.
- **Physical application.** Testing whether a Geiger counter’s counts per interval follow a Poisson distribution is a goodness-of-fit test.

S.18 Covariance & Correlation

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad r = \frac{\text{Cov}(X, Y)}{s_x s_y} = \frac{1}{n-1} \sum_{i=1}^n z_{x_i} z_{y_i} \quad -1 \leq r \leq 1$$

STATED-EQUATION The covariance of X and Y equals one over n minus one times the sum of x_i minus $\bar{x}$, times y_i minus $\bar{y}$. r equals the covariance over $s_x s_y$, which equals one over n minus one times the sum of z_x times z_y . r is between minus one and one.

MEMORY AID Correlation is not causation.

TERMS

$\bar{x}, \bar{y}$ – means [S.1]; s_x, s_y SDs [S.2]

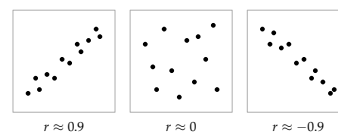
Covariance – direction of joint variation; depends on units

r – Pearson correlation: unit-free linear association

z – z -scores [S.4]

Geometry – r is the cosine similarity of centered vectors [T.12]

Caution – r measures linear association; correlation $\neq$ causation



CONTEXT

- **Origin.** Galton (1888) and Pearson (1896).
- **Use.** Measuring how two variables move together.
- **Data science.** Correlation matrices and heatmaps in EDA, detecting multicollinearity, covariance matrices in PCA and portfolio risk.
- **Physical application.** Air temperature and air density are strongly negatively correlated at fixed pressure.

S.19 Least-Squares Linear Regression & R^2

$$\hat{y} = b_0 + b_1 x \quad b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = r \frac{s_y}{s_x} \quad b_0 = \bar{y} - b_1 \bar{x}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad \hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$$

STATED-EQUATION $\hat{y}$ equals b naught plus b one x . b one equals the sum of x_i minus $\bar{x}$ times y_i minus $\bar{y}$, over the sum of x_i minus $\bar{x}$ squared, which equals r times s_y over s_x . b naught equals $\bar{y}$ minus b one $\bar{x}$. R squared equals one minus the residual sum of squares over the total sum of squares. In matrix form, $\hat{\beta}$ equals X transpose X inverse, X transpose $\mathbf{y}$.

MEMORY AID The regression line always passes through $(\bar{x}, \bar{y})$.

TERMS

$\hat{y}$ – predicted value; $e_i = y_i - \hat{y}_i$ residual

b_1 – slope [A1.4]: change in $\hat{y}$ per unit of x

b_0 – intercept; the line passes through $(\bar{x}, \bar{y})$

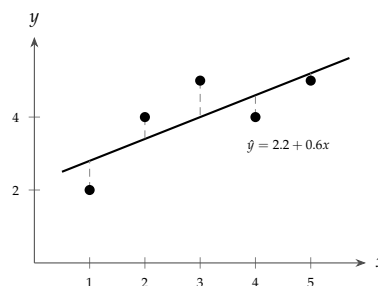
r – correlation [S.18]; in simple regression $R^2 = r^2$

Least squares – minimizes $\sum e_i^2$ by setting derivatives to 0 [C1.13]

Matrix form – normal equations [A2.13], [A2.14]

CONTEXT

- **Origin.** Legendre (1805) and Gauss (1809); “regression” coined by Galton (1886).
- **Use.** Prediction and quantifying relationships.
- **Data science.** The baseline model for everything; interpretable coefficients for stakeholders; R^2 is the share of variance explained, and the same MSE loss drives neural-net regression.
- **Physical application.** Calibrating a spring: fit force versus stretch by least squares; the slope is the spring constant k .



S.20 Continuous Distributions: Uniform & Exponential

$$\text{Uniform}(a, b) : f(x) = \frac{1}{b-a}, \mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12} \quad \text{Exponential}(\lambda) : f(x) = \lambda e^{-\lambda x}, F(x) = 1 - e^{-\lambda x}, \mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

STATED-EQUATION Uniform on a to b : f of x equals one over b minus a ; the mean is a plus b over two; the variance is b minus a squared over twelve. Exponential with rate λ : f of x equals λ times e to the minus λ times x ; F of x equals one minus e to the minus λ times x ; the mean is one over λ and the variance one over λ squared.

MEMORY AID Exponential: mean = $1/\lambda$; “memoryless” — the past doesn’t matter.

TERMS

a, b — endpoints of the interval; every value equally likely

λ — rate (events per unit time), same as Poisson [S.11]

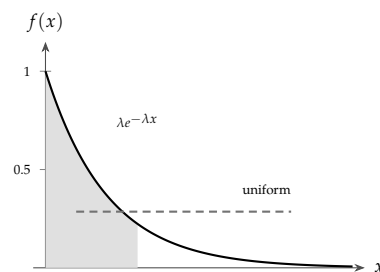
$F(x)$ — CDF = $\int_0^x f$ [C1.18], [S.12]

Memoryless — $P(X > s+t | X > s) = P(X > t)$ [S.6]

Mean — $E[X]$ [S.9], computed by parts [C2.1]

CONTEXT

- **Origin.** The exponential arises as the waiting time between Poisson events (Erlang, early 1900s).
- **Use.** Waiting times, lifetimes of components, random numbers.
- **Data science.** Uniform draws power random search and Monte Carlo; exponential models time-to-churn, inter-arrival times in queueing, and survival analysis baselines.
- **Physical application.** The waiting time until the next radioactive decay is exponential; its memorylessness is why a half-life never “resets.”



S.21 Maximum Likelihood Estimation (MLE)

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta) \quad \ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i | \theta) \quad \hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta) : \frac{d\ell}{d\theta} = 0$$

STATED-EQUATION The likelihood L of θ is the product over the data of f of x_i given θ . The log-likelihood, ℓ of θ , is the sum of the logs. The maximum likelihood estimate $\hat{\theta}$ is the value that maximizes ℓ , found by setting $d\ell/d\theta$ equal to zero.

MEMORY AID “Which parameter makes the data I saw most probable?” Take logs to turn products into sums.

TERMS

$f(x | \theta)$ — pmf or pdf with parameter θ [S.10]–[S.12]

Product — data assumed independent [S.6]

Log — monotone, so same maximizer; sums are easier [A2.7]

Solve — derivative = 0 [C1.12] or numerically [O.9]

Known results — Bernoulli $\hat{p} = \bar{x}$; Poisson $\hat{\lambda} = \bar{x}$; exponential $\hat{\lambda} = 1/\bar{x}$; normal $\hat{\mu} = \bar{x}$

CONTEXT

- **Origin.** R. A. Fisher (1912–1922); earlier ideas from Gauss and Lagrange.
- **Use.** The standard way to fit parameters of probability models.
- **Data science.** Minimizing MSE is MLE under Gaussian noise; minimizing log-loss is MLE for Bernoulli labels [ML.2]; most `fit()` methods are MLE or regularized MLE.
- **Physical application.** Estimating the decay rate of a radioactive sample from measured decay times.

11

Optimization & Simulation

Choosing the best decision subject to constraints, and estimating what is too hard to compute exactly. Linear, integer, and nonlinear programming decide schedules, routes, and portfolios; Monte Carlo methods turn randomness into answers.

0.1 Anatomy of an Optimization Problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p \quad \max f = -\min(-f)$$

STATED-EQUATION Minimize f of x over x in $\mathbb{R}^n$, subject to g_i of x less than or equal to zero and h_j of x equal to zero. Maximizing f is the same as minimizing minus f .

MEMORY AID Three parts: decisions, objective, constraints.

TERMS

$\mathbf{x}$ – decision variables (what you control)
 $f(\mathbf{x})$ – objective: cost, loss, or –profit
 g_i, h_j – inequality and equality constraints
Feasible region – all $\mathbf{x}$ satisfying every constraint (compare [A1.15])
Classes – LP: linear f, g, h [0.2]; MIP: some x_k integer [0.4]; NLP: non-linear [0.9]
Convex problem – convex f and feasible set [0.8]; every local optimum is global

CONTEXT

- **Origin.** Calculus of variations (Euler, Lagrange, 1700s); operations research born in WWII logistics.
- **Use.** Scheduling, routing, pricing, portfolio allocation, engineering design.
- **Data science.** Training a model is an optimization problem: $\min_{\theta}$ loss + penalty [ML.1]; hyperparameter tuning is optimization over a black-box function.
- **Physical application.** Choosing the shape of a beam to minimize weight subject to a maximum allowed deflection is a constrained optimization problem.

0.2 Linear Programming (LP)

$$\max_{\mathbf{x}} z = \mathbf{c}^T \mathbf{x} \quad \text{subject to} \quad A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \quad \text{Corner-point theorem: an optimum (if one exists) occurs at a vertex}$$

STATED-EQUATION Maximize z equals $\mathbf{c}$ transpose $\mathbf{x}$, subject to $A\mathbf{x}$ less than or equal to $\mathbf{b}$ and $\mathbf{x}$ greater than or equal to zero. An optimum, if one exists, occurs at a vertex of the feasible region.

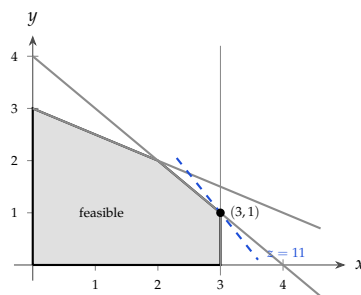
MEMORY AID “Check the corners.”

TERMS

$\mathbf{c}$ – objective coefficients (profit per unit)
 $A, \mathbf{b}$ – constraint matrix and right-hand sides [A2.13]
Feasible region – convex polygon/polyhedron formed by half-planes [A1.15]
Iso-profit line – $\mathbf{c}^T \mathbf{x} = k$, slid until it last touches the region [A1.5]
Outcomes – unique optimum, multiple optima, infeasible, or unbounded
Solvers – simplex [0.3] and interior-point methods

CONTEXT

- **Origin.** Kantorovich (1939) and Dantzig (1947); Nobel Prize in economics 1975.
- **Use.** Production mix, diets, blending, transportation, staffing.
- **Data science.** Budget allocation across ad channels, supply-chain planning, L1 (least absolute deviation) regression and quantile regression are solvable as LPs.
- **Physical application.** Blending ores or fuels to meet chemical specifications at minimum cost is a classic LP.



0.3 Simplex Method & LP Duality

$$\begin{array}{ll} \text{Primal: } \max \mathbf{c}^\top \mathbf{x} \text{ s.t. } A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0} & \text{Dual: } \min \mathbf{b}^\top \mathbf{y} \text{ s.t. } A^\top \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0} \\ \text{Weak: } \mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y} & \text{Strong: } \mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^* \\ \text{Complementary slackness: } y_i^* (b_i - (A\mathbf{x}^*)_i) = 0 \end{array}$$

STATED-EQUATION *Primal: maximize c transpose x subject to A x at most b, x at least zero. Dual: minimize b transpose y subject to A transpose y at least c, y at least zero. Weak duality: c transpose x is at most b transpose y. Strong duality: at the optimum they are equal. Complementary slackness: y i star times the slack of constraint i equals zero.*

MEMORY AID Each primal constraint becomes a dual variable; max becomes min.

TERMS

Simplex – walks from vertex to adjacent vertex [02], improving z each pivot

Slack variable – $s_i = b_i - (A\mathbf{x})_i \geq 0$ turns $\leq$ into $=$

y – dual variables = shadow prices: value of one more unit of resource b_i

$A^\top$ – transpose [A2.13]

Binding constraint – $s_i = 0$; only these can have $y_i > 0$

CONTEXT

► **Origin.** Dantzig's simplex (1947); duality theory by von Neumann (1947) and Gale–Kuhn–Tucker (1951).

► **Use.** Pricing scarce resources; certifying that a solution is optimal.

► **Data science.** Shadow prices tell a planner which capacity to expand; duality is the same idea behind the SVM dual and kernel trick [ML.13] and Lagrangian relaxation [0.10].

► **Physical application.** Shadow prices tell a power-grid operator how much an extra megawatt of line capacity is worth.

0.4 Integer & Mixed-Integer Programming (MIP)

$$\max \mathbf{c}^\top \mathbf{x} \text{ s.t. } A\mathbf{x} \leq \mathbf{b}, x_j \in \mathbb{Z} (j \in I), y_k \in \{0,1\} \quad z_{\text{MIP}} \leq z_{\text{LP relax}} \quad \text{big-M: } x \leq M y$$

STATED-EQUATION *Maximize c transpose x subject to A x at most b, with x j integer for j in I and y k binary. The MIP optimum is at most the LP-relaxation optimum. Big-M: x at most M times y.*

TERMS

I – index set of integer variables (the rest are continuous)

Binary y – yes/no decisions: open a warehouse, pick an item

LP relaxation – drop integrality; gives an upper bound (for max) [02]

Branch and bound – split on a fractional x_j ($x_j \leq \lfloor v \rfloor$ or $x_j \geq \lceil v \rceil$) and prune by bounds

Big-M – large constant that switches a constraint on or off

Hardness – NP-hard in general; rounding the LP can be infeasible or far from optimal

CONTEXT

► **Origin.** Gomory cutting planes (1958); branch and bound by Land and Doig (1960).

► **Use.** Facility location, crew scheduling, production with setup costs.

► **Data science.** Optimal assignment of promotions under budget, shift scheduling, sparse (best-subset) regression, and decision optimization after an ML forecast (“predict, then optimize”).

► **Physical application.** Deciding which power plants to switch on each hour (unit commitment) is a mixed-integer program.

0.5 Knapsack Problem

$$\max \sum_{i=1}^n v_i x_i \quad \text{s.t.} \quad \sum_{i=1}^n w_i x_i \leq W, \quad x_i \in \{0, 1\} \quad V(i, w) = \max \{V(i-1, w), v_i + V(i-1, w - w_i)\}$$

STATED-EQUATION Maximize the sum of $v_i x_i$ subject to the sum of $w_i x_i$ at most W , with each x_i zero or one. The recurrence: V of i, w equals the maximum of V of i minus one, w , and v_i plus V of i minus one, w minus w_i .

MEMORY AID Each item: “skip it or take it” — keep the better.

TERMS

v_i, w_i – value and weight of item i

W – capacity (budget)

x_i – binary: take item i or not [0.4]

$V(i, w)$ – best value using items $1..i$ with capacity w (dynamic programming [0.12])

Cost – $O(nW)$ table vs 2^n subsets [S.8]

Fractional version – greedy by v_i/w_i is optimal; gives the LP bound

CONTEXT

- **Origin.** Studied by Dantzig (1957); named by Tobias Dantzig; one of Karp’s 21 NP-complete problems (1972).
- **Use.** Capital budgeting, cargo loading, cutting stock.
- **Data science.** Selecting features or experiments under a compute budget, choosing which campaigns to fund for maximum expected lift, and packing jobs onto GPUs.
- **Physical application.** Packing a spacecraft’s limited mass budget with the most valuable instruments is a knapsack problem.

0.6 Traveling Salesman Problem (TSP)

$$\min \sum_{i \neq j} c_{ij} x_{ij} \quad \text{s.t.} \quad \sum_j x_{ij} = 1 \quad \forall i, \quad \sum_i x_{ij} = 1 \quad \forall j, \quad \sum_{i,j \in S} x_{ij} \leq |S| - 1 \quad \forall S \subset V, \quad x_{ij} \in \{0, 1\} \quad \frac{(n-1)!}{2} \text{ tours}$$

STATED-EQUATION Minimize the sum of $c_{ij} x_{ij}$ subject to: each city is left once, each city is entered once, no subtours on any subset S , and x_{ij} binary. There are n minus one factorial over two distinct tours.

MEMORY AID $(n-1)!/2$: fix the start, and a tour and its reverse are the same.

TERMS

c_{ij} – distance or cost from city i to j [G.6]

x_{ij} – 1 if the tour goes $i \rightarrow j$ [0.4]

Degree constraints – leave and enter each city once

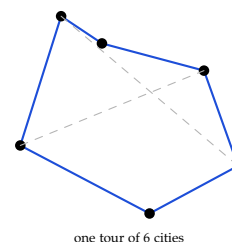
Subtour elimination – forbids disconnected loops on subsets S

$(n-1)!/2$ – distinct symmetric tours [S.8]

Heuristics – nearest neighbor, 2-opt, simulated annealing [0.13]

CONTEXT

- **Origin.** Posed by Menger (1930s); Dantzig–Fulkerson–Johnson solved 49 cities (1954); Concorde now solves tens of thousands.
- **Use.** Delivery routes, drilling circuit boards, genome sequencing.
- **Data science.** Last-mile delivery and ride-pooling routing (vehicle routing problems), warehouse pick paths, and ordering items to minimize changeover.
- **Physical application.** Routing a laser or drill head over thousands of points on a circuit board is a TSP.



0.7 Shortest Paths & Network Flow

$$d(s) = 0, \quad d(v) = \min_{(u,v) \in E} [d(u) + w(u,v)] \quad \max \sum_v f_{sv} \text{ s.t. } 0 \leq f_{uv} \leq c_{uv}, \sum_u f_{uv} = \sum_w f_{vw} \quad \max \text{ flow} = \min \text{ cut}$$

STATED-EQUATION d of s equals zero, and d of v equals the minimum over edges $u \rightarrow v$ of d of u plus w of $u \rightarrow v$. Max flow: maximize the flow out of s subject to capacities and flow conservation at each node. Max flow equals min cut.

MEMORY AID A chain is only as strong as its weakest link: max flow = min cut.

TERMS

$d(v)$ – shortest distance from source s to node v

$w(u, v)$ – edge weight (length, time, cost)

Dijkstra – greedy, needs $w \geq 0$; Bellman–Ford handles negative edges

f_{uv}, c_{uv} – flow and capacity on edge $u \rightarrow v$

Conservation – flow in = flow out at every non-terminal node

Cut – set of edges separating s from t ; its capacity bounds the flow (duality [0.3])

CONTEXT

► **Origin.** Dijkstra (1956), Bellman (1958); max-flow min-cut by Ford and Fulkerson (1956).

► **Use.** GPS navigation, network routing, pipeline and traffic capacity.

► **Data science.** Graph analytics (shortest paths for centrality), matching riders to drivers (bipartite flow), image segmentation by min cut, and the Bellman equation in RL [0.12].

► **Physical application.** Water-distribution and traffic networks: the maximum throughput equals the capacity of the tightest bottleneck cut.

0.8 Convex Sets & Convex Functions

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}), \quad \lambda \in [0, 1] \quad f''(x) \geq 0 \text{ or } \nabla^2 f(\mathbf{x}) \succeq 0 \quad \text{local min} = \text{global min}$$

STATED-EQUATION f of λx plus one minus λy is at most λf of x plus one minus λf of y , for λ between zero and one. Equivalently, f double prime at least zero, or the Hessian is positive semidefinite. For convex problems a local minimum is a global minimum.

MEMORY AID Convex looks like a bowl: any chord lies above the curve.

TERMS

Convex set – contains the segment between any two of its points

λ – mixing weight; $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ traces the segment

Chord condition – the graph lies below every chord

$f'' \geq 0$ – concave up [C1.12]

Hessian $\nabla^2 f$ – matrix of second partials; $\succeq 0$ means all eigenvalues ≥ 0 [A2.14]

Closed under – sums, positive scaling, max, and composition with affine maps

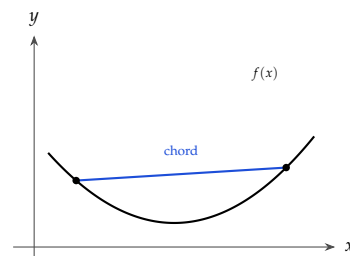
CONTEXT

► **Origin.** Jensen's inequality (1906); modern convex optimization by Rockafellar, Boyd and Vandenberghe.

► **Use.** Guarantees that algorithms find the true optimum efficiently.

► **Data science.** MSE, log-loss, and hinge loss are convex in linear-model parameters, so linear/logistic regression and SVMs have one global optimum; deep networks are non-convex.

► **Physical application.** Potential energy of a stretched spring $\frac{1}{2} kx^2$ is convex: the equilibrium is the unique minimum.



0.9 Nonlinear Programming & Newton's Method

$$\text{First order: } \nabla f(\mathbf{x}^*) = 0 \quad \text{Second order: } \nabla^2 f(\mathbf{x}^*) \succ 0 \quad x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} \quad \mathbf{x}_{k+1} = \mathbf{x}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

STATED-EQUATION First order: the gradient at x star equals zero. Second order: the Hessian at x star is positive definite. Newton's method: x k plus one equals x k minus f prime of x k over f double prime of x k ; in several variables, x k plus one equals x k minus the Hessian inverse times the gradient.

TERMS

∇f – gradient; zero at a stationary point (compare [C1.12])
 $\nabla^2 f$ – Hessian; positive definite $\Rightarrow$ strict local min [O.8]
 Newton step – minimizes the 2nd-order Taylor model [C2.13]
 Gradient descent – first-order alternative $\mathbf{x} - \eta \nabla f$ [C1.13]
 Quasi-Newton – BFGS/L-BFGS approximate $[\nabla^2 f]^{-1}$ cheaply
 Convergence – Newton is quadratic near the optimum, but each step costs a linear solve

CONTEXT

- **Origin.** Newton (1669) and Raphson (1690) for roots; applied to optimization by Simpson (1740).
- **Use.** Engineering design, curve fitting, maximum-likelihood estimation.
- **Data science.** Logistic regression is fit by Newton's method (IRLS); `scikit-learn` defaults to L-BFGS; XGBoost uses gradient and Hessian of the loss.
- **Physical application.** Finding the equilibrium shape of a hanging chain by minimizing potential energy is an NLP.

0.10 Lagrange Multipliers

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c) \quad \nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \quad g(\mathbf{x}) = c \quad \lambda = \frac{\partial f^*}{\partial c}$$

STATED-EQUATION The Lagrangian $\mathcal{L}$ of x and λ equals f of x minus λ times g of x minus c . At the optimum, the gradient of f equals λ times the gradient of g , and g of x equals c . λ equals the partial derivative of the optimal value with respect to c .

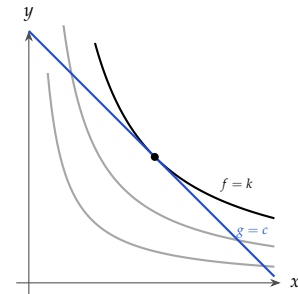
MEMORY AID At the optimum the level curve kisses (is tangent to) the constraint.

TERMS

f – objective; $g(\mathbf{x}) = c$ equality constraint
 λ – multiplier: rate the optimum changes as c changes (a shadow price [O.3])
 Geometry – at the optimum, the level curve of f is tangent to the constraint, so gradients are parallel [T.12]
 $\mathcal{L}$ – Lagrangian; set all partial derivatives to 0
 Inequalities – generalize to KKT [O.11]

CONTEXT

- **Origin.** Joseph-Louis Lagrange, *Mécanique analytique* (1788).
- **Use.** Constrained design, economics (utility under a budget), physics.
- **Data science.** Derives maximum-entropy distributions (the normal and softmax), PCA as $\max \mathbf{v}^\top S \mathbf{v}$ s.t. $\|\mathbf{v}\| = 1$ [ML.19], and regularization as a constraint $\|\beta\| \leq t$ [ML.5].
- **Physical application.** Designing a cylindrical tank of fixed volume with minimum surface area uses a Lagrange multiplier.



0.11 KKT Conditions

$$\min f(\mathbf{x}) \text{ s.t. } g_i(\mathbf{x}) \leq 0 : \quad \nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}, \quad g_i(\mathbf{x}^*) \leq 0, \quad \mu_i \geq 0, \quad \mu_i g_i(\mathbf{x}^*) = 0$$

STATED-EQUATION Minimize f subject to g_i at most zero: the gradient of f plus the sum of μ_i times the gradient of g_i equals zero; each g_i is at most zero; each μ_i is at least zero; and μ_i times g_i equals zero.

MEMORY AID Complementary slackness: either the constraint is tight or its multiplier is zero.

TERMS

Stationarity – Lagrange condition [0.10] with inequality multipliers μ_i
 Primal feasibility – $g_i \leq 0$
 Dual feasibility – $\mu_i \geq 0$
 Complementary slackness – either the constraint is active ($g_i = 0$) or $\mu_i = 0$ [0.3]
 Sufficiency – for convex problems [0.8], KKT points are global optima

CONTEXT

- **Origin.** Karush (1939 thesis) and Kuhn–Tucker (1951).
- **Use.** The optimality test built into every constrained NLP solver.
- **Data science.** In SVMs, the training points with $\mu_i > 0$ are exactly the support vectors [ML.13]; KKT also explains why lasso sets coefficients exactly to zero [ML.5].
- **Physical application.** A ball resting against a wall: the wall pushes back (multiplier > 0) only when the ball touches it (constraint active).

0.12 Dynamic Programming & the Bellman Equation

$$V(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right] \quad \text{optimal substructure + overlapping subproblems} \quad \sum_{t=0}^{\infty} \gamma^t r = \frac{r}{1 - \gamma}$$

STATED-EQUATION V of s equals the maximum over actions a of r of s , a plus gamma times the sum over s prime of P of s prime given s , a times V of s prime. It requires optimal substructure and overlapping subproblems. The discounted sum of a constant reward r is r over one minus gamma.

MEMORY AID “Solve each subproblem once and remember it.”

TERMS

$V(s)$ – value: best total reward from state s
 $a, r(s, a)$ – action and its immediate reward
 γ – discount factor, $0 \leq \gamma < 1$ (geometric series [A2.10])
 $P(s' | s, a)$ – transition probability [S.6]
 Memoization – store subproblem answers (knapsack table [0.5], shortest paths [0.7])

CONTEXT

- **Origin.** Richard Bellman (1950s) at RAND.
- **Use.** Inventory control, resource allocation, sequence alignment, speech decoding (Viterbi).
- **Data science.** Reinforcement learning (value iteration, Q-learning), edit distance for fuzzy matching, and the Viterbi algorithm in HMMs and sequence labeling.
- **Physical application.** Optimal control of a rocket’s fuel burn stage by stage is solved with dynamic programming.

0.13 Heuristics & Simulated Annealing

$$P(\text{accept}) = \begin{cases} 1, & \Delta \leq 0 \\ e^{-\Delta/T}, & \Delta > 0 \end{cases} \quad T_k = T_0 \alpha^k \quad (0 < \alpha < 1) \quad \Delta = f(\mathbf{x}_{\text{new}}) - f(\mathbf{x}_{\text{current}})$$

STATED-EQUATION The acceptance probability is one if delta is at most zero, and e to the minus delta over T if delta is positive. The temperature cools as T_k equals T naught alpha to the k , with alpha between zero and one. Delta equals the new cost minus the current cost.

MEMORY AID Hot → explore; cool → settle.

TERMS

Δ – change in cost of a proposed move (minimization)
 T – temperature: high T explores, low T exploits
 α – cooling rate (exponential decay [A2.6])
 Greedy – best immediate choice; fast but can be suboptimal
 Local search – improve by small moves (2-opt for TSP [0.6])
 Genetic algorithms – selection, crossover, mutation on a population

CONTEXT

- **Origin.** Kirkpatrick, Gelatt, Vecchi (1983), inspired by the Metropolis algorithm [0.16] and annealing of metals.
- **Use.** Near-optimal answers to NP-hard problems in reasonable time.
- **Data science.** Hyperparameter search, feature-subset search, and large routing/scheduling problems where exact MIP [0.4] is too slow.
- **Physical application.** Annealing steel: slow cooling lets atoms settle into a low-energy crystal instead of a brittle, disordered one.

0.14 Monte Carlo Estimation & Integration

$$\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^N f(X_i) \rightarrow E[f(X)] \quad \int_a^b f(x) dx \approx \frac{b-a}{N} \sum_{i=1}^N f(U_i), \quad U_i \sim \text{Unif}(a, b) \quad SE = \frac{\hat{\sigma}}{\sqrt{N}}$$

STATED-EQUATION *Theta hat N equals one over N times the sum of f of X i, which converges to the expected value of f of X. The integral from a to b of f of x d x is approximately b minus a over N times the sum of f of U i, with U i uniform on a to b. The standard error equals sigma hat over root N.*

MEMORY AID Error shrinks like $1/\sqrt{N}$: 100 times the samples for one more digit.

TERMS

X_i – independent random draws

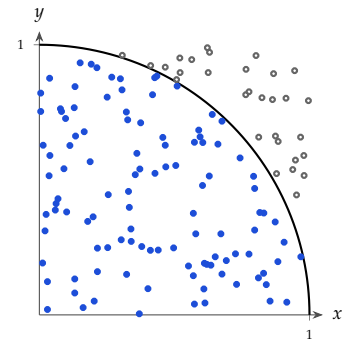
$\hat{\theta}_N$ – sample-mean estimate (law of large numbers [S.1])

$b - a$ – interval length; the formula is average value $\times$ length [C2.8]

$\hat{\sigma}$ – sample SD of $f(X_i)$ [S.2]

Error rate $\propto 1/\sqrt{N}$ regardless of dimension [S.13]

π estimate $\sim 4 \times$ fraction of random points inside the quarter circle



CONTEXT

- **Origin.** Ulam, von Neumann, and Metropolis at Los Alamos (1946); named after the casino.
- **Use.** High-dimensional integrals, physics simulation, financial risk (VaR).
- **Data science.** Simulating revenue under uncertainty, bootstrap confidence intervals, estimating expected loss, and Monte Carlo dropout for prediction uncertainty.
- **Physical application.** Physicists estimate radiation shielding by simulating millions of random particle paths.

0.15 Random Sampling Methods

$$\text{Inverse transform: } X = F^{-1}(U), \quad U \sim \text{Unif}(0, 1) \quad \text{Rejection: accept } x \sim g \text{ if } U \leq \frac{f(x)}{Mg(x)} \quad \text{Importance: } E_p[h(X)] = E_q\left[h(X) \frac{p(X)}{q(X)}\right]$$

STATED-EQUATION *Inverse transform: X equals F inverse of U, with U uniform on zero to one. Rejection sampling: draw x from g and accept it if U is at most f of x over M g of x. Importance sampling: the expectation under p of h of X equals the expectation under q of h of X times p of X over q of X.*

TERMS

F^{-1} – inverse of the CDF [A2.1], [S.12]

Exponential case – $X = -\ln(1 - U)/\lambda$ [S.20]

g, M – easy proposal density and a bound with $f \leq Mg$; acceptance rate $1/M$

q – importance (proposal) distribution; p/q are the weights

Uniform – $X = a + (b - a)U$

CONTEXT

- **Origin.** Von Neumann described inverse transform and rejection sampling (1951).
- **Use.** Generating random variates for any simulation [0.14].
- **Data science.** Synthetic data generation, off-policy evaluation of recommenders with importance weights, and correcting for sampling bias with inverse-propensity weights.
- **Physical application.** Simulating the random lifetimes of light bulbs from uniform random numbers uses the inverse CDF of the exponential.

0.16 Markov Chain Monte Carlo (Metropolis–Hastings)

$$\alpha(x \rightarrow x') = \min \left\{ 1, \frac{p(x') q(x | x')}{p(x) q(x' | x)} \right\} \quad \text{symmetric } q : \alpha = \min \left\{ 1, \frac{p(x')}{p(x)} \right\} \quad \text{accept if } U \leq \alpha$$

STATED-EQUATION The acceptance probability α from x to x prime equals the minimum of one and p of x prime q of x given x prime, over p of x q of x prime given x . For a symmetric proposal, α equals the minimum of one and p of x prime over p of x . Accept if U is at most α .

MEMORY AID Always accept uphill in probability; accept downhill sometimes, with probability equal to the ratio.

TERMS

$p(x)$ – target density, needed only up to a constant (e.g. an unnormalized posterior [S.7])

$q(x' | x)$ – proposal distribution (e.g. $x' = x + \varepsilon$, $\varepsilon \sim N(0, s^2)$ [S.12])

Markov chain – next state depends only on the current state [S.8]

Burn-in – early draws discarded before the chain reaches the target

Mixing – autocorrelation reduces the effective sample size

CONTEXT

- **Origin.** Metropolis et al. (1953); generalized by Hastings (1970); Gibbs sampling by Geman and Geman (1984).
- **Use.** Statistical physics, Bayesian inference with intractable normalizing constants.
- **Data science.** Bayesian models in PyMC and Stan (Hamiltonian MC), posterior uncertainty for A/B tests and hierarchical models.
- **Physical application.** The original 1953 Metropolis algorithm simulated atoms in a liquid at a fixed temperature.

0.17 Algorithmic Complexity (Big-O)

$$f(n) = O(g(n)) \iff \exists c, n_0 : f(n) \leq c g(n) \quad \forall n \geq n_0 \quad O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$$

STATED-EQUATION f of n is big O of g of n if and only if there are constants c and n naught such that f of n is at most c times g of n for all n at least n naught. The common orders, from fastest to slowest: constant, logarithmic, linear, $n \log n$, quadratic, exponential, factorial.

MEMORY AID Drop constants and lower-order terms: $3n^2 + 5n + 7 = O(n^2)$.

TERMS

n – input size

$O(\log n)$ – binary search: halve each step [A2.7]

$O(n \log n)$ – efficient sorting

$O(n^2)$ – all pairs (e.g. naive k -NN, distance matrices)

$O(2^n)$, $O(n!)$ – brute-force subsets and tours [O.5], [O.6]

Space – memory complexity is measured the same way

CONTEXT

- **Origin.** Bachmann (1894) and Landau (1909) for number theory; Knuth popularized it in computing (1976).
- **Use.** Predicting how runtime scales with data size.
- **Data science.** Choosing algorithms for big data: exact k -NN is $O(n)$ per query, approximate nearest-neighbor indexes are sublinear; transformer attention is $O(n^2)$ in sequence length.
- **Physical application.** Simulating gravity between n stars by all pairs is $O(n^2)$; tree codes cut it to $O(n \log n)$, making galaxy simulations feasible.

12 Machine Learning

Supervised learning (predicting a labeled target) and unsupervised learning (finding structure without labels). Every method below is assembled from earlier pieces: linear algebra, calculus, probability, and optimization.

ML.1 The Learning Problem & Empirical Risk

$$\hat{\theta} = \arg \min_{\theta} \underbrace{\frac{1}{n} \sum_{i=1}^n L(y_i, f_{\theta}(\mathbf{x}_i))}_{\text{empirical risk}} + \underbrace{\lambda R(\theta)}_{\text{regularization}} \quad \text{goal: small } E_{(\mathbf{x}, y)}[L(y, f_{\hat{\theta}}(\mathbf{x}))] \text{ on new data}$$

STATED-EQUATION $\hat{\theta}$ equals the arg min over θ of one over n times the sum of L of y_i and f_{θ} of $\mathbf{x}_i$, the empirical risk, plus λR of θ , the regularization. The goal is small expected loss on new data.

TERMS

$\mathbf{x}_i, y_i$ – features and label (target) of example i

f_{θ} – model with parameters θ

L – loss function [ML.2]

$\lambda R(\theta)$ – complexity penalty [ML.5]

arg min – solved by optimization [O.1], [ML.16]

Tasks – regression (numeric y), classification (categorical y), clustering and dimension reduction (no y)

CONTEXT

- **Origin.** Statistical learning theory of Vapnik and Chervonenkis (1970s); “machine learning” coined by Arthur Samuel (1959).
- **Use.** Prediction from data instead of hand-written rules.
- **Data science.** The template for every model in this section; held-out test data estimates the true (expected) risk [S.9].
- **Physical application.** Fitting a drag coefficient to wind-tunnel measurements is empirical-risk minimization with a physical model.

ML.2 Loss Functions

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad \text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \quad \text{log-loss} = -[y \ln p + (1 - y) \ln(1 - p)] \quad \text{hinge} = \max(0, 1 - y f(\mathbf{x}))$$

STATED-EQUATION MSE equals one over n times the sum of y_i minus $\hat{y}_i$, squared. MAE equals one over n times the sum of the absolute errors. Log-loss equals minus the quantity y natural log p plus one minus y natural log of one minus p . Hinge loss equals the maximum of zero and one minus $y f$ of $\mathbf{x}$.

MEMORY AID $\text{MSE} \rightarrow$ mean; $\text{MAE} \rightarrow$ median.

TERMS

$\hat{y}$ – prediction; $y - \hat{y}$ residual [S.19]

MSE – penalizes large errors; optimal constant prediction is the mean [S.1]

MAE – robust to outliers [A1.7]; optimal constant is the median

Log-loss – p predicted probability, $y \in \{0, 1\}$; the negative Bernoulli log-likelihood [S.10]

Hinge – $y \in \{-1, +1\}$; used by SVMs [ML.13]

Huber – quadratic near 0, linear in the tails

CONTEXT

- **Origin.** Least squares (Gauss, Legendre); log-loss from maximum likelihood (Fisher, 1920s).
- **Use.** Defines what “good” means for a model.
- **Data science.** Match the loss to the business cost: MAE for forecast error in units, log-loss for calibrated probabilities, quantile (pinball) loss for inventory buffers.
- **Physical application.** Calibrating a thermometer by minimizing squared error against a reference is least squares.

ML.3 Entropy, Cross-Entropy & KL Divergence

$$H(p) = - \sum_k p_k \log p_k \quad H(p, q) = - \sum_k p_k \log q_k \quad D_{\text{KL}}(p \parallel q) = \sum_k p_k \log \frac{p_k}{q_k} \geq 0 \quad H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$$

STATED-EQUATION Entropy H of p equals minus the sum of $p_k \log p_k$. Cross-entropy H of p, q equals minus the sum of $p_k \log q_k$. The KL divergence of p from q equals the sum of $p_k \log p_k$ over q_k , which is at least zero. Cross-entropy equals entropy plus KL divergence.

MEMORY AID KL is not symmetric: $D(p \parallel q) \neq D(q \parallel p)$.

TERMS

p – true distribution; q model's predicted distribution
 $\log$ – base 2 gives bits, base e gives nats [A2.7], [A2.8]
 $H(p)$ – uncertainty; maximal for a uniform distribution
 Cross-entropy – average code length using q when data follow p
 KL – extra cost of using q ; zero iff $p = q$; not symmetric

CONTEXT

- **Origin.** Claude Shannon, "A Mathematical Theory of Communication" (1948); KL by Kullback and Leibler (1951).
- **Use.** Data compression, communication, measuring information.
- **Data science.** Cross-entropy is the classification loss [ML.15]; decision-tree splits maximize information gain [ML.11]; KL appears in VAEs, t-SNE, and drift monitoring.
- **Physical application.** Thermodynamic entropy $S = k_B \ln W$ [R.18] is the physical ancestor of Shannon entropy.

ML.4 Bias–Variance Trade-off

$$E[(y - \hat{f}(x))^2] = \underbrace{(E[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$$

STATED-EQUATION The expected squared error equals bias squared, the square of the expected prediction minus the true value, plus variance, the variance of the prediction, plus sigma squared, the irreducible noise.

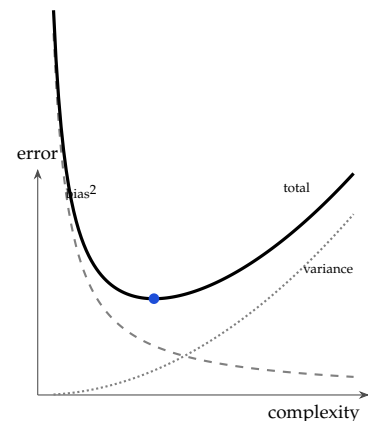
MEMORY AID Too simple $\rightarrow$ bias (underfit); too complex $\rightarrow$ variance (overfit).

TERMS

f – true function; $\hat{f}$ model fit on a random training set
 Bias – systematic error from an overly simple model (underfitting)
 Variance – sensitivity to the particular training sample (overfitting) [S.9]
 σ^2 – noise in y that no model can remove
 Diagnosis – low train error but high validation error $\Rightarrow$ variance

CONTEXT

- **Origin.** Formalized for neural networks by Geman, Bienenstock and Doursat (1992).
- **Use.** Choosing model complexity.
- **Data science.** Tune depth, k , or λ with validation data [ML.6]; bagging reduces variance [ML.12]; regularization trades a little bias for less variance [ML.5].
- **Physical application.** A very flexible curve through noisy lab measurements (high variance) predicts a new trial worse than a simple physical law.



ML.5 Regularization: Ridge, Lasso & Elastic Net

$$\text{Ridge: } \min_{\beta} \|\mathbf{y} - X\beta\|_2^2 + \lambda \|\beta\|_2^2, \quad \hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \quad \text{Lasso: } \min_{\beta} \|\mathbf{y} - X\beta\|_2^2 + \lambda \|\beta\|_1$$

$$\|\beta\|_1 = \sum_j |\beta_j| \quad \|\beta\|_2^2 = \sum_j \beta_j^2 \quad \text{Elastic net: } \lambda [\alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|_2^2]$$

STATED-EQUATION Ridge: minimize the squared norm of $\mathbf{y}$ minus $X\beta$ plus λ times the squared L2 norm of β ; $\hat{\beta}$ equals $X^\top X$ plus λ times I , inverse, $X^\top \mathbf{y}$. Lasso: the same with λ times the L1 norm. The L1 norm is the sum of absolute values; the squared L2 norm is the sum of squares. Elastic net mixes them with weight α .

MEMORY AID Lasso has a Lasso that ropes coefficients to exactly zero; ridge just shrinks.

TERMS

λ – penalty strength; $\lambda = 0$ gives OLS [S.19]
 $\|\cdot\|_2$ – Euclidean norm [T.12]; $\|\cdot\|_1$ sum of absolute values
 I – identity; λI makes $X^\top X$ invertible [A2.14]
 Ridge – shrinks all coefficients toward 0
 Lasso – sets some coefficients exactly to 0 (feature selection) [O.11]
 Scaling – standardize features first [S.4]

CONTEXT

- **Origin.** Ridge by Hoerl and Kennard (1970); lasso by Tibshirani (1996); elastic net by Zou and Hastie (2005).
- **Use.** Stable estimates with many correlated predictors.
- **Data science.** Preventing overfitting in wide data [ML.4]; weight decay in neural networks is L2; Bayesian view: ridge = Gaussian prior, lasso = Laplace prior [S.7].
- **Physical application.** Tikhonov regularization (ridge) stabilizes ill-posed inverse problems like deblurring images or CT reconstruction.

ML.6 Cross-Validation

$$CV_{(k)} = \frac{1}{k} \sum_{j=1}^k \text{Err}_j \quad \text{fold } j: \text{ train on } \frac{k-1}{k}n \text{ rows, evaluate on } \frac{n}{k} \quad \text{LOOCV: } k = n \quad SE_{CV} \approx \frac{s_{\text{Err}}}{\sqrt{k}}$$

STATED-EQUATION The k -fold CV error equals one over k times the sum of the fold errors. Each fold trains on k minus one over k of the rows and evaluates on n over k . Leave-one-out uses k equals n . The standard error is about s of the errors over root k .

TERMS

k – number of folds (typically 5 or 10)
 Err_j – validation error on fold j (any loss [ML.2])
 Mean – averaging reduces variance of the estimate [S.1], [S.13]
 Stratified – keeps class proportions in each fold
 Time series – use forward-chaining splits, never future data
 Leakage – fit scalars and feature selection inside each fold

CONTEXT

- **Origin.** Stone (1974) and Geisser (1975).
- **Use.** Estimating out-of-sample performance without wasting data.
- **Data science.** Selecting λ , depth, or k [ML.4]; comparing models with paired tests on the same folds [S.16].
- **Physical application.** Validating a heat-transfer correlation on held-out experiments, rather than the ones used to fit it.

ML.7 Logistic Regression

$$p = P(y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x} \quad \ln \frac{p}{1-p} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x} \quad \text{odds ratio} = e^{\beta_j}$$

STATED-EQUATION p , the probability y equals one given x , equals sigma of z , which is one over one plus e to the minus z , where z equals beta naught plus beta transpose x . The log-odds, natural log of p over one minus p , equals beta naught plus beta transpose x . The odds ratio for feature j is e to the beta j .

MEMORY AID Logistic regression is linear in the log-odds.

TERMS

$\sigma(z)$ – sigmoid (logistic curve [C2.16]); maps $\mathbb{R} \rightarrow (0, 1)$

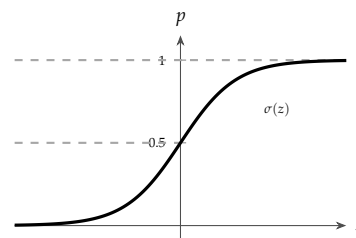
$\frac{p}{1-p}$ – odds; its log is the logit [C2.3]

β_j – change in log-odds per unit of x_j

e^{β_j} – multiplicative change in odds [A2.6]

Fitting – minimize log-loss [ML.2] (convex [0.8]) by Newton/L-BFGS [0.9]

Boundary – $p = 0.5 \iff z = 0$: a hyperplane



CONTEXT

- **Origin.** Berkson (1944); David Cox (1958).
- **Use.** Probability of a binary outcome: default, disease, purchase.
- **Data science.** The standard baseline classifier: churn, conversion, credit scoring; interpretable odds ratios; the output layer of binary neural networks.
- **Physical application.** The probability that a material fractures versus applied stress is often modeled with a logistic curve.

ML.8 Classification Metrics

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall (TPR)} = \frac{TP}{TP + FN} \quad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad \text{Accuracy} = \frac{TP + TN}{N} \quad \text{FPR} = \frac{FP}{FP + TN}$$

STATED-EQUATION Precision equals TP over TP plus FP . Recall, the true positive rate, equals TP over TP plus FN . F_1 equals two times precision times recall over precision plus recall. Accuracy equals TP plus TN over N . The false positive rate equals FP over FP plus TN .

MEMORY AID Precision: “of those I flagged, how many were right?” Recall: “of the real ones, how many did I catch?”

TERMS

TP, FP – true and false positives (FP = Type I error [S.15])

TN, FN – true and false negatives (FN = Type II error)

Precision – $P(\text{actual+} \mid \text{predicted+})$ [S.6]

Recall – $P(\text{predicted+} \mid \text{actual+})$; sensitivity

F_1 – harmonic mean of precision and recall

ROC AUC – area under TPR vs FPR as the threshold moves [G.8], [C2.14]

	pred +	pred –
actual +	TP 40	FN 20
actual –	FP 10	TN 130

CONTEXT

- **Origin.** Precision and recall from information retrieval (Kent et al., 1955); ROC from WWII radar.
- **Use.** Evaluating any classifier or diagnostic test.
- **Data science.** With rare positives (fraud, 1%), accuracy is misleading; choose thresholds by the cost of FP vs FN ; report PR curves for imbalanced data.
- **Physical application.** An earthquake early-warning system trades recall (catching quakes) against false alarms.

ML.9 *k*-Nearest Neighbors & Distance Metrics

$$d_2(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_j (x_j - z_j)^2} \quad d_1(\mathbf{x}, \mathbf{z}) = \sum_j |x_j - z_j| \quad \cos \theta = \frac{\mathbf{x} \cdot \mathbf{z}}{\|\mathbf{x}\| \|\mathbf{z}\|} \quad \hat{y} = \text{majority (or mean) of } N_k(\mathbf{x})$$

STATED-EQUATION The Euclidean distance *d* two equals the square root of the sum of *x j* minus *z j*, squared. The Manhattan distance *d* one equals the sum of the absolute differences. Cosine similarity equals *x* dot *z* over the product of the norms. Predict the majority, or mean, of the *k* nearest neighbors.

TERMS

d_2 – Euclidean distance [G.6]

d_1 – Manhattan (taxicab) distance

$\cos \theta$ – cosine similarity [T.12]

$N_k(\mathbf{x})$ – the *k* training points closest to *x*

Choosing *k* – small *k*: high variance; large *k*: high bias [ML.4]

Scaling – standardize so no feature dominates [S.4]

CONTEXT

- **Origin.** Fix and Hodges (1951); Cover and Hart (1967).
- **Use.** Simple, assumption-free classification and regression.
- **Data science.** Recommendation (“users like you”), vector search over embeddings with cosine similarity, and imputation of missing values from nearest rows.
- **Physical application.** Classifying a mineral sample by comparing its measured density and hardness to the nearest known samples.

ML.10 Naive Bayes Classifier

$$P(y | \mathbf{x}) \propto P(y) \prod_{j=1}^d P(x_j | y) \quad \hat{y} = \arg \max_y \left[\log P(y) + \sum_j \log P(x_j | y) \right] \quad \hat{P}(x_j | y) = \frac{n_{jy} + 1}{n_y + V}$$

STATED-EQUATION *P* of *y* given *x* is proportional to *P* of *y* times the product over *j* of *P* of *x j* given *y*. Predict the *y* that maximizes log *P* of *y* plus the sum of log *P* of *x j* given *y*. With Laplace smoothing, *P* hat of *x j* given *y* equals *n j y* plus one over *n y* plus *V*.

TERMS

$P(y)$ – class prior [S.7]

$P(x_j | y)$ – likelihood of feature *j* given the class

Naive assumption – features conditionally independent given *y* [A.2.7]

Logs – sum of logs avoids underflow [A.2.7]

Laplace smoothing – +1 counts; *V* is vocabulary size

Variants – multinomial (counts), Bernoulli, Gaussian

CONTEXT

- **Origin.** Applied to text classification by Maron (1961); spam filtering popularized by Paul Graham (2002).
- **Use.** Fast, strong baseline for text and categorical data.
- **Data science.** Spam and sentiment classification, document tagging, and quick baselines on high-dimensional sparse features.
- **Physical application.** Diagnosing a machine fault from several independent sensor alarms combines their likelihoods naively.

ML.11 Decision Trees: Gini, Entropy & Information Gain

$$\text{Gini} = 1 - \sum_k p_k^2 \quad H = - \sum_k p_k \log_2 p_k \quad IG = H(\text{parent}) - \sum_{c \in \text{children}} \frac{n_c}{n} H(c)$$

STATED-EQUATION Gini impurity equals one minus the sum of *p k* squared. Entropy *H* equals minus the sum of *p k* log base two of *p k*. Information gain equals the parent entropy minus the weighted sum of child entropies, *n c* over *n* times *H* of *c*.

TERMS

p_k – fraction of class *k* in a node

Gini – probability two random picks disagree; 0 for a pure node

H – entropy [ML.3]

IG – reduction in impurity; the weighted average uses $\frac{n_c}{n}$ [S.1]

Greedy growth – choose the split with largest gain at each node [S.13]

Regression trees – split to reduce variance [S.2]

CONTEXT

- **Origin.** CART by Breiman et al. (1984); ID3/C4.5 by Quinlan (1986, 1993).
- **Use.** Interpretable if-then rules; handles mixed feature types.
- **Data science.** The building block of random forests and gradient boosting [ML.12]; decision rules stakeholders can read.
- **Physical application.** A field guide’s key for identifying trees (“leaves opposite? then...”) is a decision tree.

ML.12 Ensembles: Bagging, Random Forests & Boosting

$$\text{Bagging: } \bar{f} = \frac{1}{B} \sum_{b=1}^B \hat{f}_b, \quad \text{Var}(\bar{f}) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2 \quad \text{Boosting: } F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x}) \quad \alpha_m = \frac{1}{2} \ln \frac{1 - \epsilon_m}{\epsilon_m}$$

STATED-EQUATION Bagging averages B models: $\bar{f}$ bar equals one over B times the sum of $\hat{f}$ hat b , with variance ρ sigma squared plus one minus ρ over B times sigma squared. Boosting: F_m equals F_{m-1} plus ν h_m . AdaBoost weight α_m equals one half natural log of one minus epsilon m over epsilon m .

MEMORY AID Bagging fights variance; boosting fights bias.

TERMS

B – number of models, each fit on a bootstrap sample (OOB fraction $\approx e^{-1}$ [C1.2])
 ρ – correlation between models [S.18]; random forests lower it by sampling features
 σ^2 – variance of a single model [S.9]
 h_m – weak learner fit to the current residuals (negative gradient) [C1.13]
 ν – learning rate (shrinkage)
 α_m, ϵ_m – AdaBoost weight and weighted error of learner m

CONTEXT

- **Origin.** Bagging and random forests by Breiman (1996, 2001); AdaBoost by Freund and Schapire (1997); gradient boosting by Friedman (2001).
- **Use.** State-of-the-art accuracy on tabular data.
- **Data science.** XGBoost, LightGBM, and CatBoost dominate tabular ML (churn, fraud, pricing); random forests give out-of-bag validation and feature importances for free.
- **Physical application.** Weather forecasters run ensembles of simulations with perturbed initial conditions and average them.

ML.13 Support Vector Machines

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \text{margin} = \frac{2}{\|\mathbf{w}\|} \quad K(\mathbf{x}, \mathbf{z}) = e^{-\gamma \|\mathbf{x} - \mathbf{z}\|^2}$$

STATED-EQUATION Minimize one half the squared norm of $\mathbf{w}$ plus C times the sum of the slacks ξ_i , subject to y_i times $\mathbf{w}$ transpose $\mathbf{x}_i$ plus b being at least one minus ξ_i , and ξ_i at least zero. The margin equals two over the norm of $\mathbf{w}$. The RBF kernel K of $\mathbf{x}, \mathbf{z}$ equals e to the minus gamma times the squared distance between $\mathbf{x}$ and $\mathbf{z}$.

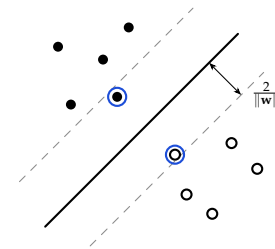
MEMORY AID Maximize the street width; only the points on the curb (support vectors) matter.

TERMS

$\mathbf{w}, b$ – normal vector and offset of the hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$ [T.12]
 $y_i \in \{-1, +1\}$ – class labels
 ξ_i – slack; equivalent to hinge loss [ML.2]
 C – trade-off between wide margin and training errors
 Support vectors – points on or inside the margin ($\mu_i > 0$ [Q.11])
 Kernel K – inner product in a feature space; RBF shown

CONTEXT

- **Origin.** Vapnik and Chervonenkis (1963); soft margin and kernels by Cortes and Vapnik (1995).
- **Use.** Robust classification in high dimensions.
- **Data science.** Text and image classification before deep learning; still strong for small, wide datasets; the kernel trick underlies Gaussian processes.
- **Physical application.** Classifying particle-detector events (signal versus background) was a classic SVM application in physics.



ML.14 Neural Networks & Backpropagation

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad \mathbf{a}^{(l)} = \phi(\mathbf{z}^{(l)}) \quad \text{ReLU}(z) = \max(0, z) \quad \sigma'(z) = \sigma(z)(1 - \sigma(z))$$

$$\delta^{(l)} = \left(\mathbf{W}^{(l+1)\top} \delta^{(l+1)} \right) \odot \phi'(\mathbf{z}^{(l)}) \quad \frac{\partial L}{\partial \mathbf{W}^{(l)}} = \delta^{(l)} \mathbf{a}^{(l-1)\top}$$

STATED-EQUATION $\mathbf{z}^{(l)}$ equals $\mathbf{W}^{(l)}$ times $\mathbf{a}^{(l-1)}$ plus $\mathbf{b}^{(l)}$, and $\mathbf{a}^{(l)}$ equals ϕ of $\mathbf{z}^{(l)}$. ReLU of z equals the maximum of zero and z . Sigma prime of z equals sigma of z times one minus sigma of z . Backpropagation: delta l equals $\mathbf{W}^{(l+1)}$ transpose times delta l plus one, elementwise times phi prime of $\mathbf{z}^{(l)}$. The gradient with respect to $\mathbf{W}^{(l)}$ equals delta l times $\mathbf{a}^{(l-1)}$ transpose.

MEMORY AID Forward pass computes; backward pass blames (chain rule, layer by layer).

TERMS

$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$ – weights and biases of layer l (matrix product [A2.13])

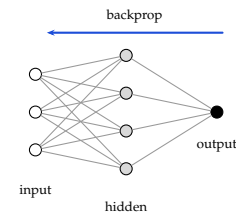
ϕ – activation (ReLU, sigmoid [ML.7], tanh)

$\mathbf{a}^{(0)}$ – input $\mathbf{x}$

$\delta^{(l)}$ – error signal $\partial L / \partial \mathbf{z}^{(l)}$, passed backward by the chain rule [C1.7]

$\odot$ – elementwise product

Universal approximation – one wide hidden layer can approximate any continuous function



CONTEXT

- **Origin.** Perceptron by Rosenblatt (1958); backpropagation popularized by Rumelhart, Hinton and Williams (1986).
- **Use.** Learning features automatically from raw data.
- **Data science.** Deep learning for images (CNNs), sequences and language (transformers), and tabular embeddings; trained with SGD/Adam [ML.16].
- **Physical application.** Neural networks now emulate expensive physics simulations (weather, fluid flow) thousands of times faster.

ML.15 Softmax & Cross-Entropy Loss

$$\hat{p}_k = \text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad L = - \sum_{k=1}^K y_k \ln \hat{p}_k \quad \frac{\partial L}{\partial \mathbf{z}} = \hat{\mathbf{p}} - \mathbf{y}$$

STATED-EQUATION $\hat{p}_k$ equals softmax of $\mathbf{z}$ at k , which is e to the z_k over the sum over j of e to the z_j . The loss L equals minus the sum of y_k natural log $\hat{p}_k$. The gradient with respect to $\mathbf{z}$ equals $\hat{\mathbf{p}}$ minus $\mathbf{y}$.

MEMORY AID Gradient is “prediction minus truth.”

TERMS

$\mathbf{z}$ – logits (unnormalized scores) from the last layer

$\hat{p}_k$ – predicted probabilities; positive and sum to 1 [S.5]

$\mathbf{y}$ – one-hot true label

L – cross-entropy [ML.3]; with $K = 2$ it is log-loss [ML.2]

Stability – subtract $\max_j z_j$ before exponentiating [A2.6]

Temperature – $e^{z_k/T}$ sharpens ($T < 1$) or flattens ($T > 1$)

CONTEXT

- **Origin.** Boltzmann distribution in statistical physics; used in multinomial logit models (Luce, 1959).
- **Use.** Multi-class classification.
- **Data science.** The output layer of image and text classifiers and language models; temperature controls randomness when an LLM samples the next token.
- **Physical application.** Softmax is the Boltzmann distribution: states with energy E_k occur with probability $\propto e^{-E_k/k_B T}$.

ML.16 SGD, Momentum & Adam

SGD: $\theta \leftarrow \theta - \eta \nabla_{\theta} L_{\text{batch}}$ Momentum: $\mathbf{v} \leftarrow \beta \mathbf{v} + \nabla L$, $\theta \leftarrow \theta - \eta \mathbf{v}$

Adam: $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1) \mathbf{g}$, $\mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1 - \beta_2) \mathbf{g}^2$, $\hat{\mathbf{m}} = \frac{\mathbf{m}}{1 - \beta_1^t}$, $\hat{\mathbf{v}} = \frac{\mathbf{v}}{1 - \beta_2^t}$, $\theta \leftarrow \theta - \eta \frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}} + \epsilon}}$

STATED-EQUATION SGD: θ becomes θ minus η times the gradient of the mini-batch loss. Momentum: $\mathbf{v}$ becomes β times $\mathbf{v}$ plus the gradient, and θ becomes θ minus η times $\mathbf{v}$. Adam: $\mathbf{m}$ becomes β_1 times $\mathbf{m}$ plus $(1 - \beta_1)$ times $\mathbf{g}$; $\mathbf{v}$ becomes β_2 times $\mathbf{v}$ plus $(1 - \beta_2)$ times $\mathbf{g}^2$; $\hat{\mathbf{m}}$ equals $\mathbf{m}$ over $1 - \beta_1$ to the t ; $\hat{\mathbf{v}}$ equals $\mathbf{v}$ over $1 - \beta_2$ to the t ; θ becomes θ minus η times $\hat{\mathbf{m}}$ over the square root of $\hat{\mathbf{v}}$ plus ϵ .

MEMORY AID Adam = momentum ($\mathbf{m}$) + per-parameter step sizes ($\mathbf{v}$).

TERMS

η – learning rate [C1.13]

Mini-batch – random subset of rows; gradient is an unbiased Monte

Carlo estimate [O.14]

β, β_1 – momentum decay (≈ 0.9); $\beta_2 \approx 0.999$

$\mathbf{m}, \mathbf{v}$ – running means of the gradient and its square (exponential moving averages [A2.10])

Bias correction – divides by $1 - \beta^t$ since averages start at 0

Epoch – one pass through the data: $\lceil n/\text{batch} \rceil$ steps

CONTEXT

► **Origin.** Robbins–Monro (1951); momentum by Polyak (1964); Adam by Kingma and Ba (2014).

► **Use.** Optimizing huge models where exact gradients are too expensive.

► **Data science.** Default optimizers for deep learning (Adam, AdamW); step-size schedules satisfy convergence conditions [C2.10].

► **Physical application.** A heavy ball rolling down a bumpy hill (momentum) coasts through small dips that would trap a massless particle.

ML.17 k -Means Clustering

$$J = \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad \text{assign: } c_i = \arg \min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad \text{update: } \boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$$

STATED-EQUATION J equals the sum over clusters k of the sum over points in C_k of the squared distance from $\mathbf{x}_i$ to $\boldsymbol{\mu}_k$. Assign each point to the nearest centroid; update each centroid to the mean of its points.

MEMORY AID Assign, update, repeat.

TERMS

K – number of clusters (chosen by elbow or silhouette [ML.18])

C_k – set of points in cluster k

$\boldsymbol{\mu}_k$ – centroid: the mean [S.1]

J – within-cluster sum of squares (inertia), distance [ML.9]

Lloyd’s algorithm – alternate assign and update; J never increases

Caveats – local optima (use k -means++ init), scale features [S.4]

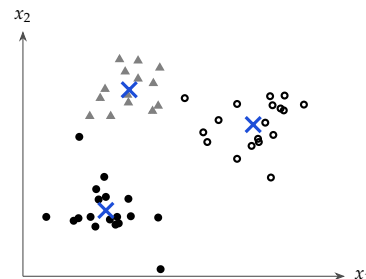
CONTEXT

► **Origin.** Lloyd (1957, published 1982); MacQueen named it (1967).

► **Use.** Grouping similar items; vector quantization.

► **Data science.** Customer segmentation, color quantization of images, and clustering embeddings to discover topics or intents.

► **Physical application.** Grouping weather stations with similar climate records into climate zones.



ML.18 Hierarchical Clustering & Silhouette

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1] \quad d_{\text{single}} = \min_{\mathbf{x} \in A, \mathbf{z} \in B} d(\mathbf{x}, \mathbf{z}) \quad d_{\text{complete}} = \max_{\mathbf{x} \in A, \mathbf{z} \in B} d(\mathbf{x}, \mathbf{z}) \quad d_{\text{avg}} = \text{mean } d$$

STATED-EQUATION The silhouette s of i equals b of i minus a of i , over the maximum of a of i and b of i , between minus one and one. Single linkage uses the minimum pairwise distance, complete linkage the maximum, and average linkage the mean.

TERMS

$a(i)$ – mean distance from point i to its own cluster
 $b(i)$ – mean distance to the nearest other cluster
 $s(i)$ – near 1: well placed; near 0: on a boundary; < 0: likely misassigned
 Agglomerative – start with singletons and repeatedly merge the closest pair
 Linkage – rule for the distance between clusters [ML.9]
 Dendrogram – tree of merges; cut it at a height to get clusters

CONTEXT

- **Origin.** Hierarchical methods from numerical taxonomy (Sokal and Sneath, 1963); silhouette by Rousseeuw (1987).
- **Use.** Taxonomies, phylogenetic trees, choosing the number of clusters.
- **Data science.** Gene-expression heatmaps, grouping products into a hierarchy, and choosing K for k -means [ML.17] by average silhouette.
- **Physical application.** Building a family tree of species from DNA distances (phylogenetics) uses hierarchical clustering.

ML.19 Principal Component Analysis (PCA)

$$S = \frac{1}{n-1} X_c^\top X_c \quad S \mathbf{v}_k = \lambda_k \mathbf{v}_k, \lambda_1 \geq \lambda_2 \geq \dots \quad \text{explained variance ratio} = \frac{\lambda_k}{\sum_j \lambda_j} \quad \mathbf{z} = V_k^\top (\mathbf{x} - \bar{\mathbf{x}})$$

STATED-EQUATION S equals one over n minus one times X c transpose X c . $S \mathbf{v}_k$ equals $\lambda_k \mathbf{v}_k$, with eigenvalues in decreasing order. The explained variance ratio equals λ_k over the sum of all λ_j . The scores $\mathbf{z}$ equal V k transpose times $\mathbf{x}$ minus $\bar{\mathbf{x}}$.

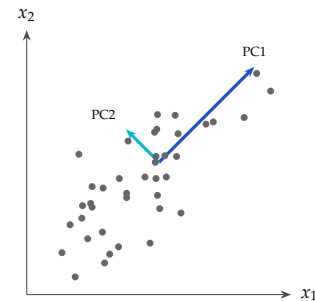
MEMORY AID PC1 points along the longest direction of the data cloud.

TERMS

X_c – centered data matrix (subtract column means [S.1])
 S – covariance matrix [S.18]
 $\mathbf{v}_k, \lambda_k$ – eigenvectors (directions) and eigenvalues (variances) [A2.14]
 PC1 – direction of maximum variance: $\max \mathbf{v}^\top S \mathbf{v}, \|\mathbf{v}\| = 1$ [O.10]
 $\mathbf{z}$ – coordinates after projecting onto the top k components (dot products [T.12])
 SVD – $X_c = U \Sigma V^\top$ gives the same V , with $\lambda = \sigma^2 / (n-1)$

CONTEXT

- **Origin.** Karl Pearson (1901); Harold Hotelling (1933).
- **Use.** Dimension reduction, visualization, and decorrelation.
- **Data science.** Compressing hundreds of correlated features, 2-D plots of embeddings, noise reduction, and removing multicollinearity before regression.
- **Physical application.** The principal axes of a rigid body's inertia tensor are its eigenvectors, the same math as PCA.



ML.20 Gaussian Mixtures & the EM Algorithm

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \sigma_k^2) \quad \text{E-step: } \gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i | \mu_j, \sigma_j^2)} \quad \text{M-step: } \mu_k = \frac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}, \quad \pi_k = \frac{N_k}{n}$$

STATED-EQUATION p of x equals the sum over k of π_k times the normal density with mean μ_k and variance σ_k^2 . E-step: the responsibility γ_{ik} equals $\pi_k \mathcal{N}(x_i | \mu_k, \sigma_k^2)$ given component k , over the sum over j of the same. M-step: μ_k equals the sum of $\gamma_{ik} x_i$ over the sum of γ_{ik} , and π_k equals N_k over n .

MEMORY AID E-step: guess memberships; M-step: refit the Gaussians.

TERMS

π_k – mixing weights, $\sum \pi_k = 1$

$\mathcal{N}$ – normal density [S.12]

γ_{ik} – responsibility: posterior $P(\text{component } k | x_i)$ by Bayes [S.7]

N_k – effective count $\sum_i \gamma_{ik}$; M-step uses weighted means [S.1]

EM – alternates E and M; never decreases the likelihood

Relation – soft version of k -means [ML.17]

CONTEXT

- **Origin.** Dempster, Laird and Rubin (1977); mixtures studied by Pearson (1894).
- **Use.** Density estimation and soft clustering; missing-data problems.
- **Data science.** Anomaly detection by low density, soft customer segments with membership probabilities, and speaker identification.
- **Physical application.** Separating overlapping peaks in a spectrometer reading is fitting a Gaussian mixture.

ML.21 ROC, AUC & Precision–Recall Curves

$$\text{ROC: } (\text{FPR}(\tau), \text{TPR}(\tau)) \quad \text{AUC} = \int_0^1 \text{TPR} d(\text{FPR}) = P(s(\mathbf{x}_+) > s(\mathbf{x}_-)) \quad \text{PR: } (\text{Recall}(\tau), \text{Precision}(\tau)) \quad \text{PR baseline} = \pi_+$$

STATED-EQUATION The ROC curve plots the false positive rate against the true positive rate as the threshold τ varies. The AUC, the area under it, equals the probability that a random positive gets a higher score than a random negative. The precision-recall curve plots precision against recall. The precision-recall baseline equals the prevalence of positives, π_+ .

MEMORY AID Lower the threshold $\Rightarrow$ flag more $\Rightarrow$ recall up, precision (usually) down. Raise it $\Rightarrow$ the reverse.

TERMS

τ – decision threshold on the score $s(\mathbf{x})$

TPR, FPR – [ML.8]

AUC – threshold-free ranking quality; area [C1.17], parametric curve [C2.14]

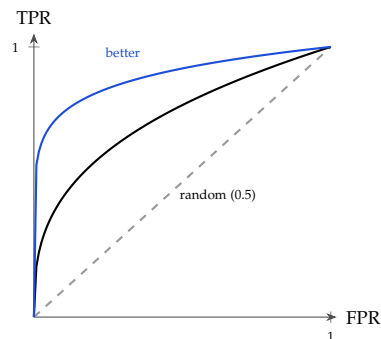
Diagonal – random guessing, AUC = 0.5

π_+ – fraction of positives; a random classifier's precision

Imbalanced data – prefer PR curves: ROC can look good while precision is poor

CONTEXT

- **Origin.** ROC analysis from WWII radar operators; adopted in medicine (1970s) and machine learning (1990s).
- **Use.** Comparing classifiers and choosing operating thresholds.
- **Data science.** Pick the threshold by business cost: fraud screening favors high recall, marketing lists favor high precision; report PR-AUC when positives are rare.
- **Physical application.** Setting the sensitivity of a smoke detector: more sensitive catches every fire but triggers on burnt toast.



ML.22 Regression Metrics

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \quad \text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \quad \text{MAPE} = \frac{100\%}{n} \sum_i \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$$

STATED-EQUATION RMSE is the square root of the mean squared error. MAE is the mean absolute error. MAPE is one hundred percent over n times the sum of the absolute errors divided by the actual values. Adjusted R squared equals one minus the quantity one minus R squared, times n minus one over n minus p minus one.

MEMORY AID RMSE punishes big misses; MAE treats all misses equally; both are in the units of y .

TERMS

$y, \hat{y}$ – actual and predicted values

RMSE – same units as y ; $\text{RMSE} \geq \text{MAE}$ always

MAPE – scale-free percentage; undefined when $y = 0$

R^2 – share of variance explained [S.19]

p – number of predictors; adjusted R^2 penalizes adding useless ones

Losses – [ML.2]

CONTEXT

► **Origin.** Least squares (Gauss); adjusted R^2 by Ezekiel (1930).

► **Use.** Evaluating forecasts and continuous predictions.

► **Data science.** Report RMSE or MAE in business units (“off by \$12 on average”); compare to a naive baseline such as predicting the mean or last value.

► **Physical application.** Evaluating a weather model’s temperature forecasts: an MAE of 1.5°C is useful, 8°C is not.

ML.23 Reading Metrics: Bad, Good & Perfect

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad \text{Brier} = \frac{1}{n} \sum_i (p_i - y_i)^2 \quad \text{baseline log-loss (balanced)} = \ln 2 \approx 0.693$$

STATED-EQUATION The Matthews correlation coefficient equals TP times TN minus FP times FN , over the square root of the product of the four sums. The Brier score is the mean squared difference between predicted probability and outcome. A balanced binary classifier that always predicts one half has log-loss natural log of two, about zero point six nine three.

MEMORY AID A metric means nothing without its baseline: always ask “compared to what?”

Metric	Range	Bad / baseline	Good (rule of thumb)	Perfect
Accuracy	0 to 1	majority-class rate	well above the baseline	1
Precision	0 to 1	prevalence (random)	high when false alarms are costly	1 (no false alarms)
Recall	0 to 1	fraction flagged at random	high when misses are costly	1 (no misses)
F_1	0 to 1	near 0	> 0.7 common target	1
ROC AUC	0 to 1	0.5 (random); < 0.5 inverted	0.7–0.8 fair, 0.8–0.9 good, > 0.9 excellent	1
PR AUC	0 to 1	prevalence π_+	many times π_+	1
Log-loss	0 to ∞	0.693 (balanced, $p = 0.5$)	well below baseline	0
Brier	0 to 1	$\pi_+(1 - \pi_+)$; 0.25 balanced	well below baseline	0
MCC	–1 to 1	0 (random); –1 inverted	> 0.5 strong	1
R^2	– ∞ to 1	0 (predict the mean); < 0 worse	0.5–0.9 depends on field	1
RMSE, MAE	0 to ∞	SD of y (mean baseline)	small vs. SD and business tolerance	0
Silhouette	–1 to 1	≤ 0 overlapping/misassigned	> 0.5 reasonable structure	1

TERMS

Baseline – simplest possible model (majority class, mean, last value)

MCC – balanced summary of the confusion matrix, robust to imbalance [ML.8]

Brier – MSE of probabilities; rewards calibration [ML.2]

Calibration – predicted 70% should happen about 70% of the time

Context – “good” depends on the field: $R^2 = 0.3$ can be excellent in social science, poor in physics

CONTEXT

► **Origin.** MCC by Brian Matthews (1975); Brier score by Glenn Brier for weather forecasts (1950).

► **Use.** Judging whether a model is useful, not just whether a number is high.

► **Data science.** A 95%-accurate fraud model is worthless if 95% of transactions are legitimate; always report metrics next to the baseline and with confidence intervals [S.14].

► **Physical application.** Weather services score rain forecasts with the Brier score against climatology (the historical rain rate) as the baseline.

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13

Electrical Engineering

Circuits, AC power, signals, and electromagnetism. Electrical engineering applies algebra, complex numbers, differential equations, and Fourier analysis to the devices and grids that power and connect the world, including the hardware your models run on.

EE.1 Charge, Current, Voltage, Power & Energy

$$I = \frac{dQ}{dt} \quad V = \frac{dW}{dQ} \quad P = VI \quad E = \int P dt = Pt \text{ (constant } P) \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

STATED-EQUATION Current I equals the rate of flow of charge, dQ/dt . Voltage V equals energy per unit charge, dW/dQ . Power P equals V times I . Energy equals the integral of power over time, which is Pt for constant power. One kilowatt-hour equals three point six million joules.

MEMORY AID Water analogy: voltage = pressure, current = flow rate, resistance = pipe narrowness, power = pressure $\times$ flow.

TERMS

Q – charge in coulombs (C); electron charge 1.602×10^{-19} C
 I – current in amperes ($A = C/s$) [C1.4]
 V – voltage (potential difference) in volts ($V = J/C$)
 P – power in watts (W)
 E – energy in joules or kilowatt-hours [P.4]
 Passive sign convention – $P > 0$ means the element absorbs power

CONTEXT

- **Origin.** Volta’s battery (1800); Ampère (1820s); the watt named for James Watt.
- **Use.** Sizing wiring and power supplies; reading utility bills.
- **Data science.** Estimating the energy and carbon cost of training runs: GPUs $\times$ watts $\times$ hours; data-center power budgeting.
- **Physical application.** A 1500 W space heater run for 3 hours uses 4.5 kWh, about \$0.68 at \$0.15/kWh.

EE.2 Kirchhoff’s Laws

$$\text{KCL (node): } \sum_k I_k = 0 \Leftrightarrow \sum I_{\text{in}} = \sum I_{\text{out}} \quad \text{KVL (loop): } \sum_k V_k = 0$$

STATED-EQUATION Kirchhoff’s current law: at any node the currents sum to zero, so current in equals current out. Kirchhoff’s voltage law: around any closed loop the voltages sum to zero.

MEMORY AID KCL: “what flows in must flow out” (conservation of charge). KVL: “every round trip ends where it started” (conservation of energy).

TERMS

Node – junction where two or more elements meet
 Loop – closed path through the circuit
 Sign – rises and drops take opposite signs around a loop
 Ohm’s law – $V = IR$ for each resistor [P.20]
 Basis of – nodal and mesh analysis [EE.5]

CONTEXT

- **Origin.** Gustav Kirchhoff (1845), as a student.
- **Use.** Analyzing any circuit.
- **Data science.** KCL is flow conservation, the same constraint as network flow [O.7] and graph-based supply models.
- **Physical application.** In a house, the current into the breaker panel equals the sum of the currents to all the running appliances.

EE.3 Voltage & Current Dividers

$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2} \quad I_1 = I_{\text{total}} \frac{R_2}{R_1 + R_2} \quad (\text{two resistors in parallel})$$

STATED-EQUATION The output voltage of a divider equals the input voltage times R_2 over R_1 plus R_2 . In a current divider with two parallel resistors, the current through R_1 equals the total current times R_2 over R_1 plus R_2 : the other resistor on top.

MEMORY AID Voltage divider: “my resistor on top”; current divider: “the other resistor on top.”

TERMS

R_1, R_2 – series resistors; V_{out} is across R_2 [P20]
 Loading – attaching a load in parallel with R_2 changes V_{out}
 Current divider – more current takes the smaller resistance
 Weighted average – a proportion [A1.14]

CONTEXT

- **Origin.** Follows from Ohm’s and Kirchhoff’s laws [EE.2].
- **Use.** Sensor interfaces, level shifting, biasing, potentiometers.
- **Data science.** Reading analog sensors (thermistors, light sensors) into an ADC [EE.18] for IoT data pipelines.
- **Physical application.** A volume knob (potentiometer) is an adjustable voltage divider.

EE.4 Thévenin & Norton Equivalents; Maximum Power Transfer

$$V_{\text{th}} = V_{\text{open circuit}} \quad R_{\text{th}} = \frac{V_{\text{oc}}}{I_{\text{sc}}} \quad I_N = \frac{V_{\text{th}}}{R_{\text{th}}} \quad R_L = R_{\text{th}} \Rightarrow P_{\text{max}} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$$

STATED-EQUATION The Thevenin voltage is the open-circuit voltage. The Thevenin resistance equals the open-circuit voltage over the short-circuit current. The Norton current equals V_{th} over R_{th} . Maximum power transfer: when the load resistance equals R_{th} , the load power is V_{th} squared over four R_{th} .

MEMORY AID Any linear circuit, seen from two terminals, is “one battery plus one resistor.”

TERMS

$V_{\text{th}}, R_{\text{th}}$ – equivalent source and series resistance
 R_{th} shortcut – turn off sources (voltage $\rightarrow$ short, current $\rightarrow$ open), then combine resistors
 Norton – current source I_N in parallel with R_{th}
 Max power – found by setting $dP/dR_L = 0$ [C1.13]
 Efficiency – only 50% at maximum power

CONTEXT

- **Origin.** Léon Charles Thévenin (1883); Edward Norton (1926); Jacobi’s maximum power theorem (1840).
- **Use.** Simplifying circuits, matching amplifiers and antennas.
- **Data science.** A “black-box” surrogate model: replace a complex subsystem by a simple equivalent that reproduces its input–output behavior.
- **Physical application.** Audio amplifiers are matched to speaker impedance (e.g. $8\ \Omega$) to deliver maximum power.

EE.5 Nodal & Mesh Analysis

$$\text{Nodal: } \sum_{\text{branches}} \frac{V_n - V_m}{R_{nm}} = I_{\text{source}} \Rightarrow G\mathbf{v} = \mathbf{i} \quad \text{Mesh: } \sum_{\text{loop}} R I = \sum V_{\text{source}} \Rightarrow R\mathbf{i} = \mathbf{v}$$

STATED-EQUATION Nodal analysis: at each node, the sum over branches of the voltage difference over resistance equals the source current, giving the matrix equation $G\mathbf{v} = \mathbf{i}$. Mesh analysis: around each loop, the sum of R times I equals the source voltages, giving $R\mathbf{i} = \mathbf{v}$.

TERMS

G – conductance matrix ($1/R$ entries), symmetric
 $\mathbf{v}$ – unknown node voltages (relative to ground)
 Mesh currents – loop currents in each window of a planar circuit
 Solving – Gaussian elimination [LA.5], Cramer [A2.14]
 SPICE – circuit simulators assemble and solve these systems

CONTEXT

- **Origin.** Formalized with matrix methods in the 20th century; SPICE simulator from Berkeley (1973).
- **Use.** Solving any size of linear circuit systematically.
- **Data science.** The graph Laplacian of a network is exactly a conductance matrix; “effective resistance” between nodes is used in graph ML and network centrality.
- **Physical application.** Power-grid operators solve huge nodal systems every few minutes to schedule generation.

EE.6 Capacitors & Inductors

$$Q = CV \quad i = C \frac{dv}{dt} \quad v = L \frac{di}{dt} \quad E_C = \frac{1}{2} CV^2 \quad E_L = \frac{1}{2} LI^2 \quad C_{\parallel} = C_1 + C_2, \quad \frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2}$$

STATED-EQUATION Charge equals capacitance times voltage. Capacitor current equals C times $d v$ $d t$. Inductor voltage equals L times $d i$ $d t$. Stored energy is one half $C V$ squared for a capacitor and one half $L I$ squared for an inductor. Capacitors in parallel add; in series, their reciprocals add.

MEMORY AID Capacitors combine like resistors “upside down.” A capacitor resists sudden voltage changes; an inductor resists sudden current changes.

TERMS

C – capacitance in farads (F); typically μF or pF

L – inductance in henries (H)

$dv/dt, di/dt$ – derivatives [C1.4]

DC steady state – capacitor $\rightarrow$ open circuit; inductor $\rightarrow$ short circuit

Energy – compare $\frac{1}{2}mv^2$ and $\frac{1}{2}kx^2$ [E4]

CONTEXT

- **Origin.** Leyden jar (1745); Faraday’s induction (1831); Joseph Henry.
- **Use.** Energy storage, filtering, timing, power conversion.
- **Data science.** Supercapacitors and battery buffers smooth GPU power spikes in data centers; capacitive touchscreens sense changes in C .
- **Physical application.** A camera flash charges a capacitor slowly, then dumps its energy in a millisecond.

EE.7 RC & RL Transients

$$\text{charging: } v_C(t) = V_s(1 - e^{-t/\tau}) \quad \text{discharging: } v_C(t) = V_0 e^{-t/\tau} \quad \tau_{RC} = RC \quad \tau_{RL} = \frac{L}{R}$$

STATED-EQUATION While charging, the capacitor voltage equals V_s times one minus e to the minus t over τ . While discharging, it equals V naught e to the minus t over τ . The time constant is $R C$ for an RC circuit and L over R for an RL circuit.

MEMORY AID After 1τ : 63% of the way; after 5τ : over 99% (“five tau and done”).

TERMS

τ – time constant (s)

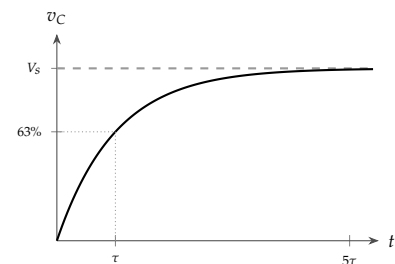
Origin – first-order ODE $RC \frac{dv}{dt} + v = V_s$ [C2.16]

$e^{-t/\tau}$ – exponential decay [A2.6]

Same math as – Newton’s cooling [P.14]

CONTEXT

- **Origin.** Solved with the differential equations of Kirchhoff and Heaviside (1880s).
- **Use.** Timers, debouncing buttons, sensor filtering, pulse shaping.
- **Data science.** An exponential moving average is a discrete RC low-pass filter: $s_t = \alpha x_t + (1 - \alpha)s_{t-1}$.
- **Physical application.** A car’s interior light fades out slowly after you close the door, thanks to an RC circuit.



EE.8 AC Signals & RMS Values

$$v(t) = V_m \sin(\omega t + \phi) \quad \omega = 2\pi f \quad T = \frac{1}{f} \quad V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2 dt} = \frac{V_m}{\sqrt{2}} \text{ (sine)}$$

STATED-EQUATION v of t equals V_m sine of ωt plus ϕ . ω equals two pi f . The period equals one over f . The RMS voltage is the square root of the average of v squared over one period, which for a sine wave equals V_m over root two.

MEMORY AID RMS = Root of the Mean of the Square; for sine waves, peak $\approx 1.414 \times$ RMS.

TERMS

V_m – peak (amplitude) [T.9]

f – frequency (Hz); ω angular frequency [T.2]

ϕ – phase

RMS – equivalent DC value for heating; average of $\sin^2$ is $\frac{1}{2}$ [C.2.2], [C.2.8]

Mains – US 120 V_{rms} at 60 Hz; Europe 230 V at 50 Hz

CONTEXT

- **Origin.** AC systems championed by Tesla and Westinghouse (1880s “war of the currents”).
- **Use.** Power grids, audio, communications.
- **Data science.** RMS is the same as the standard deviation of a zero-mean signal [S.2]; RMSE is RMS of errors [ML.22].
- **Physical application.** A US outlet labeled 120 V actually swings to about ± 170 V peak.

EE.9 Phasors & Impedance

$$Z_R = R \quad Z_L = j\omega L \quad Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C} \quad \tilde{V} = \tilde{I}Z \quad Z = R + jX, \quad |Z| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

STATED-EQUATION The impedance of a resistor is R ; of an inductor, $j\omega L$; of a capacitor, one over $j\omega C$, which equals minus j over ωC . Phasor voltage equals phasor current times impedance. Z equals R plus jX , with magnitude the square root of R squared plus X squared and angle the inverse tangent of X over R .

MEMORY AID ELI the ICE man: in an inductor (L), voltage (E) leads current (I); in a capacitor (C), current (I) leads voltage (E).

TERMS

j – imaginary unit ($j^2 = -1$); engineers use j because i is current [A.2.3]

Phasor – $V_m e^{j\phi}$, a sinusoid as a complex number [T.13]

X – reactance: ωL (inductive) or $-\frac{1}{\omega C}$ (capacitive)

Combining – impedances combine like resistors [P.20]

Polar form – $|Z|\angle\theta$ [T.11]

CONTEXT

- **Origin.** Charles Steinmetz (1893) introduced phasors, turning differential equations into algebra.
- **Use.** AC circuit analysis, power systems, RF design.
- **Data science.** Complex-valued features from FFTs [EE.20] carry magnitude and phase exactly like phasors.
- **Physical application.** Motors are mostly inductive, so their current lags the voltage.

EE.10 AC Power & Power Factor

$$P = V_{\text{rms}} I_{\text{rms}} \cos \theta \quad Q = V_{\text{rms}} I_{\text{rms}} \sin \theta \quad S = \tilde{V} \tilde{I}^* = P + jQ, \quad |S|^2 = P^2 + Q^2 \quad \text{pf} = \cos \theta = \frac{P}{|S|}$$

STATED-EQUATION Real power P equals V_{rms} times I_{rms} times cosine θ . Reactive power Q equals V_{rms} times I_{rms} times sine θ . Complex power S equals V times the conjugate of I , which is P plus jQ , and the magnitude of S squared equals P squared plus Q squared. The power factor equals cosine θ , which is P over the magnitude of S .

MEMORY AID The power triangle: P is the “beer” (useful), Q the “foam,” $|S|$ the whole glass.

TERMS

P – real power (W), does useful work

Q – reactive power (var), sloshes back and forth

$|S|$ – apparent power (VA); sizes wires and transformers

θ – angle between voltage and current [EE.9]

Right triangle – Pythagorean [G.3], trig ratios [T.1]

Correction – capacitors cancel inductive Q to raise pf

CONTEXT

- **Origin.** Steinmetz and early utility engineers (1890s–1900s).
- **Use.** Utility billing, motor sizing, grid efficiency.
- **Data science.** Industrial energy analytics flag low power factor from smart-meter data to cut penalties.
- **Physical application.** Factories are charged extra when their power factor is low, because the grid must carry the extra current.

EE.11 RLC Circuits & Resonance

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_{\text{series}} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0 L}{R} \quad \text{BW} = \frac{\omega_0}{Q}$$

STATED-EQUATION The resonant angular frequency ω naught equals one over the square root of $L C$, so f naught equals one over two π root $L C$. The quality factor of a series RLC equals one over R times the square root of L over C , which is ω naught L over R . The bandwidth equals ω naught over Q .

MEMORY AID At resonance the inductor and capacitor reactances cancel: $\omega L = \frac{1}{\omega C}$.

TERMS

ω_0 – natural frequency (compare $\sqrt{k/m}$ [E.7])

Q – sharpness of resonance; high Q = narrow, tall peak

BW – width of the -3 dB band [EE.12]

Series resonance – impedance minimum, $Z = R$

Second-order ODE – $L\ddot{q} + R\dot{q} + q/C = v(t)$: underdamped, critical, overdamped

CONTEXT

- **Origin.** Resonance studied by Hertz (1887) and Tesla; radio tuning circuits from Marconi's era.
- **Use.** Radio tuning, oscillators, wireless charging, filters.
- **Data science.** Damped oscillator models describe seasonal and business-cycle dynamics; the same second-order equation underlies momentum SGD [ML.16].
- **Physical application.** Turning a radio dial changes C so f_0 matches the station's frequency.

EE.12 Filters, Frequency Response & Decibels

$$H(j\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{1 + j\omega RC} \quad |H| = \frac{1}{\sqrt{1 + (f/f_c)^2}} \quad f_c = \frac{1}{2\pi RC} \quad |H|_{\text{dB}} = 20 \log_{10} |H|$$

STATED-EQUATION The transfer function of an RC low-pass filter, V_{out} over V_{in} , equals one over one plus j ω $R C$. Its magnitude equals one over the square root of one plus f over f_c squared. The cutoff frequency f_c equals one over two π $R C$. In decibels, the gain is twenty log base ten of the magnitude.

MEMORY AID At the cutoff, $|H| = \frac{1}{\sqrt{2}}$, i.e. -3 dB (half power). Past it, -20 dB per decade for each RC stage.

TERMS

$H(j\omega)$ – transfer function (a ratio of phasors [EE.9])

Low-pass / high-pass – pass low / high frequencies

Band-pass – pass a range (RLC [EE.11])

dB – logarithmic scale [A2.7], [EE.19]

Bode plot – $|H|_{\text{dB}}$ vs. $\log f$

CONTEXT

- **Origin.** Filter theory by Campbell and Wagner (1915); Bode plots (1930s).
- **Use.** Audio equalizers, anti-aliasing, noise removal, communications.
- **Data science.** Smoothing time-series data (moving averages are low-pass filters); convolution layers in CNNs learn filters [ML.14].
- **Physical application.** A subwoofer's crossover is a low-pass filter that sends only bass to the big speaker.

EE.13 Operational Amplifiers

$$\text{ideal (with feedback): } V_+ = V_-, \quad I_+ = I_- = 0 \quad \text{inverting: } \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_{\text{in}}} \quad \text{non-inverting: } \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_f}{R_g}$$

STATED-EQUATION For an ideal op-amp with negative feedback, the two input voltages are equal and no current flows into the inputs. The inverting amplifier's gain equals minus R_f over R_{in} . The non-inverting amplifier's gain equals one plus R_f over R_g .

MEMORY AID The two golden rules: “the inputs are at the same voltage” and “no current enters the inputs.”

TERMS

V_+, V_- – non-inverting and inverting inputs

R_f – feedback resistor

Follower – $R_f = 0$: gain 1, used as a buffer

Summing amplifier – $V_{\text{out}} = -R_f \sum V_i / R_i$ (a weighted sum!)

Limits – output clipped at the supply rails

CONTEXT

- **Origin.** Harold Black's negative-feedback amplifier (1927); “operational” amplifiers in analog computers (1940s).
- **Use.** Sensor amplification, filters, analog computation.
- **Data science.** Analog in-memory AI chips compute neural-network weighted sums with op-amp style summing circuits.
- **Physical application.** A guitar pickup's tiny signal is amplified by op-amp circuits in the pedal board.

EE.14 Diodes, LEDs & Transistors

$$I = I_s \left(e^{V/(nV_T)} - 1 \right) \quad V_T = \frac{kT}{q} \approx 25.9 \text{ mV at } 300 \text{ K} \quad R_{\text{LED}} = \frac{V_s - V_f}{I_{\text{LED}}}$$

STATED-EQUATION Shockley's diode equation: I equals I_s times e to the V over $n V_T$, minus one. The thermal voltage V_T equals $k T$ over q , about twenty-five point nine millivolts at room temperature. An LED's series resistor equals the supply voltage minus the LED forward voltage, over the LED current.

MEMORY AID Silicon diode “turns on” near 0.7 V; the current grows about $10\times$ for every 60 mV more (with $n = 1$).

TERMS

I_s – saturation current (tiny)

n – ideality factor (1 to 2)

V_f – LED forward voltage (red ≈ 2 V, blue ≈ 3 V)

Exponential – [A2.6](#); semilog plot is a straight line

Transistor – a current- or voltage-controlled switch/amplifier; billions form a GPU

CONTEXT

- **Origin.** Shockley (1949); the transistor invented at Bell Labs by Bardeen, Brattain and Shockley (1947).
- **Use.** Rectifiers, LEDs, solar cells, all digital electronics.
- **Data science.** Moore's law, the doubling of transistor counts roughly every two years, is an exponential growth trend fitted on a log scale [A2.6](#).
- **Physical application.** Solar panels are large diodes: light generates current across the junction.

EE.15 Transformers & Three-Phase Power

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} \quad P_{3\phi} = \sqrt{3} V_L I_L \cos \theta \quad P_{\text{loss}} = I^2 R$$

STATED-EQUATION For an ideal transformer, the secondary voltage over the primary voltage equals the turns ratio, N_s over N_p , which also equals the primary current over the secondary current. Three-phase power equals root three times the line voltage times the line current times the power factor. Line losses equal I squared R .

MEMORY AID “Step up voltage, step down current”: power in = power out (ideal).

TERMS

N_p, N_s – turns on the primary and secondary coils

Ideal – no losses, $V_p I_p = V_s I_s$

High-voltage transmission – lower I means much lower $I^2 R$ loss

V_L, I_L – line voltage and current

$\sqrt{3}$ – from phases 120° apart [T.8](#)

CONTEXT

- **Origin.** Faraday's induction (1831); practical transformers by Gaulard, Stanley (1880s); three-phase by Tesla and Dolivo-Dobrovolsky.
- **Use.** Power transmission, chargers, industrial motors.
- **Data science.** Grid load forecasting and transformer failure prediction are major utility ML applications.
- **Physical application.** Your phone charger steps 120 V down to 5 V with a small transformer (or switching converter).

EE.16 Electric & Magnetic Fields

$$F = k \frac{q_1 q_2}{r^2} \quad \mathbf{E} = \frac{\mathbf{F}}{q} \quad B = \frac{\mu_0 I}{2\pi r} \quad \mathcal{E} = -N \frac{d\Phi_B}{dt} \quad \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

STATED-EQUATION Coulomb's law: the force equals $k q$ one q two over r squared. The electric field equals force per unit charge. The magnetic field around a long wire equals μ naught I over two πr . Faraday's law: the induced $\mathcal{E}$ $M F$ equals minus N times the rate of change of magnetic flux. The Lorentz force equals q times the quantity $\mathbf{E}$ plus $\mathbf{v}$ cross $\mathbf{B}$.

MEMORY AID Faraday: "change the flux, get a voltage"; the minus sign (Lenz) means the induced current opposes the change.

TERMS

$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$; inverse-square like gravity [P.6]

$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m}/\text{A}$

B – magnetic field (tesla); Earth $\approx 50 \mu\text{T}$

Φ_B – magnetic flux $\iint \mathbf{B} \cdot \mathbf{n} dA$ [V.12]

Maxwell's equations – divergence and curl of $\mathbf{E}$ and $\mathbf{B}$ [V.10], [V.13]

CONTEXT

- **Origin.** Coulomb (1785); Ørsted and Ampère (1820); Faraday (1831); Maxwell (1865); Lorentz (1895).
- **Use.** Motors, generators, MRI, wireless communication.
- **Data science.** Magnetometer and accelerometer fusion in phones; MRI image reconstruction with deep learning.
- **Physical application.** Induction cooktops heat pans with a rapidly changing magnetic field that induces currents in the metal.

EE.17 Digital Logic, Binary & Boolean Algebra

$$N = \sum_{i=0}^{n-1} b_i 2^i \quad n \text{ bits} \rightarrow 2^n \text{ values} \quad A \cdot B \text{ (AND)}, \quad A + B \text{ (OR)}, \quad \bar{A} \text{ (NOT)} \quad \overline{A \cdot B} = \bar{A} + \bar{B}, \quad \overline{A + B} = \bar{A} \cdot \bar{B}$$

STATED-EQUATION A binary number equals the sum of each bit b_i times two to the i . n bits represent two to the n values. A dot B is AND, A plus B is OR, A bar is NOT. De Morgan's laws: NOT of A AND B equals NOT A OR NOT B , and NOT of A OR B equals NOT A AND NOT B .

MEMORY AID De Morgan: "break the bar, change the operator" (same as for sets [A1.16]).

TERMS

Bit – 0 or 1; byte = 8 bits

Hexadecimal – base 16, one digit per 4 bits (0xFF = 255)

Gates – AND, OR, NOT, NAND, NOR, XOR

Truth table – all input combinations, 2^n rows [S.8]

Two's complement – standard representation of negative integers

CONTEXT

- **Origin.** George Boole (1854); Claude Shannon linked Boolean algebra to switching circuits (1937).
- **Use.** All digital computers, controllers, and memory.
- **Data science.** Bit masks, hashing, and quantized models (8-bit and 4-bit weights) reduce memory for large neural networks.
- **Physical application.** A two-switch hallway light (either switch toggles it) is an XOR circuit.

EE.18 Sampling, Nyquist & Quantization

$$f_s > 2f_{\max} \text{ (Nyquist)} \quad f_{\text{alias}} = |f - kf_s| \quad \Delta = \frac{V_{\text{ref}}}{2^N} \quad \text{SQNR} \approx 6.02N + 1.76 \text{ dB}$$

STATED-EQUATION The sampling rate f_s must exceed twice the highest frequency, the Nyquist criterion. A tone above half the sampling rate appears at the alias frequency, the absolute value of f minus kf_s . The resolution of an N -bit converter is V_{ref} over two to the N . The signal-to-quantization-noise ratio is about six point oh two N plus one point seven six decibels.

MEMORY AID "Sample at least twice as fast as the fastest wiggle"; each extra bit buys about 6 dB.

TERMS

f_s – sampling rate (samples/s)

$f_s/2$ – Nyquist frequency

Aliasing – undersampled high frequencies masquerade as low ones

N – ADC bit depth; 2^N levels [EE.17]

dB – [EE.19]

CONTEXT

- **Origin.** Nyquist (1928), Shannon (1949), Kotelnikov (1933).
- **Use.** Digital audio, sensors, imaging, telecommunications.
- **Data science.** Resampling time series and sensor streams; choosing logging frequency for IoT data; aliasing in downsampled images.
- **Physical application.** CDs sample at 44.1 kHz because human hearing tops out near 20 kHz.

EE.19 Decibels, SNR & Channel Capacity

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = 20 \log_{10} \frac{V_2}{V_1} \quad \text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad C = B \log_2 (1 + \text{SNR})$$

STATED-EQUATION A power ratio in decibels is ten log base ten of P_2 over P_1 , which equals twenty log base ten of the voltage ratio. The signal-to-noise ratio is signal power over noise power. Shannon's capacity: C equals the bandwidth B times log base two of one plus the SNR.

MEMORY AID +3 dB $\approx 2\times$ power, +10 dB = $10\times$ power, +20 dB = $100\times$ power ($10\times$ voltage).

TERMS

dB – logarithmic ratio [A2.7]

$20 \log$ – for voltages because $P \propto V^2$ [P20]

dBm – dB relative to 1 mW

C – maximum error-free data rate (bits/s)

B – bandwidth (Hz)

Entropy link – [ML3]

CONTEXT

- **Origin.** The bel named for Alexander Graham Bell (1920s); Shannon–Hartley theorem (1948).
- **Use.** Audio levels, Wi-Fi and cellular link budgets, amplifier specs.
- **Data science.** SNR thinking applies to features and metrics: a small effect in a noisy metric needs a larger sample [S.13].
- **Physical application.** Wi-Fi drops to a slower rate as you move away because SNR falls and so does capacity.

EE.20 Fourier Series, the DFT & FFT

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N} \quad \Delta f = \frac{f_s}{N} \quad \text{DFT } O(N^2) \rightarrow \text{FFT } O(N \log N)$$

STATED-EQUATION A periodic signal x of t equals the sum of coefficients c_n times e to the $j n$ omega naught t . The discrete Fourier transform: X_k equals the sum from n equals zero to N minus one of x_n times e to the minus j two pi $k n$ over N . The frequency resolution equals f_s over N . The FFT computes the DFT in order $N \log N$ instead of N squared operations.

MEMORY AID Every signal is a sum of spinning arrows (sinusoids) [T.13]; the FFT finds how much of each.

TERMS

c_n – Fourier coefficient (amplitude and phase of harmonic n)

ω_0 – fundamental angular frequency

X_k – frequency content at bin k (complex [EE.9])

N – number of samples; f_s sampling rate [EE.18]

Square wave – odd harmonics with amplitudes $\frac{4}{n\pi}$

Complexity – [O.17]

CONTEXT

- **Origin.** Fourier (1807, heat conduction [P.13]); FFT by Cooley and Tukey (1965), anticipated by Gauss (1805).
- **Use.** Audio compression (MP3), spectrum analyzers, image compression (JPEG), vibration analysis.
- **Data science.** Spectral features for time series and audio (spectrograms feed CNNs); detecting seasonality; FFT-based fast convolution.
- **Physical application.** Noise-canceling headphones analyze and invert the frequency content of ambient noise in real time.

EE.21 Batteries & Energy Storage

$$E \text{ (Wh)} = V_{\text{nominal}} \times \text{capacity (Ah)} \quad t_{\text{run}} \approx \frac{\text{capacity (Ah)}}{I \text{ (A)}} \quad \text{C-rate} = \frac{I}{\text{capacity}} \quad \eta_{\text{round trip}} = \frac{E_{\text{out}}}{E_{\text{in}}}$$

STATED-EQUATION Stored energy in watt-hours equals the nominal voltage times the capacity in amp-hours. Runtime is approximately capacity in amp-hours over current in amps. The C-rate equals current over capacity. Round-trip efficiency equals energy out over energy in.

MEMORY AID “mAh is charge, Wh is energy”: multiply by voltage to compare batteries fairly.

TERMS

Ah, mAh – charge capacity [EE.1](#)

Wh – energy capacity

1C – current that empties the battery in one hour

State of charge – fraction of capacity remaining

Real runtime – shorter at high C-rate and low temperature

CONTEXT

- **Origin.** Volta’s pile (1800); lead-acid (Planté, 1859); lithium-ion (Goodenough, Whittingham, Yoshino; Nobel 2019).
- **Use.** Phones, laptops, EVs, grid storage.
- **Data science.** Battery state-of-health and remaining-useful-life prediction from charge-discharge curves is a common time-series ML task.
- **Physical application.** A 75 kWh EV battery using 250 Wh per mile has about a 300-mile range.

EE.22 Schematic Symbols: Passive Components & Switches

$$\text{resistor: } V = IR \quad \text{capacitor: } i = C \frac{dv}{dt} \quad \text{inductor: } v = L \frac{di}{dt} \quad \text{transformer: } \frac{V_s}{V_p} = \frac{N_s}{N_p} \quad \text{fuse: opens when } I > I_{\text{rated}}$$

STATED-EQUATION A resistor obeys V equals $I R$. A capacitor’s current equals C times $d v$ $d t$. An inductor’s voltage equals L times $d i$ $d t$. A transformer’s voltage ratio equals its turns ratio, N_s over N_p . A fuse opens when the current exceeds its rating.

MEMORY AID Zigzag = resistor (friction); two parallel plates = capacitor (a gap that stores charge); coils = inductor (a wound wire).



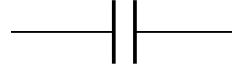
SY.1 Resistor

Zigzag (US) or rectangle (IEC);
 $V = IR$.



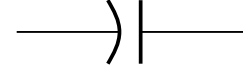
SY.2 Potentiometer

Resistor with an arrow wiper; volume knobs.



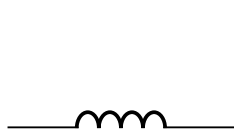
SY.3 Capacitor

Two parallel plates; blocks DC, passes AC.



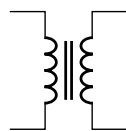
SY.4 Polarized Capacitor

Curved plate is negative; large values.



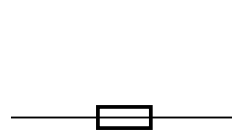
SY.5 Inductor

Coil; resists changes in current.



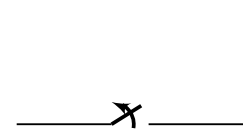
SY.6 Transformer

Coupled coils; steps AC voltage up/down.



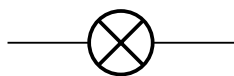
SY.7 Fuse

Overcurrent protection (one-time).



SY.8 Switch (SPST)

Opens or closes a single path.



SY.9 Lamp

Circle with an X; a resistive load.

TERMS

US vs. IEC – US zigzag resistor; IEC (Europe) uses a plain rectangle

Values – written beside the symbol (e.g. 4.7 kΩ, 10 μF, 1 mH)

Passive – absorbs or stores energy; cannot amplify [EE.6](#)

Polarity – polarized parts must be installed the right way round

CONTEXT

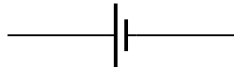
- **Origin.** Standardized symbols from IEEE 315 (1975) and IEC 60617.
- **Use.** Reading and drawing circuit diagrams (schematics).
- **Data science.** Hardware and IoT data teams read schematics to know what a sensor channel measures and where noise enters.
- **Physical application.** A light-dimmer wall switch is a potentiometer (or its electronic equivalent) controlling a lamp.

EE.23 Schematic Symbols: Sources, Meters & Ground

ideal voltage source: $V = \text{const}$ for any I ideal current source: $I = \text{const}$ for any V voltmeter ($R \rightarrow \infty$) in parallel ammeter ($R \rightarrow 0$) in series

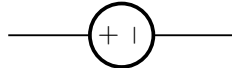
STATED-EQUATION An ideal voltage source holds a constant voltage for any current. An ideal current source holds a constant current for any voltage. A voltmeter has very high resistance and is connected in parallel. An ammeter has very low resistance and is connected in series.

MEMORY AID “Volts across, amps through”: voltmeter across the part, ammeter in the path.



SY.10 Battery

Long plate is +; DC voltage.



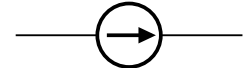
SY.11 DC Voltage Source

Circle with +/−; fixed voltage.



SY.12 AC Voltage Source

Circle with a sine wave; mains, signals.



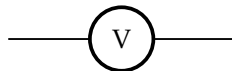
SY.13 Current Source

Circle with an arrow; fixed current.



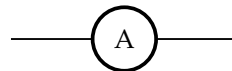
SY.14 Ground

Stacked lines; the 0 V reference.



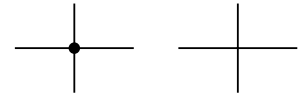
SY.15 Voltmeter

Circle with V; connect across.



SY.16 Ammeter

Circle with A; connect in series.



SY.17 Junction vs. Crossing

Dot = connected; no dot = crossing.

TERMS

Independent source – value set by itself [EE.1](#)

Dependent source – diamond shape; value controlled by another voltage or current

Ground – reference node for nodal analysis [EE.5](#)

Node – all points joined by wire and dots share one voltage [EE.2](#)

Meters – ideal meters do not disturb the circuit

CONTEXT

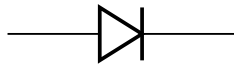
- **Origin.** Symbols evolved from telegraph and power engineering drawings (late 1800s).
- **Use.** Wiring measurements and power supplies correctly.
- **Data science.** Understanding where a data-acquisition system measures voltage vs. current prevents misinterpreted sensor data.
- **Physical application.** Using a multimeter: measure battery voltage across its terminals; measure current by breaking the circuit and inserting the meter.

EE.24 Schematic Symbols: Semiconductors & Op-Amps

diode: $I = I_s(e^{V/(nV_T)} - 1)$, conducts along the arrow BJT: $I_C \approx \beta I_B$ MOSFET: gate voltage controls I_D op-amp: $V_{out} = A(V_+ - V_-)$

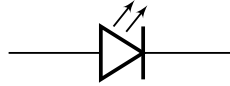
STATED-EQUATION A diode's current equals I_s times e to the V over $n V T$ minus one, and it conducts in the direction of its arrow. A bipolar transistor's collector current is about β times the base current. In a MOSFET, the gate voltage controls the drain current. An op-amp's output equals its open-loop gain A times the difference between its plus and minus inputs.

MEMORY AID The diode's arrow points the way conventional current flows (anode $\rightarrow$ cathode; the bar is the cathode).



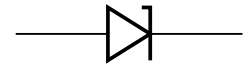
SY.18 Diode

Arrow into a bar; one-way current.



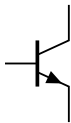
SY.19 LED

Diode with outgoing arrows (light).



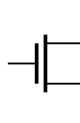
SY.20 Zener Diode

Bent bar; holds a fixed reverse voltage.



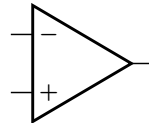
SY.21 NPN Transistor (BJT)

Base, collector, emitter (arrow out).



SY.22 MOSFET (n-channel)

Gate, drain, source; digital switch.



SY.23 Op-Amp

Triangle with + and - inputs.

TERMS

Anode / cathode – diode terminals; the bar marks the cathode [\[EE.14\]](#)

β – current gain of a BJT (often 50–300)

Gate, drain, source – MOSFET terminals

Op-amp rules – [\[EE.13\]](#)

Active device – can amplify (needs a power supply)

CONTEXT

- **Origin.** Diode (Fleming valve 1904; semiconductor diodes 1940s); BJT (1947); MOSFET (Atalla and Kahng, 1959).
- **Use.** Rectification, switching, amplification, all digital logic.
- **Data science.** A GPU contains tens of billions of MOSFETs switching billions of times per second to run neural networks.
- **Physical application.** A phone charger rectifies AC to DC with a bridge of four diodes.

EE.25 Schematic Symbols: Logic Gates

AND: $Y = A \cdot B$

OR: $Y = A + B$

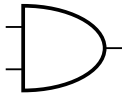
NOT: $Y = \bar{A}$

NAND: $Y = \overline{A \cdot B}$

XOR: $Y = A \oplus B = A\bar{B} + \bar{A}B$

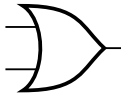
STATED-EQUATION AND: *Y equals A and B.* OR: *Y equals A or B.* NOT: *Y equals not A.* NAND: *Y equals not of A and B.* XOR: *Y equals A exclusive-or B, which is A and not B, or not A and B.*

MEMORY AID A small circle (“bubble”) on an output means NOT. AND has a flat back, OR has a curved back, XOR adds an extra curved line.



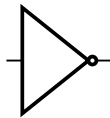
SY.24 AND Gate

Flat back, round front.



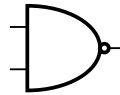
SY.25 OR Gate

Curved back, pointed front.



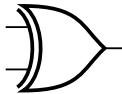
SY.26 NOT Gate (Inverter)

Triangle with an output bubble.



SY.27 NAND Gate

AND with a bubble.



SY.28 XOR Gate

OR with an extra back curve.

A	B	AND	OR	NAND	XOR	NOT A
0	0	0	0	1	0	1
0	1	0	1	1	1	1
1	0	0	1	1	1	0
1	1	1	1	0	0	0

TERMS

Truth table – output for every input combination [EE.17](#)
Bubble – inversion (NOT)
Universal gates – NAND (or NOR) alone can build any logic
Boolean algebra – De Morgan [EE.17](#), [A1.16](#)

CONTEXT

- **Origin.** Boole (1854); Shannon (1937); standardized gate shapes in ANSI/IEEE 91.
- **Use.** Designing digital circuits, CPUs, and controllers.
- **Data science.** Boolean filters in SQL and pandas masks (&, |, ~) are the software form of these gates.
- **Physical application.** A car’s “door open AND key in ignition ⇒ chime” warning is an AND gate.

14 Finance

The time value of money, valuation, risk and return, fixed income, and derivatives. Finance is applied exponentials, series, probability, and optimization: every loan, bond, portfolio, and option below is built from earlier concepts.

FN.1 Simple vs. Compound Interest

$$\text{simple: } A = P(1 + rt) \quad \text{compound: } A = P \left(1 + \frac{r}{n}\right)^{nt} \quad \text{continuous: } A = Pe^{rt}$$

STATED-EQUATION Simple interest: A equals P times one plus r t . Compound interest: A equals P times one plus r over n , raised to the n t . Continuous compounding: A equals P e to the r t .

MEMORY AID Compounding is “interest on interest”; “Pert” for continuous.

TERMS

P – principal (present amount)
 r – annual nominal rate (decimal)
 n – compounding periods per year
 t – years
 Growth – linear (simple) vs. exponential (compound) [A1.5](#), [A2.6](#)
 Limit – $n \rightarrow \infty$ gives e^{rt} [C1.2](#)

CONTEXT

- **Origin.** Compound interest tables in Babylon (~2000 BCE); Jacob Bernoulli discovered e studying continuous compounding (1683).
- **Use.** Savings, loans, credit cards, inflation.
- **Data science.** Customer lifetime value and growth projections compound monthly retention and revenue rates.
- **Physical application.** Bacterial and radioactive growth follow the same compounding law as money.

FN.2 APR vs. Effective Annual Rate

$$\text{EAR} = \left(1 + \frac{\text{APR}}{n}\right)^n - 1 \quad \text{continuous: } \text{EAR} = e^r - 1$$

STATED-EQUATION The effective annual rate equals one plus the APR over n , raised to the n , minus one. With continuous compounding, the effective annual rate equals e to the r minus one.

MEMORY AID APR is the “sticker” rate; EAR is what you actually pay or earn. $\text{EAR} \geq \text{APR}$ whenever $n > 1$.

TERMS

APR – annual percentage rate (nominal)
 EAR / APY – effective rate after compounding
 n – compounding frequency (12 monthly, 365 daily)
 Comparison – only compare offers using EAR

CONTEXT

- **Origin.** US Truth in Lending Act (1968) required APR disclosure.
- **Use.** Comparing loans, credit cards and savings accounts.
- **Data science.** Normalizing rates quoted with different conventions before feeding them into pricing or credit-risk models.
- **Physical application.** Growth rates quoted per year vs. per month (e.g. population or reaction rates) must be converted the same way.

FN.3 Doubling Time & the Rule of 72

$$t_{\text{double}} = \frac{\ln 2}{\ln(1+r)} \quad t_{\text{double}} \approx \frac{72}{r \text{ (in \%)}}$$

STATED-EQUATION The exact doubling time equals the natural log of two over the natural log of one plus r . The rule of seventy-two: doubling time is approximately seventy-two divided by the rate in percent.

MEMORY AID Rule of 72 (use 69.3 for continuous compounding, since $\ln 2 \approx 0.693$).

TERMS

r – growth rate per period

$\ln 2$ – from solving $(1+r)^t = 2$ [A2.8]

Why 72 – close to 69.3 and divisible by 2, 3, 4, 6, 8, 9, 12

Tripling – rule of 114

CONTEXT

- **Origin.** Luca Pacioli, *Summa de arithmetica* (1494).
- **Use.** Mental math for investments, inflation and debt.
- **Data science.** Quick sanity checks of growth metrics: 10% monthly user growth doubles in about 7 months.
- **Physical application.** Moore's law and bacterial growth doubling times use the same logarithmic calculation.

FN.4 Present Value & Discounting

$$PV = \frac{FV}{(1+r)^t} \quad FV = PV(1+r)^t \quad \text{discount factor} = \frac{1}{(1+r)^t} = e^{-rt} \text{ (continuous)}$$

STATED-EQUATION Present value equals future value over one plus r , to the t . Future value equals present value times one plus r to the t . The discount factor is one over one plus r to the t , or e to the minus r t with continuous compounding.

MEMORY AID "A dollar today is worth more than a dollar tomorrow."

TERMS

PV – value today

FV – value at time t

r – discount rate (opportunity cost of capital)

Inverse – discounting undoes compounding [FN.1], [A2.1]

CONTEXT

- **Origin.** Fibonacci's *Liber Abaci* (1202) included present-value problems; formalized by Irving Fisher (1930).
- **Use.** Valuing any future payment: pensions, settlements, projects.
- **Data science.** Discount factors appear in reinforcement learning (γ [0.12]) and in customer-lifetime-value models.
- **Physical application.** Energy-efficiency upgrades (insulation, solar panels) are evaluated by discounting future energy savings.

FN.5 Net Present Value & Internal Rate of Return

$$NPV = \sum_{t=0}^T \frac{CF_t}{(1+r)^t} \quad \text{IRR solves } \sum_{t=0}^T \frac{CF_t}{(1+\text{IRR})^t} = 0 \quad \text{accept if } NPV > 0 \text{ (} \Leftrightarrow \text{IRR} > r \text{ for normal projects)}$$

STATED-EQUATION Net present value equals the sum over time of each cash flow CF_t divided by one plus r to the t . The internal rate of return is the rate that makes that sum equal zero. Accept the project if the NPV is positive, which for normal projects means the IRR exceeds r .

MEMORY AID NPV answers "how much value?"; IRR answers "what rate?" When they disagree, trust NPV.

TERMS

CF_t – cash flow at time t (negative = investment)

r – hurdle rate / cost of capital

IRR – a root of a polynomial in $\frac{1}{1+r}$, solved numerically [C1.20]

Pitfalls – multiple IRRs with sign-changing cash flows

Sum – finite series [A2.11]

CONTEXT

- **Origin.** Irving Fisher (1907, 1930); widespread corporate use after Joel Dean (1951).
- **Use.** Capital budgeting: factories, software projects, acquisitions.
- **Data science.** Valuing a data-science project: upfront build cost vs. discounted future savings; payback of an ML platform investment.
- **Physical application.** Evaluating a solar installation: pay now, save on electricity for 25 years.

FN.6 Annuities & Perpetuities

$$PV_{\text{annuity}} = C \frac{1 - (1 + r)^{-n}}{r} \quad FV_{\text{annuity}} = C \frac{(1 + r)^n - 1}{r} \quad PV_{\text{perpetuity}} = \frac{C}{r} \quad PV_{\text{growing perpetuity}} = \frac{C_1}{r - g} \quad (r > g)$$

STATED-EQUATION The present value of an annuity equals C times one minus one plus r to the minus n , over r . The future value of an annuity equals C times one plus r to the n minus one, over r . A perpetuity is worth C over r . A growing perpetuity is worth C one over r minus g , when r exceeds g .

MEMORY AID A perpetuity is an annuity with $n \rightarrow \infty$: the $(1 + r)^{-n}$ term vanishes.

TERMS

C – payment per period (end of period)

n – number of payments

r – rate per period

g – growth rate of payments

Derivation – geometric series [A2.10](#), [C2.10](#)

CONTEXT

- **Origin.** Annuity valuation by Edmond Halley and de Moivre (1690s–1720s) for life insurance.
- **Use.** Retirement income, leases, pensions, loan payments.
- **Data science.** Subscription revenue with churn is a growing (or shrinking) perpetuity: $CLV \approx \frac{m}{r + \text{churn}}$.
- **Physical application.** The present value of a hydroelectric dam's steady output is valued like a perpetuity.

FN.7 Loan Amortization & Mortgages

$$M = P \frac{i(1+i)^n}{(1+i)^n - 1} \quad i = \frac{\text{APR}}{12}, \quad n = 12 \times \text{years} \quad \text{interest}_k = i \cdot \text{balance}_{k-1}, \quad \text{principal}_k = M - \text{interest}_k$$

STATED-EQUATION The monthly payment M equals the principal P times i times one plus i to the n , over one plus i to the n minus one, where i is the APR over twelve and n is twelve times the number of years. Each month, interest equals i times the previous balance, and the rest of the payment reduces principal.

MEMORY AID Early payments are mostly interest; late payments are mostly principal.

TERMS

P – loan amount

i – monthly rate

n – number of payments

Derivation – set $P = PV_{\text{annuity}}$ and solve for C [FN.6](#)

Amortization schedule – table of interest, principal, balance

CONTEXT

- **Origin.** Amortized mortgages became standard in the US after the 1930s (FHA).
- **Use.** Mortgages, car loans, student loans.
- **Data science.** Prepayment and default models (credit risk) simulate amortization schedules for millions of loans.
- **Physical application.** Financing a home solar or heat-pump system: the monthly payment vs. monthly energy savings.

FN.8 Inflation & Real Returns

$$(1 + i_{\text{nominal}}) = (1 + r_{\text{real}})(1 + \pi) \quad r_{\text{real}} = \frac{1 + i}{1 + \pi} - 1 \approx i - \pi \quad \text{real value} = \frac{\text{nominal}}{(1 + \pi)^t}$$

STATED-EQUATION The Fisher equation: one plus the nominal rate equals one plus the real rate times one plus inflation. So the real rate equals one plus i over one plus π , minus one, approximately i minus π . A future nominal amount's real value equals it divided by one plus π to the t .

MEMORY AID "Real $\approx$ nominal minus inflation."

TERMS

i – nominal interest or return

π – inflation rate (here π is not 3.14!)

r – real (inflation-adjusted) rate

CPI – consumer price index used to measure π

Deflating – same as discounting [FN.4](#)

CONTEXT

- **Origin.** Irving Fisher, *The Theory of Interest* (1930).
- **Use.** Comparing salaries, returns and prices across years.
- **Data science.** Always deflate multi-year revenue and price series to real terms before modeling trends.
- **Physical application.** Comparing energy costs over decades requires inflation-adjusted prices.

FN.9 Returns: Simple, Log & CAGR

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad r_t = \ln \frac{P_t}{P_{t-1}} \quad \text{CAGR} = \left(\frac{V_T}{V_0} \right)^{1/T} - 1 \quad \bar{R}_{\text{geo}} = \left[\prod (1 + R_t) \right]^{1/T} - 1 \leq \bar{R}_{\text{arith}}$$

STATED-EQUATION The simple return R_t equals price change over the previous price. The log return r_t equals the natural log of P_t over P_{t-1} minus one. The compound annual growth rate equals ending value over starting value, to the one over T , minus one. The geometric mean return, the product of one plus each return to the one over T minus one, is at most the arithmetic mean.

MEMORY AID Log returns add over time; simple returns add across assets. +50% then -50% loses 25%.

TERMS

P_t – price at time t (with dividends for total return)

Log return – additive over time [A2.7]

CAGR – constant rate giving the same growth [FN.1]

Volatility drag – geometric $\approx$ arithmetic $-\frac{1}{2}\sigma^2$

Means – [S.1]

CONTEXT

- **Origin.** Compounding math is ancient; log returns popularized by Osborne (1959) and Black–Scholes (1973).
- **Use.** Reporting investment performance honestly.
- **Data science.** Use log returns as model targets (roughly symmetric, additive); report CAGR for growth metrics like revenue or users.
- **Physical application.** Population growth over decades is summarized by its compound annual growth rate.

FN.10 Volatility & Risk

$$\sigma = \sqrt{\frac{1}{n-1} \sum_t (r_t - \bar{r})^2} \quad \sigma_{\text{annual}} = \sigma_{\text{daily}} \sqrt{252} \quad \sigma_{\text{horizon } T} = \sigma \sqrt{T} \quad \mu_{\text{annual}} = 252 \mu_{\text{daily}}$$

STATED-EQUATION Volatility sigma is the sample standard deviation of returns. Annual volatility equals daily volatility times the square root of two hundred fifty-two trading days. Over a horizon T , volatility scales with the square root of T , while the mean scales linearly with T .

MEMORY AID “Risk grows with the square root of time; expected return grows with time.”

TERMS

r_t – periodic (usually log) returns [FN.9]

σ – standard deviation [S.2]

$\sqrt{T}$ – variances of independent returns add [S.9]

252 – trading days per year (12 months, 52 weeks)

Assumption – i.i.d. returns; real markets show volatility clustering

CONTEXT

- **Origin.** Markowitz (1952) defined risk as variance.
- **Use.** Measuring and comparing investment risk.
- **Data science.** GARCH and ML models forecast volatility; $\sqrt{T}$ scaling is the same as the standard error law [S.13].
- **Physical application.** Random-walk diffusion distance also grows like $\sqrt{t}$ [R.19].

FN.11 Portfolio Return, Variance & Diversification

$$E[R_p] = \sum_i w_i E[R_i] \quad \sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{two assets: } \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

STATED-EQUATION Expected portfolio return equals the weighted sum of expected asset returns. Portfolio variance equals $\mathbf{w}$ transpose Sigma $\mathbf{w}$. For two assets: $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$.

MEMORY AID “Diversification is the only free lunch”: with $\rho < 1$, portfolio risk is less than the weighted average risk.

TERMS

w_i – portfolio weights, $\sum w_i = 1$

Σ – covariance matrix [S.18], a quadratic form [LA.11]

ρ – correlation between assets

Linearity – expected values add [S.9]; variances need covariances

Efficient frontier – minimum variance for each target return (a QP [O.11])

CONTEXT

- **Origin.** Harry Markowitz, “Portfolio Selection” (1952); Nobel Prize 1990.
- **Use.** Asset allocation, fund construction, risk budgeting.
- **Data science.** Ensemble averaging of models reduces variance by the same formula when errors are not perfectly correlated [ML.12].
- **Physical application.** Grid operators combine wind farms in different regions: uncorrelated wind makes total output steadier.

FN.12 Sharpe Ratio & Risk-Adjusted Return

$$S = \frac{E[R_p] - R_f}{\sigma_p} \quad \text{Sortino} = \frac{E[R_p] - R_f}{\sigma_{\text{downside}}} \quad S_{\text{annual}} = S_{\text{daily}} \sqrt{252}$$

STATED-EQUATION The Sharpe ratio equals the expected portfolio return minus the risk-free rate, over the portfolio's volatility. The Sortino ratio uses only downside volatility in the denominator. An annual Sharpe ratio equals the daily Sharpe ratio times the square root of two hundred fifty-two.

MEMORY AID Sharpe = "excess return per unit of risk." Rough guide: < 0.5 poor, ≈ 1 good, > 2 excellent (and suspicious).

TERMS

R_f – risk-free rate (Treasury bills)

Excess return – $R_p - R_f$

σ_p – volatility [\[FN.10\]](#)

Downside deviation – SD of returns below a target

Similar to – a z-score / t-statistic of returns [\[S.4\]](#), [\[S.16\]](#)

CONTEXT

- **Origin.** William Sharpe (1966); Nobel Prize 1990.
- **Use.** Comparing funds and strategies with different risk levels.
- **Data science.** Backtests of trading models report Sharpe; a high backtest Sharpe from many tried strategies is often overfitting [\[ML.4\]](#).
- **Physical application.** Engineering analog: signal-to-noise ratio of returns [\[EE.19\]](#).

FN.13 CAPM & Beta

$$E[R_i] = R_f + \beta_i(E[R_m] - R_f) \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{im} \frac{\sigma_i}{\sigma_m} \quad \alpha_i = R_i - [R_f + \beta_i(R_m - R_f)]$$

STATED-EQUATION The capital asset pricing model: the expected return on asset i equals the risk-free rate plus beta i times the market risk premium. Beta equals the covariance of the asset with the market over the market's variance, which equals rho times sigma i over sigma m . Alpha is the actual return minus the CAPM-predicted return.

MEMORY AID $\beta > 1$: amplifies the market; $\beta < 1$: dampens it; β is just a regression slope.

TERMS

$E[R_m] - R_f$ – market risk premium

β – systematic risk; slope of R_i on R_m [\[S.19\]](#)

Cov, Var – [\[S.18\]](#), [\[S.2\]](#)

α – excess return not explained by the market (intercept)

Idiosyncratic risk – diversifiable [\[FN.11\]](#)

CONTEXT

- **Origin.** Sharpe (1964), Lintner (1965), Mossin (1966), Treynor.
- **Use.** Cost of equity, performance attribution.
- **Data science.** Factor models (Fama–French) are multiple regressions of returns on factors; "alpha" is the intercept.
- **Physical application.** Like measuring how strongly a building sways (β) relative to ground motion.

FN.14 Bond Pricing & Yield

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n} = C \frac{1 - (1+y)^{-n}}{y} + \frac{F}{(1+y)^n} \quad \text{zero-coupon: } P = \frac{F}{(1+y)^n}$$

STATED-EQUATION A bond's price equals the sum of its discounted coupons C over one plus y to the t , plus the discounted face value F over one plus y to the n . That equals a coupon annuity plus the discounted face. A zero-coupon bond's price is F over one plus y to the n .

MEMORY AID Bond prices and yields move in opposite directions ("see-saw"). Coupon > yield $\Rightarrow$ premium; coupon < yield $\Rightarrow$ discount.

TERMS

C – coupon payment (= coupon rate $\times F$)

F – face (par) value

y – yield to maturity: the IRR of the bond [\[FN.5\]](#)

n – periods to maturity

Annuity – [\[FN.6\]](#)

CONTEXT

- **Origin.** Government bonds date to 12th-century Venice; modern yield math by Macaulay (1938).
- **Use.** Pricing government and corporate debt, interest-rate risk.
- **Data science.** Yield-curve modeling (Nelson–Siegel fits) and credit-spread prediction from company data.
- **Physical application.** Infrastructure projects (bridges, power plants) are often financed with long-term bonds.

FN.15 Duration & Convexity

$$D_{\text{Mac}} = \frac{1}{P} \sum_t t \frac{CF_t}{(1+y)^t} \quad D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y} = -\frac{1}{P} \frac{dP}{dy} \quad \frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} Cx (\Delta y)^2$$

STATED-EQUATION Macaulay duration is the present-value-weighted average time of the cash flows. Modified duration equals Macaulay duration over one plus y , which equals minus one over P times dP/dy . The percent price change is approximately minus modified duration times the yield change, plus one half convexity times the yield change squared.

MEMORY AID Duration $\approx$ “the percent price drop for a 1-point rise in rates.”

TERMS

D_{Mac} – weighted mean of payment times (years) [S.1]

D_{mod} – price sensitivity: a derivative [C1.4]

Convexity – second derivative term; a Taylor expansion [C2.13]

Zero-coupon – $D_{\text{Mac}} = n$

Longer maturity – higher duration, more rate risk

CONTEXT

- **Origin.** Frederick Macaulay (1938); Redington’s immunization (1952).
- **Use.** Managing interest-rate risk for banks, pensions and insurers.
- **Data science.** Sensitivity analysis: duration is a first-order Taylor sensitivity, the same idea as gradients and feature attributions.
- **Physical application.** Like linearizing a spring’s force for small displacements, adding a curvature correction for larger ones.

FN.16 Stock Valuation & Multiples

$$P_0 = \frac{D_1}{r-g} \text{ (Gordon growth)} \quad \frac{P}{E} = \frac{\text{price}}{\text{earnings per share}} \quad \text{EV} = \text{market cap} + \text{debt} - \text{cash} \quad \text{DCF: } V = \sum \frac{FCF_t}{(1+WACC)^t} + \frac{TV}{(1+WACC)^T}$$

STATED-EQUATION The Gordon growth model: today’s price equals next year’s dividend over r minus g . The P/E ratio equals price over earnings per share. Enterprise value equals market capitalization plus debt minus cash. A discounted cash flow valuation sums discounted free cash flows plus a discounted terminal value.

MEMORY AID Price is the present value of all future cash; growth near the discount rate makes values explode.

TERMS

D_1 – next year’s dividend

r – required return (e.g. from CAPM [FN.13])

g – constant growth rate, $g < r$ [FN.6]

EPS – net income per share

WACC – weighted average cost of capital

TV – terminal value, often a growing perpetuity

CONTEXT

- **Origin.** John Burr Williams (1938); Myron Gordon (1959).
- **Use.** Equity research, M&A, startup valuation.
- **Data science.** Sensitivity of valuations to r and g is a classic Monte Carlo exercise [O.14].
- **Physical application.** Valuing a mine or oil field by discounting the cash from future extraction.

FN.17 Options: Payoffs & Put–Call Parity

$$\text{call payoff} = \max(S_T - K, 0) \quad \text{put payoff} = \max(K - S_T, 0) \quad \text{profit} = \text{payoff} - \text{premium} \quad C - P = S_0 - Ke^{-rT}$$

STATED-EQUATION A call’s payoff is the maximum of S_T minus K and zero. A put’s payoff is the maximum of K minus S_T and zero. Profit equals payoff minus premium. Put-call parity: call price minus put price equals the stock price minus the discounted strike, Ke^{-rT} .

MEMORY AID “Call up, put down”: a call profits when price goes up, a put when it goes down. The call payoff is a ReLU [ML.14]!

TERMS

S_T – underlying price at expiry

K – strike price

C, P – call and put prices (premiums)

European – exercise only at expiry

Parity – no-arbitrage relation linking calls, puts, stock and cash [FN.4]

CONTEXT

- **Origin.** Options traded in 17th-century Amsterdam; parity formalized by Stoll (1969).
- **Use.** Hedging, speculation, employee stock options.
- **Data science.** The call payoff $\max(0, S - K)$ is exactly a ReLU; option-like payoffs appear in pricing models for insurance and cloud reserved capacity.
- **Physical application.** A home generator is like a put option on grid reliability: you pay a premium to be protected when power fails.

FN.18 Black–Scholes Option Pricing

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} \quad d_2 = d_1 - \sigma\sqrt{T} \quad P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

STATED-EQUATION The Black-Scholes call price equals S naught times N of d one, minus K e to the minus r T times N of d two. d one equals the natural log of S naught over K , plus r plus sigma squared over two times T , all over sigma root T . d two equals d one minus sigma root T . The put price equals K e to the minus r T N of minus d two, minus S naught N of minus d one.

MEMORY AID $N(d_2) \approx$ risk-neutral probability the call finishes in the money; $N(d_1)$ is the hedge ratio (delta).

TERMS

$N(\cdot)$ – standard normal CDF Φ [S.12]

σ – volatility [FN.10]

r – risk-free rate (continuous)

T – time to expiry (years)

Assumes – geometric Brownian motion [FN.22], constant σ , no arbitrage

Origin – a heat equation [P.19] in disguise

CONTEXT

- **Origin.** Black, Scholes (1973) and Merton; Nobel Prize 1997.
- **Use.** Pricing and hedging options; implied volatility.
- **Data science.** Implied-volatility surfaces are fit and forecast with ML; neural networks are trained to approximate pricing functions quickly.
- **Physical application.** The Black–Scholes PDE transforms into the heat equation, the same one that governs diffusion of heat.

FN.19 Value at Risk (VaR) & Expected Shortfall

$$\text{VaR}_\alpha = z_\alpha \sigma \sqrt{t} V \text{ (parametric, zero mean)} \quad \text{ES}_\alpha = E[\text{loss} \mid \text{loss} > \text{VaR}_\alpha] \quad z_{0.95} = 1.645, \quad z_{0.99} = 2.326$$

STATED-EQUATION Parametric value at risk at confidence alpha equals z alpha times sigma times the square root of t times the portfolio value V . Expected shortfall is the expected loss given that the loss exceeds VaR . z at ninety-five percent is one point six four five; at ninety-nine percent, two point three two six.

MEMORY AID VaR: “the loss you won’t exceed α of the time”; ES: “how bad it gets when you do.”

TERMS

α – confidence level (95% or 99%)

z_α – normal quantile [S.12]

$\sigma\sqrt{t}$ – horizon volatility [FN.10]

Historical VaR – a percentile of past losses [S.3]

Monte Carlo VaR – simulated losses [Q.14]

Fat tails – normal VaR underestimates extreme losses

CONTEXT

- **Origin.** J.P. Morgan’s RiskMetrics (1994); Basel accords adopted VaR and later ES.
- **Use.** Bank capital requirements, risk limits.
- **Data science.** Quantile regression and extreme-value models estimate tail risk; the same idea sets SLOs on p99 latency.
- **Physical application.** Engineers design levees for the “100-year flood,” a 1% annual exceedance level, like a 99% VaR.

FN.20 Financial Ratios & DuPont Analysis

$$\text{ROE} = \frac{\text{net income}}{\text{equity}} = \underbrace{\frac{\text{NI}}{\text{sales}}}_{\text{margin}} \cdot \underbrace{\frac{\text{sales}}{\text{assets}}}_{\text{turnover}} \cdot \underbrace{\frac{\text{assets}}{\text{equity}}}_{\text{leverage}} \quad \text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}} \quad \frac{D}{E} = \frac{\text{debt}}{\text{equity}}$$

STATED-EQUATION Return on equity equals net income over equity, which DuPont analysis splits into profit margin times asset turnover times financial leverage. The current ratio equals current assets over current liabilities. Debt-to-equity equals debt over equity.

MEMORY AID Accounting identity: assets = liabilities + equity.

TERMS

NI – net income (profit after all expenses and taxes)

Equity – assets minus liabilities (book value)

Margin, turnover, leverage – three drivers of ROE

Liquidity – current ratio > 1 covers short-term bills

Ratios – [A1.14]

CONTEXT

- **Origin.** DuPont Corporation’s analysts (1920s).
- **Use.** Credit analysis, equity research, management dashboards.
- **Data science.** Financial ratios are standard features for credit-scoring and bankruptcy-prediction models (e.g. Altman Z-score).
- **Physical application.** Energy intensity (energy per unit of output) is the engineering counterpart of asset turnover.

FN.21 Expected Value of Bets & the Kelly Criterion

$$EV = p \cdot W - q \cdot L \quad f^* = \frac{bp - q}{b} = p - \frac{q}{b}, \quad q = 1 - p \quad \text{maximizes } E[\ln(\text{wealth})]$$

STATED-EQUATION The expected value of a bet equals the probability of winning times the win, minus the probability of losing times the loss. The Kelly fraction f^* equals bp minus q , over b , which equals p minus q over b , where q is one minus p . It maximizes the expected log of wealth.

MEMORY AID Never bet on negative EV; even with an edge, bet a fraction (many use “half Kelly”).

TERMS

p, q – win and loss probabilities
 b – net odds: profit per \$1 staked on a win
 f^* – fraction of bankroll to wager
 Log utility – maximizes long-run growth (geometric mean [\[FN.9\]](#))
 EV – expected value [\[S.9\]](#)

CONTEXT

- **Origin.** John Kelly at Bell Labs (1956), linked to Shannon’s information theory [\[ML.3\]](#).
- **Use.** Position sizing, betting, portfolio growth.
- **Data science.** Budget allocation to experiments or campaigns with uncertain returns; multi-armed bandits balance exploration and exploitation.
- **Physical application.** Choosing how much of a limited energy reserve to spend on a risky maneuver is the same growth-versus-ruin trade-off.

FN.22 Geometric Brownian Motion

$$dS = \mu S dt + \sigma S dW_t \quad S_T = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T \right] \quad E[S_T] = S_0 e^{\mu T} \quad \text{median}(S_T) = S_0 e^{(\mu - \sigma^2/2)T}$$

STATED-EQUATION Geometric Brownian motion: dS equals $\mu S dt$ plus $\sigma S dW$. The solution: S_T equals S_0 times e to the quantity μ minus σ^2 squared over two, times T , plus σW_T . The expected value of S_T equals S_0 times e to the μT , while the median is S_0 times e to the μ minus σ^2 squared over two, times T .

MEMORY AID Mean > median: a few huge winners pull the average up (log-normal skew).

TERMS

μ – drift (expected return)
 σ – volatility
 W_t – Brownian motion: $W_T \sim N(0, T)$ [\[S.12\]](#)
 $\ln S_T$ – normally distributed, so S_T is log-normal
 Simulation – Euler steps with random shocks [\[C2.17\]](#), [\[O.14\]](#)

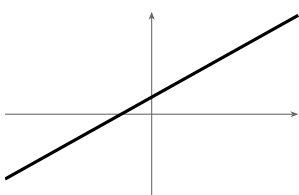
CONTEXT

- **Origin.** Bachelier (1900) modeled prices with Brownian motion; Samuelson (1965) proposed the geometric form; Itô calculus (1944).
- **Use.** Option pricing [\[FN.18\]](#), risk simulation, retirement planning.
- **Data science.** Monte Carlo simulations of portfolios and revenue paths; stochastic differential equations underlie diffusion generative models.
- **Physical application.** Brownian motion was first observed as pollen jiggling in water (Brown, 1827) and explained by Einstein (1905).

15 Graph Gallery

Recognize functions and probability distributions by shape. Each tile below is also a flashcard: the front shows only the graph, the back gives the name and formula.

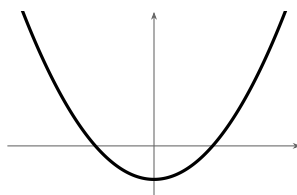
Functions



F.1 Linear

$$y = mx + b$$

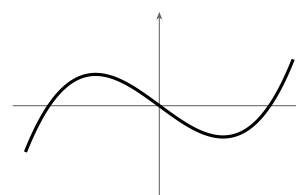
slope m , intercept b



F.2 Quadratic

$$y = ax^2 + bx + c$$

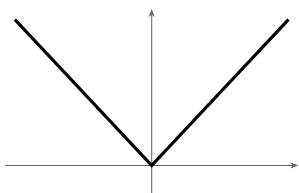
parabola; vertex at $x = -\frac{b}{2a}$



F.3 Cubic

$$y = ax^3 + bx^2 + cx + d$$

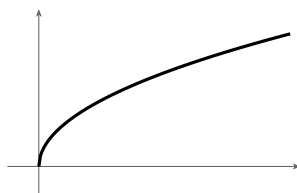
odd end behavior; up to 2 turns



F.4 Absolute Value

$$y = |x|$$

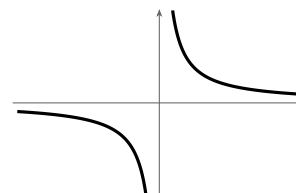
V shape, vertex at $(0,0)$



F.5 Square Root

$$y = \sqrt{x}$$

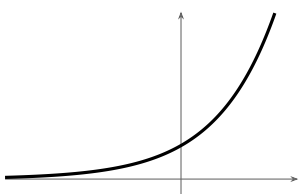
domain $x \geq 0$



F.6 Reciprocal

$$y = \frac{1}{x}$$

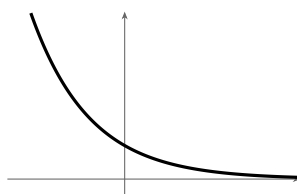
asymptotes $x = 0$, $y = 0$



F.7 Exponential Growth

$$y = e^x \text{ (} b^x, b > 1 \text{)}$$

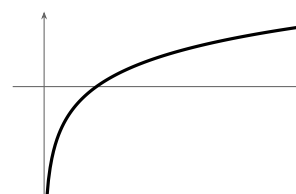
y -intercept 1; HA $y = 0$



F.8 Exponential Decay

$$y = e^{-x} \text{ (} b^x, 0 < b < 1 \text{)}$$

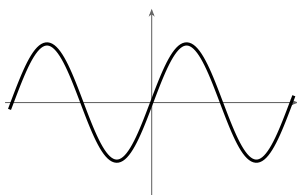
half-life $\frac{\ln 2}{k}$



F.9 Natural Logarithm

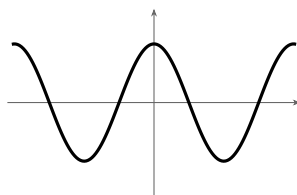
$$y = \ln x$$

domain $x > 0$; VA $x = 0$; $\ln 1 = 0$

**F.10 Sine**

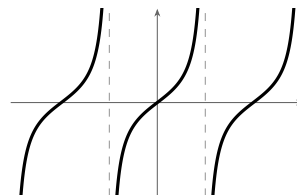
$$y = \sin x$$

period 2π ; $\sin 0 = 0$

**F.11 Cosine**

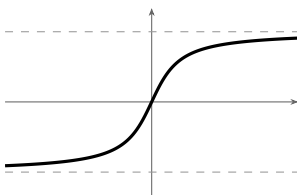
$$y = \cos x$$

period 2π ; $\cos 0 = 1$

**F.12 Tangent**

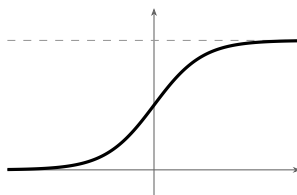
$$y = \tan x = \frac{\sin x}{\cos x}$$

period π ; VA at $x = \pm \frac{\pi}{2}$

**F.13 Arctangent**

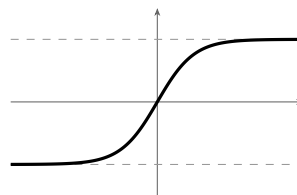
$$y = \arctan x$$

HA $y = \pm \frac{\pi}{2}$

**F.14 Logistic Sigmoid**

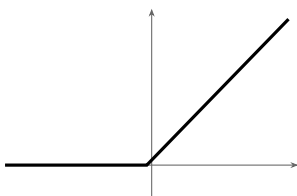
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

range $(0, 1)$; $\sigma(0) = \frac{1}{2}$

**F.15 Hyperbolic Tangent**

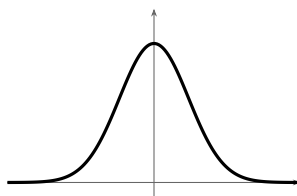
$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

range $(-1, 1)$; odd

**F.16 ReLU**

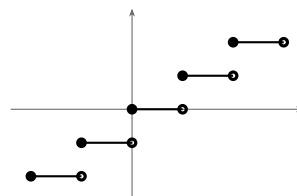
$$\text{ReLU}(x) = \max(0, x)$$

kink at 0

**F.17 Gaussian (Bell Curve)**

$$y = e^{-x^2}$$

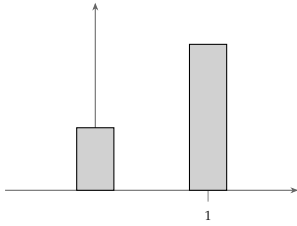
even; maximum 1 at $x = 0$

**F.18 Floor (Step) Function**

$$y = \lfloor x \rfloor$$

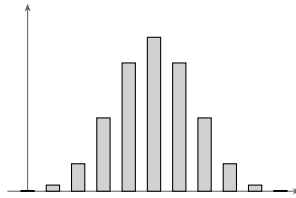
greatest integer $\leq x$

Probability Distributions

**D.1 Bernoulli**

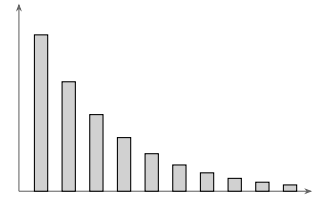
$$P(X=1) = p, \quad P(X=0) = 1-p$$

$$\mu = p, \quad \sigma^2 = p(1-p)$$

**D.2 Binomial**

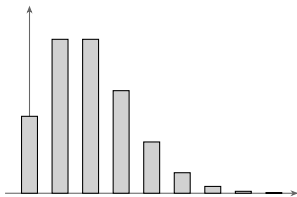
$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \quad \sigma^2 = np(1-p)$$

**D.3 Geometric**

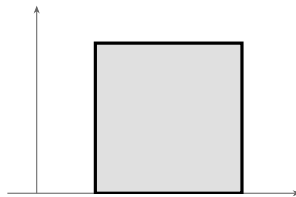
$$P(X=k) = (1-p)^{k-1} p, \quad k=1, 2, \dots$$

trials to first success; $\mu = \frac{1}{p}, \quad \sigma^2 = \frac{1-p}{p^2}$

**D.4 Poisson**

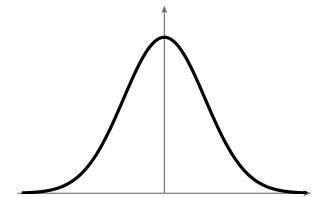
$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\mu = \sigma^2 = \lambda$$

**D.5 Uniform (Continuous)**

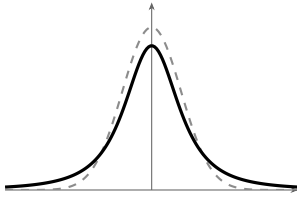
$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$

**D.6 Normal**

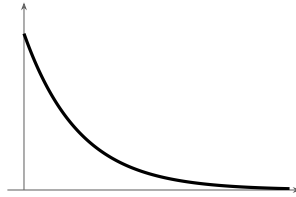
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

symmetric bell; 68-95-99.7

**D.7 Student's t (vs. normal)**

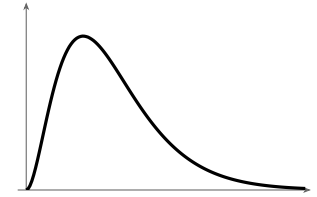
$$f(t) \propto \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

heavier tails; $t \rightarrow N(0,1)$ as $\nu \rightarrow \infty$

**D.8 Exponential**

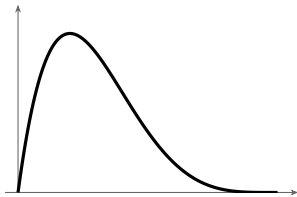
$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$\mu = \frac{1}{\lambda}; \text{ memoryless}$$

**D.9 Gamma**

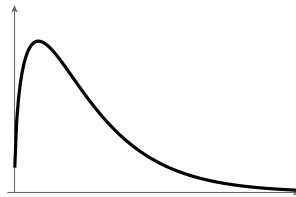
$$f(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k) \theta^k}$$

$$\mu = k\theta, \quad \sigma^2 = k\theta^2$$

**D.10 Beta**

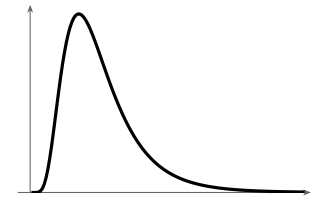
$$f(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1$$

$$\mu = \frac{\alpha}{\alpha + \beta}$$

**D.11 Chi-Square**

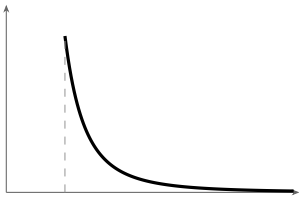
$$\chi_k^2 = \sum_{i=1}^k Z_i^2$$

$$\mu = k, \quad \sigma^2 = 2k; \text{ right-skewed}$$

**D.12 Log-Normal**

$$\ln X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

**D.13 Pareto (Power Law)**

$$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m$$

heavy tail; mean finite iff $\alpha > 1$

16

Glossary & Reference Cards

The Greek alphabet, common machine-learning algorithms, acronyms, and key terms not covered elsewhere. Every entry is also a flashcard: the front shows the name, the back gives the definition.

Greek Alphabet

GR.1 $\text{A } \alpha$ Alpha α : significance level, learning rate, angle, Beta/Gamma parameter.	GR.12 $\text{M } \mu$ Mu μ : mean, viscosity, friction coefficient, micro-
GR.2 $\text{B } \beta$ Beta β : regression coefficient, Type II error, momentum, thermal expansion.	GR.13 $\text{N } \nu$ Nu ν : degrees of freedom, kinematic viscosity, frequency.
GR.3 $\text{Γ } \gamma$ Gamma Γ : gamma function; γ : discount factor, RBF kernel width.	GR.14 $\Xi \zeta$ Xi ζ : slack variable (SVM), random variable.
GR.4 $\Delta \delta$ Delta Δ : change, discriminant, Laplacian; δ : small change, error signal.	GR.15 $\text{O } o$ Omicron Rarely used (looks like o); little- <i>o</i> notation for “grows slower than”.
GR.5 $\text{E } \epsilon$ Epsilon ϵ, ε : small quantity, error term, emissivity.	GR.16 $\Pi \pi$ Pi $\pi \approx 3.14159$; π_k mixing weights; Π : product.
GR.6 $\text{Z } \zeta$ Zeta ζ : Riemann zeta function, damping ratio.	GR.17 $\text{P } \rho$ Rho ρ : correlation, density, spherical radius.
GR.7 $\text{H } \eta$ Eta η : learning rate, efficiency.	GR.18 $\Sigma \sigma$ Sigma Σ : summation, covariance matrix; σ : standard deviation, sigmoid, singular value.
GR.8 $\Theta \theta$ Theta θ : angle, model parameters; Θ : parameter space, tight bound.	GR.19 $\text{T } \tau$ Tau τ : time constant, threshold, shear stress, 2π .
GR.9 $\text{I } \iota$ Iota ι : rarely used; “not one iota” means not the smallest amount.	GR.20 $\text{Υ } \upsilon$ Upsilon υ : rarely used; sometimes velocity.
GR.10 $\text{K } \kappa$ Kappa κ : condition number, Cohen’s kappa, curvature.	GR.21 $\Phi \phi$ Phi Φ : standard normal CDF; ϕ, φ : angle, activation, potential, golden ratio.
GR.11 $\Lambda \lambda$ Lambda λ : eigenvalue, rate (Poisson, exponential), regularization strength, wavelength.	GR.22 χ Chi χ^2 : chi-square distribution and test.
	GR.23 $\Psi \psi$ Psi ψ : wave function (physics), digamma function.
	GR.24 $\Omega \omega$ Omega Ω : ohm, sample space, lower bound; ω : angular frequency, outcome.

Algorithms & Where They Are Used

AL.1 Linear Regression <i>Supervised, regression.</i> Fits $\hat{y} = X\beta$ by least squares. Used for: pricing, forecasting, effect sizes.	dent features. Used for: spam filtering, text tagging.
AL.2 Logistic Regression <i>Supervised, classification.</i> Sigmoid of a linear score. Used for: churn, credit risk, conversion.	AL.6 Decision Tree <i>Supervised.</i> Splits data by if-then rules. Used for: interpretable rules, triage.
AL.3 Ridge / Lasso <i>Supervised, regularized regression.</i> L2 / L1 penalties. Used for: many correlated features, feature selection.	AL.7 Random Forest <i>Supervised.</i> Averages many decorrelated trees (bagging). Used for: robust tabular baselines, feature importance.
AL.4 k-Nearest Neighbors <i>Supervised, classification or regression.</i> Vote of the k closest points. Used for: recommendations, simple baselines, imputation.	AL.8 Gradient Boosting (XGBoost, LightGBM) <i>Supervised.</i> Adds trees that fix previous errors. Used for: top accuracy on tabular data (fraud, pricing, ranking).
AL.5 Naive Bayes <i>Supervised, classification.</i> Bayes’ rule with indepen-	AL.9 Support Vector Machine <i>Supervised, classification.</i> Maximum-margin boundary, kernels. Used for: small high-dimensional data,

text.

- AL-10 Neural Network (MLP)** *Supervised*. Layers of linear maps and nonlinearities. Used for: flexible function fitting, embeddings.
- AL-11 CNN** *Supervised (deep learning)*. Convolutional filters share weights across space. Used for: images, medical scans, audio spectrograms.
- AL-12 RNN / LSTM** *Supervised (deep learning)*. Recurrent state over a sequence. Used for: time series, speech (largely replaced by transformers).
- AL-13 Transformer** *Self-supervised / supervised*. Attention over all tokens. Used for: LLMs, translation, vision transformers.
- AL-14 k-Means** *Unsupervised, clustering*. Minimizes squared distance to K centroids. Used for: customer segmentation, quantization.
- AL-15 DBSCAN** *Unsupervised, clustering*. Groups dense regions and labels sparse points as noise; no K needed. Used for: geospatial hot spots, anomaly detection, irregular-shaped clusters.
- AL-16 Hierarchical Clustering** *Unsupervised*. Merges clusters into a dendrogram. Used for: taxonomies, gene expression.
- AL-17 Gaussian Mixture Model** *Unsupervised*. Soft clustering with EM. Used for: density estimation, soft segments.
- AL-18 PCA** *Unsupervised, dimension reduction*. Projects onto directions of maximum variance. Used for: compression, decorrelation, visualization.
- AL-19 t-SNE / UMAP** *Unsupervised, nonlinear embedding*. Preserves lo-

cal neighborhoods in 2-D. Used for: visualizing embeddings and single-cell data (not for distances).

- AL-20 Isolation Forest** *Unsupervised, anomaly detection*. Anomalies are isolated in few random splits. Used for: fraud, sensor faults, intrusion detection.
- AL-21 Autoencoder** *Unsupervised (self-supervised)*. Network compresses and reconstructs input. Used for: denoising, anomaly detection by reconstruction error.
- AL-22 Apriori / Association Rules** *Unsupervised*. Finds frequent itemsets with support, confidence, lift. Used for: market-basket analysis.
- AL-23 Matrix Factorization** *Unsupervised / collaborative filtering*. Low-rank user $\times$ item factors. Used for: recommender systems.
- AL-24 LDA (Topic Model)** *Unsupervised*. Latent Dirichlet Allocation: documents as mixtures of topics. Used for: topic discovery in text.
- AL-25 ARIMA** *Supervised time series*. AutoRegressive Integrated Moving Average. Used for: demand and sales forecasting.
- AL-26 Q-Learning** *Reinforcement learning*. Learns action values via the Bellman equation. Used for: games, control, recommendations over time.
- AL-27 Spectral Clustering** *Unsupervised*. Clusters using eigenvectors of a graph Laplacian. Used for: non-convex clusters, community detection.
- AL-28 GAN** *Unsupervised generative*. Generator vs. discriminator game. Used for: synthetic images and data augmentation.

Acronyms

- AC-1 CNN** Convolutional Neural Network: learns local filters (convolutions) shared across an image or signal.
- AC-2 RNN** Recurrent Neural Network: processes sequences with a hidden state carried forward.
- AC-3 LSTM** Long Short-Term Memory: an RNN with gates that preserve long-range information.
- AC-4 GRU** Gated Recurrent Unit: a simpler gated RNN.
- AC-5 GAN** Generative Adversarial Network: a generator and a discriminator trained against each other.
- AC-6 VAE** Variational Autoencoder: probabilistic autoencoder that learns a latent distribution.
- AC-7 LLM** Large Language Model: a very large transformer trained to predict text.
- AC-8 NLP** Natural Language Processing: computing with human language.
- AC-9 RAG** Retrieval-Augmented Generation: an LLM answers using documents retrieved by search.
- AC-10 MLP** Multilayer Perceptron: a fully connected feed-forward neural network.
- AC-11 SGD** Stochastic Gradient Descent: gradient steps on random mini-batches.
- AC-12 ReLU** Rectified Linear Unit: activation $\max(0, x)$.
- AC-13 ROC / AUC** Receiver Operating Characteristic curve / Area Under the Curve.
- AC-14 MSE / RMSE / MAE** Mean Squared Error / Root Mean Squared Error / Mean Absolute Error.
- AC-15 PCA** Principal Component Analysis.
- AC-16 SVD** Singular Value Decomposition: $A = U\Sigma V^T$.
- AC-17 SVM** Support Vector Machine.
- AC-18 k-NN** k -Nearest Neighbors.
- AC-19 EDA** Exploratory Data Analysis: summarizing and plotting data before modeling.
- AC-20 ETL** Extract, Transform, Load: moving and reshaping data into a warehouse.
- AC-21 TF-IDF** Term Frequency-Inverse Document Frequency: weights words that are frequent in a document but rare overall.
- AC-22 BERT** Bidirectional Encoder Representations from Transformers: an encoder language model.
- AC-23 GPT** Generative Pre-trained Transformer: a decoder language

model.

AC.24 OLS Ordinary Least Squares.

AC.25 MLE / MAP Maximum Likelihood Estimate / Maximum A Posteriori estimate (likelihood $\times$ prior).

AC.26 MCMC Markov Chain Monte Carlo.

AC.27 KL Kullback–Leibler divergence.

AC.28 i.i.d. Independent and Identically Distributed.

AC.29 PDF / PMF / CDF Probability Density Function / Probability Mass Function / Cumulative Distribution Function.

AC.30 IQR Interquartile Range, $Q_3 - Q_1$.

AC.31 CI Confidence Interval (also Continuous Integration in software).

AC.32 RL Reinforcement Learning: learning actions from rewards.

AC.33 LP / MIP / NLP Linear / Mixed-Integer / Nonlinear Programming (in optimization, NLP means nonlinear programming).

AC.34 ODE / PDE Ordinary / Partial Differential Equation.

AC.35 GPU Graphics Processing Unit: parallel hardware used to train neural networks.

AC.36 MLOps Machine Learning Operations: deploying, monitoring, and maintaining models.

Key Terms

KT.1 Domain & Range Domain: all allowed inputs of a function. Range: all outputs it produces.

KT.2 Monotonic Always increasing or always decreasing.

KT.3 One-to-One (Injective) Different inputs give different outputs; needed for an inverse.

KT.4 Even & Odd Function Even: $f(-x) = f(x)$ (symmetric about the y -axis). Odd: $f(-x) = -f(x)$ (symmetric about the origin).

KT.5 Asymptote A line a graph approaches but does not reach (vertical, horizontal, or slant).

KT.6 Identity vs. Equation An identity is true for all values; an equation is true only for its solutions.

KT.7 Theorem, Lemma, Corollary Theorem: major proven result. Lemma: helper result. Corollary: follows directly from a theorem.

KT.8 Necessary vs. Sufficient A is necessary for B if $B \Rightarrow A$; sufficient if $A \Rightarrow B$.

KT.9 Population vs. Sample Population: every unit of interest. Sample: the subset you observe.

KT.10 Parameter vs. Statistic Parameter describes a population (μ); statistic describes a sample ($\bar{x}$).

KT.11 Bias (Statistical) Systematic difference between an estimator's average and the truth.

KT.12 Confounder A variable that affects both the cause and the outcome, creating a spurious association.

KT.13 Stationarity A time series whose mean, variance and autocorrelation do not change over time.

KT.14 Heteroscedasticity Residual variance that changes with the predictors.

KT.15 Multicollinearity Predictors that are highly correlated, making coefficients unstable.

KT.16 Feature & Label Feature: an input variable. Label (target): the value to predict.

KT.17 Training, Validation, Test Sets Fit on training; tune on validation;

report once on the untouched test set.

KT.18 Overfitting & Underfitting Overfitting: memorizes noise (great train, poor test). Underfitting: too simple for both.

KT.19 Hyperparameter A setting chosen before training (learning rate, depth, k), not learned from data.

KT.20 Epoch & Batch Epoch: one full pass over the training data. Batch: the subset used for one gradient step.

KT.21 Data Leakage Information from outside the training data (e.g. the future or the test set) sneaks into training.

KT.22 One-Hot Encoding Representing a category as a vector with a single 1.

KT.23 Normalization vs. Standardization Normalization rescales to $[0, 1]$; standardization gives mean 0, SD 1.

KT.24 Imputation Filling in missing values (mean, median, model-based).

KT.25 Embedding A dense vector representing an item (word, user, image) so that similar items are close.

KT.26 Tensor A multi-dimensional array (scalar, vector, matrix, and higher).

KT.27 Loss vs. Metric Loss: what training minimizes. Metric: what you report to judge usefulness.

KT.28 Class Imbalance One class is much rarer; use PR curves, resampling, or class weights.

KT.29 Calibration Predicted probabilities match observed frequencies.

KT.30 Ensemble Combining several models to improve accuracy or stability.

KT.31 Transfer Learning & Fine-Tuning Reusing a pre-trained model and adapting it to a new task.

KT.32 Attention Weighting inputs by relevance: $\text{softmax}(QK^\top / \sqrt{d})V$.

KT.33 Supervised vs. Unsupervised Supervised learns from labeled examples; unsupervised finds structure without labels.

KT.34 Self-Supervised Learning Labels are created from the data itself (e.g. predict the next word).

KT.35 Latent Variable A hidden quantity inferred from data, such as a cluster or topic.

KT.36 Convex vs. Non-Convex Convex problems have one global minimum; non-convex ones can have many local minima.

Electrical Engineering Terms

ET.1 AC vs. DC Alternating current reverses direction periodically (grid power); direct current flows one way (batteries, electronics).

ET.2 Ground The reference node defined as 0 V; also a safety connection to earth.

ET.3 Short Circuit A near-zero-resistance path; current can become dangerously large.

ET.4 Open Circuit A break in the path; no current flows.

ET.5 Series vs. Parallel Series: same current through each element. Parallel: same voltage across each element.

ET.6 Impedance AC generalization of resistance, $Z = R + jX$, in ohms.

ET.7 Reactance The imaginary part of impedance from inductors (+) and capacitors (−).

ET.8 Conductance Reciprocal of resistance, $G = 1/R$, in siemens.

ET.9 Bandwidth The range of frequencies a system passes (Hz); in data links, the achievable data rate.

ET.10 Gain Output-to-input ratio of an amplifier, often in dB.

ET.11 Feedback Feeding part of the output back to the input; negative

feedback stabilizes, positive feedback oscillates.

ET.12 ADC / DAC Analog-to-Digital / Digital-to-Analog Converter.

ET.13 PWM Pulse-Width Modulation: controlling average power by switching on and off at a varying duty cycle.

ET.14 Semiconductor A material (e.g. silicon) whose conductivity can be controlled by doping; basis of diodes and transistors.

ET.15 MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor: the voltage-controlled switch in all modern chips.

ET.16 Breaker & Fuse Devices that interrupt current when it exceeds a safe level.

ET.17 Duty Cycle Fraction of each period that a signal is on.

ET.18 Harmonics Frequency components at integer multiples of the fundamental frequency.

ET.19 Microcontroller A small computer on one chip with CPU, memory and I/O, used in embedded devices.

ET.20 Kilowatt-hour Energy of 1 kW for 1 hour (3.6 MJ); the unit on electricity bills.

Finance Terms

FT.1 Time Value of Money Money now is worth more than the same amount later, because it can earn a return.

FT.2 Liquidity How quickly an asset can be sold near its fair price.

FT.3 Arbitrage A riskless profit from price differences; its absence underlies pricing models.

FT.4 Hedge A position taken to reduce the risk of another position.

FT.5 Leverage Using borrowed money to magnify returns (and losses).

FT.6 Bid-Ask Spread Difference between the highest buy and lowest sell price; a trading cost.

FT.7 Market Capitalization Share price times shares outstanding.

FT.8 Dividend A cash payment from a company's profits to shareholders.

FT.9 EBITDA Earnings Before Interest, Taxes, Depreciation and Amortization; a proxy for operating cash profit.

FT.10 Free Cash Flow Operating cash flow minus capital expenditures.

FT.11 Balance Sheet Snapshot of assets, liabilities and equity at a point in time.

FT.12 Income Statement Revenue, expenses and profit over a period.

FT.13 Yield Curve Plot of bond yields vs. maturity; an inverted curve often precedes recessions.

FT.14 Basis Point One hundredth of a percent (0.01%).

FT.15 Bull vs. Bear Market Rising market (bull) vs. falling market, typically −20% or more (bear).

FT.16 Index Fund / ETF A fund that tracks a market index; ETFs trade like stocks.

FT.17 Short Selling Selling borrowed shares to profit if the price falls.

FT.18 Drawdown Decline from a peak to a trough in portfolio value.

FT.19 Implied Volatility The volatility that makes an option-pricing model match the market price.

FT.20 Credit Score / Default Credit score: estimate of a borrower's reliability. Default: failure to make required payments.

FT.21 Efficient Market Hypothesis Prices already reflect available information, so consistent excess returns are hard to earn.

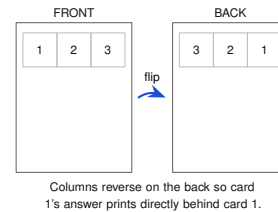
FT.22 Backtesting Testing a strategy on historical data; prone to overfitting and look-ahead bias.

Flashcards

Printing. The flashcard sheets begin on the next odd-numbered page. Each sheet is a pair of pages: a **front** (concept ID and name) followed by its **back** (formula).

- Print only the flashcard page range, double-sided.
- Scale: **100% / Actual size** (do not “fit to page”).
- Duplex: **flip on long edge** (the 11-inch side).
- The back layout is mirrored column-by-column, and the grid is centered on the sheet, so every back aligns with its front.
- Cut along the dashed lines: 3 columns $\times$ 4 rows of $2.5'' \times 2.5''$ cards.

Studying. Read the front, recall the formula and what each symbol means, flip to check. Every card’s back also repeats the concept ID so you can jump to the full entry.



A1.1

ALGEBRA I

Properties of Real Numbers & Order of Operations

STATE THE FORMULA/CONCEPT

A1.2

ALGEBRA I

Exponent Rules

STATE THE FORMULA/CONCEPT

A1.3

ALGEBRA I

Radicals & Rational Exponents

STATE THE FORMULA/CONCEPT

A1.4

ALGEBRA I

Slope

STATE THE FORMULA/CONCEPT

A1.5

ALGEBRA I

Equations of a Line

STATE THE FORMULA/CONCEPT

A1.6

ALGEBRA I

Parallel & Perpendicular Lines

STATE THE FORMULA/CONCEPT

A1.7

ALGEBRA I

Absolute Value Equations & Inequalities

STATE THE FORMULA/CONCEPT

A1.8

ALGEBRA I

Systems of Linear Equations

STATE THE FORMULA/CONCEPT

A1.9

ALGEBRA I

Polynomial Products & Special Products

STATE THE FORMULA/CONCEPT

A1.10

ALGEBRA I

Factoring

STATE THE FORMULA/CONCEPT

A1.11

ALGEBRA I

Quadratic Formula & Discriminant

STATE THE FORMULA/CONCEPT

A1.12

ALGEBRA I

Vertex Form of a Quadratic

STATE THE FORMULA/CONCEPT

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

$$\sqrt{ab} = \sqrt{a} \sqrt{b}$$

“Power over root”: in $a^{m/n}$ the top is the power, the bottom is the root.

A1.3 Radicals & Rational Exponents

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn} \quad a^0 = 1 \quad a^{-n} = \frac{1}{a^n}$$

Multiply → add exponents; power of a power → multiply exponents.

A1.2 Exponent Rules

$$a(b+c) = ab+ac$$

$$P \rightarrow E \rightarrow M/D \rightarrow A/S$$

PEMDAS: “Please Excuse My Dear Aunt Sally” (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

A1.1 Properties of Real Numbers & Order of Operations

$$\parallel: m_1 = m_2$$

$$\perp: m_1 m_2 = -1 \iff m_2 = -\frac{1}{m_1}$$

Perpendicular slopes: “flip it and switch the sign.”

A1.6 Parallel & Perpendicular Lines

$$y = mx + b$$

$$y - y_1 = m(x - x_1)$$

$$Ax + By = C$$

A1.5 Equations of a Line

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

“Rise over run” — y changes on top, like rising up.

A1.4 Slope

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$(a+b)(a-b) = a^2 - b^2$$

FOIL: First, Outer, Inner, Last.

A1.9 Polynomial Products & Special Products

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

one, none, or infinitely many solutions

A1.8 Systems of Linear Equations

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$|x - a| < r \iff a - r < x < a + r$$

“Less than” is a sandwich (and); “greater than” goes two ways (or).

A1.7 Absolute Value Equations & Inequalities

$$y = a(x - h)^2 + k$$

$$h = -\frac{b}{2a}, \quad k = f(h)$$

A1.12 Vertex Form of a Quadratic

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sung to “Pop Goes the Weasel”: “x equals negative b, plus or minus the square root, of b squared minus four a c, all over two a.”

A1.11 Quadratic Formula & Discriminant

$$x^2 + (p+q)x + pq = (x+p)(x+q)$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

Sum/difference of cubes signs: SOAP — Same, Opposite, Always Positive.

A1.10 Factoring

A1.13

ALGEBRA I

Direct & Inverse Variation

STATE THE FORMULA/CONCEPT

A1.14

ALGEBRA I

Ratios, Proportions & Percent Change

STATE THE FORMULA/CONCEPT

A1.15

ALGEBRA I

Linear Inequalities

STATE THE FORMULA/CONCEPT

A1.16

ALGEBRA I

Sets, Set Operations & Interval Notation

STATE THE FORMULA/CONCEPT

A2.1

ALGEBRA II

Functions: Composition & Inverses

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A2.2

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Transformations of Functions

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Complex Numbers

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A2.4

ALGEBRA II

Remainder, Factor & Rational Root Theorems

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Rational Functions & Asymptotes

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A2.6

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Exponential Growth & Decay

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A2.7

ALGEBRA II

Logarithms: Definition & Properties

STATE THE FORMULA/CONCEPT

A2.8

ALGEBRA II

Change of Base & Exponential Equations

STATE THE FORMULA/CONCEPT

$$a < b \Rightarrow a + c < b + c$$

$$c > 0 : ac < bc \quad c < 0 : ac > bc$$

"Multiply or divide by a negative, flip the sign."

A1.15 Linear Inequalities

$$\% \Delta = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

Percent change: "difference over original."

A1.14 Ratios, Proportions & Percent Change

$$y = kx \quad (\text{direct})$$

$$y = \frac{k}{x} \quad (\text{inverse})$$

A1.13 Direct & Inverse Variation

$$y = a f(b(x - h)) + k$$

Inside the parentheses acts on x and does the opposite (minus h shifts right); outside acts on y and does what it says.

A2.2 Transformations of Functions

$$(f \circ g)(x) = f(g(x))$$

$$f^{-1}(f(x)) = x$$

Composition works inside-out: apply g first, then f.

A2.1 Functions: Composition & Inverses

$$A \cup B, A \cap B, A^c, A \setminus B$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$(A \cup B)^c = A^c \cap B^c$$

$$[a, b) = \{x : a \leq x < b\}$$

Square bracket = "included" (closed); parenthesis = "not included" (open). De Morgan: "break the bar, change the sign."

A1.16 Sets, Set Operations & Interval Notation

$$f(x) = \frac{P(x)}{Q(x)}$$

$$\text{VA: } Q(c) = 0, P(c) \neq 0$$

$$\deg P < \deg Q \Rightarrow \text{HA } y = 0$$

$$\deg P = \deg Q \Rightarrow \text{HA } y = \frac{a_n}{b_n}$$

Degrees: "Bigger on Bottom, y = 0; Exactly Even, divide the leads; Bigger on Top by one, slant."

A2.5 Rational Functions & Asymptotes

$$P(x) = (x - c)Q(x) + P(c)$$

$$P(c) = 0 \iff (x - c) \text{ is a factor}$$

$$\text{rational roots } \pm \frac{p}{q} : p \mid a_0, q \mid a_n$$

A2.4 Remainder, Factor & Rational Root Theorems

$$i^2 = -1 \quad z = a + bi$$

$$\bar{z} = a - bi \quad |z| = \sqrt{a^2 + b^2}$$

A2.3 Complex Numbers

$$\log_b x = \frac{\ln x}{\ln b}$$

$$b^x = c \Rightarrow x = \frac{\ln c}{\ln b}$$

A2.8 Change of Base & Exponential Equations

$$\log_b x = y \iff b^y = x$$

$$\log(xy) = \log x + \log y$$

$$\log \frac{x}{y} = \log x - \log y \quad \log x^p = p \log x$$

"A log is an exponent": $\log_b x$ asks "b to what power gives x?"

A2.7 Logarithms: Definition & Properties

$$A = P \left(1 + \frac{r}{n}\right)^{nt} \quad A = Pe^{rt}$$

$$t_{1/2} = \frac{\ln 2}{|k|}$$

"Pert": continuous compounding is $A = Pe^{rt}$.

A2.6 Exponential Growth & Decay

A2.9

ALGEBRA II

Arithmetic Sequences & Series

STATE THE FORMULA/CONCEPT

A2.10

ALGEBRA II

Geometric Sequences & Series

STATE THE FORMULA/CONCEPT

A2.11

ALGEBRA II

Sigma Notation & Summation Formulas

STATE THE FORMULA/CONCEPT

A2.12

ALGEBRA II

Binomial Theorem

STATE THE FORMULA/CONCEPT

A2.13

ALGEBRA II

Matrix Operations

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A2.14

ALGEBRA II

Determinants, Inverses & Cramer's Rule

STATE THE FORMULA/CONCEPT

A2.15

ALGEBRA II

Conic Sections

STATE THE FORMULA/CONCEPT

G.1

GEOMETRY

Angle Relationships

STATE THE FORMULA/CONCEPT

G.2

GEOMETRY

Triangle Angle Sum & Triangle Inequality

STATE THE FORMULA/CONCEPT

G.3

GEOMETRY

Pythagorean Theorem

STATE THE FORMULA/CONCEPT

G.4

GEOMETRY

Special Right Triangles

STATE THE FORMULA/CONCEPT

G.5

GEOMETRY

Similar Triangles & Scale Factor

STATE THE FORMULA/CONCEPT

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum (ca_i + b_i) = c \sum a_i + \sum b_i$$

A2.11 Sigma Notation & Summation Formulas

$$a_n = a_1 r^{n-1}$$

$$S_n = \frac{a_1(1-r^n)}{1-r} \quad S_\infty = \frac{a_1}{1-r}, |r| < 1$$

A2.10 Geometric Sequences & Series

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{n(a_1 + a_n)}{2}$$

Arithmetic sum: "average of first and last, times the number of terms" (Gauss's trick).

A2.9 Arithmetic Sequences & Series

$$\det A = ad - bc$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$x = \frac{\det A_x}{\det A}$$

Two-by-two inverse: "swap the diagonal, negate the off-diagonal, divide by the determinant."

A2.14 Determinants, Inverses & Cramer's Rule

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

"Row times column": slide across the row of A and down the column of B.

A2.13 Matrix Operations

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Coefficients come from Pascal's triangle: 1; 1 1; 1 2 1; 1 3 3 1; ...

A2.12 Binomial Theorem

$$A + B + C = 180^\circ$$

exterior = sum of remote interiors

$$|a - b| < c < a + b$$

G.2 Triangle Angle Sum & Triangle Inequality

$$\alpha + \beta = 90^\circ \text{ (complementary)}$$

$$\alpha + \beta = 180^\circ \text{ (supplementary)}$$

vertical, corresponding, alternate interior: equal

Complementary makes a Corner (90°);
Supplementary makes a Straight line (180°).

G.1 Angle Relationships

circle: $(x-h)^2 + (y-k)^2 = r^2$

ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

hyperbola: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

Ellipse subtracts ($c^2 = a^2 - b^2$),
hyperbola adds ($c^2 = a^2 + b^2$).

A2.15 Conic Sections

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k$$

area ratio k^2 , volume ratio k^3

Length k , area k^2 , volume k^3 .

G.5 Similar Triangles & Scale Factor

$$45^\circ\text{-}45^\circ\text{-}90^\circ : x : x : x\sqrt{2}$$

$$30^\circ\text{-}60^\circ\text{-}90^\circ : x : x\sqrt{3} : 2x$$

30–60–90: "1, root 3, 2" — the short leg is half the hypotenuse.

G.4 Special Right Triangles

$$a^2 + b^2 = c^2$$

$$c^2 < a^2 + b^2 : \text{acute} \quad c^2 > a^2 + b^2 : \text{obtuse}$$

Common triples: 3–4–5, 5–12–13, 8–15–17, 7–24–25.

G.3 Pythagorean Theorem

G.6

GEOMETRY

Distance & Midpoint Formulas

STATE THE FORMULA/CONCEPT

G.7

GEOMETRY

Polygon Angle Sums & Diagonals

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GEOMETRY

Area of Plane Figures

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G.9

GEOMETRY

Circles: Circumference, Area, Arcs & Sectors

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GEOMETRY

Circle Theorems

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GEOMETRY

Prisms & Cylinders

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GEOMETRY

Pyramids, Cones & Spheres

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GEOMETRY

Coordinate Transformations

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T.1

TRIGONOMETRY

Right-Triangle Trigonometric Ratios

STATE THE FORMULA/CONCEPT

T.2

TRIGONOMETRY

Radians, Degrees & Angular Speed

STATE THE FORMULA/CONCEPT

T.3

TRIGONOMETRY

Unit Circle & Exact Values

STATE THE FORMULA/CONCEPT

T.4

TRIGONOMETRY

Pythagorean & Fundamental Identities

STATE THE FORMULA/CONCEPT

$$A_{\triangle} = \frac{1}{2}bh \quad A_{\text{trap}} = \frac{1}{2}(b_1 + b_2)h$$

Heron: $A = \sqrt{s(s-a)(s-b)(s-c)}$

G.8 Area of Plane Figures

$$\Sigma_{\text{int}} = (n-2) \cdot 180^\circ$$

$$\Sigma_{\text{ext}} = 360^\circ$$

$$\text{diagonals} = \frac{n(n-3)}{2}$$

G.7 Polygon Angle Sums & Diagonals

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum (p_i - q_i)^2}$$

G.6 Distance & Midpoint Formulas

$$V = Bh$$

$$V_{\text{cyl}} = \pi r^2 h$$

$$SA_{\text{cyl}} = 2\pi r^2 + 2\pi rh$$

G.11 Prisms & Cylinders

$$\angle_{\text{inscribed}} = \frac{1}{2} \angle_{\text{central}}$$

$$\text{angle in a semicircle} = 90^\circ$$

$$\text{tangent} \perp \text{radius} \quad ab = cd$$

G.10 Circle Theorems

$$C = 2\pi r \quad A = \pi r^2$$

$$s = r\theta \quad A_{\text{sector}} = \frac{1}{2}r^2\theta$$

*"Cherry pie is delicious" ($C = \pi d$);
"apple pies are too" ($A = \pi r^2$).*

G.9 Circles: Circumference, Area, Arcs & Sectors

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc \theta = \frac{1}{\sin \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

SOH-CAH-TOA: Sine = Opposite/Hypotenuse,
Cosine = Adjacent/Hypotenuse,
Tangent = Opposite/Adjacent.

T.1 Right-Triangle Trigonometric Ratios

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$(x, y) \rightarrow (x + a, y + b)$$

reflect in $y = x : (x, y) \rightarrow (y, x)$

G.13 Coordinate Transformations

$$V_{\text{pyr}} = \frac{1}{3}Bh \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

$$V_{\text{sph}} = \frac{4}{3}\pi r^3 \quad SA_{\text{sph}} = 4\pi r^2$$

*"A cone is one-third of the cylinder"
with the same base and height.*

G.12 Pyramids, Cones & Spheres

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(-\theta) = -\sin \theta \quad \cos(-\theta) = \cos \theta$$

Divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ to
get the other two. Cosine is the only even one.

T.4 Pythagorean & Fundamental Identities

$$(\cos \theta, \sin \theta)$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

Sine values are $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cosine is
the reverse. Signs by quadrant: "All Students
Take Calculus" (All, Sin, Tan, Cos positive).

T.3 Unit Circle & Exact Values

$$\pi \text{ rad} = 180^\circ$$

$$\theta_{\text{rad}} = \theta^\circ \cdot \frac{\pi}{180^\circ}$$

$$s = r\theta \quad v = r\omega$$

Degrees to radians: multiply by
 $\frac{\pi}{180}$ ("pi on top to go to radians").

T.2 Radians, Degrees & Angular Speed

T.5

TRIGONOMETRY

Sum & Difference Formulas

STATE THE FORMULA/CONCEPT

T.6

TRIGONOMETRY

Double-Angle & Half-Angle Formulas

STATE THE FORMULA/CONCEPT

T.7

TRIGONOMETRY

Law of Sines & Triangle Area

STATE THE FORMULA/CONCEPT

T.8

TRIGONOMETRY

Law of Cosines

STATE THE FORMULA/CONCEPT

T.9

TRIGONOMETRY

Graphs of Sinusoids

STATE THE FORMULA/CONCEPT

T.10

TRIGONOMETRY

Inverse Trigonometric Functions

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T.11

TRIGONOMETRY

Polar Coordinates

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T.12

TRIGONOMETRY

Vectors, Dot Product & Cosine Similarity

STATE THE FORMULA/CONCEPT

T.13

TRIGONOMETRY

Euler's Formula & De Moivre's Theorem

STATE THE FORMULA/CONCEPT

C1.1

CALCULUS I

Limits & Limit Laws

STATE THE FORMULA/CONCEPT

C1.2

CALCULUS I

Special Limits & the Number e

STATE THE FORMULA/CONCEPT

C1.3

CALCULUS I

Continuity & the Intermediate Value Theorem

STATE THE FORMULA/CONCEPT

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{Area} = \frac{1}{2}ab \sin C$$

T.7 Law of Sines & Triangle Area

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}} \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

T.6 Double-Angle & Half-Angle Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Sine keeps the sign and mixes (sin cos + cos sin); cosine keeps the pairs and flips the sign.

T.5 Sum & Difference Formulas

$$y = \arcsin x \iff \sin y = x, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\arccos x \in [0, \pi] \quad \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

T.10 Inverse Trigonometric Functions

$$y = A \sin(B(x - C)) + D$$

amplitude $|A|$ period $\frac{2\pi}{|B|}$

phase shift C midline $y = D$

T.9 Graphs of Sinusoids

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

It is the Pythagorean theorem with a correction: when $C = 90^\circ$, $\cos C = 0$.

T.8 Law of Cosines

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$z = re^{i\theta} \quad z^n = r^n e^{in\theta}$$

$$e^{i\pi} + 1 = 0$$

T.13 Euler's Formula & De Moivre's Theorem

$$\|\mathbf{v}\| = \sqrt{\sum v_i^2}$$

$$\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

T.12 Vectors, Dot Product & Cosine Similarity

$$x = r \cos \theta \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2} \quad \tan \theta = \frac{y}{x}$$

T.11 Polar Coordinates

$$f \text{ continuous at } a \iff \lim_{x \rightarrow a} f(x) = f(a)$$

$$\text{IVT: } N \text{ between } f(a), f(b) \Rightarrow \exists c : f(c) = N$$

C1.3 Continuity & the Intermediate Value Theorem

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \quad \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n = e^r$$

C1.2 Special Limits & the Number e

$$\lim_{x \rightarrow a} f(x) = L$$

$$\lim(f \pm g) = \lim f \pm \lim g \quad \lim \frac{f}{g} = \frac{\lim f}{\lim g}$$

$$\lim_{x \rightarrow a} f \text{ exists} \iff \lim_{x \rightarrow a^-} f = \lim_{x \rightarrow a^+} f$$

C1.1 Limits & Limit Laws

C1.4

CALCULUS I

Definition of the Derivative

STATE THE FORMULA/CONCEPT

C1.5

CALCULUS I

Power, Constant & Sum Rules

STATE THE FORMULA/CONCEPT

C1.6

CALCULUS I

Product & Quotient Rules

STATE THE FORMULA/CONCEPT

C1.7

CALCULUS I

Chain Rule

STATE THE FORMULA/CONCEPT

C1.8

CALCULUS I

Derivatives of Trig, Exponential & Log Functions

STATE THE FORMULA/CONCEPT

C1.9

CALCULUS I

Implicit Differentiation

STATE THE FORMULA/CONCEPT

C1.10

CALCULUS I

Related Rates

STATE THE FORMULA/CONCEPT

C1.11

CALCULUS I

Tangent Lines & Linear Approximation

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C1.12

CALCULUS I

Critical Points, Extrema & Concavity

STATE THE FORMULA/CONCEPT

C1.13

CALCULUS I

Optimization & Gradient Descent

STATE THE FORMULA/CONCEPT

C1.14

CALCULUS I

Mean Value Theorem

STATE THE FORMULA/CONCEPT

C1.15

CALCULUS I

L'Hôpital's Rule

STATE THE FORMULA/CONCEPT

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Quotient: "low d-high minus high d-low, over the square of what's below."

C1.6 Product & Quotient Rules

$$\frac{d}{dx}c = 0 \quad \frac{d}{dx}x^n = nx^{n-1}$$

$$(cf)' = cf' \quad (f \pm g)' = f' \pm g'$$

Power rule: "bring the power down, subtract one."

C1.5 Power, Constant & Sum Rules

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{dy}{dx}$$

C1.4 Definition of the Derivative

$$\frac{d}{dx}[y^n] = ny^{n-1} \frac{dy}{dx}$$

differentiate both sides, solve for $\frac{dy}{dx}$

C1.9 Implicit Differentiation

$$(\sin x)' = \cos x \quad (\cos x)' = -\sin x \quad (\tan x)' = \sec^2 x$$

$$(e^x)' = e^x \quad (a^x)' = a^x \ln a \quad (\ln x)' = \frac{1}{x}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

Derivatives of "co" functions
(cos, cot, csc) get a minus sign.

C1.8 Derivatives of Trig, Exponential & Log Functions

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

"Derivative of the outside (inside left alone) times derivative of the inside."

C1.7 Chain Rule

$$f'(c) = 0 \text{ or undefined}$$

$$f''(c) > 0 \Rightarrow \min \quad f''(c) < 0 \Rightarrow \max$$

$$f'' \text{ changes sign} \Rightarrow \text{inflection}$$

Concave up holds water like a cup (a min);
concave down spills like a frown (a max).

C1.12 Critical Points, Extrema & Concavity

$$y - f(a) = f'(a)(x - a)$$

$$f(x) \approx f(a) + f'(a)(x - a)$$

C1.11 Tangent Lines & Linear Approximation

$$\frac{d}{dt}[\text{relation}]$$

e.g. $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

C1.10 Related Rates

If $\lim f = \lim g = 0$ or $\pm \infty$:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Differentiate top and bottom
separately — not the quotient rule.

C1.15 L'Hôpital's Rule

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\text{Rolle: } f(a) = f(b) \Rightarrow f'(c) = 0$$

C1.14 Mean Value Theorem

$$f'(x^*) = 0$$

$$x_{k+1} = x_k - \eta f'(x_k)$$

$$\theta \leftarrow \theta - \eta \nabla L(\theta)$$

C1.13 Optimization & Gradient Descent

C1.16

CALCULUS I

Antiderivatives & Basic Integrals

STATE THE FORMULA/CONCEPT

C1.17

CALCULUS I

Riemann Sums & the Definite Integral

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C1.18

CALCULUS I

Fundamental Theorem of Calculus

STATE THE FORMULA/CONCEPT

C1.19

CALCULUS I

Integration by u -Substitution

STATE THE FORMULA/CONCEPT

C1.20

CALCULUS I

Newton's Method for Roots

STATE THE FORMULA/CONCEPT

C2.1

CALCULUS II

Integration by Parts

STATE THE FORMULA/CONCEPT

C2.2

CALCULUS II

Trig Integrals & Trig Substitution

STATE THE FORMULA/CONCEPT

C2.3

CALCULUS II

Partial Fractions

STATE THE FORMULA/CONCEPT

C2.4

CALCULUS II

Improper Integrals

STATE THE FORMULA/CONCEPT

C2.5

CALCULUS II

Area Between Curves

STATE THE FORMULA/CONCEPT

C2.6

CALCULUS II

Volumes of Revolution

STATE THE FORMULA/CONCEPT

C2.7

CALCULUS II

Arc Length

STATE THE FORMULA/CONCEPT

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

$$\int_a^b f(x) dx = F(b) - F(a)$$

C1.18 Fundamental Theorem of Calculus

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

$$\Delta x = \frac{b-a}{n}$$

C1.17 Riemann Sums & the Definite Integral

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln |x| + C \quad \int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C \quad \int \cos x dx = \sin x + C$$

"Add one to the power, divide by the new power"; don't forget +C.

C1.16 Antiderivatives & Basic Integrals

$$\int u dv = uv - \int v du$$

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

LIATE picks u : Log, Inverse trig, Algebraic, Trig, Exponential. "Ultraviolet voodoo": $uv - \int v du$.

C2.1 Integration by Parts

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

fails if $f'(x_n) \approx 0$

"Follow the tangent line down to the axis, then repeat."

C1.20 Newton's Method for Roots

$$\int f(g(x))g'(x) dx = \int f(u) du$$

$$u = g(x), \quad du = g'(x) dx$$

Look for an "inside function" whose derivative is also present.

C1.19 Integration by u -Substitution

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$\int_1^\infty \frac{dx}{x^p} = \frac{1}{p-1} \quad (p > 1)$$

diverges if $p \leq 1$

p -test: " p bigger than one converges."

C2.4 Improper Integrals

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

repeated: $\frac{A}{x-a} + \frac{B}{(x-a)^2}$

C2.3 Partial Fractions

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sqrt{a^2 - x^2} : x = a \sin \theta$$

$$\sqrt{a^2 + x^2} : x = a \tan \theta$$

$$\sqrt{x^2 - a^2} : x = a \sec \theta$$

Match the root to the identity: $a^2 - x^2 \rightarrow \sin$, $a^2 + x^2 \rightarrow \tan$, $x^2 - a^2 \rightarrow \sec$.

C2.2 Trig Integrals & Trig Substitution

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

C2.7 Arc Length

Disk: $V = \pi \int_a^b [R(x)]^2 dx$

Washer: $V = \pi \int_a^b (R^2 - r^2) dx$

Shell: $V = 2\pi \int_a^b x h(x) dx$

Washer = "big disk minus small disk"; shell = "circumference $\times$ height $\times$ thickness."

C2.6 Volumes of Revolution

$$A = \int_a^b [f(x) - g(x)] dx$$

$$f(x) \geq g(x) \text{ on } [a, b]$$

"Top minus bottom" (or "right minus left").

C2.5 Area Between Curves

C2.8

CALCULUS II

Average Value of a Function

STATE THE FORMULA/CONCEPT

C2.9

CALCULUS II

Sequences

STATE THE FORMULA/CONCEPT

C2.10

CALCULUS II

Series & Basic Convergence Tests

STATE THE FORMULA/CONCEPT

C2.11

CALCULUS II

Ratio, Root & Alternating Series Tests

STATE THE FORMULA/CONCEPT

C2.12

CALCULUS II

Power Series & Radius of Convergence

STATE THE FORMULA/CONCEPT

C2.13

CALCULUS II

Taylor & Maclaurin Series

STATE THE FORMULA/CONCEPT

C2.14

CALCULUS II

Parametric Equations

STATE THE FORMULA/CONCEPT

C2.15

CALCULUS II

Polar Area & Calculus

STATE THE FORMULA/CONCEPT

C2.16

CALCULUS II

Differential Equations: Separable, Growth & Logistic

STATE THE FORMULA/CONCEPT

C2.17

CALCULUS II

Euler's Method for Differential Equations

STATE THE FORMULA/CONCEPT

C2.18

CALCULUS II

Numerical Integration: Trapezoid & Simpson's Rules

STATE THE FORMULA/CONCEPT

LA.1

LINEAR ALGEBRA

Vectors & Linear Combinations

STATE THE FORMULA/CONCEPT

$$S = \sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} S_N$$

$$\lim a_n \neq 0 \Rightarrow \text{diverges}$$

$$\sum \frac{1}{n^p} \text{ converges } \iff p > 1$$

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad |r| < 1$$

"If the terms don't go to zero, the series won't go."

C2.10 Series & Basic Convergence Tests

$$\lim_{n \rightarrow \infty} a_n = L$$

$$r^n \rightarrow 0 \text{ if } |r| < 1$$

bounded + monotone $\Rightarrow$ convergent

Growth order: logs $\ll$ powers
 $\ll$ exponentials $\ll$ factorials.

C2.9 Sequences

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

$$\exists c : f(c) = f_{\text{avg}}$$

"Average = total over length."

C2.8 Average Value of a Function

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

$$e^x = \sum \frac{x^n}{n!} \quad \sin x = \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$|R_n| \leq \frac{M|x-a|^{n+1}}{(n+1)!}$$

sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

C2.13 Taylor & Maclaurin Series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

converges for $|x-a| < R$

$$R = \lim \left| \frac{c_n}{c_{n+1}} \right|$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

C2.12 Power Series & Radius of Convergence

$$L = \lim \left| \frac{a_{n+1}}{a_n} \right| \quad \text{or} \quad \lim \sqrt[n]{|a_n|}$$

$L < 1$ conv. $L > 1$ div. $L = 1$?

$$\sum (-1)^n b_n : b_n \downarrow 0 \Rightarrow \text{conv.}$$

$$|S - S_N| \leq b_{N+1}$$

C2.11 Ratio, Root & Alternating Series Tests

$$\frac{dy}{dx} = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x) dx$$

$$\frac{dy}{dt} = ky \Rightarrow y = y_0 e^{kt}$$

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) \Rightarrow P = \frac{K}{1 + Ae^{-rt}}$$

Separable: "get all the y's with dy, all the x's with dx, then integrate both sides."

C2.16 Differential Equations: Separable, Growth & Logistic

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

Polar area is a sum of thin pizza slices: $\frac{1}{2} r^2 d\theta$.

C2.15 Polar Area & Calculus

$$x = f(t), \quad y = g(t)$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

C2.14 Parametric Equations

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$$

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots) \quad c\mathbf{v} = (cv_1, \dots)$$

Add tip-to-tail; scaling stretches (and a negative scalar flips).

LA.1 Vectors & Linear Combinations

$$T = \frac{h}{2} [f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n]$$

$$S = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \cdots + 4f_{n-1} + f_n]$$

errors $O(h^2)$, $O(h^4)$

Simpson's weights: 1, 4, 2, 4, 2, ..., 4, 1.

C2.18 Numerical Integration: Trapezoid & Simpson's Rules

$$y_{n+1} = y_n + h f(t_n, y_n)$$

$$t_{n+1} = t_n + h$$

global error $O(h)$

"New value = old value + step $\times$ slope."

C2.17 Euler's Method for Differential Equations

LA.2

LINEAR ALGEBRA

Span, Linear Independence, Basis & Dimension

STATE THE FORMULA/CONCEPT

LA.3

LINEAR ALGEBRA

Change of Basis

STATE THE FORMULA/CONCEPT

LA.4

LINEAR ALGEBRA

Linear Transformations & Their Matrices

STATE THE FORMULA/CONCEPT

LA.5

LINEAR ALGEBRA

Gaussian Elimination & Row Reduction

STATE THE FORMULA/CONCEPT

LA.6

LINEAR ALGEBRA

Rank, Column Space, Null Space & Rank–Nullity

STATE THE FORMULA/CONCEPT

LA.7

LINEAR ALGEBRA

Orthogonality, Projections & Least Squares

STATE THE FORMULA/CONCEPT

LA.8

LINEAR ALGEBRA

Gram–Schmidt & QR Factorization

STATE THE FORMULA/CONCEPT

LA.9

LINEAR ALGEBRA

Eigenvalues & Eigenvectors

STATE THE FORMULA/CONCEPT

LA.10

LINEAR ALGEBRA

Diagonalization & Matrix Powers

STATE THE FORMULA/CONCEPT

LA.11

LINEAR ALGEBRA

Symmetric, Positive Definite Matrices & Quadratic Forms

STATE THE FORMULA/CONCEPT

LA.12

LINEAR ALGEBRA

Singular Value Decomposition (SVD)

STATE THE FORMULA/CONCEPT

LA.13

LINEAR ALGEBRA

Norms, Trace & Condition Number

STATE THE FORMULA/CONCEPT

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

$$A = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$$

$$T(\mathbf{x}) = A\mathbf{x}$$

"Columns are where the basis vectors land."

LA.4 Linear Transformations & Their Matrices

$$\mathbf{v} = P[\mathbf{v}]_B \quad [\mathbf{v}]_B = P^{-1}\mathbf{v}$$

$$P = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots]$$

$$[A]_B = P^{-1}AP$$

"P takes you from the new basis to the old (standard) one; P^{-1} takes you back."

LA.3 Change of Basis

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \Rightarrow \text{all } c_i = 0$$

basis = independent + spanning

$$\dim = \# \text{ basis vectors}$$

A basis is a "minimal complete set": enough vectors to reach everything, none redundant. n vectors in $\mathbb{R}^n$ form a basis $\iff \det \neq 0$.

LA.2 Span, Linear Independence, Basis & Dimension

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$$

$$\hat{\mathbf{b}} = A(A^\top A)^{-1}A^\top \mathbf{b}$$

$$A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}$$

$$Q^\top Q = I$$

Least squares: the error $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to every column of A .

LA.7 Orthogonality, Projections & Least Squares

$$\text{rank } A = \dim \text{Col } A$$

$$\text{Null } A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

$$\text{rank } A + \text{nullity } A = n$$

LA.6 Rank, Column Space, Null Space & Rank-Nullity

$$[A \mid \mathbf{b}] \rightarrow \text{row echelon form}$$

swap, scale, add a multiple

unique / none / infinitely many

Solutions: a pivot in every column $\rightarrow$ unique; a row $[0 \cdots 0 \mid c \neq 0] \rightarrow$ none; a free variable $\rightarrow$ infinitely many.

LA.5 Gaussian Elimination & Row Reduction

$$A = PDP^{-1}$$

$$A^k = PD^kP^{-1}$$

symmetric: $A = Q\Lambda Q^\top$

In the eigenbasis, a matrix just stretches each axis; powers become powers of numbers.

LA.10 Diagonalization & Matrix Powers

$$A\mathbf{v} = \lambda\mathbf{v}, \mathbf{v} \neq \mathbf{0}$$

$$\det(A - \lambda I) = 0$$

$$\text{tr } A = \sum \lambda_i \quad \det A = \prod \lambda_i$$

"Eigen" is German for "own": an eigenvector keeps its own direction; λ is the stretch factor.

LA.9 Eigenvalues & Eigenvectors

$$\mathbf{u}_1 = \mathbf{v}_1$$

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j < k} \text{proj}_{\mathbf{u}_j} \mathbf{v}_k$$

$$\mathbf{q}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|} \quad A = QR$$

"Subtract the shadows, then normalize."

LA.8 Gram-Schmidt & QR Factorization

$$\|\mathbf{x}\|_1 = \sum |x_i| \quad \|\mathbf{x}\|_2 = \sqrt{\sum x_i^2} \quad \|\mathbf{x}\|_\infty = \max |x_i|$$

$$\|A\|_F = \sqrt{\sum a_{ij}^2} = \sqrt{\text{tr}(A^\top A)}$$

$$\kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

Large κ = "wobbly": tiny changes in the data cause big changes in the answer.

LA.13 Norms, Trace & Condition Number

$$A = U\Sigma V^\top$$

$$\sigma_i = \sqrt{\lambda_i(A^\top A)}$$

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \text{ (best rank-}k\text{)}$$

Every matrix is "rotate, stretch, rotate" ($V^\top, \Sigma, U$).

LA.12 Singular Value Decomposition (SVD)

$$\mathbf{x}^\top A \mathbf{x} > 0 \quad \forall \mathbf{x} \neq \mathbf{0}$$

$$\iff \text{all } \lambda_i > 0$$

$$\iff \text{leading principal minors} > 0$$

Positive definite = a bowl (unique minimum); indefinite = a saddle vs.

LA.11 Symmetric, Positive Definite Matrices & Quadratic Forms

LA.14

LINEAR ALGEBRA

Matrix Algebra Rules & Identities

STATE THE FORMULA/CONCEPT

LA.15

LINEAR ALGEBRA

Special Matrices

STATE THE FORMULA/CONCEPT

LA.16

LINEAR ALGEBRA

Inner, Outer, Hadamard, Kronecker & Block Products

STATE THE FORMULA/CONCEPT

LA.17

LINEAR ALGEBRA

Matrix Calculus Identities

STATE THE FORMULA/CONCEPT

LA.18

LINEAR ALGEBRA

Pseudoinverse & Inverse-Update Identities

STATE THE FORMULA/CONCEPT

V.1

MULTIVARIABLE & VECTOR CALCULUS

Partial Derivatives

STATE THE FORMULA/CONCEPT

V.2

MULTIVARIABLE & VECTOR CALCULUS

Gradient & Directional Derivative

STATE THE FORMULA/CONCEPT

V.3

MULTIVARIABLE & VECTOR CALCULUS

Multivariable Chain Rule

STATE THE FORMULA/CONCEPT

V.4

MULTIVARIABLE & VECTOR CALCULUS

The Jacobian

STATE THE FORMULA/CONCEPT

V.5

MULTIVARIABLE & VECTOR CALCULUS

Hessian & the Second-Derivative Test

STATE THE FORMULA/CONCEPT

V.6

MULTIVARIABLE & VECTOR CALCULUS

Double & Triple Integrals

STATE THE FORMULA/CONCEPT

V.7

MULTIVARIABLE & VECTOR CALCULUS

Polar, Cylindrical & Spherical Coordinates

STATE THE FORMULA/CONCEPT

$\mathbf{u}^\top \mathbf{v}$ (scalar) $\mathbf{u}\mathbf{v}^\top$ (rank-1 matrix)

$$(A \odot B)_{ij} = a_{ij}b_{ij}$$

$$A \otimes B = [a_{ij}B]$$

$$A\mathbf{x} = \sum_j x_j \mathbf{a}_j$$

Inner product “collapses” ($1 \times n$ times $n \times 1$ is 1×1); outer product “expands” ($n \times 1$ times $1 \times m$ is $n \times m$).

LA.16 Inner, Outer, Hadamard, Kronecker & Block Products

diagonal, triangular, symmetric ($A = A^\top$)

orthogonal ($Q^\top Q = I$, $Q^{-1} = Q^\top$)

idempotent ($P^2 = P$) nilpotent ($N^k = 0$)

Orthogonal matrices are “rigid”: they rotate or reflect without stretching, so their inverse is just the transpose.

LA.15 Special Matrices

$AI = IA = A$ $AB \neq BA$ (in general)

$$(AB)^\top = B^\top A^\top \quad (AB)^{-1} = B^{-1}A^{-1}$$

$$\det(AB) = \det A \det B$$

$$\text{tr}(AB) = \text{tr}(BA)$$

“Socks and shoes”: to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

LA.14 Matrix Algebra Rules & Identities

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

$$f_{xy} = f_{yx} \text{ (Clairaut)}$$

“Freeze the other variables, then differentiate as usual.”

V.1 Partial Derivatives

$$A^+ = (A^\top A)^{-1} A^\top \text{ (full column rank)}$$

$$A^+ = V\Sigma^+U^\top$$

$$(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1}\mathbf{u}}$$

A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

LA.18 Pseudoinverse & Inverse-Update Identities

$$\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$$

$$\nabla_{\mathbf{x}}(\mathbf{x}^\top A \mathbf{x}) = (A + A^\top)\mathbf{x}$$

$$\nabla_{\beta} \|\mathbf{y} - X\beta\|^2 = -2X^\top(\mathbf{y} - X\beta)$$

$$\frac{\partial \ln \det X}{\partial X} = X^{-\top}$$

Matrix calculus mirrors scalar calculus: $\frac{d}{dx}(ax) = a$, $\frac{d}{dx}(ax^2) = 2ax$; just watch transposes and shapes.

LA.17 Matrix Calculus Identities

$$J = \left[\frac{\partial f_i}{\partial x_j} \right]$$

$$\mathbf{f}(\mathbf{x} + \mathbf{h}) \approx \mathbf{f}(\mathbf{x}) + J\mathbf{h}$$

$$dA = |\det J| du dv$$

Rows are outputs, columns are inputs; $|\det J|$ is the local area-stretch factor.

V.4 The Jacobian

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

Sum over every path from the input to the output: multiply along each path, add the paths.

V.3 Multivariable Chain Rule

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}, \quad \|\mathbf{u}\| = 1$$

$$\max D_{\mathbf{u}}f = \|\nabla f\|$$

The gradient points straight uphill; go the opposite way to descend [\[C1.13\]](#).

V.2 Gradient & Directional Derivative

$$dA = r dr d\theta$$

$$dV = r dz dr d\theta$$

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

“Don’t forget the extra r ”: it is the Jacobian [\[V.4\]](#).

V.7 Polar, Cylindrical & Spherical Coordinates

$$\iint_R f dA = \int_a^b \int_c^d f(x, y) dy dx$$

Fubini: order can be swapped

$$\iiint_E f dV$$

Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

V.6 Double & Triple Integrals

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

$$D = f_{xx}f_{yy} - f_{xy}^2$$

$$D > 0, f_{xx} > 0: \text{min}; \quad D > 0, f_{xx} < 0: \text{max}$$

$$D < 0: \text{saddle}$$

Hessian positive definite $\rightarrow$ bowl (min); negative definite $\rightarrow$ dome (max); mixed $\rightarrow$ saddle [\[LA.11\]](#).

V.5 Hessian & the Second-Derivative Test

V.8

MULTIVARIABLE & VECTOR CALCULUS

Vector Fields & Line Integrals

STATE THE FORMULA/CONCEPT

V.9

MULTIVARIABLE & VECTOR CALCULUS

Conservative Fields & Potentials

STATE THE FORMULA/CONCEPT

V.10

MULTIVARIABLE & VECTOR CALCULUS

Divergence & Curl

STATE THE FORMULA/CONCEPT

V.11

MULTIVARIABLE & VECTOR CALCULUS

Green's Theorem

STATE THE FORMULA/CONCEPT

V.12

MULTIVARIABLE & VECTOR CALCULUS

Flux & the Divergence Theorem

STATE THE FORMULA/CONCEPT

V.13

MULTIVARIABLE & VECTOR CALCULUS

Stokes' Theorem

STATE THE FORMULA/CONCEPT

V.14

MULTIVARIABLE & VECTOR CALCULUS

The Laplacian & Classic PDEs

STATE THE FORMULA/CONCEPT

P.1

PHYSICS & NATURAL PHENOMENA

Units & Dimensional Analysis

STATE THE FORMULA/CONCEPT

P.2

PHYSICS & NATURAL PHENOMENA

Kinematics: Constant Acceleration

STATE THE FORMULA/CONCEPT

P.3

PHYSICS & NATURAL PHENOMENA

Newton's Laws & Friction

STATE THE FORMULA/CONCEPT

P.4

PHYSICS & NATURAL PHENOMENA

Work, Energy & Power

STATE THE FORMULA/CONCEPT

P.5

PHYSICS & NATURAL PHENOMENA

Momentum & Impulse

STATE THE FORMULA/CONCEPT

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$\nabla \times \mathbf{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$$

Divergence = “spreading out” (source > 0, sink < 0); curl = “spinning” (right-hand rule).

V.10 Divergence & Curl

$$\mathbf{F} = \nabla \varphi$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A)$$

2-D test: $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

The Fundamental Theorem of Calculus for line integrals: only the endpoints matter [C1.18].

V.9 Conservative Fields & Potentials

$$\mathbf{F} = (P, Q, R)$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

V.8 Vector Fields & Line Integrals

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$$

Right-hand rule: curl your fingers along C, your thumb gives $\mathbf{n}$.

V.13 Stokes' Theorem

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \nabla \cdot \mathbf{F} dV$$

What flows out through the skin = what is produced inside.

V.12 Flux & the Divergence Theorem

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\text{Area} = \frac{1}{2} \oint_C (x dy - y dx)$$

Circulation around the boundary = total spin inside. Counterclockwise is positive.

V.11 Green's Theorem

$$v = \frac{dx}{dt}, \quad a = \frac{dv}{dt}$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a \Delta x$$

“Velocity is the slope of position; acceleration is the slope of velocity” (and area under v - t is distance).

P.2 Kinematics: Constant Acceleration

$$[F] = \text{kg} \cdot \text{m}/\text{s}^2 = \text{N}$$

both sides must have the same dimensions

Buckingham π : #groups = $n - k$

“Units first”: if the units of both sides do not match, the equation is wrong.

P.1 Units & Dimensional Analysis

$$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}$$

heat: $u_t = \alpha \nabla^2 u$

wave: $u_{tt} = c^2 \nabla^2 u$

Laplace: $\nabla^2 u = 0$

The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

V.14 The Laplacian & Classic PDEs

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{J} = \mathbf{F}\Delta t = \Delta \mathbf{p}$$

$$\sum \mathbf{p}_{\text{before}} = \sum \mathbf{p}_{\text{after}}$$

P.5 Momentum & Impulse

$$W = Fd \cos \theta$$

$$KE = \frac{1}{2}mv^2 \quad PE = mgh$$

$$KE_1 + PE_1 = KE_2 + PE_2$$

$$P = \frac{W}{t} = Fv$$

P.4 Work, Energy & Power

$$\sum \mathbf{F} = m\mathbf{a}$$

$$W = mg$$

$$f_k = \mu_k N$$

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

“F equals m a” — every problem starts with a free-body diagram: draw all forces, then sum them.

P.3 Newton's Laws & Friction

P.6

PHYSICS & NATURAL PHENOMENA

Circular Motion & Gravitation

STATE THE FORMULA/CONCEPT

P.7

PHYSICS & NATURAL PHENOMENA

Oscillations & Waves

STATE THE FORMULA/CONCEPT

P.8

PHYSICS & NATURAL PHENOMENA

Pressure, Hydrostatics & Buoyancy

STATE THE FORMULA/CONCEPT

P.9

PHYSICS & NATURAL PHENOMENA

Continuity & Bernoulli's Equation

STATE THE FORMULA/CONCEPT

P.10

PHYSICS & NATURAL PHENOMENA

Viscosity & the Reynolds Number

STATE THE FORMULA/CONCEPT

P.11

PHYSICS & NATURAL PHENOMENA

Drag & Terminal Velocity

STATE THE FORMULA/CONCEPT

P.12

PHYSICS & NATURAL PHENOMENA

Heat, Temperature & Phase Change

STATE THE FORMULA/CONCEPT

P.13

PHYSICS & NATURAL PHENOMENA

Conduction: Fourier's Law & Thermal Resistance

STATE THE FORMULA/CONCEPT

P.14

PHYSICS & NATURAL PHENOMENA

Convection & Newton's Law of Cooling

STATE THE FORMULA/CONCEPT

P.15

PHYSICS & NATURAL PHENOMENA

Biot Number & Lumped Capacitance

STATE THE FORMULA/CONCEPT

P.16

PHYSICS & NATURAL PHENOMENA

Natural Convection: Nu, Pr, Gr & Ra

STATE THE FORMULA/CONCEPT

P.17

PHYSICS & NATURAL PHENOMENA

Thermal Radiation

STATE THE FORMULA/CONCEPT

$$P = \frac{F}{A}$$

$$P = P_0 + \rho gh$$

$$F_B = \rho_{\text{fluid}} g V_{\text{disp}}$$

Archimedes: "the fluid pushes up with the weight of the fluid you push aside."

P.8 Pressure, Hydrostatics & Buoyancy

$$x(t) = A \cos(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}}, T = 2\pi \sqrt{\frac{m}{k}}$$

$$T_{\text{pend}} = 2\pi \sqrt{\frac{L}{g}}$$

$$v = f\lambda$$

P.7 Oscillations & Waves

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$F = \frac{Gm_1 m_2}{r^2}$$

$$g = \frac{GM}{R^2} \quad v_{\text{orbit}} = \sqrt{\frac{GM}{r}}$$

Gravity is an inverse-square law: double the distance, one quarter the force.

P.6 Circular Motion & Gravitation

$$F_D = \frac{1}{2} \rho C_D A v^2$$

$$v_t = \sqrt{\frac{2mg}{\rho C_D A}}$$

Stokes ($\text{Re} \ll 1$): $F_D = 6\pi\mu r v$

P.11 Drag & Terminal Velocity

$$\tau = \mu \frac{du}{dy}$$

$$\text{Re} = \frac{\rho v L}{\mu} = \frac{v L}{\nu}$$

pipe: laminar < 2300 , turbulent > 4000

$$Q = \frac{\pi r^4 \Delta P}{8\mu L}$$

"Re is big, flow goes wild" — high Reynolds means turbulence; low means smooth, syrupy flow.

P.10 Viscosity & the Reynolds Number

$$A_1 v_1 = A_2 v_2 \quad (Q = Av)$$

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{const}$$

Torricelli: $v = \sqrt{2gh}$

"Fast fluid, low pressure" — squeeze the pipe, speed goes up, pressure goes down.

P.9 Continuity & Bernoulli's Equation

$$\dot{Q} = hA(T_s - T_\infty)$$

$$T(t) = T_\infty + (T_0 - T_\infty)e^{-t/\tau}$$

$$\tau = \frac{\rho V c}{hA}$$

"Hot things cool fastest when they are hottest": the rate is proportional to the temperature gap.

P.14 Convection & Newton's Law of Cooling

$$q = -k \frac{dT}{dx}$$

$$\dot{Q} = \frac{kA\Delta T}{L}$$

$$R = \frac{L}{kA}, \quad \dot{Q} = \frac{\Delta T}{R_{\text{total}}}$$

Heat flows like current: ΔT is the "voltage," R the "resistor" **[P.20]**.

P.13 Conduction: Fourier's Law & Thermal Resistance

$$Q = mc \Delta T$$

$$Q = mL$$

$$T_K = T_C + 273.15$$

$$T_F = \frac{9}{5} T_C + 32$$

Water: $c \approx 4186 \text{ J/(kg}\cdot\text{K)}$; melting ice takes as much heat as warming the water by about 80°C .

P.12 Heat, Temperature & Phase Change

$$P = \epsilon \sigma A T^4$$

$$P_{\text{net}} = \epsilon \sigma A (T^4 - T_{\text{surr}}^4)$$

$$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

"Fourth power": double the absolute temperature, sixteen times the radiation. Always use kelvin.

P.17 Thermal Radiation

$$\text{Nu} = \frac{hL}{k_f} \quad \text{Pr} = \frac{\nu}{\alpha}$$

$$\text{Gr} = \frac{g\beta\Delta T L^3}{\nu^2}$$

$$\text{Ra} = \text{Gr} \cdot \text{Pr} = \frac{g\beta\Delta T L^3}{\nu\alpha}$$

$$\text{Ra}_c \approx 1708$$

Ra is the "Reynolds number of buoyancy": above ≈ 1708 , a heated layer starts to roll into convection cells.

P.16 Natural Convection: Nu, Pr, Gr & Ra

$$\text{Bi} = \frac{hL_c}{k}, \quad L_c = \frac{V}{A_s}$$

$$\text{Bi} < 0.1 \Rightarrow \text{uniform internal } T$$

$$\text{Fo} = \frac{\alpha t}{L_c^2}$$

"Biot small, same temperature all" — below 0.1, the inside keeps up with the surface.

P.15 Biot Number & Lumped Capacitance

P.18

PHYSICS & NATURAL PHENOMENA

Ideal Gas Law & Thermodynamics

STATE THE FORMULA/CONCEPT

P.19

PHYSICS & NATURAL PHENOMENA

Diffusion & the Heat Equation

STATE THE FORMULA/CONCEPT

P.20

PHYSICS & NATURAL PHENOMENA

Electric Circuits: Ohm's Law & Power

STATE THE FORMULA/CONCEPT

S.1

STATISTICS & PROBABILITY

Measures of Center

STATE THE FORMULA/CONCEPT

S.2

STATISTICS & PROBABILITY

Variance & Standard Deviation

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S.3

STATISTICS & PROBABILITY

Quartiles, IQR & Outliers

STATE THE FORMULA/CONCEPT

S.4

STATISTICS & PROBABILITY

z-Scores & Standardization

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S.5

STATISTICS & PROBABILITY

Basic Probability Rules

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S.6

STATISTICS & PROBABILITY

Conditional Probability & Independence

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S.7

STATISTICS & PROBABILITY

Bayes' Theorem

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S.8

STATISTICS & PROBABILITY

Counting: Permutations & Combinations

STATE THE FORMULA/CONCEPT

S.9

STATISTICS & PROBABILITY

Random Variables: Expected Value & Variance

STATE THE FORMULA/CONCEPT

$$V = IR \quad P = IV = I^2 R = \frac{V^2}{R}$$

$$R_{\text{series}} = R_1 + R_2$$

$$\frac{1}{R_{\text{par}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

"VIR" triangle: cover the one you want
 $- V = IR, I = V/R, R = V/I.$

P.20 Electric Circuits: Ohm's Law & Power

$$J = -D \frac{dC}{dx}$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$L_{\text{diff}} \approx \sqrt{2Dt}$$

Diffusion distance grows like $\sqrt{t}$: to go twice as far takes four times as long.

P.19 Diffusion & the Heat Equation

$$PV = nRT$$

$$\Delta U = Q - W$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

"You can't win, you can't break even, you can't get out of the game" (first, second, third laws).

P.18 Ideal Gas Law & Thermodynamics

$$\text{IQR} = Q_3 - Q_1$$

$$\text{outlier: } x < Q_1 - 1.5 \text{ IQR}$$

$$\text{or } x > Q_3 + 1.5 \text{ IQR}$$

$$\text{min, } Q_1, \text{ med, } Q_3, \text{ max}$$

S.3 Quartiles, IQR & Outliers

$$\sigma^2 = \frac{1}{N} \sum (x_i - \mu)^2$$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$\sigma = \sqrt{\sigma^2}, \quad s = \sqrt{s^2}$$

Divide by $n-1$ for a Sample ("s for sample, subtract one").

S.2 Variance & Standard Deviation

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \mu = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

median: middle value

mode: most frequent

S.1 Measures of Center

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A | B) P(B)$$

$$\text{independent} \iff P(A \cap B) = P(A)P(B)$$

"Given" shrinks the sample space to B.

S.6 Conditional Probability & Independence

$$0 \leq P(A) \leq 1 \quad P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{disjoint: } P(A \cup B) = P(A) + P(B)$$

"Or" adds (minus the overlap); "at least one" = $1 - \text{"none."}$

S.5 Basic Probability Rules

$$z = \frac{x - \mu}{\sigma} \quad z = \frac{x - \bar{x}}{s}$$

$$x = \mu + z\sigma$$

"How many SDs from the mean?"

S.4 z-Scores & Standardization

$$E[X] = \sum x p(x) \quad E[X] = \int x f(x) dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[aX + b] = aE[X] + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2 .

S.9 Random Variables: Expected Value & Variance

$$n! = n(n-1) \cdots 1$$

$${}_n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_n C_r = \frac{n!}{r!(n-r)!}$$

Permutation: Position matters. Combination: Committee (order doesn't matter).

S.8 Counting: Permutations & Combinations

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

$$P(B) = \sum_i P(B | A_i) P(A_i)$$

Posterior $\propto$ likelihood $\times$ prior.

S.7 Bayes' Theorem

S.10

STATISTICS & PROBABILITY

Binomial Distribution

STATE THE FORMULA/CONCEPT

S.11

STATISTICS & PROBABILITY

Poisson Distribution

STATE THE FORMULA/CONCEPT

S.12

STATISTICS & PROBABILITY

Normal Distribution & the CDF

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S.13

STATISTICS & PROBABILITY

Central Limit Theorem & Standard Error

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STATISTICS & PROBABILITY

Confidence Intervals

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S.15

STATISTICS & PROBABILITY

Hypothesis Testing, p -Values & Errors

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S.16

STATISTICS & PROBABILITY

t -Tests

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STATISTICS & PROBABILITY

Chi-Square Tests

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STATISTICS & PROBABILITY

Covariance & Correlation

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S.19

STATISTICS & PROBABILITY

Least-Squares Linear Regression & R^2

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S.20

STATISTICS & PROBABILITY

Continuous Distributions: Uniform & Exponential

STATE THE FORMULA/CONCEPT

S.21

STATISTICS & PROBABILITY

Maximum Likelihood Estimation (MLE)

STATE THE FORMULA/CONCEPT

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$$

$$68\% - 95\% - 99.7\%$$

68–95–99.7: within 1, 2, 3 SDs.

S.12 Normal Distribution & the CDF

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\mu = \sigma^2 = \lambda$$

Poisson: mean = variance = λ .

S.11 Poisson Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np \quad \sigma^2 = np(1-p)$$

BINS conditions: Binary outcomes,
Independent trials, fixed Number, Same p .

S.10 Binomial Distribution

$$H_0 \text{ vs } H_a \quad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

reject H_0 if $p < \alpha$

$$\alpha = P(\text{Type I}) \quad \beta = P(\text{Type II})$$

$$\text{power} = 1 - \beta$$

“If p is low, the null must go.” Type I = false alarm; Type II = missed detection.

S.15 Hypothesis Testing, p -Values & Errors

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \quad \bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$n = \left(\frac{z^* \sigma}{E} \right)^2$$

Estimate $\pm$ (critical value) $\times$ (standard error).

S.14 Confidence Intervals

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

“Root-n law”: four times the data halves the error.

S.13 Central Limit Theorem & Standard Error

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$r = \frac{\text{Cov}(X, Y)}{s_x s_y}$$

$$-1 \leq r \leq 1$$

Correlation is not causation.

S.18 Covariance & Correlation

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\text{GOF: } df = k - 1$$

$$E = \frac{\text{row} \cdot \text{col}}{\text{total}}$$

$$\text{indep.: } df = (r-1)(c-1)$$

S.17 Chi-Square Tests

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (\text{Welch})$$

paired: one-sample t on differences

S.16 t -Tests

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta)$$

$$\ell(\theta) = \sum \ln f(x_i | \theta)$$

$$\hat{\theta} : \frac{d\ell}{d\theta} = 0$$

“Which parameter makes the data I saw most probable?” Take logs to turn products into sums.

S.21 Maximum Likelihood Estimation (MLE)

$$\text{Uniform: } f = \frac{1}{b-a}$$

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$

$$\text{Exp.: } f = \lambda e^{-\lambda x}, \quad F = 1 - e^{-\lambda x}$$

$$\mu = \frac{1}{\lambda}, \quad \sigma^2 = \frac{1}{\lambda^2}$$

Exponential: mean = $1/\lambda$;
“memoryless” — the past doesn’t matter.

S.20 Continuous Distributions: Uniform & Exponential

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = r \frac{s_y}{s_x} \quad b_0 = \bar{y} - b_1 \bar{x}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

The regression line always passes through $(\bar{x}, \bar{y})$.

S.19 Least-Squares Linear Regression & R^2

0.1

OPTIMIZATION & SIMULATION

Anatomy of an Optimization Problem

STATE THE FORMULA/CONCEPT

0.2

OPTIMIZATION & SIMULATION

Linear Programming (LP)

STATE THE FORMULA/CONCEPT

0.3

OPTIMIZATION & SIMULATION

Simplex Method & LP Duality

STATE THE FORMULA/CONCEPT

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OPTIMIZATION & SIMULATION

Integer & Mixed-Integer Programming (MIP)

STATE THE FORMULA/CONCEPT

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OPTIMIZATION & SIMULATION

Knapsack Problem

STATE THE FORMULA/CONCEPT

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OPTIMIZATION & SIMULATION

Traveling Salesman Problem (TSP)

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OPTIMIZATION & SIMULATION

Shortest Paths & Network Flow

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OPTIMIZATION & SIMULATION

Convex Sets & Convex Functions

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OPTIMIZATION & SIMULATION

Nonlinear Programming & Newton's Method

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OPTIMIZATION & SIMULATION

Lagrange Multipliers

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OPTIMIZATION & SIMULATION

KKT Conditions

STATE THE FORMULA/CONCEPT

0.12

OPTIMIZATION & SIMULATION

Dynamic Programming & the Bellman Equation

STATE THE FORMULA/CONCEPT

Primal: $\max \mathbf{c}^\top \mathbf{x}, A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0$

Dual: $\min \mathbf{b}^\top \mathbf{y}, A^\top \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq 0$

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$$

optimal: $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$

Each primal constraint becomes a dual variable; max becomes min.

O.3 Simplex Method & LP Duality

$$\max_{\mathbf{x}} \mathbf{c}^\top \mathbf{x}$$

s.t. $A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0$

optimum at a vertex (corner point)

"Check the corners."

O.2 Linear Programming (LP)

$$\min_{\mathbf{x}} f(\mathbf{x})$$

s.t. $g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0$

LP / MIP / NLP

Three parts: decisions, objective, constraints.

O.1 Anatomy of an Optimization Problem

$$\min \sum_{i \neq j} c_{ij} x_{ij}$$

$$\sum_j x_{ij} = 1, \sum_i x_{ij} = 1, x_{ij} \in \{0, 1\}$$

+ subtour elimination

$$\frac{(n-1)!}{2} \text{ tours}$$

$(n-1)!/2$: fix the start, and a tour and its reverse are the same.

O.6 Travelling Salesman Problem (TSP)

$$\max \sum v_i x_i \text{ s.t. } \sum w_i x_i \leq W, x_i \in \{0, 1\}$$

$$V(i, w) = \max \{V(i-1, w), v_i + V(i-1, w-w_i)\}$$

$O(nW)$

Each item: "skip it or take it" — keep the better.

O.5 Knapsack Problem

$$\max \mathbf{c}^\top \mathbf{x}, A\mathbf{x} \leq \mathbf{b}$$

$x_j \in \mathbb{Z}$ (some or all), $x_j \in \{0, 1\}$

LP relaxation bounds the MIP

big-M : $x \leq My$

O.4 Integer & Mixed-Integer Programming (MIP)

$$\nabla f(\mathbf{x}^*) = \mathbf{0}, \nabla^2 f(\mathbf{x}^*) \succ 0$$

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\nabla^2 f]^{-1} \nabla f$$

O.9 Nonlinear Programming & Newton's Method

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

$$f''(x) \geq 0 \text{ or } \nabla^2 f \succeq 0$$

convex $\Rightarrow$ local min = global min

Convex looks like a bowl: any chord lies above the curve.

O.8 Convex Sets & Convex Functions

$$d(v) = \min_u [d(u) + w(u, v)]$$

Dijkstra: $w \geq 0$; Bellman-Ford: any w

max flow = min cut

A chain is only as strong as its weakest link: max flow = min cut.

O.7 Shortest Paths & Network Flow

$$V(s) = \max_a [r(s, a) + \gamma V(s')]$$

optimal substructure

+ overlapping subproblems

$$\sum_{t=0}^{\infty} \gamma^t r = \frac{r}{1-\gamma}$$

"Solve each subproblem once and remember it."

O.12 Dynamic Programming & the Bellman Equation

$$\min f \text{ s.t. } g_i \leq 0$$

$$\nabla f + \sum \mu_i \nabla g_i = \mathbf{0}$$

$$g_i \leq 0, \mu_i \geq 0, \mu_i g_i = 0$$

Complementary slackness: either the constraint is tight or its multiplier is zero.

O.11 KKT Conditions

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c)$$

$$\nabla f = \lambda \nabla g, g(\mathbf{x}) = c$$

$$\lambda = \frac{\partial f^*}{\partial c}$$

At the optimum the level curve kisses (is tangent to) the constraint.

O.10 Lagrange Multipliers

O.13

OPTIMIZATION & SIMULATION

Heuristics & Simulated Annealing

STATE THE FORMULA/CONCEPT

O.14

OPTIMIZATION & SIMULATION

Monte Carlo Estimation & Integration

STATE THE FORMULA/CONCEPT

O.15

OPTIMIZATION & SIMULATION

Random Sampling Methods

STATE THE FORMULA/CONCEPT

O.16

OPTIMIZATION & SIMULATION

Markov Chain Monte Carlo (Metropolis–Hastings)

STATE THE FORMULA/CONCEPT

O.17

OPTIMIZATION & SIMULATION

Algorithmic Complexity (Big- O)

STATE THE FORMULA/CONCEPT

ML.1

MACHINE LEARNING

The Learning Problem & Empirical Risk

STATE THE FORMULA/CONCEPT

ML.2

MACHINE LEARNING

Loss Functions

STATE THE FORMULA/CONCEPT

ML.3

MACHINE LEARNING

Entropy, Cross-Entropy & KL Divergence

STATE THE FORMULA/CONCEPT

ML.4

MACHINE LEARNING

Bias–Variance Trade-off

STATE THE FORMULA/CONCEPT

ML.5

MACHINE LEARNING

Regularization: Ridge, Lasso & Elastic Net

STATE THE FORMULA/CONCEPT

ML.6

MACHINE LEARNING

Cross-Validation

STATE THE FORMULA/CONCEPT

ML.7

MACHINE LEARNING

Logistic Regression

STATE THE FORMULA/CONCEPT

$$X = F^{-1}(U), U \sim \text{Unif}(0,1)$$

exponential: $X = -\frac{\ln(1-U)}{\lambda}$

rejection: accept if $U \leq \frac{f(x)}{Mg(x)}$

$$E_p[h] = E_q\left[h \frac{p}{q}\right]$$

O.15 Random Sampling Methods

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N f(X_i) \rightarrow E[f(X)]$$

$$\int_a^b f dx \approx \frac{b-a}{N} \sum f(U_i)$$

$$SE = \frac{\sigma}{\sqrt{N}}$$

Error shrinks like $1/\sqrt{N}$: 100 times the samples for one more digit.

O.14 Monte Carlo Estimation & Integration

$$P(\text{accept worse}) = e^{-\Delta/T}$$

$$T_k = T_0 \alpha^k$$

greedy, local search, 2-opt, GA

Hot $\rightarrow$ explore; cool $\rightarrow$ settle.

O.13 Heuristics & Simulated Annealing

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(y_i, f_{\theta}(\mathbf{x}_i))$$

$$+ \lambda R(\theta)$$

supervised: $(\mathbf{x}, y)$

unsupervised: $\mathbf{x}$ only

ML.1 The Learning Problem & Empirical Risk

$$f(n) = O(g(n)) \iff f \leq c g \text{ for large } n$$

$$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$$

Drop constants and lower-order terms: $3n^2 + 5n + 7 = O(n^2)$.

O.17 Algorithmic Complexity (Big-O)

$$\alpha = \min \left\{ 1, \frac{p(x') q(x | x')}{p(x) q(x' | x)} \right\}$$

symmetric q :

$$\alpha = \min \left\{ 1, \frac{p(x')}{p(x)} \right\}$$

Always accept uphill in probability; accept downhill sometimes, with probability equal to the ratio.

O.16 Markov Chain Monte Carlo (Metropolis-Hastings)

$$E[(y - \hat{f})^2] = \text{Bias}^2 + \text{Var} + \sigma^2$$

$$\text{Bias} = E[\hat{f}] - f$$

underfit $\leftrightarrow$ overfit

Too simple $\rightarrow$ bias (underfit); too complex $\rightarrow$ variance (overfit).

ML.4 Bias-Variance Trade-off

$$H(p) = -\sum_k p_k \log p_k$$

$$H(p, q) = -\sum_k p_k \log q_k$$

$$D_{\text{KL}}(p \parallel q) = \sum_k p_k \log \frac{p_k}{q_k} \geq 0$$

$$H(p, q) = H(p) + D_{\text{KL}}$$

KL is not symmetric: $D(p \parallel q) \neq D(q \parallel p)$.

ML.3 Entropy, Cross-Entropy & KL Divergence

$$\text{MSE} = \frac{1}{n} \sum (y - \hat{y})^2$$

$$\text{MAE} = \frac{1}{n} \sum |y - \hat{y}|$$

$$\text{log-loss} = -[y \ln p + (1-y) \ln(1-p)]$$

$$\text{hinge} = \max(0, 1 - y f(\mathbf{x}))$$

MSE $\rightarrow$ mean; MAE $\rightarrow$ median.

ML.2 Loss Functions

$$p = \sigma(\beta_0 + \beta^\top \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top \mathbf{x})}}$$

$$\ln \frac{p}{1-p} = \beta_0 + \beta^\top \mathbf{x}$$

odds ratio = e^{β_j}

Logistic regression is linear in the log-odds.

ML.7 Logistic Regression

$$CV_{(k)} = \frac{1}{k} \sum_{j=1}^k \text{Err}_j$$

each fold: train on $\frac{k-1}{k}n$, test on $\frac{n}{k}$

LOOCV: $k = n$

ML.6 Cross-Validation

Ridge: $\min \|\mathbf{y} - X\beta\|^2 + \lambda \|\beta\|_2^2$

$$\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$$

Lasso: $\min \|\mathbf{y} - X\beta\|^2 + \lambda \|\beta\|_1$

Lasso has a Lasso that ropes coefficients to exactly zero; ridge just shrinks.

ML.5 Regularization: Ridge, Lasso & Elastic Net

ML.8

MACHINE LEARNING

Classification Metrics

STATE THE FORMULA/CONCEPT

ML.9

MACHINE LEARNING

k -Nearest Neighbors & Distance Metrics

STATE THE FORMULA/CONCEPT

ML.10

MACHINE LEARNING

Naive Bayes Classifier

STATE THE FORMULA/CONCEPT

ML.11

MACHINE LEARNING

Decision Trees: Gini, Entropy & Information Gain

STATE THE FORMULA/CONCEPT

ML.12

MACHINE LEARNING

Ensembles: Bagging, Random Forests & Boosting

STATE THE FORMULA/CONCEPT

ML.13

MACHINE LEARNING

Support Vector Machines

STATE THE FORMULA/CONCEPT

ML.14

MACHINE LEARNING

Neural Networks & Backpropagation

STATE THE FORMULA/CONCEPT

ML.15

MACHINE LEARNING

Softmax & Cross-Entropy Loss

STATE THE FORMULA/CONCEPT

ML.16

MACHINE LEARNING

SGD, Momentum & Adam

STATE THE FORMULA/CONCEPT

ML.17

MACHINE LEARNING

k -Means Clustering

STATE THE FORMULA/CONCEPT

ML.18

MACHINE LEARNING

Hierarchical Clustering & Silhouette

STATE THE FORMULA/CONCEPT

ML.19

MACHINE LEARNING

Principal Component Analysis (PCA)

STATE THE FORMULA/CONCEPT

$$\hat{y} = \arg \max_y P(y) \prod_j P(x_j | y)$$

$$\text{Laplace: } \hat{P}(x_j | y) = \frac{n_{jy} + 1}{n_y + V}$$

ML.10 Naive Bayes Classifier

$$d_2 = \sqrt{\sum (x_j - z_j)^2} \quad d_1 = \sum |x_j - z_j|$$

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{z}}{\|\mathbf{x}\| \|\mathbf{z}\|}$$

$\hat{y}$ = majority / mean of k nearest

ML.9 k-Nearest Neighbors & Distance Metrics

$$\text{Prec} = \frac{TP}{TP + FP} \quad \text{Rec} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

$$\text{Acc} = \frac{TP + TN}{N} \quad \text{Spec} = \frac{TN}{TN + FP}$$

Precision: "of those I flagged, how many were right?" Recall: "of the real ones, how many did I catch?"

ML.8 Classification Metrics

$$\min \frac{1}{2} \|\mathbf{w}\|^2 + C \sum \xi_i$$

$$\text{s.t. } y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i$$

$$\text{margin} = \frac{2}{\|\mathbf{w}\|}$$

$$K(\mathbf{x}, \mathbf{z}) = e^{-\gamma \|\mathbf{x} - \mathbf{z}\|^2}$$

Maximize the street width; only the points on the curb (support vectors) matter.

ML.13 Support Vector Machines

$$\text{Var}(\bar{f}) = \rho \sigma^2 + \frac{1 - \rho}{B} \sigma^2$$

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x})$$

$$\alpha_m = \frac{1}{2} \ln \frac{1 - \epsilon_m}{\epsilon_m}$$

Bagging fights variance; boosting fights bias.

ML.12 Ensembles: Bagging, Random Forests & Boosting

$$\text{Gini} = 1 - \sum_k p_k^2$$

$$H = - \sum_k p_k \log_2 p_k$$

$$IG = H(\text{parent}) - \sum_c \frac{n_c}{n} H(c)$$

ML.11 Decision Trees: Gini, Entropy & Information Gain

$$\theta \leftarrow \theta - \eta \nabla L_{\text{batch}}$$

$$\mathbf{v} \leftarrow \beta \mathbf{v} + \nabla L, \theta \leftarrow \theta - \eta \mathbf{v}$$

$$\text{Adam: } \hat{\mathbf{m}} = \frac{\mathbf{m}}{1 - \beta_1^t}, \hat{\mathbf{v}} = \frac{\mathbf{v}}{1 - \beta_2^t}$$

$$\theta \leftarrow \theta - \eta \frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}} + \epsilon}}$$

Adam = momentum (m) + per-parameter step sizes (v).

ML.16 SGD, Momentum & Adam

$$\text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

$$L = - \sum_k y_k \ln \hat{p}_k$$

$$\frac{\partial L}{\partial \mathbf{z}} = \hat{\mathbf{p}} - \mathbf{y}$$

Gradient is "prediction minus truth."

ML.15 Softmax & Cross-Entropy Loss

$$\mathbf{a}^{(l)} = \phi(W^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$$

$$\text{ReLU}(z) = \max(0, z) \quad \sigma' = \sigma(1 - \sigma)$$

$$\delta^{(l)} = (W^{(l+1)\top} \delta^{(l+1)}) \odot \phi'(\mathbf{z}^{(l)})$$

Forward pass computes; backward pass blames (chain rule, layer by layer).

ML.14 Neural Networks & Backpropagation

$$S = \frac{1}{n-1} X_c^\top X_c$$

$$S \mathbf{v}_k = \lambda_k \mathbf{v}_k$$

$$\text{explained ratio} = \frac{\lambda_k}{\sum_j \lambda_j}$$

$$\mathbf{z} = V_k^\top (\mathbf{x} - \bar{\mathbf{x}})$$

PC1 points along the longest direction of the data cloud.

ML.19 Principal Component Analysis (PCA)

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

single: $\min d$

complete: $\max d$

average: $\text{mean } d$

ML.18 Hierarchical Clustering & Silhouette

$$J = \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \mu_k\|^2$$

$$\text{assign to nearest } \mu_k; \mu_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$$

Assign, update, repeat.

ML.17 k-Means Clustering

ML.20

MACHINE LEARNING

Gaussian Mixtures & the EM Algorithm

STATE THE FORMULA/CONCEPT

ML.21

MACHINE LEARNING

ROC, AUC & Precision–Recall Curves

STATE THE FORMULA/CONCEPT

ML.22

MACHINE LEARNING

Regression Metrics

STATE THE FORMULA/CONCEPT

ML.23

MACHINE LEARNING

Reading Metrics: Bad, Good & Perfect

STATE THE FORMULA/CONCEPT

EE.1

ELECTRICAL ENGINEERING

Charge, Current, Voltage, Power & Energy

STATE THE FORMULA/CONCEPT

EE.2

ELECTRICAL ENGINEERING

Kirchhoff's Laws

STATE THE FORMULA/CONCEPT

EE.3

ELECTRICAL ENGINEERING

Voltage & Current Dividers

STATE THE FORMULA/CONCEPT

EE.4

ELECTRICAL ENGINEERING

Thévenin & Norton Equivalents; Maximum Power Transfer

STATE THE FORMULA/CONCEPT

EE.5

ELECTRICAL ENGINEERING

Nodal & Mesh Analysis

STATE THE FORMULA/CONCEPT

EE.6

ELECTRICAL ENGINEERING

Capacitors & Inductors

STATE THE FORMULA/CONCEPT

EE.7

ELECTRICAL ENGINEERING

RC & RL Transients

STATE THE FORMULA/CONCEPT

EE.8

ELECTRICAL ENGINEERING

AC Signals & RMS Values

STATE THE FORMULA/CONCEPT

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2} \quad \text{MAE} = \frac{1}{n} \sum |y - \hat{y}|$$

$$\text{MAPE} = \frac{100\%}{n} \sum \left| \frac{y - \hat{y}}{y} \right|$$

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$$

RMSE punishes big misses; MAE treats all misses equally; both are in the units of y .

ML.22 Regression Metrics

ROC: TPR vs FPR over thresholds

$$\text{AUC} = P(s_+ > s_-)$$

PR curve: precision vs recall

PR baseline = prevalence

Lower the threshold $\Rightarrow$ flag more $\Rightarrow$ recall up, precision (usually) down. Raise it $\Rightarrow$ the reverse.

ML.21 ROC, AUC & Precision-Recall Curves

$$p(x) = \sum_k \pi_k N(x | \mu_k, \sigma_k^2)$$

$$E: \gamma_{ik} = \frac{\pi_k N(x_i | \mu_k, \sigma_k^2)}{\sum_j \pi_j N(x_i | \mu_j, \sigma_j^2)}$$

$$M: \mu_k = \frac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}, \quad \pi_k = \frac{N_k}{n}$$

E-step: guess memberships;
M-step: refit the Gaussians.

ML.20 Gaussian Mixtures & the EM Algorithm

$$\text{KCL: } \sum I_{\text{in}} = \sum I_{\text{out}}$$

$$\text{KVL: } \sum_{\text{loop}} V = 0$$

KCL: "what flows in must flow out" (conservation of charge). KVL: "every round trip ends where it started" (conservation of energy).

EE.2 Kirchhoff's Laws

$$I = \frac{dQ}{dt} \quad V = \frac{W}{Q}$$

$$P = VI \quad E = Pt$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Water analogy: voltage = pressure, current = flow rate, resistance = pipe narrowness, power = pressure $\times$ flow.

EE.1 Charge, Current, Voltage, Power & Energy

AUC: 0.5 random, 1 perfect

R^2 : 0 = mean baseline, 1 perfect

log-loss: $\ln 2 \approx 0.693$ coin flip, 0 perfect

always compare to a baseline

A metric means nothing without its baseline: always ask "compared to what?"

ML.23 Reading Metrics: Bad, Good & Perfect

KCL at each node $\rightarrow G\mathbf{v} = \mathbf{i}$
KVL around each mesh $\rightarrow R\mathbf{i} = \mathbf{v}$

EE.5 Nodal & Mesh Analysis

$$V_{\text{th}} = V_{\text{open}} \quad R_{\text{th}} = \frac{V_{\text{oc}}}{I_{\text{sc}}}$$

$$I_N = \frac{V_{\text{th}}}{R_{\text{th}}}$$

$$R_L = R_{\text{th}} \Rightarrow P_{\text{max}} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$$

Any linear circuit, seen from two terminals, is "one battery plus one resistor."

EE.4 Thévenin & Norton Equivalents; Maximum Power Transfer

$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

$$I_1 = I \frac{R_2}{R_1 + R_2}$$

Voltage divider: "my resistor on top";
current divider: "the other resistor on top."

EE.3 Voltage & Current Dividers

$$v(t) = V_m \sin(\omega t + \phi)$$

$$\omega = 2\pi f \quad T = \frac{1}{f}$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

RMS = Root of the Mean of the Square;
for sine waves, peak $\approx 1.414 \times$ RMS.

EE.8 AC Signals & RMS Values

$$v_C(t) = V_s(1 - e^{-t/\tau}), \quad \tau = RC$$

$$i_L(t) = \frac{V_s}{R}(1 - e^{-t/\tau}), \quad \tau = \frac{L}{R}$$

$5\tau \approx$ fully settled

After 1τ : 63% of the way; after 5τ : over 99% ("five tau and done").

EE.7 RC & RL Transients

$$Q = CV \quad i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$E_C = \frac{1}{2} CV^2 \quad E_L = \frac{1}{2} LI^2$$

$$C_{\text{par}} = C_1 + C_2 \quad \frac{1}{C_{\text{ser}}} = \frac{1}{C_1} + \frac{1}{C_2}$$

Capacitors combine like resistors "upside down."
A capacitor resists sudden voltage changes;
an inductor resists sudden current changes.

EE.6 Capacitors & Inductors

EE.9

ELECTRICAL ENGINEERING

Phasors & Impedance

STATE THE FORMULA/CONCEPT

EE.10

ELECTRICAL ENGINEERING

AC Power & Power Factor

STATE THE FORMULA/CONCEPT

EE.11

ELECTRICAL ENGINEERING

RLC Circuits & Resonance

STATE THE FORMULA/CONCEPT

EE.12

ELECTRICAL ENGINEERING

Filters, Frequency Response & Decibels

STATE THE FORMULA/CONCEPT

EE.13

ELECTRICAL ENGINEERING

Operational Amplifiers

STATE THE FORMULA/CONCEPT

EE.14

ELECTRICAL ENGINEERING

Diodes, LEDs & Transistors

STATE THE FORMULA/CONCEPT

EE.15

ELECTRICAL ENGINEERING

Transformers & Three-Phase Power

STATE THE FORMULA/CONCEPT

EE.16

ELECTRICAL ENGINEERING

Electric & Magnetic Fields

STATE THE FORMULA/CONCEPT

EE.17

ELECTRICAL ENGINEERING

Digital Logic, Binary & Boolean Algebra

STATE THE FORMULA/CONCEPT

EE.18

ELECTRICAL ENGINEERING

Sampling, Nyquist & Quantization

STATE THE FORMULA/CONCEPT

EE.19

ELECTRICAL ENGINEERING

Decibels, SNR & Channel Capacity

STATE THE FORMULA/CONCEPT

EE.20

ELECTRICAL ENGINEERING

Fourier Series, the DFT & FFT

STATE THE FORMULA/CONCEPT

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$Q_{\text{series}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$\text{BW} = \frac{\omega_0}{Q}$$

At resonance the inductor and capacitor reactances cancel: $\omega L = \frac{1}{\omega C}$.

EE.11 RLC Circuits & Resonance

$$P = V_{\text{rms}} I_{\text{rms}} \cos \theta$$

$$Q = V_{\text{rms}} I_{\text{rms}} \sin \theta$$

$$|S|^2 = P^2 + Q^2 \quad \text{pf} = \cos \theta$$

The power triangle: P is the “beer” (useful), Q the “foam,” $|S|$ the whole glass.

EE.10 AC Power & Power Factor

$$Z_R = R \quad Z_L = j\omega L \quad Z_C = \frac{1}{j\omega C}$$

$$\tilde{V} = \tilde{I}Z \quad |Z| = \sqrt{R^2 + X^2}$$

$$\theta = \tan^{-1} \frac{X}{R}$$

ELI the ICE man: in an inductor (L), voltage (E) leads current (I); in a capacitor (C), current (I) leads voltage (E).

EE.9 Phasors & Impedance

$$I = I_s (e^{V/(nV_T)} - 1)$$

$$V_T = \frac{kT}{q} \approx 25.9 \text{ mV}$$

$$R_{\text{LED}} = \frac{V_s - V_f}{I}$$

Silicon diode “turns on” near 0.7 V; the current grows about $10\times$ for every 60 mV more (with $n = 1$).

EE.14 Diodes, LEDs & Transistors

ideal: $V_+ = V_-$, $I_{\text{in}} = 0$

inverting: $A = -\frac{R_f}{R_{\text{in}}}$

non-inverting: $A = 1 + \frac{R_f}{R_g}$

The two golden rules: “the inputs are at the same voltage” and “no current enters the inputs.”

EE.13 Operational Amplifiers

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$f_c = \frac{1}{2\pi RC}$$

$$|H|_{\text{dB}} = 20 \log_{10} |H|$$

−3 dB at f_c ; −20 dB/decade

At the cutoff, $|H| = \frac{1}{\sqrt{2}}$, i.e. −3 dB (half power). Past it, −20 dB per decade for each RC stage.

EE.12 Filters, Frequency Response & Decibels

$$N = \sum b_i 2^i \quad n \text{ bits} \rightarrow 2^n \text{ values}$$

$$\overline{A \cdot B} = \bar{A} \cdot \bar{B}$$

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

De Morgan: “break the bar, change the operator” (same as for sets [A1.16](#)).

EE.17 Digital Logic, Binary & Boolean Algebra

$$F = k \frac{q_1 q_2}{r^2}, \quad k = 8.99 \times 10^9$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$\mathcal{E} = -N \frac{d\Phi}{dt}$$

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Faraday: “change the flux, get a voltage”; the minus sign (Lenz) means the induced current opposes the change.

EE.16 Electric & Magnetic Fields

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

$$P_{3\phi} = \sqrt{3} V_L I_L \text{ pf}$$

losses = $I^2 R$

“Step up voltage, step down current”: power in = power out (ideal).

EE.15 Transformers & Three-Phase Power

$$x(t) = \sum_n c_n e^{jn\omega_0 t}$$

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}$$

$$\Delta f = \frac{f_s}{N}$$

FFT: $O(N \log N)$

Every signal is a sum of spinning arrows (sinusoids) [T.13](#); the FFT finds how much of each.

EE.20 Fourier Series, the DFT & FFT

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = 20 \log_{10} \frac{V_2}{V_1}$$

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

$$C = B \log_2(1 + \text{SNR})$$

+3 dB $\approx 2\times$ power, +10 dB = $10\times$ power, +20 dB = $100\times$ power ($10\times$ voltage).

EE.19 Decibels, SNR & Channel Capacity

$$f_s > 2f_{\text{max}}$$

$$\text{resolution} = \frac{V_{\text{ref}}}{2^N}$$

$$\text{SQNR} \approx 6.02N + 1.76 \text{ dB}$$

“Sample at least twice as fast as the fastest wiggle”; each extra bit buys about 6 dB.

EE.18 Sampling, Nyquist & Quantization

EE.21

ELECTRICAL ENGINEERING

Batteries & Energy Storage

STATE THE FORMULA/CONCEPT

EE.22

ELECTRICAL ENGINEERING

Schematic Symbols: Passive Components & Switches

STATE THE FORMULA/CONCEPT

SY.1

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.2

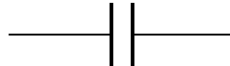
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.3

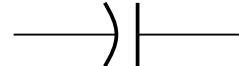
EE SYMBOLS



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SY.4

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.5

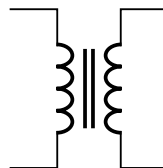
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SY.6

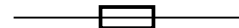
EE SYMBOLS



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SY.7

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.8

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.9

EE SYMBOLS



NAME IT / STATE ITS FORMULA

EE.23

ELECTRICAL ENGINEERING

Schematic Symbols: Sources, Meters & Ground

STATE THE FORMULA/CONCEPT

Resistor

Limits current
 $V = IR$, unit Ω

SY.1

resistor: $V = IR$

capacitor: $i = C \frac{dv}{dt}$

inductor: $v = L \frac{di}{dt}$

transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

Zigzag = resistor (friction); two parallel plates = capacitor (a gap that stores charge); coils = inductor (a wound wire).

EE.22 Schematic Symbols: Passive Components & Switches

$E(\text{Wh}) = V \times \text{Ah}$

$t = \frac{\text{capacity (Ah)}}{I}$

$\text{C-rate} = \frac{I}{\text{capacity}}$

"mAh is charge, Wh is energy": multiply by voltage to compare batteries fairly.

EE.21 Batteries & Energy Storage

Polarized Capacitor

Electrolytic capacitor
curved plate = negative; observe polarity

SY.4

Capacitor

Stores charge / electric energy

$Q = CV$, $i = C \frac{dv}{dt}$

SY.3

Potentiometer

Adjustable resistor with a wiper
(adjustable voltage divider)

SY.2

Fuse

Melts to open the circuit
when current exceeds its rating

SY.7

Transformer

Two coupled coils on a core

$\frac{V_s}{V_p} = \frac{N_s}{N_p}$

SY.6

Inductor

Stores magnetic energy

$v = L \frac{di}{dt}$, $E = \frac{1}{2} LI^2$

SY.5

ideal voltage source: fixed V
ideal current source: fixed I
voltmeter: parallel; ammeter: series
ground = 0 V

"Volts across, amps through": voltmeter across the part, ammeter in the path.

EE.23 Schematic Symbols: Sources, Meters & Ground

Lamp

Incandescent lamp / bulb

a resistive load: $P = \frac{V^2}{R}$

SY.9

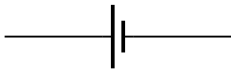
Switch (SPST)

Single-pole single-throw switch
open = no current, closed = short

SY.8

SY.10

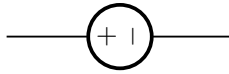
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.11

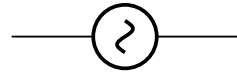
EE SYMBOLS



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SY.12

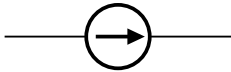
EE SYMBOLS



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SY.13

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.14

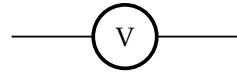
EE SYMBOLS



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SY.15

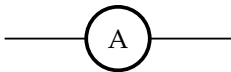
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.16

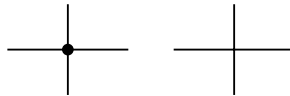
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.17

EE SYMBOLS



NAME IT / STATE ITS FORMULA

EE.24

ELECTRICAL ENGINEERING

Schematic Symbols: Semiconductors & Op-Amps

STATE THE FORMULA/CONCEPT

SY.18

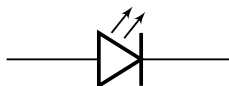
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.19

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.20

EE SYMBOLS



NAME IT / STATE ITS FORMULA

AC Voltage Source

Sinusoidal source
 $v(t) = V_m \sin(\omega t + \phi)$

SY.12

DC Voltage Source

Ideal independent voltage source
 $V = \text{constant}$

SY.11

Battery

DC source (cell)
long plate = +, short plate = -

SY.10

Voltmeter

Measures voltage
connect in parallel; $R_{\text{in}} \rightarrow \infty$

SY.15

Ground

Reference node = 0 V
(earth or common return)

SY.14

Current Source

Ideal independent current source
 $I = \text{constant}$, arrow gives direction

SY.13

Junction vs. Crossing

diode: current flows in the arrow direction

$$I = I_s(e^{V/nV_T} - 1)$$

op-amp: $V_{\text{out}} = A(V_+ - V_-)$

*The diode's arrow points the way
conventional current flows (anode
→ cathode; the bar is the cathode).*

EE.24 Schematic Symbols: Semiconductors & Op-Amps

Dot = wires connected (a node)
no dot = wires cross, not connected

SY.17

Ammeter

Measures current
connect in series; $R_{\text{in}} \rightarrow 0$

SY.16

Zener Diode

Conducts in reverse at V_Z
used as a voltage reference / regulator

SY.20

LED

Light-emitting diode
needs a series resistor: $R = \frac{V_s - V_f}{I}$

SY.19

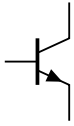
Diode

One-way valve for current
anode → cathode (bar); ≈ 0.7 V drop

SY.18

SY.21

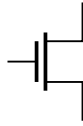
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.22

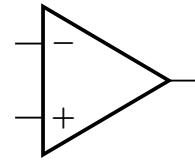
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.23

EE SYMBOLS



NAME IT / STATE ITS FORMULA

EE.25

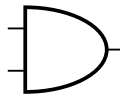
ELECTRICAL ENGINEERING

Schematic Symbols: Logic Gates

STATE THE FORMULA/CONCEPT

SY.24

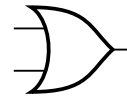
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.25

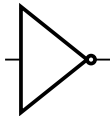
EE SYMBOLS



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SY.26

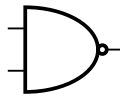
EE SYMBOLS



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SY.27

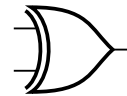
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.28

EE SYMBOLS



NAME IT / STATE ITS FORMULA

FN.1

FINANCE

Simple vs. Compound Interest

STATE THE FORMULA/CONCEPT

FN.2

FINANCE

APR vs. Effective Annual Rate

STATE THE FORMULA/CONCEPT

FN.3

FINANCE

Doubling Time & the Rule of 72

STATE THE FORMULA/CONCEPT

Op-Amp

Operational amplifier

$$V_{\text{out}} = A(V_+ - V_-); \text{ ideal: } V_+ = V_-$$

SY.23

MOSFET (n-channel)

Voltage-controlled switch / amplifier
gate, drain, source; basis of CPUs and GPUs

SY.22

NPN Transistor (BJT)

Bipolar junction transistor

$$I_C \approx \beta I_B; \text{ emitter arrow points out}$$

SY.21

OR Gate

$$Y = A + B$$

1 if any input is 1

SY.25

AND Gate

$$Y = A \cdot B$$

1 only if all inputs are 1

SY.24

$$\text{AND: } A \cdot B \quad \text{OR: } A + B \quad \text{NOT: } \bar{A}$$

$$\text{NAND: } \overline{A \cdot B} \quad \text{XOR: } A \oplus B$$

A small circle ("bubble") on an output means NOT. AND has a flat back, OR has a curved back, XOR adds an extra curved line.

EE.25 Schematic Symbols: Logic Gates

XOR Gate

$$Y = A \oplus B$$

1 if the inputs differ

SY.28

NAND Gate

$$Y = \overline{A \cdot B}$$

universal: builds every other gate

SY.27

NOT Gate (Inverter)

$$Y = \bar{A}$$

flips the input

SY.26

$$t_{\text{double}} = \frac{\ln 2}{\ln(1+r)}$$
$$t \approx \frac{72}{r(\%)}$$

Rule of 72 (use 69.3 for continuous compounding, since $\ln 2 \approx 0.693$).

FN.3 Doubling Time & the Rule of 72

$$\text{EAR} = \left(1 + \frac{\text{APR}}{n}\right)^n - 1$$

continuous: $\text{EAR} = e^r - 1$

*APR is the "sticker" rate; EAR is what you actually pay or earn.
 $\text{EAR} \geq \text{APR}$ whenever $n > 1$.*

FN.2 APR vs. Effective Annual Rate

$$I = Prt \text{ (simple)}$$

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$A = Pe^{rt} \text{ (continuous)}$$

Compounding is "interest on interest"; "Pert" for continuous.

FN.1 Simple vs. Compound Interest

FN.4

FINANCE

Present Value & Discounting

STATE THE FORMULA/CONCEPT

FN.5

FINANCE

Net Present Value & Internal Rate of Return

STATE THE FORMULA/CONCEPT

FN.6

FINANCE

Annuities & Perpetuities

STATE THE FORMULA/CONCEPT

FN.7

FINANCE

Loan Amortization & Mortgages

STATE THE FORMULA/CONCEPT

FN.8

FINANCE

Inflation & Real Returns

STATE THE FORMULA/CONCEPT

FN.9

FINANCE

Returns: Simple, Log & CAGR

STATE THE FORMULA/CONCEPT

FN.10

FINANCE

Volatility & Risk

STATE THE FORMULA/CONCEPT

FN.11

FINANCE

Portfolio Return, Variance & Diversification

STATE THE FORMULA/CONCEPT

FN.12

FINANCE

Sharpe Ratio & Risk-Adjusted Return

STATE THE FORMULA/CONCEPT

FN.13

FINANCE

CAPM & Beta

STATE THE FORMULA/CONCEPT

FN.14

FINANCE

Bond Pricing & Yield

STATE THE FORMULA/CONCEPT

FN.15

FINANCE

Duration & Convexity

STATE THE FORMULA/CONCEPT

$$PV_{\text{ann}} = C \frac{1 - (1+r)^{-n}}{r}$$

$$FV_{\text{ann}} = C \frac{(1+r)^n - 1}{r}$$

$$PV_{\text{perp}} = \frac{C}{r} \quad PV_{\text{growing}} = \frac{C_1}{r-g}$$

A perpetuity is an annuity with $n \rightarrow \infty$: the $(1+r)^{-n}$ term vanishes.

FN.6 Annuities & Perpetuities

$$NPV = \sum_{t=0}^T \frac{CF_t}{(1+r)^t}$$

$$\text{IRR: } \sum \frac{CF_t}{(1+\text{IRR})^t} = 0$$

accept if $NPV > 0$ ($\text{IRR} > r$)

NPV answers "how much value?"; IRR answers "what rate?" When they disagree, trust NPV.

FN.5 Net Present Value & Internal Rate of Return

$$PV = \frac{FV}{(1+r)^t}$$

$$FV = PV(1+r)^t$$

$$\text{discount factor} = \frac{1}{(1+r)^t}$$

"A dollar today is worth more than a dollar tomorrow."

FN.4 Present Value & Discounting

$$R = \frac{P_1 - P_0}{P_0} \quad r = \ln \frac{P_1}{P_0}$$

$$\text{CAGR} = \left(\frac{V_T}{V_0} \right)^{1/T} - 1$$

geometric $\leq$ arithmetic mean

Log returns add over time; simple returns add across assets. +50% then -50% loses 25%.

FN.9 Returns: Simple, Log & CAGR

$$(1+i_{\text{nom}}) = (1+r_{\text{real}})(1+\pi)$$

$$r_{\text{real}} \approx i_{\text{nom}} - \pi$$

$$\text{purchasing power} = \frac{1}{(1+\pi)^t}$$

"Real $\approx$ nominal minus inflation."

FN.8 Inflation & Real Returns

$$M = P \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$i = \frac{\text{APR}}{12}, \quad n = 12 \times \text{years}$$

Early payments are mostly interest; late payments are mostly principal.

FN.7 Loan Amortization & Mortgages

$$S = \frac{R_p - R_f}{\sigma_p}$$

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_{\text{down}}}$$

$$S_{\text{annual}} = S_{\text{daily}} \sqrt{252}$$

Sharpe = "excess return per unit of risk." Rough guide: < 0.5 poor, ≈ 1 good, > 2 excellent (and suspicious).

FN.12 Sharpe Ratio & Risk-Adjusted Return

$$E[R_p] = \sum w_i E[R_i]$$

$$\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

"Diversification is the only free lunch": with $\rho < 1$, portfolio risk is less than the weighted average risk.

FN.11 Portfolio Return, Variance & Diversification

$$\sigma = \sqrt{\frac{1}{n-1} \sum (r_t - \bar{r})^2}$$

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \sqrt{252}$$

$$\sigma_T = \sigma \sqrt{T}$$

"Risk grows with the square root of time; expected return grows with time."

FN.10 Volatility & Risk

$$D_{\text{Mac}} = \frac{1}{P} \sum t \frac{CF_t}{(1+y)^t}$$

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y}$$

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} Cx (\Delta y)^2$$

Duration $\approx$ "the percent price drop for a 1-point rise in rates."

FN.15 Duration & Convexity

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}$$

$$y \uparrow \Rightarrow P \downarrow$$

$$\text{zero: } P = \frac{F}{(1+y)^n}$$

Bond prices and yields move in opposite directions ("see-saw"). Coupon $>$ yield $\Rightarrow$ premium; coupon $<$ yield $\Rightarrow$ discount.

FN.14 Bond Pricing & Yield

$$E[R_i] = R_f + \beta_i (E[R_m] - R_f)$$

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

$\beta > 1$: amplifies the market; $\beta < 1$: dampens it; β is just a regression slope.

FN.13 CAPM & Beta

FN.16

FINANCE

Stock Valuation & Multiples

STATE THE FORMULA/CONCEPT

FN.17

FINANCE

Options: Payoffs & Put-Call Parity

STATE THE FORMULA/CONCEPT

FN.18

FINANCE

Black-Scholes Option Pricing

STATE THE FORMULA/CONCEPT

FN.19

FINANCE

Value at Risk (VaR) & Expected Shortfall

STATE THE FORMULA/CONCEPT

FN.20

FINANCE

Financial Ratios & DuPont Analysis

STATE THE FORMULA/CONCEPT

FN.21

FINANCE

Expected Value of Bets & the Kelly Criterion

STATE THE FORMULA/CONCEPT

FN.22

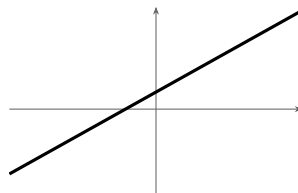
FINANCE

Geometric Brownian Motion

STATE THE FORMULA/CONCEPT

F.1

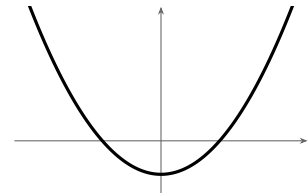
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F.2

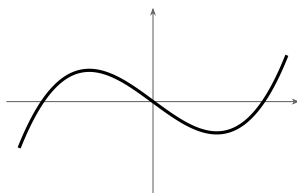
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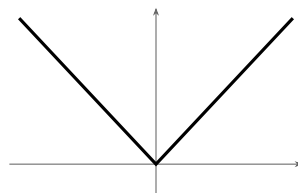
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F.4

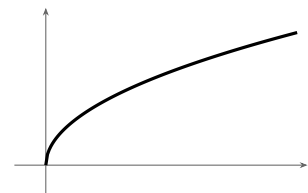
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F.5

PROBABILITY DISTRIBUTIONS



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$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$N(d_2) \approx$ risk-neutral probability the call finishes in the money; $N(d_1)$ is the hedge ratio (delta).

FN.18 Black-Scholes Option Pricing

$$\text{call} = \max(S_T - K, 0)$$

$$\text{put} = \max(K - S_T, 0)$$

$$C - P = S_0 - Ke^{-rT}$$

"Call up, put down": a call profits when price goes up, a put when it goes down. The call payoff is a ReLU [ML.14]!

FN.17 Options: Payoffs & Put-Call Parity

$$P_0 = \frac{D_1}{r - g} \text{ (Gordon)}$$

$$P/E = \frac{\text{price}}{\text{EPS}}$$

$$\text{EV} = \text{market cap} + \text{debt} - \text{cash}$$

Price is the present value of all future cash; growth near the discount rate makes values explode.

FN.16 Stock Valuation & Multiples

$$\text{EV} = pW - qL$$

$$f^* = \frac{bp - q}{b}, q = 1 - p$$

Never bet on negative EV; even with an edge, bet a fraction (many use "half Kelly").

FN.21 Expected Value of Bets & the Kelly Criterion

$$\text{ROE} = \frac{\text{NI}}{\text{equity}} = \frac{\text{NI}}{\text{sales}} \cdot \frac{\text{sales}}{\text{assets}} \cdot \frac{\text{assets}}{\text{equity}}$$

$$\text{current ratio} = \frac{\text{CA}}{\text{CL}}$$

$$D/E = \frac{\text{debt}}{\text{equity}}$$

Accounting identity: assets = liabilities + equity.

FN.20 Financial Ratios & DuPont Analysis

$$\text{VaR}_\alpha = z_\alpha \sigma \sqrt{t} V$$

$$\text{ES} = E[\text{loss} \mid \text{loss} > \text{VaR}]$$

$$z_{0.95} = 1.645, z_{0.99} = 2.326$$

VaR: "the loss you won't exceed α of the time"; ES: "how bad it gets when you do."

FN.19 Value at Risk (VaR) & Expected Shortfall

Quadratic

$$y = ax^2 + bx + c$$

parabola; vertex at $x = -\frac{b}{2a}$

F.2

Linear

$$y = mx + b$$

slope m , intercept b

F.1

$$dS = \mu S dt + \sigma S dW$$

$$S_T = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T \right]$$

$$E[S_T] = S_0 e^{\mu T}$$

Mean > median: a few huge winners pull the average up (log-normal skew).

FN.22 Geometric Brownian Motion

Square Root

$$y = \sqrt{x}$$

domain $x \geq 0$

F.5

Absolute Value

$$y = |x|$$

V shape, vertex at (0, 0)

F.4

Cubic

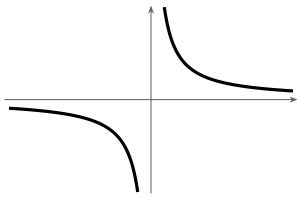
$$y = ax^3 + bx^2 + cx + d$$

odd end behavior; up to 2 turns

F.3

F.6

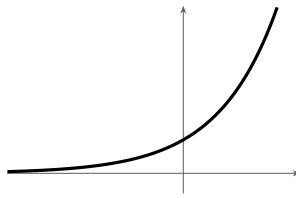
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F.7

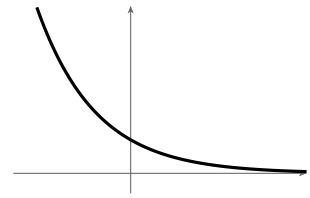
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F.8

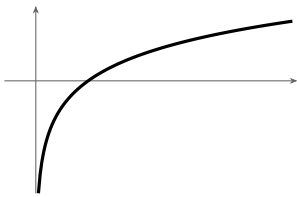
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F.9

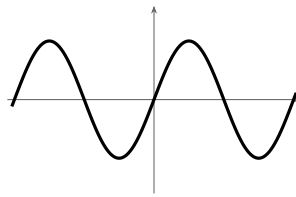
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F.10

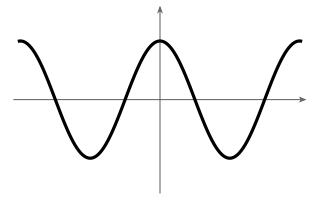
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F.11

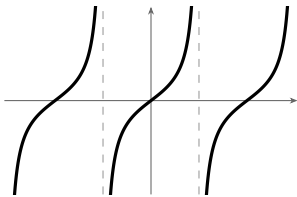
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F.12

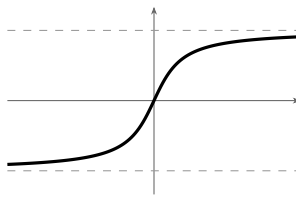
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F.13

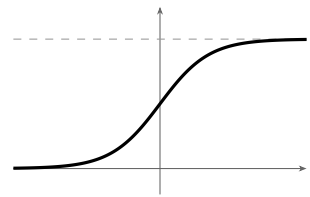
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F.14

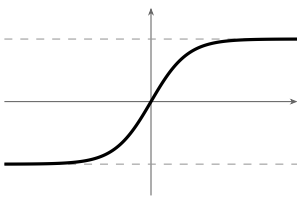
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F.15

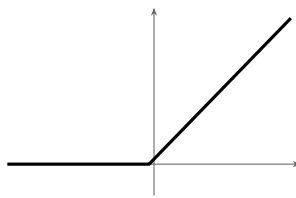
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F.16

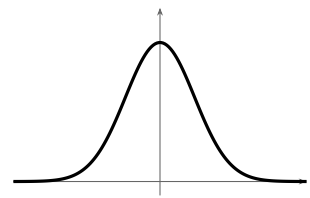
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F.17

PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

Exponential Decay

$$y = e^{-x} (b^x, 0 < b < 1)$$

half-life $\frac{\ln 2}{k}$

F:8

Exponential Growth

$$y = e^x (b^x, b > 1)$$

y-intercept 1; HA $y = 0$

F:7

Reciprocal

$$y = \frac{1}{x}$$

asymptotes $x = 0, y = 0$

F:6

Cosine

$$y = \cos x$$

period 2π ; $\cos 0 = 1$

F:11

Sine

$$y = \sin x$$

period 2π ; $\sin 0 = 0$

F:10

Natural Logarithm

$$y = \ln x$$

domain $x > 0$; VA $x = 0$; $\ln 1 = 0$

F:9

Logistic Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

range $(0, 1)$; $\sigma(0) = \frac{1}{2}$

F:14

Arctangent

$$y = \arctan x$$

HA $y = \pm \frac{\pi}{2}$

F:13

Tangent

$$y = \tan x = \frac{\sin x}{\cos x}$$

period π ; VA at $x = \pm \frac{\pi}{2}$

F:12

Gaussian (Bell Curve)

$$y = e^{-x^2}$$

even; maximum 1 at $x = 0$

F:17

ReLU

$$\text{ReLU}(x) = \max(0, x)$$

kink at 0

F:16

Hyperbolic Tangent

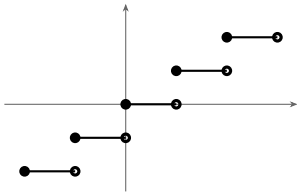
$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

range $(-1, 1)$; odd

F:15

F.18

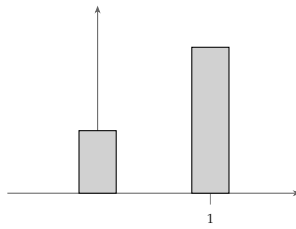
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D.1

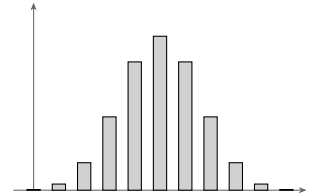
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D.2

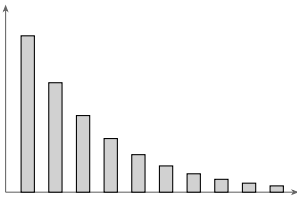
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D.3

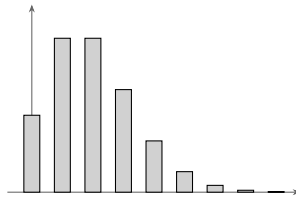
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D.4

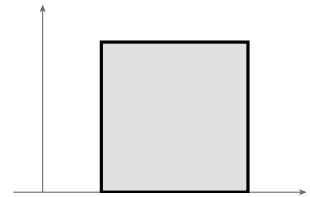
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D.5

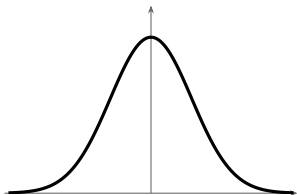
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D.6

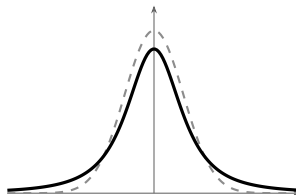
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D.7

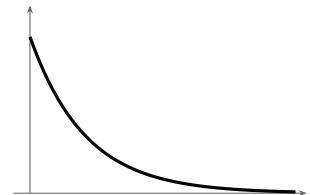
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D.8

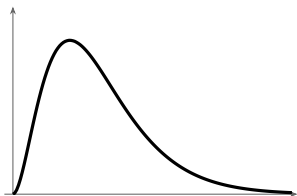
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D.9

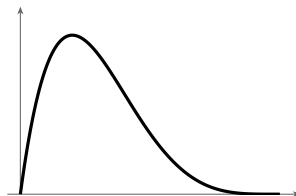
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D.10

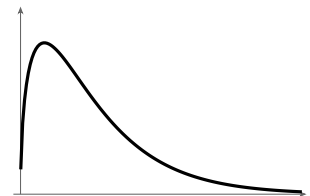
PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

D.11

PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

Binomial

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$
$$\mu = np, \sigma^2 = np(1-p)$$

D.2

Bernoulli

$$P(X = 1) = p, P(X = 0) = 1 - p$$
$$\mu = p, \sigma^2 = p(1-p)$$

D.1

Floor (Step) Function

$$y = \lfloor x \rfloor$$

greatest integer $\leq x$

F.18

Uniform (Continuous)

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$
$$\mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12}$$

D.5

Poisson

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$
$$\mu = \sigma^2 = \lambda$$

D.4

Geometric

$$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \dots$$

trials to first success; $\mu = \frac{1}{p}, \sigma^2 = \frac{1-p}{p^2}$

D.3

Exponential

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$
$$\mu = \frac{1}{\lambda}; \text{ memoryless}$$

D.8

Student's t (vs. normal)

$$f(t) \propto \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

heavier tails; $t \rightarrow N(0,1)$ as $\nu \rightarrow \infty$

D.7

Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

symmetric bell; 68-95-99.7

D.6

Chi-Square

$$\chi_k^2 = \sum_{i=1}^k Z_i^2$$
$$\mu = k, \sigma^2 = 2k; \text{ right-skewed}$$

D.11

Beta

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1$$
$$\mu = \frac{\alpha}{\alpha + \beta}$$

D.10

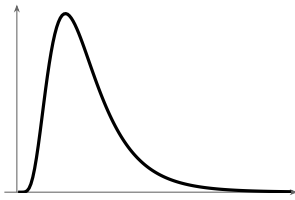
Gamma

$$f(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k) \theta^k}$$
$$\mu = k\theta, \sigma^2 = k\theta^2$$

D.9

D.12

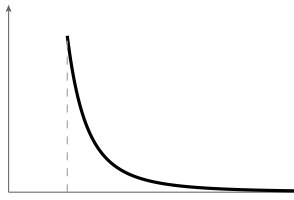
PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

D.13

PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

GR.1

GREEK ALPHABET

A α

NAME THE LETTER

GR.2

GREEK ALPHABET

B β

NAME THE LETTER

GR.3

GREEK ALPHABET

Γ γ

NAME THE LETTER

GR.4

GREEK ALPHABET

Δ δ

NAME THE LETTER

GR.5

GREEK ALPHABET

E €

NAME THE LETTER

GR.6

GREEK ALPHABET

Z ζ

NAME THE LETTER

GR.7

GREEK ALPHABET

H η

NAME THE LETTER

GR.8

GREEK ALPHABET

Θ θ

NAME THE LETTER

GR.9

GREEK ALPHABET

I ι

NAME THE LETTER

GR.10

GREEK ALPHABET

K κ

NAME THE LETTER

Alpha

A (capital), α (lowercase)

α : significance level, learning rate, angle, Beta/Gamma parameter.

GR.1 Greek Alphabet

Pareto (Power Law)

$$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m$$

heavy tail; mean finite iff $\alpha > 1$

D.13

Log-Normal

$$\ln X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

D.12

Delta

Δ (capital), δ (lowercase)

Δ : change, discriminant, Laplacian;
 δ : small change, error signal.

GR.4 Greek Alphabet

Gamma

Γ (capital), γ (lowercase)

Γ : gamma function; γ : discount factor, RBF kernel width.

GR.3 Greek Alphabet

Beta

B (capital), β (lowercase)

β : regression coefficient, Type II error, momentum, thermal expansion.

GR.2 Greek Alphabet

Eta

H (capital), η (lowercase)

η : learning rate, efficiency.

GR.7 Greek Alphabet

Zeta

Z (capital), ζ (lowercase)

ζ : Riemann zeta function, damping ratio.

GR.6 Greek Alphabet

Epsilon

E (capital), ϵ (lowercase)

ϵ, ε : small quantity, error term, emissivity.

GR.5 Greek Alphabet

Kappa

K (capital), κ (lowercase)

κ : condition number, Cohen's kappa, curvature.

GR.10 Greek Alphabet

Iota

I (capital), ι (lowercase)

ι : rarely used; "not one iota" means not the smallest amount.

GR.9 Greek Alphabet

Theta

Θ (capital), θ (lowercase)

θ : angle, model parameters; Θ : parameter space, tight bound.

GR.8 Greek Alphabet

GR.11

GREEK ALPHABET

Λ λ

NAME THE LETTER

GR.12

GREEK ALPHABET

Μ μ

NAME THE LETTER

GR.13

GREEK ALPHABET

Ν ν

NAME THE LETTER

GR.14

GREEK ALPHABET

Ξ ξ

NAME THE LETTER

GR.15

GREEK ALPHABET

Ο ο

NAME THE LETTER

GR.16

GREEK ALPHABET

Π π

NAME THE LETTER

GR.17

GREEK ALPHABET

Ρ ρ

NAME THE LETTER

GR.18

GREEK ALPHABET

Σ σ

NAME THE LETTER

GR.19

GREEK ALPHABET

Τ τ

NAME THE LETTER

GR.20

GREEK ALPHABET

Υ υ

NAME THE LETTER

GR.21

GREEK ALPHABET

Φ φ

NAME THE LETTER

GR.22

GREEK ALPHABET

Χ χ

NAME THE LETTER

Nu

N (capital), ν (lowercase)

ν : degrees of freedom,
kinematic viscosity, frequency.

GR.13 Greek Alphabet

Mu

M (capital), μ (lowercase)

μ : mean, viscosity,
friction coefficient, micro-

GR.12 Greek Alphabet

Lambda

Λ (capital), λ (lowercase)

λ : eigenvalue, rate (Poisson,
exponential), regularization
strength, wavelength.

GR.11 Greek Alphabet

Pi

Π (capital), π (lowercase)

$\pi \approx 3.14159$; π_k mixing
weights; Π : product.

GR.16 Greek Alphabet

Omicron

O (capital), o (lowercase)

Rarely used (looks like o); little- o
notation for “grows slower than”.

GR.15 Greek Alphabet

Xi

Ξ (capital), ξ (lowercase)

ξ : slack variable (SVM),
random variable.

GR.14 Greek Alphabet

Tau

T (capital), τ (lowercase)

τ : time constant,
threshold, shear stress, 2π .

GR.19 Greek Alphabet

Sigma

Σ (capital), σ (lowercase)

Σ : summation, covariance
matrix; σ : standard deviation,
sigmoid, singular value.

GR.18 Greek Alphabet

Rho

P (capital), ρ (lowercase)

ρ : correlation, density, spherical radius.

GR.17 Greek Alphabet

Chi

X (capital), χ (lowercase)

χ^2 : chi-square distribution and test.

GR.22 Greek Alphabet

Phi

Φ (capital), ϕ (lowercase)

Φ : standard normal CDF; ϕ , φ : angle,
activation, potential, golden ratio.

GR.21 Greek Alphabet

Upsilon

Y (capital), υ (lowercase)

υ : rarely used; sometimes velocity.

GR.20 Greek Alphabet

GR.23

GREEK ALPHABET

Ψ ψ

NAME THE LETTER

GR.24

GREEK ALPHABET

Ω ω

NAME THE LETTER

AL.1

ALGORITHMS

Linear Regression

DEFINE IT

AL.2

ALGORITHMS

Logistic Regression

DEFINE IT

AL.3

ALGORITHMS

Ridge / Lasso

DEFINE IT

AL.4

ALGORITHMS

k -Nearest Neighbors

DEFINE IT

AL.5

ALGORITHMS

Naive Bayes

DEFINE IT

AL.6

ALGORITHMS

Decision Tree

DEFINE IT

AL.7

ALGORITHMS

Random Forest

DEFINE IT

AL.8

ALGORITHMS

**Gradient Boosting
(XGBoost, LightGBM)**

DEFINE IT

AL.9

ALGORITHMS

Support Vector Machine

DEFINE IT

AL.10

ALGORITHMS

Neural Network (MLP)

DEFINE IT

Supervised, regression. Fits $\hat{y} = X\beta$ by least squares. Used for: pricing, forecasting, effect sizes.

AL.1 Algorithms

Omega

Ω (capital), ω (lowercase)
 Ω : ohm, sample space, lower bound;
 ω : angular frequency, outcome.

GR.24 Greek Alphabet

Psi

Ψ (capital), ψ (lowercase)
 ψ : wave function (physics),
digamma function.

GR.23 Greek Alphabet

Supervised, classification or regression.
Vote of the k closest points.
Used for: recommendations,
simple baselines, imputation.

AL.4 Algorithms

Supervised, regularized regression. L2
/ L1 penalties. Used for: many
correlated features, feature selection.

AL.3 Algorithms

Supervised, classification. Sigmoid
of a linear score. Used for:
churn, credit risk, conversion.

AL.2 Algorithms

Supervised. Averages many
decorrelated trees (bagging).
Used for: robust tabular
baselines, feature importance.

AL.7 Algorithms

Supervised. Splits data by
if-then rules. Used for:
interpretable rules, triage.

AL.6 Algorithms

Supervised, classification. Bayes' rule
with independent features. Used
for: spam filtering, text tagging.

AL.5 Algorithms

Supervised. Layers of linear maps
and nonlinearities. Used for:
flexible function fitting, embeddings.

AL.10 Algorithms

Supervised, classification.
Maximum-margin boundary,
kernels. Used for: small
high-dimensional data, text.

AL.9 Algorithms

Supervised. Adds trees that
fix previous errors. Used
for: top accuracy on tabular
data (fraud, pricing, ranking).

AL.8 Algorithms

AL.11

ALGORITHMS

CNN

DEFINE IT

AL.12

ALGORITHMS

RNN / LSTM

DEFINE IT

AL.13

ALGORITHMS

Transformer

DEFINE IT

AL.14

ALGORITHMS

k-Means

DEFINE IT

AL.15

ALGORITHMS

DBSCAN

DEFINE IT

AL.16

ALGORITHMS

Hierarchical Clustering

DEFINE IT

AL.17

ALGORITHMS

Gaussian Mixture Model

DEFINE IT

AL.18

ALGORITHMS

PCA

DEFINE IT

AL.19

ALGORITHMS

t-SNE / UMAP

DEFINE IT

AL.20

ALGORITHMS

Isolation Forest

DEFINE IT

AL.21

ALGORITHMS

Autoencoder

DEFINE IT

AL.22

ALGORITHMS

Apriori / Association Rules

DEFINE IT

Self-supervised / supervised. Attention over all tokens. Used for: LLMs, translation, vision transformers.

AL.13 Algorithms

Supervised (deep learning). Recurrent state over a sequence. Used for: time series, speech (largely replaced by transformers).

AL.12 Algorithms

Supervised (deep learning). Convolutional filters share weights across space. Used for: images, medical scans, audio spectrograms.

AL.11 Algorithms

Unsupervised. Merges clusters into a dendrogram. Used for: taxonomies, gene expression.

AL.16 Algorithms

Unsupervised, clustering. Groups dense regions and labels sparse points as noise; no K needed. Used for: geospatial hot spots, anomaly detection, irregular-shaped clusters.

AL.15 Algorithms

Unsupervised, clustering. Minimizes squared distance to K centroids. Used for: customer segmentation, quantization.

AL.14 Algorithms

Unsupervised, nonlinear embedding. Preserves local neighborhoods in 2-D. Used for: visualizing embeddings and single-cell data (not for distances).

AL.19 Algorithms

Unsupervised, dimension reduction. Projects onto directions of maximum variance. Used for: compression, decorrelation, visualization.

AL.18 Algorithms

Unsupervised. Soft clustering with EM. Used for: density estimation, soft segments.

AL.17 Algorithms

Unsupervised. Finds frequent itemsets with support, confidence, lift. Used for: market-basket analysis.

AL.22 Algorithms

Unsupervised (self-supervised). Network compresses and reconstructs input. Used for: denoising, anomaly detection by reconstruction error.

AL.21 Algorithms

Unsupervised, anomaly detection. Anomalies are isolated in few random splits. Used for: fraud, sensor faults, intrusion detection.

AL.20 Algorithms

AL.23

ALGORITHMS

Matrix Factorization

DEFINE IT

AL.24

ALGORITHMS

LDA (Topic Model)

DEFINE IT

AL.25

ALGORITHMS

ARIMA

DEFINE IT

AL.26

ALGORITHMS

Q-Learning

DEFINE IT

AL.27

ALGORITHMS

Spectral Clustering

DEFINE IT

AL.28

ALGORITHMS

GAN

DEFINE IT

AC.1

ACRONYMS

CNN

DEFINE IT

AC.2

ACRONYMS

RNN

DEFINE IT

AC.3

ACRONYMS

LSTM

DEFINE IT

AC.4

ACRONYMS

GRU

DEFINE IT

AC.5

ACRONYMS

GAN

DEFINE IT

AC.6

ACRONYMS

VAE

DEFINE IT

Supervised time series.
AutoRegressive Integrated
Moving Average. Used for:
demand and sales forecasting.

AL.25 Algorithms

Unsupervised. Latent Dirichlet
Allocation: documents as
mixtures of topics. Used
for: topic discovery in text.

AL.24 Algorithms

Unsupervised / collaborative filtering.
Low-rank user $\times$ item factors.
Used for: recommender systems.

AL.23 Algorithms

Unsupervised generative.
Generator vs. discriminator
game. Used for: synthetic
images and data augmentation.

AL.28 Algorithms

Unsupervised. Clusters using
eigenvectors of a graph
Laplacian. Used for: non-convex
clusters, community detection.

AL.27 Algorithms

Reinforcement learning. Learns
action values via the Bellman
equation. Used for: games, control,
recommendations over time.

AL.26 Algorithms

Long Short-Term Memory: an
RNN with gates that preserve
long-range information.

AC.3 Acronyms

Recurrent Neural Network:
processes sequences with a
hidden state carried forward.

AC.2 Acronyms

Convolutional Neural Network:
learns local filters (convolutions)
shared across an image or signal.

AC.1 Acronyms

Variational Autoencoder:
probabilistic autoencoder that
learns a latent distribution.

AC.6 Acronyms

Generative Adversarial Network: a
generator and a discriminator
trained against each other.

AC.5 Acronyms

Gated Recurrent Unit:
a simpler gated RNN.

AC.4 Acronyms

AC.7

ACRONYMS

LLM

DEFINE IT

AC.8

ACRONYMS

NLP

DEFINE IT

AC.9

ACRONYMS

RAG

DEFINE IT

AC.10

ACRONYMS

MLP

DEFINE IT

AC.11

ACRONYMS

SGD

DEFINE IT

AC.12

ACRONYMS

ReLU

DEFINE IT

AC.13

ACRONYMS

ROC / AUC

DEFINE IT

AC.14

ACRONYMS

MSE / RMSE / MAE

DEFINE IT

AC.15

ACRONYMS

PCA

DEFINE IT

AC.16

ACRONYMS

SVD

DEFINE IT

AC.17

ACRONYMS

SVM

DEFINE IT

AC.18

ACRONYMS

k -NN

DEFINE IT

Retrieval-Augmented Generation:
an LLM answers using
documents retrieved by search.

AC.9 Acronyms

Natural Language Processing:
computing with human language.

AC.8 Acronyms

Large Language Model:
a very large transformer
trained to predict text.

AC.7 Acronyms

Rectified Linear Unit:
activation $\max(0, x)$.

AC.12 Acronyms

Stochastic Gradient
Descent: gradient steps
on random mini-batches.

AC.11 Acronyms

Multilayer Perceptron:
a fully connected
feed-forward neural network.

AC.10 Acronyms

Principal Component Analysis.

AC.15 Acronyms

Mean Squared Error /
Root Mean Squared Error
/ Mean Absolute Error.

AC.14 Acronyms

Receiver Operating Characteristic
curve / Area Under the Curve.

AC.13 Acronyms

k -Nearest Neighbors.

AC.18 Acronyms

Support Vector Machine.

AC.17 Acronyms

Singular Value
Decomposition: $A = U\Sigma V^T$.

AC.16 Acronyms

AC.19

ACRONYMS

EDA

DEFINE IT

AC.20

ACRONYMS

ETL

DEFINE IT

AC.21

ACRONYMS

TF-IDF

DEFINE IT

AC.22

ACRONYMS

BERT

DEFINE IT

AC.23

ACRONYMS

GPT

DEFINE IT

AC.24

ACRONYMS

OLS

DEFINE IT

AC.25

ACRONYMS

MLE / MAP

DEFINE IT

AC.26

ACRONYMS

MCMC

DEFINE IT

AC.27

ACRONYMS

KL

DEFINE IT

AC.28

ACRONYMS

i.i.d.

DEFINE IT

AC.29

ACRONYMS

PDF / PMF / CDF

DEFINE IT

AC.30

ACRONYMS

IQR

DEFINE IT

Term Frequency–Inverse Document Frequency: weights words that are frequent in a document but rare overall.

AC.21 Acronyms

Extract, Transform, Load: moving and reshaping data into a warehouse.

AC.20 Acronyms

Exploratory Data Analysis: summarizing and plotting data before modeling.

AC.19 Acronyms

Ordinary Least Squares.

AC.24 Acronyms

Generative Pre-trained Transformer: a decoder language model.

AC.23 Acronyms

Bidirectional Encoder Representations from Transformers: an encoder language model.

AC.22 Acronyms

Kullback–Leibler divergence.

AC.27 Acronyms

Markov Chain Monte Carlo.

AC.26 Acronyms

Maximum Likelihood Estimate / Maximum A Posteriori estimate (likelihood $\times$ prior).

AC.25 Acronyms

Interquartile Range, $Q_3 - Q_1$.

AC.30 Acronyms

Probability Density Function / Probability Mass Function / Cumulative Distribution Function.

AC.29 Acronyms

Independent and Identically Distributed.

AC.28 Acronyms

AC.31

ACRONYMS

CI

DEFINE IT

AC.32

ACRONYMS

RL

DEFINE IT

AC.33

ACRONYMS

LP / MIP / NLP

DEFINE IT

AC.34

ACRONYMS

ODE / PDE

DEFINE IT

AC.35

ACRONYMS

GPU

DEFINE IT

AC.36

ACRONYMS

MLOps

DEFINE IT

KT.1

KEY TERMS

Domain & Range

DEFINE IT

KT.2

KEY TERMS

Monotonic

DEFINE IT

KT.3

KEY TERMS

One-to-One (Injective)

DEFINE IT

KT.4

KEY TERMS

Even & Odd Function

DEFINE IT

KT.5

KEY TERMS

Asymptote

DEFINE IT

KT.6

KEY TERMS

Identity vs. Equation

DEFINE IT

Linear / Mixed-Integer / Nonlinear Programming (in optimization, NLP means nonlinear programming).

AC.33 Acronyms

Reinforcement Learning: learning actions from rewards.

AC.32 Acronyms

Confidence Interval (also Continuous Integration in software).

AC.31 Acronyms

Machine Learning Operations: deploying, monitoring, and maintaining models.

AC.36 Acronyms

Graphics Processing Unit: parallel hardware used to train neural networks.

AC.35 Acronyms

Ordinary / Partial Differential Equation.

AC.34 Acronyms

Different inputs give different outputs; needed for an inverse.

KT.3 Key Terms

Always increasing or always decreasing.

KT.2 Key Terms

Domain: all allowed inputs of a function. Range: all outputs it produces.

KT.1 Key Terms

An identity is true for all values; an equation is true only for its solutions.

KT.6 Key Terms

A line a graph approaches but does not reach (vertical, horizontal, or slant).

KT.5 Key Terms

Even: $f(-x) = f(x)$ (symmetric about the y -axis). Odd: $f(-x) = -f(x)$ (symmetric about the origin).

KT.4 Key Terms

KT.7

KEY TERMS

Theorem, Lemma, Corollary

DEFINE IT

KT.8

KEY TERMS

Necessary vs. Sufficient

DEFINE IT

KT.9

KEY TERMS

Population vs. Sample

DEFINE IT

KT.10

KEY TERMS

Parameter vs. Statistic

DEFINE IT

KT.11

KEY TERMS

Bias (Statistical)

DEFINE IT

KT.12

KEY TERMS

Confounder

DEFINE IT

KT.13

KEY TERMS

Stationarity

DEFINE IT

KT.14

KEY TERMS

Heteroscedasticity

DEFINE IT

KT.15

KEY TERMS

Multicollinearity

DEFINE IT

KT.16

KEY TERMS

Feature & Label

DEFINE IT

KT.17

KEY TERMS

Training, Validation, Test Sets

DEFINE IT

KT.18

KEY TERMS

Overfitting & Underfitting

DEFINE IT

Population: every unit of interest.
Sample: the subset you observe.

KT.9 Key Terms

A is necessary for B if
 $B \Rightarrow A$; sufficient if $A \Rightarrow B$.

KT.8 Key Terms

Theorem: major proven result.
Lemma: helper result. Corollary:
follows directly from a theorem.

KT.7 Key Terms

A variable that affects both
the cause and the outcome,
creating a spurious association.

KT.12 Key Terms

Systematic difference between an
estimator's average and the truth.

KT.11 Key Terms

Parameter describes a population
(μ); statistic describes a sample ($\bar{x}$).

KT.10 Key Terms

Predictors that are highly correlated,
making coefficients unstable.

KT.15 Key Terms

Residual variance that
changes with the predictors.

KT.14 Key Terms

A time series whose mean,
variance and autocorrelation
do not change over time.

KT.13 Key Terms

Overfitting: memorizes
noise (great train, poor test).
Underfitting: too simple for both.

KT.18 Key Terms

Fit on training; tune on
validation; report once
on the untouched test set.

KT.17 Key Terms

Feature: an input variable. Label
(target): the value to predict.

KT.16 Key Terms

KT.19

KEY TERMS

Hyperparameter

DEFINE IT

KT.20

KEY TERMS

Epoch & Batch

DEFINE IT

KT.21

KEY TERMS

Data Leakage

DEFINE IT

KT.22

KEY TERMS

One-Hot Encoding

DEFINE IT

KT.23

KEY TERMS

Normalization vs. Standardization

DEFINE IT

KT.24

KEY TERMS

Imputation

DEFINE IT

KT.25

KEY TERMS

Embedding

DEFINE IT

KT.26

KEY TERMS

Tensor

DEFINE IT

KT.27

KEY TERMS

Loss vs. Metric

DEFINE IT

KT.28

KEY TERMS

Class Imbalance

DEFINE IT

KT.29

KEY TERMS

Calibration

DEFINE IT

KT.30

KEY TERMS

Ensemble

DEFINE IT

Information from outside the training data (e.g. the future or the test set) sneaks into training.

KT.21 Key Terms

Epoch: one full pass over the training data. Batch: the subset used for one gradient step.

KT.20 Key Terms

A setting chosen before training (learning rate, depth, k), not learned from data.

KT.19 Key Terms

Filling in missing values (mean, median, model-based).

KT.24 Key Terms

Normalization rescales to $[0, 1]$; standardization gives mean 0, SD 1.

KT.23 Key Terms

Representing a category as a vector with a single 1.

KT.22 Key Terms

Loss: what training minimizes. Metric: what you report to judge usefulness.

KT.27 Key Terms

A multi-dimensional array (scalar, vector, matrix, and higher).

KT.26 Key Terms

A dense vector representing an item (word, user, image) so that similar items are close.

KT.25 Key Terms

Combining several models to improve accuracy or stability.

KT.30 Key Terms

Predicted probabilities match observed frequencies.

KT.29 Key Terms

One class is much rarer; use PR curves, resampling, or class weights.

KT.28 Key Terms

KT.31

KEY TERMS

Transfer Learning & Fine-Tuning

DEFINE IT

KT.32

KEY TERMS

Attention

DEFINE IT

KT.33

KEY TERMS

Supervised vs. Unsupervised

DEFINE IT

KT.34

KEY TERMS

Self-Supervised Learning

DEFINE IT

KT.35

KEY TERMS

Latent Variable

DEFINE IT

KT.36

KEY TERMS

Convex vs. Non-Convex

DEFINE IT

ET.1

EE TERMS

AC vs. DC

DEFINE IT

ET.2

EE TERMS

Ground

DEFINE IT

ET.3

EE TERMS

Short Circuit

DEFINE IT

ET.4

EE TERMS

Open Circuit

DEFINE IT

ET.5

EE TERMS

Series vs. Parallel

DEFINE IT

ET.6

EE TERMS

Impedance

DEFINE IT

Supervised learns from labeled examples; unsupervised finds structure without labels.

KT.33 Key Terms

Weighting inputs by relevance:
 $\text{softmax}(QK^T / \sqrt{d})V$.

KT.32 Key Terms

Reusing a pre-trained model and adapting it to a new task.

KT.31 Key Terms

Convex problems have one global minimum; non-convex ones can have many local minima.

KT.36 Key Terms

A hidden quantity inferred from data, such as a cluster or topic.

KT.35 Key Terms

Labels are created from the data itself (e.g. predict the next word).

KT.34 Key Terms

A near-zero-resistance path; current can become dangerously large.

ET.3 EE Terms

The reference node defined as 0 V; also a safety connection to earth.

ET.2 EE Terms

Alternating current reverses direction periodically (grid power); direct current flows one way (batteries, electronics).

ET.1 EE Terms

AC generalization of resistance,
 $Z = R + jX$, in ohms.

ET.6 EE Terms

Series: same current through each element. Parallel: same voltage across each element.

ET.5 EE Terms

A break in the path; no current flows.

ET.4 EE Terms



ET.7

EE TERMS

Reactance

DEFINE IT



ET.8

EE TERMS

Conductance

DEFINE IT



ET.9

EE TERMS

Bandwidth

DEFINE IT



ET.10

EE TERMS

Gain

DEFINE IT



ET.11

EE TERMS

Feedback

DEFINE IT



ET.12

EE TERMS

ADC / DAC

DEFINE IT



ET.13

EE TERMS

PWM

DEFINE IT



ET.14

EE TERMS

Semiconductor

DEFINE IT



ET.15

EE TERMS

MOSFET

DEFINE IT



ET.16

EE TERMS

Breaker & Fuse

DEFINE IT



ET.17

EE TERMS

Duty Cycle

DEFINE IT



ET.18

EE TERMS

Harmonics

DEFINE IT

The range of frequencies a system passes (Hz); in data links, the achievable data rate.

ET.9 EE Terms

Reciprocal of resistance,
 $G = 1/R$, in siemens.

ET.8 EE Terms

The imaginary part of impedance from inductors (+) and capacitors (-).

ET.7 EE Terms

Analog-to-Digital /
Digital-to-Analog Converter.

ET.12 EE Terms

Feeding part of the output back to the input; negative feedback stabilizes, positive feedback oscillates.

ET.11 EE Terms

Output-to-input ratio of an amplifier, often in dB.

ET.10 EE Terms

Metal-Oxide-Semiconductor Field-Effect Transistor: the voltage-controlled switch in all modern chips.

ET.15 EE Terms

A material (e.g. silicon) whose conductivity can be controlled by doping; basis of diodes and transistors.

ET.14 EE Terms

Pulse-Width Modulation: controlling average power by switching on and off at a varying duty cycle.

ET.13 EE Terms

Frequency components at integer multiples of the fundamental frequency.

ET.18 EE Terms

Fraction of each period that a signal is on.

ET.17 EE Terms

Devices that interrupt current when it exceeds a safe level.

ET.16 EE Terms

ET.19

EE TERMS

Microcontroller

DEFINE IT

ET.20

EE TERMS

Kilowatt-hour

DEFINE IT

FT.1

FINANCE TERMS

Time Value of Money

DEFINE IT

FT.2

FINANCE TERMS

Liquidity

DEFINE IT

FT.3

FINANCE TERMS

Arbitrage

DEFINE IT

FT.4

FINANCE TERMS

Hedge

DEFINE IT

FT.5

FINANCE TERMS

Leverage

DEFINE IT

FT.6

FINANCE TERMS

Bid–Ask Spread

DEFINE IT

FT.7

FINANCE TERMS

Market Capitalization

DEFINE IT

FT.8

FINANCE TERMS

Dividend

DEFINE IT

FT.9

FINANCE TERMS

EBITDA

DEFINE IT

FT.10

FINANCE TERMS

Free Cash Flow

DEFINE IT

Money now is worth more than the same amount later, because it can earn a return.

FT.1 Finance Terms

Energy of 1 kW for 1 hour (3.6 MJ); the unit on electricity bills.

ET.20 EE Terms

A small computer on one chip with CPU, memory and I/O, used in embedded devices.

ET.19 EE Terms

A position taken to reduce the risk of another position.

FT.4 Finance Terms

A riskless profit from price differences; its absence underlies pricing models.

FT.3 Finance Terms

How quickly an asset can be sold near its fair price.

FT.2 Finance Terms

Share price times shares outstanding.

FT.7 Finance Terms

Difference between the highest buy and lowest sell price; a trading cost.

FT.6 Finance Terms

Using borrowed money to magnify returns (and losses).

FT.5 Finance Terms

Operating cash flow minus capital expenditures.

FT.10 Finance Terms

Earnings Before Interest, Taxes, Depreciation and Amortization; a proxy for operating cash profit.

FT.9 Finance Terms

A cash payment from a company's profits to shareholders.

FT.8 Finance Terms

FT.11

FINANCE TERMS

Balance Sheet

DEFINE IT

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Income Statement

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Yield Curve

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Basis Point

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Bull vs. Bear Market

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Index Fund / ETF

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Short Selling

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Drawdown

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Implied Volatility

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Credit Score / Default

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Efficient Market Hypothesis

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Backtesting

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Plot of bond yields vs. maturity; an inverted curve often precedes recessions.

FT.13 Finance Terms

Revenue, expenses and profit over a period.

FT.12 Finance Terms

Snapshot of assets, liabilities and equity at a point in time.

FT.11 Finance Terms

A fund that tracks a market index; ETFs trade like stocks.

FT.16 Finance Terms

Rising market (bull) vs. falling market, typically –20% or more (bear).

FT.15 Finance Terms

One hundredth of a percent (0.01%).

FT.14 Finance Terms

The volatility that makes an option-pricing model match the market price.

FT.19 Finance Terms

Decline from a peak to a trough in portfolio value.

FT.18 Finance Terms

Selling borrowed shares to profit if the price falls.

FT.17 Finance Terms

Testing a strategy on historical data; prone to overfitting and look-ahead bias.

FT.22 Finance Terms

Prices already reflect available information, so consistent excess returns are hard to earn.

FT.21 Finance Terms

Credit score: estimate of a borrower's reliability. Default: failure to make required payments.

FT.20 Finance Terms